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ORIGINAL SCIENTIFIC PAPER

Physical Performance in Highest-Level Soccer: Comparison between FIFA World Cup and UEFA Champions League Matches

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Abstract

Match running performance (RP) of Union of European Football Associations Champions League (UCL) and Fédération Internationale De Football Association World Cup (WC) players are often investigated separately; however, no study to simultaneously observed RP in both competitions. Therefore, the main objective of this study was to provide a comparative analysis of RP of professional soccer match-play across two highest-level soccer competitions: UCL and WC. The RPs (UCL; n=244, WC; n=581) of professional soccer players were analyzed using optical systems, and classified according to their playing positions as: central defenders (CD), fullbacks (FB), central midfielders (CM), wide midfielders (WM) and forward (FW). The RP variables included total distance and high intensity running (>5.5 m/s). One-way analysis of variance was used to examine position-specific differences between RP in WC and UCL, while Cohen's d was used to identify effect sizes. Results indicated that CDs, FBs, CMs, and WMs from UCL covered significantly greater total distance than their counterparts from WC (all moderate effect sizes). In addition, CDs from UCL achieved greater amount of high intensity running than CDs from WC (small effect size). This study demonstrated differences in various aspects of physical performance between two highest-level soccer competitions with variations dependent upon playing position. The findings enabled a more thorough understanding of the physical requirements for playing in UCL and WC, and may have direct consequences for specific training programs.

Keywords: match analysis, football, running performance, comparative analysis, playing positions

Introduction

The application and use of technology in soccer has grown exponentially in recent years, allowing practitioners and researchers to evaluate the physical demands of competitive match play (Doncaster, Page, White, Svenson, & Twist, 2020). The most common technique to evaluate physical demands in soccer is analysis of players' running performance (RP) such as total distance and high intensity running (Pranjic, Modric, & Uljevic, 2022). Detailed knowledge on these metrics is useful when determining performance profiles and designing sport-specific training programs which is vital to optimize soccer performance (Carling, 2011). For example, total dis-

tance and high intensity running distance covered by players in a match can be used to prescribe more specific training or to consider new ways to improve the efficiency of team training (Modric, Versic, Morgans, & Sekulic, 2023).

Studies have demonstrated that players can cover between 9 and 14 km during the matches, performing 5–15% of that distance in high intensity running (Modric, Versic, & Sekulic, 2021; De Albuquerque Freire et al., 2022). This wide ranges are determined by different variables that are associated with running performance (Jerkovic, Modric, & Versic, 2022), but primarily with playing positions due to the players' specific roles during the match (Di Salvo et al., 2007). Thus, midfielders are



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responsible for the connection between defense and attack, and such role require them to achieve greater total distance compared to the players on other playing positions (Modric, Versic, Sekulic, & Liposek, 2019). On the other hand, side players (e.g., fullbacks and wingers) cover the greatest distances in high intensity running due to their frequent involvement in attacking activities (Bradley et al., 2009). Accordingly, the position-specific load needs to be taken into consideration when designing and implementing training programs (Baptista, Johansen, Figueiredo, Rebelo, & Pettersen, 2020).

Soccer is no doubt one of the most popular sports in the world, played in more than 200 countries (Milanović, Pantelić, Čović, Sporiš, & Krustup, 2015). The highest-level soccer competitions in soccer are Union of European Football Associations Champions League (UCL) and Fédération Internationale De Football Association World Cup (WC) (Tuo, Wang, Huang, Zhang, & Liu, 2019; Modric et al., 2023). Analyzing the RP of players' competing in such competitions may help to constitute activity profiles which can be used by coaches and players for achieving the most elite game-play. Therefore, it is not surprising that in recent years, researchers have put forth great efforts to provide knowledge on the RP of UCL and WC players and factors influencing RP (Bradley, Dellal, Mohr, Castellano, & Wilkie, 2014; Miñano-Espin, Casáis, Lago-Peñas, & Gómez-Ruano, 2017; Tuo et al., 2019; Yi et al., 2019; Modric et al., 2022).

In brief, Minano-Espin et al. analyzed the high intensity running patterns of UCL players according to the quality of the opposition (Miñano-Espin et al., 2017). Modric et al. recently analyzed factors affecting RP of UCL players (Modric et al., 2022), while Bradley et al.'s focus was on the gender differences of UCL players (Bradley et al., 2014). Regarding the RP of WC players, Tuo et al. quantified the differences during the matches from different continental confederations (Tuo et al., 2019), while Yi et al. investigated the effect of playing styles on the RP of WC (Yi et al., 2019).

Evidently, RP of UCL and WC players are often investigated separately. However, to the best of our knowledge, there is no study that simultaneously observed RP in both competitions. A direct comparison of RP between competitions would be useful in determining whether physical performance is general to all highest-level competitions or if each competition has its own specific playing characteristics (Dellal et al., 2011). The findings may help soccer practitioners to tailor position-specific training programs that are suitable for each respective competition (Aquino et al., 2017). Therefore, the main objective of this study was to provide a comparative analysis of RP of professional soccer match-play across two highest-level soccer competitions: UCL and WC.

Methods

Design and sample

The RPs of players competing in UCL were analysed in 20 randomly selected matches played in group stage. The RPs of players competing in WC were analysed in all matches played both in group stage and knock-out stage ($n=64$). Due to the methodological reasons, matches that included a player dismissal (red card) or extra times were excluded from analysis. This resulted in excluding 9 matches from WC, while no matches from UCL have been excluded from initial sample. Therefore, 55 matches from WC and 20 from UCL were included for final analysis. As suggested previously, only the RPs

of those players who participated in entire matches were analyzed (Lorenzo-Martinez et al., 2021), resulting in retrieval of 244 RPs from UCL and 581 RPs from WC which were included as cases in this study. The RPs were classified automatically into five positional subsets by the data provider based on the players' role in the team as: central defenders (CD; $n=300$), fullbacks (FB; $n=189$), central midfielders (CM; $n=195$), wide midfielders (WM; $n=87$) and forward (FW; $n=54$). Players' identities were anonymized in accordance with the principles of the Declaration of Helsinki to ensure confidentiality. The investigation was approved by the Faculty of Kinesiology, University of Split (approval number: 2181-205-02-05-19-0020, 1 September 2019).

Procedures

The RP in UCL matches were collected using an optical system InStat Fitness (Instat Limited, Limerick, Republic of Ireland) with a sampling frequency of 25 Hz. The reliability of this system has been demonstrated, as the system has passed the official Fédération Internationale de Football Association (FIFA) test protocol for Electronic & Performance Tracking Systems (EPTS). The RP from WC matches were provided by official FIFA Football Data Ecosystem (publicly available on FIFA webpage: <https://www.fifatrainingcentre.com/en/fwc2022/post-match-summaries/post-match-summary-reports.php>), a complex network consisting of several optical systems. All optical systems included in this network operate at 25Hz and have the same threshold values for high intensity running (i.e., variable included in this study) as InStat Fitness does. Moreover, all of them have passed the same test protocol for EPTS as InStat Fitness does. This confirms quality of data from WC matches and justify usage in current study. The RPs variables included total distance covered (m) and high intensity running (m) (>5.5 m/s).

Statistical Analyses

The Kolmogorov-Smirnov test revealed that all data were normally distributed (all K-S $p>0.05$). Homogeneity was evaluated using Levene's test, and data are presented as means $\pm$ standard deviations. One-way analysis of variance (ANOVA) was used to examine position-specific differences between RP in WC and UCL. Cohen's d was used to identify effect sizes (ES) and interpreted as follows: trivial (<0.2), small ($\geq 0.2-0.5$), moderate ($\geq 0.5-0.8$) and large (>0.8) (Cohen, 2013). For all analyses, SPSS software Version 25 (IBM, Armonk, NY, USA) was used, and $p<0.05$ was considered statistically significant.

Results

Table 1 and 2 present descriptive statistics and differences for total distance and high intensity running during the WC and UCL matches according to the soccer-specific playing positions. Results indicated that CDs, FBs, CMs, and WMs from UCL (10201 m, 10903 m, 11880 m, and 11129 m, respectively) covered significantly greater total distance ($f=22.07$, 14.52, 11.03, and 6.36, respectively; all $p=0.01$, all moderate ES) than their counterparts from WC (9762 m, 10482 m, 11434 m, and 10547 m). In addition, CDs from UCL achieved greater amount of high intensity running than CDs from WC (638 m and 585 m, respectively; $f=4.58$, $p=0.03$, small ES). No differences between FWs' total distance covered ($f=0.05$, $p>0.05$) and high intensity running ($f=0.40$, $p>0.05$) in UCL and WC were found.

Table 1. Descriptive statistics for total distance and high intensity running during the World Cup and UEFA Champions League matches according to the soccer-specific playing positions.

		World Cup	UEFA Champions League
		Mean±SD	Mean±SD
Central defenders	TD (m)	9762±715	10201±704
	HIR (m)	585±181	638±207
Fullbacks	TD (m)	10482±722	10903±722
	HIR (m)	1005±250	1017±230
Central midfielders	TD (m)	11434±917	11880±615
	HIR (m)	843±267	914±263
Wide midfielders	TD (m)	10547±1116	11129±714
	HIR (m)	1034±230	1098±195
Forwards	TD (m)	10421±971	10479±793
	HIR (m)	849±294	900±242

TD – total distance, HIR – high intensity running

Table 2. Comparison of for total distance and high intensity running during the World Cup and UEFA Champions League matches according to the soccer-specific playing positions.

		ANOVA			Effect size
		MD	f	p	Cohen's d (95% CI)
Central defenders	TD	-439	22.07	0.01	-0.62 (-0.88; -0.35)
	HIR	-53	4.58	0.03	-0.28 (-0.54; -0.02)
Fullbacks	TD	-421	14.52	0.01	-0.58 (-0.89; -0.28)
	HIR	-12	0.10	0.75	-0.05 (-0.35; 0.25)
Central midfielders	TD	-446	11.03	0.01	-0.53 (-0.84; -0.21)
	HIR	-71	2.81	0.10	-0.27 (-0.58; 0.05)
Wide midfielders	TD	-582	6.36	0.01	-0.58 (-1.03; -0.12)
	HIR	-65	1.65	0.20	-0.29 (-0.74; 0.16)
Forwards	TD	-58	0.05	0.83	-0.06 (-0.64; 0.51)
	HIR	-52	0.40	0.53	-0.19 (-0.76; 0.39)

TD – total distance, HIR – high intensity running; MD – mean difference, CI – confidence interval

Discussion

This study aimed to provide a comparative analysis of RP of professional soccer match-play across two highest-level soccer competitions: UCL and WC. Results demonstrate that specific physical performance exist in the two highest-level soccer competitions and notably across individual positional roles. These findings may contribute to the prescription of individualized training and preparation strategies.

The various demands of elite soccer performance have been frequently described in the scientific literature (Sarmiento et al., 2014; Petrov, Michaelides, & Parpa, 2022). However, there is a lack of studies attempting to address differences in physical match performance especially between highest-level soccer competitions while using the same method of analysis and definitions of movement speed (Dellal et al., 2011). Current study therefore aimed to fulfil this research gap. Our results indicated significant differences in total distance covered during the UCL and WC for players on almost all playing positions. Specifically, while no significant differences between FWs' total distance covered in UCL and WC were found, greater total distance in UCL compared to WC was evidenced for CDs (10201 m and 9762 m, respectively), FBs (10903 m and 10482 m, respectively), CMs (11880 m and 11434 m, re-

spectively), and WMs (11129 m and 10547 m, respectively). Because total distance is an accepted indicator of the overall physical demands during competition in soccer (Rey, Costa, & Corredoir, 2019), it is possible that playing in UCL presents a greater degree of physical demand for defensive (e.g., CDs and FBs) and midfield players (e.g., CMs and WMs) than for offensive players (e.g., FWs).

It is especially important to emphasize significant differences in high intensity running during the UCL and WC for CDs. Specifically, CDs playing in UCL (638 m) achieved greater high intensity running distance than CDs playing in WC (585 m). We may speculate that CDs in UCL were more involved in offensive actions than their peers in WC. Given that this requires their deeper positioning in the opponent's half of the pitch, they consequently leave larger spaces behind their backs. Considering that after losing ball in the attack phase, a rapid offensive transition by the opposing team regularly followed (Hewitt, Greenham, & Norton, 2016), CDs should utilize higher running speeds to outperform opponent players and successfully defend this space (Modric et al., 2023). This may explain their increased high intensity running distance when playing in UCL. Irrespective of causality, since high intensity running distance has been suggested to be a specific

measure of physical performance due to its relationship with training status (Bradley et al., 2009; Rey et al., 2019), CDs' increased high intensity running distance in UCL indicates that playing in UCL may be physical more demanding for them than playing in WC.

The data on match physical performance in general aid soccer coaches and practitioners in decision making processes for structuring the elements of training (Aquino et al., 2017). Thus, evident differences in physical performances in UCL and WC matches examined in current study suggest that the same training programs are not suitable for both UCL and WC irrespective the fact that both can be considered as highest-level competition. Training programs should be tailored differently for each respective competition, considering specificities of playing positions as well. Specifically, given that a greater degree of physical demand was found for defensive and midfield players from UCL, overall conditioning status for players on these positions (e.g., CDs, FBs, CMs, and WMs) should be at higher level when playing in UCL than in WC. In addition, considering CDs' increased high intensity running distance which directly corresponds to the intensity of the anaerobic threshold (Modric, Versic, & Sekulic, 2021), CDs playing in UCL should have better developed anaerobic capacities compared to their peers playing in WC.

The present investigation has some limitations that should be considered. Firstly, although all matches were analysed with optical tracking system and the same threshold for high intensity running was applied, it should be emphasized that data provider were different which could influence the established differences between competitions to some extent. Secondly, not all matches from UCL and WC were analysed, which reduced the number of observations and may have affected RP. Thirdly, situational factors such as team and opposition quality, match location or match outcome, which may influence RP in national competitions (Castellano, Blanco-Villaseñor, & Alvarez, 2011),

were not considered in the current study. However, very recent study demonstrated small influence of such situational factors on RP in highest-level soccer (Modric et al., 2022); therefore, influence on results in current study may be negligible. Further studies with longitudinal design and larger sample are necessary to confirm the present results while controlling for several situational factors. Finally, for a more detailed understanding of physical performance during the UCL and WC, there should be some attempt to account for accelerations rates.

Despite the limitations, there are several important contributions which current study provides. Firstly, this study was the first to examine position-specific comparative analysis of RP of professional soccer match-play across two highest-level soccer competitions: UCL and WC. This enables soccer practitioners a more thorough understanding of the physical requirements for playing in UCL and WC. Secondly, the findings from current study may serve soccer coaches in decision making process of structuring specific training programs. Thirdly, the position-specific approach is important strength of current investigation as it allows "real-world" applications of the findings. In the end, although not being the final word on a problem, results will improve the knowledge in area and may initiate further research.

In conclusion, this study demonstrated differences in various aspects of physical performance between two highest-level soccer competitions with variations dependent upon playing position. Specifically, greater physical performance of defensive and midfield players playing in UCL was evidenced compared to their peers playing in WC. Based on such findings, it is recommended that overall conditioning status for defensive and midfield players should be at higher level when playing in UCL than in WC. Additionally, CDs playing in UCL should have better developed anaerobic capacities compared to their peers playing in WC due to their increased high intensity running distance evidenced in UCL.

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ORIGINAL SCIENTIFIC PAPER

Acute Effect of a Cross-Training Benchmark on Psychophysiological Factors of Cross-Training According to Performance

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Abstract

The study aims to analyze the acute effect of a Cross-Training benchmark on executive functions and physiological parameters according to performance. Thirty-two practitioners were divided according to their performance, Elite group (n=7; age: 28.9±4.7 years; practice: 50.0±13.3 months), Advanced group (n=10; age: 33.4±4.6 years; practice: 27.6±13.8 months) and Beginner group (n=15; age: 30.6±7.1 years; practice: 22.9±9.2 months). This research compares the groups and the pre-WOD and post-WOD moments for physiological (heart rate, lactate concentration, diastolic and systolic blood pressure) and neuropsychological variables (executive function); correlations between the physiological and neuropsychological effects of the benchmark. The results showed a significant difference concerning the performance of the WOD time (Elite: 177.1'±29.8' s and the difference in executive functions regarding the comparison between moments: reading (Elite:76.4±20.2 percentiles), counting (Elite: 86.4±10.7 percentiles), choice (Elite: 89.3±9.8 percentiles), shifting (Advanced: 91.0±8.4 percentiles), inhibition (Advanced:76.5±12.5 percentiles), flexibility (Advanced: 93.0±6.3 percentiles), lactate (Elite:13.1±1.8 mmol/L), heart rate (Elite: 188.0±6.6 bpm), systolic blood pressure (Elite: 149.7±11.5 mmHg), and diastolic blood pressure (Elite: 73.4±6.2 mmHg). Our findings confirm that high-intensity exercise could influence the physiological mechanisms responsible for the intervention in cognitive performance, improving executive functions.

Keywords: sport psychology, cognition, anxiety, physiology of exercise, neuroscience, sport performance

Introduction

Neurophysiological factors are why different sports agents are cited to justify obtaining specific results (Vealey, 1992; Brown & Fletcher, 2017; Slimani et al., 2017; Znazen et al., 2017; Brandt et al., 2019). The mind is often more important than any tactic, technique, or skill (Browne et al., 2016; Diamond & Ling, 2016). Understanding the cognitive aspects that improve a Cross-Training benchmark practice (Griffin et al., 2011; Wang et al., 2013; Murawska-Cialowicz

et al., 2015; Tomporowski & Pesce, 2019; Brandt et al., 2021). Conceptually, Cross-Training is a strength training and general conditioning program that enables a broad physiological adaptation for any person (Glassman, 2015). The Program is based on three exercise bases: functional movements, high intensity, and constant variation (Claudino et al., 2018). Some studies (Bellar et al., 2015; Haddock et al., 2016; Box et al., 2019) have recently indicated individual differences between Cross-Training participants in their respective primary modes



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of physical practice, determining psychologically predictive inferences for performance without assessing the physiological interaction associated with executive functions.

Several investigations (Mangine et al., 2018; Beatty & Janelle, 2019; Heinrich et al., 2020) have demonstrated that engaging in Cross-Training programs, which involve the simultaneous practice of multiple physical activities, can lead to positive changes in psychophysiological outcomes. These outcomes include cognitive function, mood regulation, stress reduction, and overall mental well-being (Coco et al., 2019; Heinrich et al., 2020). Moreover, the diverse nature of Cross-Training activities appears to promote improved mood states and stress resilience, potentially attributed to the combination of aerobic exercise, strength training, and flexibility components (Reppa et al., 2023). However, despite these promising findings, there remain notable gaps in the literature concerning the impact of Cross-Training benchmarks on psychophysiological factors.

Given the particularities of the modality and characteristics of Cross-Training (Claudino et al., 2018), it has yet to be discovered how this practice can affect executive functions associated with physiological parameters and what benefit this can have in the lives of practitioners of this modality. Therefore, it is relevant to understand the executive functions' role in achieving complex and unpredictable activities (Diamond, 2013). Thus, the present study proposes to investigate the effect of a Cross-Training benchmark on executive functions and physiological parameters and the interdependence of these factors according to the time performance of the Workout of the Day - WOD Fran.

Methods

Sample

The sample consisted of 32 Cross-training practitioners randomly chosen and, after that, stratified by level: Elite group (E=7; gender: 28.6% female and 71.4% male; age: 28.9 ± 4.7 years; body mass: 80.0 ± 10.9 kg; height: 1.72 ± 0.1 m; practice time: 50.0 ± 13.3 months; training volume: $13:34 \pm 3:54$ week hours and WOD execution time: 177.1 ± 29.9 seconds), Advanced group (A=10; gender: 50% female and 50% male; age: 33.4 ± 4.6 years; body mass: 71.7 ± 15.5 kg; height: 1.70 ± 0.1 m; practice time: 27.6 ± 13.8 months; training volume: $8:34 \pm 3:05$ week hours and WOD execution time: 314.3 ± 46.8 seconds) and Beginner group (B=15; gender: 33.3% female and 66.7% male; age: 30.6 ± 7.1 years; body mass: 72.3 ± 11.0 kg; height: 1.70 ± 0.1 m; practice time: 22.9 ± 9.2 months; training volume: $4:42 \pm 1:07$ week hours and WOD execution time: 538.2 ± 102.8 seconds). The CrossFit-affiliated boxes were located in Governador Valadares (Minas Gerais, BRA). For the sample calculation, the post-hoc statistical power ($1-\beta$) was applied using the analysis of variance (ANOVA: Repeated measures, within-between interaction), Effect size $f=0.25$, significance level $=0.05$, α err $=0.05$. Thus, the conferred statistical power present in the sample was 0.83%. G * Power software© version 3.1 was used (Krakatau Metrics, 2020).

The first group was formed by athletes who performed WOD Fran in up to 225 seconds ($n=7$), being classified as the Elite group; the second group consisted of athletes who performed the WOD Fran between 240 to 393 seconds ($n=10$), and was considered the Advanced group; lastly, the third group was formed by athletes who performed the WOD Fran over

394 seconds ($n=15$), being considered the Beginner group.

The following inclusion criteria were considered for participants: participants should have ≥ 18 years old; regular training routine (minimum three times a week) and minimum experience of 12 continuous months of activity; belonging to boxes affiliated with the CrossFit Program; any women doing hormonal control.

The following exclusion criteria were applied: Athletes who reported the use of drugs that may alter psychophysiological characteristics in the last three months according to the list of substances considered doping by the World Anti-Doping Agency (Heuberger & Cohen, 2019); not completing one of the proposed tests, and participating in two or more physical activity programs simultaneously. Before the study, athletes attended a briefing meeting. They signed an informed consent document to ensure they understood the testing procedures and the risks and benefits associated with the research. No interferences were made in athletes' training, nutritional, or hydration status. Following the WMA's Declaration of Helsinki, the local Ethics and Research Committee previously approved this research (n°13846919.8.0000.5257).

Procedures

Participants were familiar with WOD. The pre-test was before the execution of the WOD Fran, and the post-tests were immediately after completing the WOD Fran. All collected data occurred in air-conditioned boxes between 18:30 and 21:30 at a range temperature between 24.5o-26.5oC. During the intervention, the athletes were instructed to reach the condition of exhaustion, with continuous heart rate monitoring and voice encouragement as behavior modulation to motivate the participant to execute a clean WOD without quitting and finishing at the appropriate time. A single evaluator was used per parameter, training, and alignment, with a pilot model to make the adjustments. All subjects were instructed to maintain their usual lifestyle and regular diet before and during the study so that there was no interference from uncontrolled variables during the WOD. Women were instructed to maintain hormonal control. This protocol reproduced the actual Cross-training activity, and coaches controlled the data collection to guarantee that all athletes realized validated techniques.

Measures and Instruments

Executive Function Evaluation – Five Digit Test pre and post-WOD

The Five Digit Test (Sedó et al., 2015) is an instrument used to evaluate the effect of attentional interference using conflicting information about numbers and quantities, the task of assessing executive capacity, mainly inhibitory control, and cognitive flexibility (Campos et al., 2016). The four main variables of the test were used as measures (reading, counting, choosing, and Shifting times) and two executive indices (inhibition and flexibility). In the four test situations, these last two indices provide information about some mental processes such as processing speed (reading and counting times); inhibitory control/selective attention (choice and inhibition times); cognitive flexibility/alternating attention (shifting times and flexibility) (Sedó et al., 2015; Campos et al., 2016).

Scores are generated from the execution times in different stages: inhibition, calculated by the time difference of the choice/reading step; flexibility, calculated by the time difference of the alternation/counting step. The cut scoring guide-

lines are provided with the recommendation that the scores be adjusted based on the characteristics of the participants and the purpose of use (Sedó et al., 2015; Campos et al., 2016; de Paula et al., 2017). This test is private to the psychologist.

Assessments of Physiological Parameters pre and post-WOD Fran

Heart rate: Measured with a chest monitor and wristwatch receiver (Polar ProTrainer 5, USA) used during and after the intervention for monitoring (Maté-Muñoz et al., 2018).

Blood pressure: Systolic and diastolic blood pressure were recorded before and shortly after the intervention using an aneroid sphygmomanometer, and a stethoscope (Premium, Duque de Caxias – Brazil) calibrated and with the appropriate cuff size (Shaw et al., 2015; Malachias et al., 2016).

Blood lactate: Blood lactate concentrations were measured with the Accutrend® analyzer (GC/GCT, USA) before and after WOD in a blood sample taken from the finger (Zebrowska et al., 2019).

Benchmark Intervention – WOD Fran

The acute intervention was performed using a benchmark - WOD Fran, a three-round workout with a repetition scheme (Perform the 21 Thrusters and 21 Pull-Ups, then 15 Thrusters and 15 Pull-Ups, then 9 Thrusters and 9 Pull-Ups). This type of WOD aims to complete the prescribed exercises and repetitions as quickly as possible. This benchmark was chosen because it is a classic reference training, a performance marker in Cross-

Training, and stimulates the three energy systems (Glassman, 2015; McArdle et al., 2016). The total load in the Thruster exercise was 95lb for men and 65lb for women (Glassman, 2015). There was a standardized 5-minute warm-up which consisted of running around the box and simulating movements at low intensities (~60% of the maximum heart rate). The execution of WOD Fran started after 5 minutes of rest.

Statistical Analysis

The data are described as mean (M) and standard deviation (SD), with a calculation of the 95% confidence interval for the difference (CI), with $p \leq 0.05$ as the significance criterion. ANOVA with independent factor was performed to compare groups, and repeated measures ANOVA was applied to compare intra-conditions to compare the dimensions of the executive function and the physiological parameters in the pre and post-WOD moments. The size of the variance effect was calculated by the eta squared (η^2) (Cohen, 1992). Pearson's correlation was used for interdependence between executive function and physiological variables (Schober et al., 2018). All analyses were applied using the Statistical Package for Social Sciences (SPSS 22.0) for Windows.

Results

The performance of participants in the executive function testing stages and the assessment of physiological parameters are described in Table 1 as mean and standard deviation.

Table 1. Evaluation of cognition and physiological parameters pre and post intervention with WOD Fran.

Variables	M±SD					
	Elite (n=7)		Advanced (n=10)		Beginner (n=15)	
	pre	post	pre	post	pre	post
Cognition						
Total Five Digits	17.29±4.11	17.43±3.61*	17.70±2.54	19.20±3.12*	16.43±3.52	18.86±3.59*
Reading (P_k)	92.14±7.56	76.43±20.15*	88.50±14.91	75.50±28.91*	85.71±16.74	74.64±22.32*
Counting (P_k)	77.86±7.56	86.43±10.69*	79.00±8.43	77.50±20.58*	67.50±24.86	81.07±21.14*
Choice (P_k)	77.14±15.24	89.29±9.76*	71.00±23.07	89.00±9.66*	55.00±27.46	74.64±22.83*
Shifting (P_k)	60.00±23.09	86.43±10.69*	49.50±22.42	91.00±8.43*	49.64±23.24	76.43±21.70*
Inhibitory (P_k)	55.71±29.07	76.43±20.15*	59.00±27.87	76.50±12.48*	46.43±26.12	68.57±18.34*
Flexibility (P_k)	60.00±23.09	89.29±9.76*	49.50±22.42	93.00±6.32*	53.57±24.21	86.07±11.14*
Physiological Parameters						
HR (bpm)	100.14±17.69	188.00±6.63*	87.90±13.31	174.10±16.06*	92.07±9.13	185.13±8.93*
SBP (mmHg)	127.14±7.65	149.71±11.46*	127.00±7.96	151.00±9.25*	129.60±8.66	152.53±8.05*
DBP (mmHg)	81.43±2.51	73.43±6.19*	83.00±4.45	72.80±9.58*	80.93±6.13	69.33±7.08*
LAC (mmol/L)	3.57±0.97	13.07±1.81*	3.10±0.99	15.14±3.21*	3.61±0.61	16.05±3.69*

Legend: *Difference between pre and post of the three groups is significant $p \leq 0.05$. Note: M – Mean; SD – Standard Deviation; HR - Heart Rate; SBP - Systolic Blood Pressure; DBP - Diastolic Blood Pressure; Lac - Blood Lactate.

The statistical analysis for executive functions showed a significant difference between pre and post-WOD Fran in all of them, with a total of five digits ($F_{1,29}=8.04$; $p=0.008$; $\eta^2=0.217$), reading factor ($F_{1,29}=7.25$; $p=0.012$; $\eta^2=0.200$), counting factor ($F_{1,29}=4.87$; $p=0.035$; $\eta^2=0.144$), choice factor ($F_{1,29}=21.18$; $p=0.001$; $\eta^2=0.422$), shifting factor ($F_{1,29}=74.71$; $p=0.001$; $\eta^2=0.720$), inhibition factor ($F_{1,29}=13.82$; $p=0.001$; $\eta^2=0.323$), and flexibility factor ($F_{1,29}=50.73$; $p=0.001$; $\eta^2=0.636$).

The statistical analysis for the physiological parameters identified a difference in the comparison between the pre and post-measurements in all groups for heart rate ($F_{1,29}=1091.07$; $p=0.001$; $\eta^2=0.97$), systolic blood pressure ($F_{1,29}=138.47$; $p=0.001$; $\eta^2=0.83$), diastolic blood pressure ($F_{1,29}=31.35$; $p=0.001$; $\eta^2=0.52$) and blood lactate ($F_{1,29}=318.19$; $p=0.001$; $\eta^2=0.92$).

The correlations of the physiological parameters and executive functions of the Elite, Advanced, and Intermediate group participants are presented in Table 2.

Table 2. Pearson's correlation for physiological parameters and cognition.

		Elite Group								
Variables		TEMPO WOD	HR		SBP		DBP		LAC	
			pre	post	pre	post	pre	post	pre	post
Total Five Digits	pre	-0.022	-0.292	0.287	-0.108	-0.748	-0.661	0.125	0.834*	-0.417
	Post	0.154	0.339	0.209	0.112	-0.304	-0.19	0.642	0.756*	-0.569
Reading	pre	-0.471	0.153	0.133	0.297	0.451	0.603	0.387	-0.217	0.253
	Post	0.316	0.020	0.168	0.236	0.385	0.052	-0.5	-0.672	0.294
Counting	pre	0.027	0.769*	-0.332	-0.066	0.396	0.452	0.326	-0.258	-0.156
	Post	0.569	-0.063	-0.188	-0.186	0.412	0.036	-0.49	-0.729	0.571
Choice	pre	-0.476	-0.151	0.067	-0.548	-0.689	-0.52	0.107	0.553	-0.091
	Post	0.472	0.372	-0.824*	-0.881**	-0.256	-0.156	0.379	0.399	-0.191
Shifting	pre	0.322	0.141	0.310	0.038	-0.17	-0.403	-0.152	0.054	-0.151
	Post	0.516	0.149	-0.329	-0.839*	-0.296	-0.462	0.115	0.211	0.177
Inhibitory	pre	0.398	0.233	-0.315	-0.469	-0.089	-0.268	-0.248	-0.165	0.004
	Post	0.239	0.156	-0.723	-0.824*	-0.576	-0.377	0.222	0.628	-0.419
Flexibility	pre	0.322	0.141	0.31	0.038	-0.17	-0.403	-0.152	0.054	-0.151
	Post	0.072	-0.805*	0.721	0.013	-0.375	-0.701	-0.505	0.032	0.505
Advanced Group										
Total Five Digits	pre	-0.524	0.689*	0.692*	-0.181	0.648*	-0.403	-0.437	0.524	0.178
	Post	-0.517	0.429	-0.071	0.17	0.724*	-0.432	-0.527	0.669*	0.462
Reading	pre	0.282	-0.037	0.527	-0.407	-0.109	-0.427	0.196	0.291	-0.631
	Post	-0.142	0.497	0.873**	-0.254	0.272	0.065	-0.266	0.049	0.051
Counting	pre	-0.217	0.123	0.374	0.132	0.456	-0.237	-0.154	0.21	-0.018
	Post	-0.178	0.577	0.214	-0.071	0.493	-0.212	-0.344	0.553	0.118
Choice	pre	0.018	0.386	0.817**	-0.478	0.26	-0.401	-0.235	0.283	-0.023
	Post	0.009	0.219	0.319	-0.029	0.323	-0.466	-0.134	0.619	-0.186
Shifting	pre	-0.323	0.396	0.410	-0.103	0.512	-0.362	-0.153	0.231	0.403
	Post	-0.498	0.491	-0.144	-0.132	0.342	-0.593	-0.011	0.329	0.624
Inhibitory	pre	-0.308	0.686*	0.547	-0.225	0.526	-0.511	-0.425	0.471	0.424
	Post	0.085	-0.404	-0.386	0.576	-0.101	0.47	0.147	-0.356	0.089
Flexibility	pre	-0.323	0.396	0.410	-0.103	0.512	-0.362	-0.153	0.231	0.403
	Post	-0.073	-0.029	-0.26	-0.486	-0.342	-0.395	0.616	-0.32	0.333
Beginner Group										
Total Five Digits	pre	-0.255	0.042	-0.108	-0.045	-0.125	0.043	-0.096	0.28	-0.277
	Post	0.011	-0.414	-0.171	0.191	-0.031	-0.128	0.076	0.336	0.169
Reading	pre	-0.008	0.174	0.303	-0.041	-0.109	-0.006	-0.363	-0.018	0.154
	Post	-0.557*	0.076	-0.024	-0.169	0.064	0.152	-0.361	0.242	0.034
Counting	pre	-0.375	0.259	-0.419	-0.078	-0.124	0.301	-0.433	-0.184	-0.293
	Post	-0.473	0.137	-0.236	-0.207	-0.123	0.187	-0.249	0.081	-0.083
Choice	pre	-0.064	-0.137	-0.315	-0.047	-0.247	0.32	0.095	0.578*	0.074
	Post	-0.136	-0.066	-0.225	0.039	-0.293	0.276	-0.089	0.229	0.082
Shifting	pre	-0.144	0.152	-0.228	0.077	-0.233	0.491	-0.006	0.476	-0.098
	Post	-0.218	-0.045	-0.296	-0.174	-0.102	0.196	-0.254	0.159	0.221
Inhibitory	pre	-0.191	-0.126	-0.443	0.013	-0.174	0.285	0.299	0.422	-0.111
	Post	0.12	0.159	-0.022	0.276	-0.401	0.14	0.347	-0.072	-0.024
Flexibility	pre	0.176	-0.132	0.157	-0.011	0.024	0.311	0.256	0.731**	0.142
	Post	0.298	-0.343	-0.079	0.345	0.050	0.302	0.283	0.176	0.641**

Legend: **The correlation is significant at the level $p \leq 0.01$. *The correlation is significant at the level $p \leq 0.05$. Note: HR – Heart Rate; SBP – Systolic Blood Pressure; DBP – Diastolic Blood Pressure; LAC – Lactate Blood

The analysis in the Elite group showed that there is a strong positive correlation between the counting factor and heart rate ($r=0.769$) and a negative and strong correlation between the choice factor and heart rate ($r=-0.824$), choice factor and systolic blood pressure ($r=-0.881$), shifting factor and systolic blood pressure ($r=-0.839$), inhibition factor and systolic blood pressure ($r=-0.824$), flexibility factor and heart rate ($r=-0.805$).

The Advanced group demonstrated a strong and positive correlation between the reading factor and heart rate ($r=0.873$), choices factor and heart rate ($r=0.817$), total five digits and systolic blood pressure ($r=0.724$), as well as a moderate and positive correlation for inhibition and heart rate ($r=0.686$), total five digits and heart rate ($r=0.689$), total five digits and systolic blood pressure ($r=0.648$), total five digits and lactate ($r=0.669$).

The Beginner group identified a strong correlation between flexibility factor and blood lactate ($r=0.731$); a moderate and positive correlation between flexibility factor and blood lactate ($r=0.641$), choice factor and lactate ($r=0.578$); and a moderate and negative correlation between reading factor and WOD execution time ($r=-0.557$).

Discussion

There are currently no studies addressing the acute effect on executive functions related to physiological parameters in Cross-Training (Claudino et al., 2018; Brander Löf & Lindblom, 2019). However, there are studies separately analyzing the processes of attention and lactate (Perciavalle et al., 2016), cognitive functions and lactate (Coco et al., 2019), and memory and lactate (Perciavalle et al., 2015) in Cross-Training practitioners. We assume that the psychological aspects are multifactorial and fundamental in the athlete's performance and encompass a series of combined factors that can explain different effects on performance according to the competitive level (Crust, 2007; Crust & Azadi, 2010; Basso & Suzuki, 2017).

In contrast to our study, we found a study by Mangine et al. (2018), who developed normative values for five benchmark exercises (Fran, Grace, Helen, Filthy-50, and Fight-Gone-Bad) using the performance data of 133,857 male and female profiles located on a publicly available website and classified by gender and competitive age. This sample was randomized and stratified according to the level of performance in the benchmark execution time - WOD Fran, to form the subgroups. However, they did not control essential variables to verify the physiological and psychological effects on the practitioner. They did not consider the reliability of the evaluators since such secondary data presents descriptive elements associated with competitive events.

Elite group participants completed the Benchmark in 53% less time than the Advanced group, and this percentage increased to 84% of the participants compared to the Beginner group. Furthermore, 61% of the participants in the advanced group completed the Benchmark in less time than the Beginner group. These differences can be explained by the technical skill and specific physical aptitude acquired by the practice time of the modality and the training time, which facilitates performing movements and consequently improves the final time (Bellar et al., 2015; Glassman, 2015). Regarding the practice time in the modality, 60% of the Elite group had a longer time than the other groups considering the weekly training volume, which was almost twice as voluminous as the other groups. This information helps create normative

data for Cross-Training considering the execution time of the Benchmark - WOD Fran, according to the competitive-level classification.

The findings suggest that a WOD Fran session improves the performance of executive functions, especially concerning cognitive flexibility and inhibitory control. These results can be of great importance in elucidating the influence of exercise on the efficiency of executive functions and therefore contribute to improving athletic performance (Yanagisawa et al., 2010; Li et al., 2014). Athletes with better-developed cognitive processes tend to achieve higher performance levels (Vestberg et al., 2017). We found that cognitive performance after acute exercise seems to be linked to exercise intensity, as demonstrated in a meta-analysis (Chang et al., 2012). The exercise intensity had a significant influence when <50% of HRMax was prescribed, showing a result that had a significant negative effect with magnitude (Cohen's $d=-0.113$) on the cognitive performance; furthermore, the results were positive when prescribed above 60% of FCMax, with an effect and magnitude of ($d=-0.202$) and ($d=-0.268$).

The experimental study by Lambrick et al. (2016) suggests that an acute exercise of just 15 minutes in duration, whether of an intermittent or continuous nature, is sufficient to cause significant improvements in executive functions. These effects can be maintained for up to 30 minutes after the end of the activity. Most studies use the Stroop test and point out that the higher the intensity, the better the cognitive performance (Yanagisawa et al., 2010; Li et al., 2014; Vazan et al., 2017). This study obtained a similar result to the abovementioned studies in performing an acute intervention with high intensity and short duration exercise (WOD Fran), varying between athletes from 145 seconds to 763 seconds. It was observed that a single exercise session could promote the performance of executive functions using the five-digit test (Campos et al., 2016). Thus, the ability to inhibit dominant responses or actions in progress is important for successful performance, as well as greater control of behavior, attention, thought, and emotion, enabling the inhibition of behaviors or automatic routines and the execution of controlled or conscious routines in favor of what is most appropriate or precise (Huijgen et al., 2015).

There is still a discussion about the physiological mechanisms that guide executive functions' acute brain adaptations (Chang et al., 2012; Li et al., 2014; Perciavalle et al., 2016; Coco et al., 2019). The existing literature highlights a positive relationship if the exercise is of submaximal intensity, while the effects of exhaustive exercises seem negative (Perciavalle et al., 2015, 2017; Strömmer et al., 2020). Possible hypotheses to explain improved post-exercise executive function include the acute effect of exercise intensity on cognitive response or increased cerebral blood flow generated by exercise effort, noted in post-exercise cognitive performance.

The present results of the physiological parameters corroborate the findings in the study by previous authors, as Perciavalle et al. (2016), Hall et al. (2016), and Fernández et al. (2015), which compared the Fran and Cindy post-WOD routine in healthy adults aged 30 ± 4.2 years. The results revealed that both WODs could be characterized as high-intensity exercises, reaching acute physiological responses and representing 90-95% HRmax. We noticed in our study that there was hypotension right after the intervention; after a training session, the body produces physiological responses classified into immediate treble pre and post-intervention (Materko et

al., 2020). Several studies, such as that by Rezk et al. (2006) and Ferrari et al. (2014), report that the DBP decreased at 15 and 30 minutes after the exercise session, justifying hypotension. Most findings indicate an increase in SBP after effort, which corroborates the results obtained in this study (Tibana et al., 2017; Fonseca et al., 2018; Materko et al., 2020). We agree with a hypothesis raised by Coco (2019) that high levels of lactate in the blood induced by exercise can affect some factors of executive functions; however, our study shows significant improvement in cognitive flexibility and processing speed, in contrast to some studies which note the relation to the functions supported by the prefrontal cortex, such as processing speed (Strömmer et al., 2020), cognitive flexibility (Coco et al., 2019) and resistance to interference (Laurent et al., 2020) which seem to be more affected, while the functions supported by more posterior cortical areas such as visual attention and task changes are not affected (Uehara et al., 2019).

This study is the first to analyze the effect of a Cross-Training WOD Fran on the physiological variables and executive functions according to the performance of the WOD. However, limitations must be pointed out for a better analysis of these findings: Absence of collection time after the recovery

period, which would enable inferences about the periods necessary for the return to base values; equivalent distribution of both genders for each group. All of this compromises the physical activity performance level measurement on these functions. We suggest that future studies explore randomized experiments with effects after one hour, eight hours, 12 hours, 24 hours, and chronic effects.

Conclusion

The present study is essential for physical training by scientifically investigating the psychophysiological effects of an acute Cross-training intervention. Our research demonstrated a positive effect on the executive function associated with a WOD Fran session. The outcomes suggest that a WOD Fran session improves the performance of executive functions, especially concerning cognitive flexibility and inhibitory control. Professionals working in this area can use this information to improve the quality of performance of their athletes. Our findings confirm that high-intensity exercise presented a strategy that sharply influenced the physiological mechanisms responsible for intervening in cognitive performance, efficiently improving executive functions.

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Conflict of Interest

The authors declare that they have no competing interests.

Authors' contributions

MAB, JRF, PHBC and BM participated in the study design, collection and data analysis, manuscript preparation, funding acquisition; CJB, EAM, DASS participated in the data analysis and manuscript preparation.

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ORIGINAL SCIENTIFIC PAPER

Differences in Aerobic and Anaerobic Capacity between Special Police Force Members and Elite Basketball Players

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Abstract

The goal of this research is to evaluate and compare the functional abilities of the special anti-terrorist unit from Serbia and professional basketball players. The idea is to determine whether members of the special unit police have high functional abilities and at what level are they compared to professional athletes. The sample of participants, for the purposes of this research, consisted of 162 respondents with an average age of 29.75 ± 8.12 years who were classified into two groups. One group consisted of members of the SAF (special anti-terrorist force) ($n=101$) of average age 33.74 ± 6.19 years and in other group were elite level professional basketball players from Serbia ($n=57$), of average age 22.39 ± 4.25 from three Serbian clubs. To measure cardiorespiratory fitness, a cardiopulmonary exercises test (CPET) was performed on a treadmill (HP-COSMOS®). Maximal oxygen consumption (VO_{2max}), as a measure of aerobic capacity, was determined using the Quark CPET system (Cosmed®) by direct monitoring of gas exchange (oxygen and carbon dioxide). Aerobic ability of members of the Serbian SAF (special anti-terrorist force) can be considered excellent and almost close to the professional athletes, but statistically lower ($p < 0.05$). Anaerobic capacity is also high and there is no difference in relation to basketball players ($p > 0.05$). These results can be explained by the selection of people who are chosen for this unit, which can be called very rigorous and equated with sports selection. The results of our study point to the fact that a high level of cardiorespiratory fitness is not only required by professional sports, but also by certain professions such as SAF.

Keywords: *VO₂ max, pulmonary ventilation, cardiopulmonary testing, anaerobic threshold*

Introduction

The work of professional police officers, members of the special anti-terrorist unit is one of the most physically demanding jobs that require good functional abilities. High value of these abilities stands for a base for developing and perfecting both motor and technical-tactical abilities for special anti-terrorist police unit because their job requires performing complex and high-risk physically demanding tasks for the security and protection of the Republic of Serbia and its citizens (Radovanović, 2020). That is why the training of elite force is more demanding than the training of members of the ordinary police (Ricciardi, Deuster, & Talbot, 2007). At the same

time, members of the police must be highly aware of maintaining their level of health and physical ability in order to be able to perform tasks at a high level of effort during their careers (Poncio, 2020). A high level of aerobic capacity is considered more important than strength and power in elite police officers (Robinson et al., 2018). Furthermore, a high level of functional and cardiorespiratory capacity has a large the potential of protecting the physical and mental health of police officers in their daily work (Schilling et al., 2020). At the same time, better functional capacity and cardiorespiratory response to exercise are associated with reduced cardiometabolic risk (Schilling et al., 2019). The selection process itself for mem-



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bers of the elite unit is quite competitive and requires people with good cardiorespiratory abilities (Poirier et al., 2022).

For assessment of functional abilities (aerobic and anaerobic), as well as long endurance parameters, generally are evaluated maximum oxygen consumption (VO₂max), pulmonary ventilation (VE max), aerobic and anaerobic thresholds (Westra et al., 2020). Maximal oxygen consumption is defined as the largest amount of oxygen consumed per unit time during progressively increasing work intensity involving large muscle groups that continues despite exhaustion (Kohozuki et al., 2018). On the other hand, pulmonary ventilation is important because stands for the process of renewing oxygen in the lungs, which is used for muscle work, and is conducted through the processes of inhalation (inspiration) and exhalation (expiration) and is expressed in liters per minute (Đorđević Saranović et al., 2019).

In regular police officers, the average value of VO₂max is 38.7 ml/kg/min (Marins, David, & Del Vecchio, 2019). At the same time, in elite units, maximum oxygen consumption was recorded between 45 and 57.4±4.3 ml/kg/min. Professional basketball players must have a high level of aerobic capacity due to recovery from anaerobic activities during the game itself. Maximum oxygen consumption in basketball players is in the range of 45 to 65 ml/kg/min (Midgley, McNaughton, & Marchant, 2007). These values, of course, also depend on the level at which senior basketball players compete. In addition to aerobic abilities, functional abilities include anaerobic abilities, which take place in conditions when there is no oxygen available. When we say anaerobic capacity, it primarily refers to the ability to tolerate the accumulated products of anaerobic work (Živković, 2016). When the concentration of lactate reaches a level at which the body cannot remove them, the aerobic threshold is reached. In the literature, the level of around 2 mmol/l is most often mentioned (Kenney, Willmore & Costill, 2012), and the terms are different compared to different authors such as “lactate threshold”, “first ventilatory threshold” or “anaerobic threshold” (Binder et al., 2008).

This implies that the level of physical activity is so high that aerobic and anaerobic metabolism intermix. When the body uses up its buffer reserves, a period of unstable lactate state occurs, or in sports literature, the anaerobic threshold is often mentioned. The lactate concentration at this threshold is about 4 mmol/L. Above the anaerobic threshold, the organism creates energy in anaerobic conditions without the presence of oxygen, and physical activity is highly intense and leads to fatigue and cessation of exercise (Wasserman, Cox, & Sietsema, 2014). The organism uses the respiratory system to try to eliminate the accumulated carbon dioxide (CO₂) and thereby compensates for the resulting acidity of the organism. In functional diagnostics, this moment is called the respiratory exchange ratio (respiratory exchange ratio = CO₂/O₂ – RER) or anaerobic threshold. At the maximum effort, the RER values are 1.10 and above, which represent a realistic measure of the achieved metabolic fatigue and the ability of the body to tolerate elevated levels of lactate in muscles and blood, as well as exercise in exclusively anaerobic conditions, which stands for a highly trained person.

Previous research comparing the functional abilities of police officers and athletes is very limited. The authors searched the literature and found a similar study that showed that aerobic and anaerobic capacity is significantly less developed in police officers compared to elite athletes from most Olympic disciplines. But, compared to basketball players in the same study, it was significantly higher (VO₂max - 52.4 ml/kg/min

vs 46.5 ml/kg/min) (Zwingman et al., 2021). At the same time, Vuckovic and Koropanovski (2007) compared the motor skills of members of the police academy and elite karate fighters and concluded that they differ statistically significantly.

The lack of research in the existing literature indicates that more similar studies are needed to conduct in order to contribute to a better understanding of the functional capacity and design of standards in the training program of elite police forces, in order to meet the demands of the profession. Given all the above, the aim of this research is to evaluate the functional abilities of elite police officers and compares their level in relation to professional athletes.

Methods

Procedures

Pre-participation medical examination was a prerequisite for taking part in the study before performing cardiopulmonary exercise test (CPET) on a treadmill. The aim of the sports medical examination was to determine the general health ability of police officers and basketball athletes for their profession/sport. CPET was performed to evaluate VO₂max of participants. Prior to being conducted, the ethics committee of University Singidunum of Serbia approved this study. University Ethics Committee (no. 36-2 decision made on January 24th, 2022). Each participant voluntarily supplied written informed consent before taking part in the study. The conducted research does not violate the rights of the examined players, according to the ethical standards of the Helsinki Declaration of the Committee on Human Rights (WMA Declaration of Helsinki, 2013).

Participants

The sample of respondents, for the purposes of this research, consisted of 162 participants with an average age of 29.75±8.12 years. Study participants were classified into two groups. One group consisted of members of the special anti-terrorist unit of the police (n=104) of average age 33.74±6.19 years. The second group was made by elite professional basketball players from Serbia (n=58) of average age 22.39±4.25 from three top rank Serbian teams. The cardiopulmonary exercise test was performed by trained medical doctors in the outpatient sports medicine clinic “Vita Maxima” and all subjects gave their written consent to perform the test. In addition, the CPET was conducted in order to check the health status of members of the police and elite basketball players.

Data collection

A sample of variables for research purposes presented the parameters of the functional abilities of members of the special anti-terrorist unit and professional basketball players. Heart rate after maximum effort (HR max), heart rate at the second anaerobic threshold, heart rate recovery during first three minutes after CPET, maximal pulmonary ventilation (VE), VO₂max and respiratory exchange ratio (RER) were evaluated during CPET.

Test protocol

Electrocardiogram of the heart at rest was performed using a 12-channel ECG (Fukuda). CPET was performed on a treadmill (HP-COSMOS®) and VO₂max, as a measure of aerobic ability, was determined by using the Quark CPET system (Cosmed®) by direct monitoring of gas exchange (oxygen and carbon dioxide). A modified Nowacki protocol (Nowacki

& Preuhs, 2003) was used to conduct the test, with an initial speed of 5 km/h and an elevation of 0°. After the introductory part of the test, which serves as a warm-up, the speed of the treadmill increased to 9 km/h and did not change during the test, but the elevation of the treadmill was raised by 2° every one minute to achieve the maximum load. Subjects had a face mask and a mobile ECG device (Quarck® T 12x, Wireless 12-lead ECG) on their back to directly monitor gas exchange and heart rate during exertion. Test was maximal if at least three of four below mentioned criteria were achieved: the value of the achieved heart rate of 90% or more of the predicted theoretical maximum heart rate for gender and age, which is calculated based on the formula: $220 - \text{number of years}$; respiratory exchange ratio (RER) >1.10 ; plateau in maximal oxygen consumption despite increasing load (differences in VO_2max values less than 150 mL/min near the end of CPET); subjective feeling of exhaustion.

Before CPET tests the calibration according to the so-called STPD criteria (ST - standard temperature/standard gas temperature: 0°; P - pressure/pressure: 760 mmHg; D - dry equivalent/dry air) of the Quark CPET system (Cosmed®) was performed after every fifth test to adequately

determination of measured parameters.

Statistical analysis

To describe parameters of importance, depending on their nature, the following were used: frequency, percentages, mean value, median, standard deviation, rank and 95% confidence intervals. To test the normality of the distribution, the Kolmogorov Smirnov test was used and the graphics: histogram and normal QQ plot. To test the differences in the functional characteristics of police officers and basketball players, the T Test of independent samples and the Mann Withney U test were used. Statistical data processing was performed in the statistical package SPSS 20.0 for Windows. Differences were considered significant when the p value was less than 0.05.

Results

Descriptive statistics of anthropometric characteristics, body composition and functional abilities of all participants in the study are presented in Table 1.

Comparative statistics of maximum oxygen consumption between members of the special police unit and basketball players in different positions are given in Tables 2, 3, 4.

Table 1. Descriptive statistic of antropometric varibales and functional ability of all participants in study

	Basketaball	SAF	T value	p value
Variables	X±SD	X±SD	/	/
N (number)	57	105	/	/
Age	22.39±4.25	33.791±6.80	-11.42	p<0.001
Height (cm)	200.65±7.79	179.61±4.63	17.42	p<0.001
Weight (kg)	97.37±9.59	87.39±11.45	5.69	p<0.001
BMI (kg/m ²)	24.18±1.84	26.87±2.87	-7.27	p<0.001
FFM kg	88.9±7.92	74.17±7.034	12.24	p<0.001
Fat %	10.65±4.63	15.85±4.85	-6.65	p<0.001
VO ₂ max ml/kg/min	55.40±5.45	51.40±7.26	3.64	p<0.001
RER	1.10±0.18	1.10±0.89	2.60	p>0.05
Ve Max	165.5±28.02	142.4±21.37	5.88	p<0.001
HR rest	60.04 ±10.35	64.10±12.45	-2.26	p<0.05
HR VT2	176.54±8.73	173.15±6.73	0.88	p=0.377
HR max	186.05±7.87	185.21±10.04	0.45	p=0.651
HR recovery 1. min	159.42±11.52	155.86±13.95	1.56	p=0.119
HR recovery 2. min	134.04±14.43	130.86±15.73	1.17	p=0.243
HR recovery 3. min	121.56±14.03	119.57±15.73	0.71	p=0.479

Abbervations: BMI (body mass index), FFM (free fat mass), VO₂ max (maximal oxygen consumption), RER (respiratory exchange ratio), VE max (maximal ventilation), HR at VT2 (heart rate at the second ventilatory threshold), HR max (maximal heart rate), HR recovery (heart rate recovery in first, second, and third minute after test).

Table 2. Comparative statistics VO₂ max and RER of the special police unit and centers

Variable	SAF	Centers	T value	P value
VO ₂ max mil/kg/min	51.41±7.23	52.05±5.51	2.385	p=0.65
RER	1.07±0.39	1.09±0.55	0.375	p=0.69

Table 3. Comparative statistics VO₂ max and RER of the special police unit and forwards

Variable	SAF	Forwards	T value	P value
VO ₂ max mil/kg/min	51.41±7.23	57.40 ±4.08	3.588	p<0.001
RER	1.07±0.39	1.08±0.03	1.630	p=0.112

Table 4. Comparative statistics V02 max and RER of the special police unit and guards

Variable	SAF	Guards	T value	P value
VO2max ml/kg/min	51.41±7.23	57.00±5.16	3.891	p<0.001
RER	1.07±0.39	1.08±0.23	1.916	p=0.064

In general, a statistically significant difference between the two groups was found in the parameters of maximum oxygen consumption and parameters of maximum pulmonary ventilation ($p<0.05$) (Figure 1 and Figure 2).

VO2max - Maximal oxygen consumption

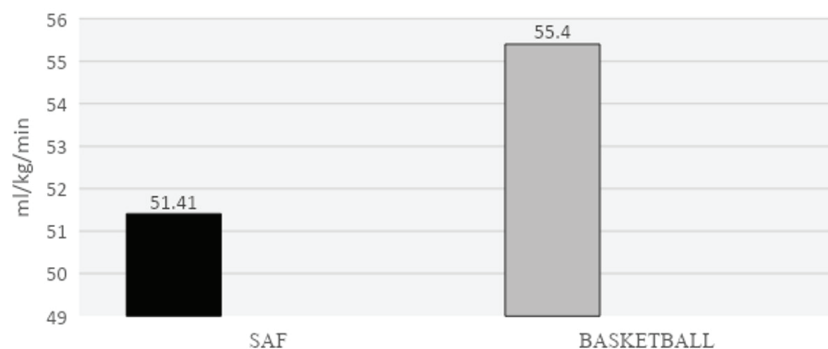


FIGURE 1. Results of maximum oxygen consumption (VO2max) for policemen and basketball players; *A statistically significant difference between the groups was found in the parameters of maximum oxygen consumption ($p<0.05$). Abbreviation: SAF- special anti-terrorist force

VE max - pulmonary ventilation

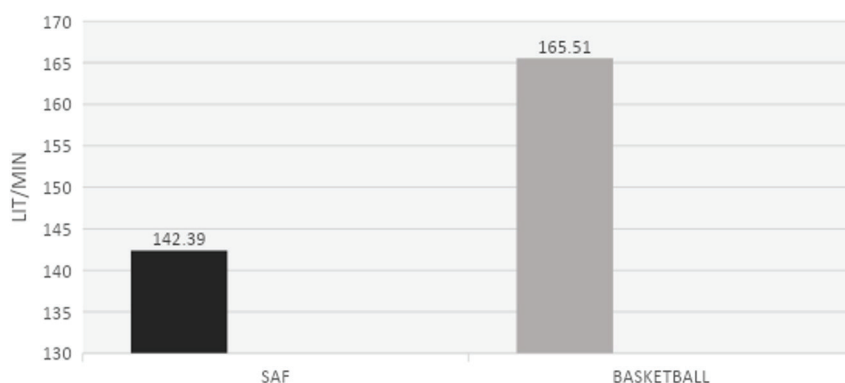


FIGURE 2. Results of maximal pulmonary ventilation for police officers and basketball players; *A statistically significant difference between the groups was found in the parameters of maximal pulmonary ventilation ($P<0.05$).

RER - Respiratory exchange ratio

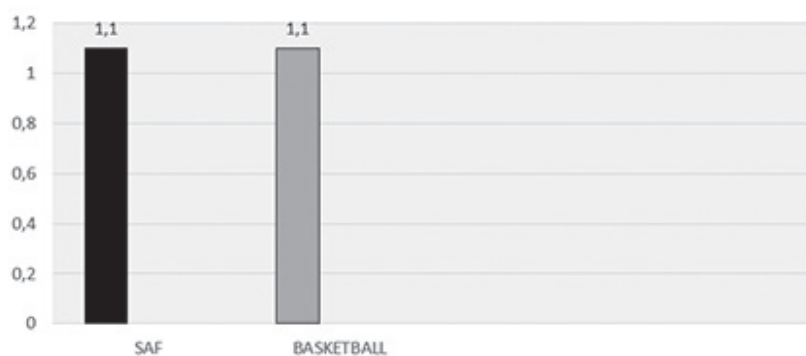


FIGURE 3. Results of anaerobic ability and respiratory gas exchange in police officers and basketball players; Abbreviation: SAF - special antiterrorist force

In the anaerobic capacity shown through the RER (respiratory exchange ratio), i.e. the capacity of the body to tolerate lactate in the anaerobic mode, no difference was found between police officers and professional basketball players ($p>0.05$) (Figure 3).

There was no statistically significant difference in the

heart rate at rest or in the maximum heart rate, and there was also no difference in the heart rate at the second anaerobic threshold (Figure 4). There was no difference in the heart rate during recovery in both groups in the first three minutes after the intensive physical activities (Figure 5).

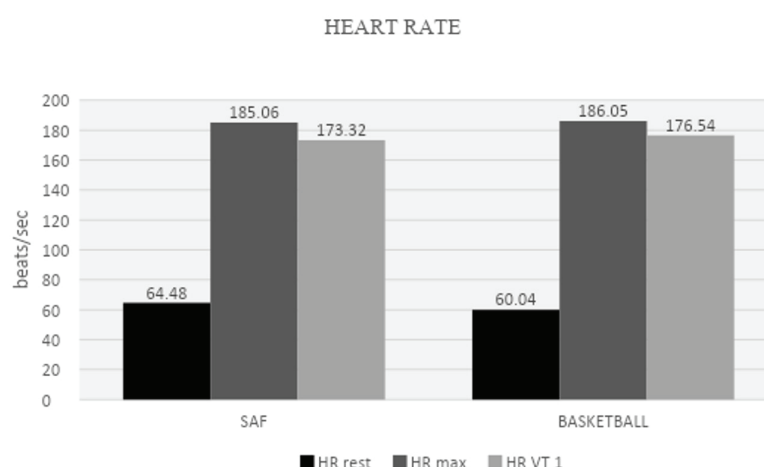


FIGURE 4. Results of heart rate frequency in police officers and basketball players

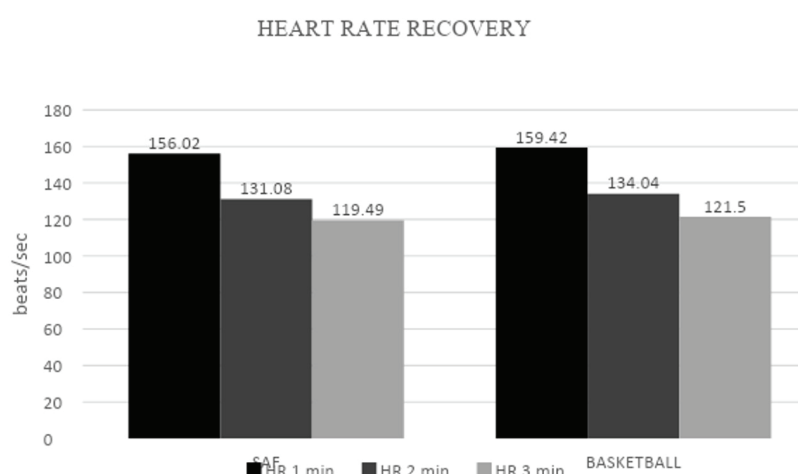


FIGURE 5. Results of heart rate recovery during three minutes after intense physical activity; Abbreviation: SAF - special antiterrorist force

Discussion

The results of this study aimed to determine the functional status of the special police unit and evaluate its difference in relative to elite basketball players. In general, the purpose was to determine whether regular training of members of the special anti-terrorist unit affects the level of aerobic abilities and at what level they are compared to elite athletes. In both groups, functional abilities and body composition represent an important aspect of the work they do. The results of this research support the fact that there is a statistically significant difference in functional abilities between the two groups, but only in maximum oxygen consumption and pulmonary ventilation (VE max) for the benefit of elite athletes. Basketball players, especially of the elite level due to the nature of the sport and professionalism, have a greater volume and intensity of training, as well as intense matches 2 times a week, compared to members of the police, who train once a day 5 times a week, and do also strength training. In terms of aerobic capacity, the elite basketball players from this study had a maximum oxygen consumption (55.04 ml/kg/min), which

is within the range of results described in earlier research on elite basketball players (Midgley et al., 2007). It should be noted that the research was conducted in the post-season, that is, before the start of preparations for the new season, so the slightly lower results of the elite basketball players from Serbia are in support of this. At the same time, members of the special anti-terrorist unit showed a high level of aerobic capacity (VO_{2max} - 51.41 ml/kg/min), which is above the reference values for gender and age people from the general population (Maupin et al., 2018). However, in basketball there are different positions in the team in relation to height and motor characteristics, so the fact that guards and forwards had better results than centers is in favor of better oxygen consumption in basketball players (Stojmenović, Trunić, & Stojmenović, 2022). But, if we compared VO_{2max} values of policemen and players in the center position it can be established that policemen have a better aerobic capacity than centers, according to previous study (Stojmenovic et al., 2022). But, in this research police officers have slightly lower aerobic capacity according to center (51.41 vs 52.04 ml/kg/min) with no statistical meanings.

At the same time, SAF members in this study had statistically significantly lower values of VO₂max compared to forwards and guards, but not RER values (respiratory exchange ratio). Besides aerobic capacity special police unit are above the ability of the regular police (Marins et al., 2019) who perform regular daily duties on the field. At the same time, members of the Serbian elite police unit had better results compared to police officers from Montenegro (Spalevic et al., 2021). Furthermore, Koropanovski and Janković (2007) showed that students of the Police Academy, at the beginning of their education, had a lower level of motor skills from the aspect of maximum isometric force, as well as dynamic and repetitive strength of the whole body, in relation to well-trained persons of the same age type of strength training. At the same time, it was shown that the motor skills of police academy students differ from elite karate players (Vuckovic, & Koropanovski, 2007). Serbian members of special forces have a good aerobic capacity, which is within the limits of the values of members of special units from other countries, which were described in earlier studies (Sperlich et al., 2011; Maupin et al., 2018; Araújo et al., 2019). Also, it was shown that Serbian special forces have a higher level of aerobic fitness than members of the American SWAT team, but also lower values than their colleagues from Slovenia (Šimenko et al., 2014). Compared to members of the fire department, where high levels of functional ability are also required, this study showed that elite police officers from Serbia have better aerobic capacity than firefighters (Donovan et al., 2009). Police work also requires good recruitment of personnel, who will become elite police officers, so this is one of the conditions for selecting candidates. Police officers from Serbia have good aerobic abilities, which are little lower than the criteria for filling police personnel in Canada (Poirier et al., 2022). In the area of anaerobic capabilities, statistic difference between groups it is not observed in the RER variable (Respiratory exchange ratio), which explains lactate tolerance in anaerobic conditions. Basketball players had a slightly higher RER, but no statistically significant compared to members of the SAF, which indicates that the job belong to the special unit of the police often practiced anaerobic work zone. In the pulmonary ventilation variable, we found statistically significant difference ($p < 0.05$) which proves that police officers have a good lung volume of breathing, but lower according to basketball players. Furthermore, there was no

difference between the groups in heart rate parameters, first of all in the values of the maximum heart rate as well as the heart rate at the second anaerobic (ventilatory) threshold, which indicates that police officers have a good tolerance to lactate and that they later enter total anaerobes. The sametime, there was no difference between the groups in the recovery heart rate in the first, second and third minutes, after maximum work, which also indicates a good level of recovery from maximal efforts. The limitations of this study are that there was an unequal number of respondents in these two research groups (105 vs 57), as well as different ages and morphological characteristics. Namely, members of the SAF were statistically significantly older than professional basketball players (33.74 vs 22.39), significantly shorter, and they also had a higher percentage of fat, and body mass index, which can be a limiting factor in functional abilities. At the same time, professional basketball players have a greater number of training sessions on a weekly basis, as well as competitive matches, which contributes a lot to the improvement of functional abilities.

Nevertheless, despite the limitations, this study is important because it provides insight into the aerobic and anaerobic abilities of SAF as well as professional basketball players and contributes to a better understanding of determining the direction and important parameters of the training process in both groups. At the same time, the study provides guidelines in which direction they could be improved.

Conclusion

Functional abilities are an important factor for the performance of the tasks of the members of the special anti-terrorist unit. Their profession demands performing complex and high-risk physically demanding tasks for the security of the state and citizens. The level of aerobic and anaerobic capacity of members of the Serbian SAF can be considered excellent and almost approximate to professional athletes. A big role in all parameters is played by the selection of people who are chosen for this unit, which can be called very rigorous and equated with sports selection. The assumption is that the level of aerobic capacity of police officers would be much higher if the volume of training during the week were increased. However, their training requires other types as technical-tactical trainings that are also important, which takes time.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Effects of Self-Management Behavior of Closed Sports Athletes on Concentration and Competition State Anxiety

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Abstract

Athletes are required to strive to demonstrate their best performance not only in physical training but also in mental toughness. This study aims to understand the effects of self-management behaviors of athletes competing in closed-skill sports on maintaining concentration and resolving competition state anxiety. An offline survey was conducted with 301 participants, shooting athletes selected using convenience sampling. This questionnaire consists of 14 items, and 11 sub-factors. All items were measured using a five-point Likert scale (1 = not at all, 5 = very much). Results revealed that self-management behaviors (training, mental power, interpersonal, intrinsic, and physical management) have significant positive effects on athletes' concentration (attention, confidence, immersion, and goal setting). Additionally, self-management behaviors (training and mental power) positively affect competition state anxiety (somatic and cognitive). The results of this study confirm that managing athletes' mental health could play an important role in the concentration and anxiety that affect athletic performance. This study provides important data for athletes who need to demonstrate their best performance in competitions.

Keywords: training, mental power, interpersonal, intrinsic, physical, psychological state

Introduction

It is important for athletes to manage their individual training time and daily life activities to achieve physical and mental stability and optimal performance. Recently, athletes have emphasized the mental aspects of improving and maintaining their abilities (Heo, 2003). Shooting competitions can be divided into rifle, pistol, and clay competitions according to the type of bullet and firearm used; however, most shooting competitions are conducted in the same pattern and are representative of closed sports not affected by external factors such as opponents or referees. Shooting competitions are also classified as closed sports because athletes are required to maintain a firing posture without moving or making physical contact with their competitors. Given these conditions, overcoming psychological obstacles such as maintaining extreme concentration and resolving anxiety,

becomes the deciding factor in this sport (Jun, 2019).

Until recently, studies in this field have clarified that proper and clear self-management by athletes, regardless of sport, is a crucial factor in maintaining a certain level of athletic performance, including strategies, tactics, skills, and the physical strength required to maintain the best performance (Ahn, 2016). Particularly, in closed sports, the information processing and nervous systems are very important, as are the physical factors such as appropriate muscle strength, flexibility, muscle endurance, and coordination (Kim, 2011). In addition, concentration through psychological stability must precede the stabilization of near-sightedness and information processing systems (Shin, 2015). Concentration has long been discussed as a positive factor in improving performance. Discussions have explored methods to suppress anxiety as much as possible to improve performance. Additionally, players'



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self-management factors have been discussed as mediating factors for concentration and anxiety. However, most Korean studies have followed the objective perspective (Jang, 2012; Moon, Sung, & Jang, 2017) or rarely show qualitative perspectives (Lee, Park, Kim, & Nam, 2015; Park, Park, & Jeon, 2017).

Previous studies evaluating the self-management of athletes playing closed sports mainly focused on archery, golf, and gymnastics (Moon & Park, 2008; Yeo, 2010; Shin & Yoon, 2018). This study will focus on shooting. The study will produce practical discourses that will help shooting athletes improve their performance and will also provide useful information to support their professional growth, such as developing coaching and training methods. Furthermore, the results of this study will provide baseline data for forming new discourses aimed at improving athletic performance in fixed-target sports such as archery, shooting, golf putting, and basketball free throws. Therefore, this

study aims to identify the factors affecting concentration maintenance and competition state anxiety in practice by analyzing the self-management of shooting athletes. These athletes are relatively understudied even though shooting is a representative closed sport in which psychological factors such as concentration and competitive anxiety are crucial to athletes' success.

Research hypotheses and study design

H1-1, 2, 3, 4. Athletes' self-management behaviors have a significant effect on concentration (attention, confidence, immersion, and goal setting).

H2-1, 2. Athletes' self-management behaviors have a significant effect on competition state anxiety (both somatic and cognitive).

To verify the research hypotheses, the study was designed as follows (Figure 1).

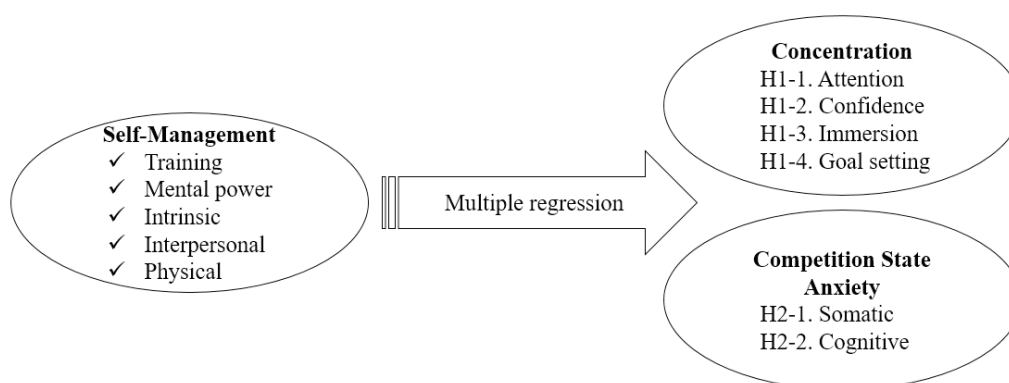


FIGURE 1. Study design

Self-management behavior

Self-management has been used in various fields, and the sports field is no exception (Kim & Cruz, 2021). Self-management, which has also been described as the concept of self-regulation or self-control (Suh, 2005), is defined as the ability to manage oneself physically and mentally to achieve goals (Weinberg & Gould, 2015). Self-regulation in everyday life plays an important role in athletes' self-management, and self-regulation acts as motivation to control oneself in sports (Lee, 2010). A study on the mental preparation and psychological strategies undertaken by national team players to maintain and exercise their best performance highlights the significance of self-regulation in athletic performance (Yu, 1996). Self-management in sports and exercise involves mentally preparing for obstacles and overcoming them through self-restraint (Heo, 2001). Athletes' self-management includes various internal, external, and environmental factors (physical and mental) that affect their goal-achievement (Yu, 1996).

Concentration

Concentration is the ability to focus on one's mind or attention. This is also considered a state of complete psychological and emotional immersion. In highly competitive sports, concentration is vital to maximizing performance. The four elements of concentration required for sports performance are selectivity (analytical skill), flexibility (conversion ability), persistence (concentration duration), and capacity (total amount of energy) (Etzel, 1979; Lee, 2020). Concentration can be considered a key prerequisite for successful sports performance. Kim (2005) presented goal-setting, attention control, repetitive practice before performance, faith, and attention as psychologi-

cal techniques that help achieve the highest concentration states psychologically, emotionally, and physically. According to Kirk and Chalfant (1984), concentration is the ability to bring one's abilities together, and the mental power to selectively accept and concentrate on the external stimuli one needs. In addition, Lee and Kim (2011) and Reid and Hresko (1981) argued that the ability to control consciousness from internal or external stimuli is concentration. In the end, concentration is described as the ability to select specific information from among various external information, focus on and pay attention to that information, and the process of information processing that can be continuously focused while adjusting the amount and assigning attention according to the information you choose.

Competition state anxiety

Anxiety is a state of depression, tension, and excitement (oversensitivity) accompanied by discomfort or irritation as a reaction to a subjective and imagined threat (Martens, 1977). The psychological state of anxiety is closely related to the concept of fear of failure (Gustafsson, Sagar, & Stenling, 2017). Fear of failure occurs due to shame and embarrassment caused by failure in evaluation situations and has been defined as a concept of social self-evaluation, resulting from scenarios created entirely by individuals (Sagar & Lavalley, 2010). The most obvious characteristic of sports situations in which competition is an essential element is competition state anxiety, and the anxiety in competitive situation may occur various consequences; the psychological, physiological, biochemical, and behavioral symptoms (Singer, 2002). In addition, the competition state anxiety by fear of failure is accompanied by negative results such as stress (Gustafsson et al., 2017), decline in performance

(Sagar, Lavalley, & Spray, 2009), and dropout (Sagar, Lavalley, & Spray, 2007). Therefore, Competition anxiety, an inevitable psychological state of athletes, needs to be properly managed.

Methodology

Data collection procedure and participants

Shooting athletes registered with the Korea Shooting Federation were selected using convenience sampling, a non-probability sampling method. The study participants responded to a self-administered questionnaire after being informed of the study's objective, and data from 301 partici-

pants were analyzed. All participants participated voluntarily and were not penalized for withdrawing from the study. Additionally, this study did not collect any sensitive personal information from survey participants, allowing it exemption from IRB review. This is in accordance with the Helsinki Declaration. All participants were informed of the details of this study and their rights as survey respondents. Finally, all survey respondents reported demographic information, such as sex, age, athletic career, daily training hours, and level of performance. The descriptive statistics of the survey participants are presented in Table 1.

Table 1. Descriptive statistics of study participants

Variables	Categories	N (%)
Gender	Male	157 (52.2%)
	Female	144 (47.8%)
Age	Early 20s	105 (34.9%)
	Mid to late 20s	116 (38.5%)
	Over 30s	80 (26.5%)
Athletic career	4 – 8 years	90 (29.9%)
	9 – 15 years	140 (46.5%)
	Over 16 years	71 (23.6%)
Training hours per week	Less than three hours	55 (18.3%)
	Four hours	87 (28.9%)
	Five hours	86 (28.6%)
	More than 6 hours	73 (24.3%)
Perceived performance level	Low	25 (8.3%)
	Medium	184 (49.2%)
	High	92 (30.6%)
	Very high	36 (12.0%)
Total		301 (100.0%)

Instruments

The Questionnaire to Measure Athletes' Self-Management Behaviors developed by Kim (2003) was used to measure the self-management of participants in this study. This tool was also used by Kim (2018) and Kang and Yoo (2018). This questionnaire consists of 20 items and five sub-factors: four items on training management, five items on mental power management, four items on intrinsic management, three items on interpersonal management, and four items on physical management.

Additionally, this study employed a modified and supplemented version of the Athletic Concentration Measurement Scale, developed by Park and Kim (2007) and used by Chun (2017) and Kim and Park (2016), to measure the concentration demonstrated by the study subjects in this study. The survey comprised 14 items and four sub-factors: four items on attention, four items on confidence, three items on immersion, and three items on goal setting.

Finally, the Competition State Anxiety scale, developed by Martens et al. (1990), was used to measure the competition state anxiety of study subjects. Recently, Kim et al. (2016) used it to measure competitive state anxiety in athletes. It was modified to meet the present study's objectives. This questionnaire consists of 14 items and two sub-factors: seven items on somatic anxiety and seven on cognitive anxiety. All items were measured using a five-point Likert scale (1 = not at all, 5 = very much).

Data analysis

This study used SPSS 23.0 for statistical data analyses (IBM

SPSS Statistics for Windows, version 23.0. Armonk, NY: IBM Corp.). First, descriptive statistics of participants' demographic information was obtained using frequency. The instruments used for data collection have been verified to have satisfactory validity and reliability in previous studies. However, as this study modified and supplemented the instruments for the research topic and purpose, three exploratory factor analyses (EFA) and Cronbach's alpha coefficients were calculated for each factor to ensure adequate statistical clarity. Finally, to determine the relationship between the variables, multivariate regression analyses were implemented with a statistical significance level of .05.

Results

Scale validity and reliability

Based on the results of the EFAs with Varimax rotation, eigenvalues greater than 1.0 and factor loadings greater than 4.0 were retained as follows: (a) self-management behavior (training, mental power, intrinsic, and physical), (b) concentration (attention, confidence, immersion, and goal setting), and (c) competition state anxiety (somatic and cognitive anxiety). Additionally, the reliability of the items was tested using Cronbach's alpha. The results were finalized with a cutoff value of 0.70 as a satisfactory internal consistency for reliability (Nunnally & Bernstein, 1994): (a) training management, $\alpha=0.810$; (b) mental power management, $\alpha=0.865$; (c) intrinsic management, $\alpha=0.797$; (d) interpersonal management, $\alpha=0.853$; (e) physical management, $\alpha=0.783$; (f)

attention, $\alpha=0.794$; (g) confidence, $\alpha=0.865$; (h) immersion, $\alpha=0.857$; (i) goal setting, $\alpha=0.881$; (j) somatic anxiety, $\alpha=0.914$; and (k) cognitive anxiety, $\alpha=0.937$. All measure-

ment tools showed satisfactory statistical reliability. The detailed results of the EFAs and Cronbach's alpha are presented in Table 2.

Table 2. Results of three exploratory factor analyses and Cronbach's alphas coefficients

Items	1	2	3	4	5	Items	6	7	8	9	Items	10	11
A1	.813	.176	.075	.025	.127	B1	.801	.147	.158	.210	C1	.808	.358
A2	.808	.073	.242	.247	.115	B2	.746	.344	.129	.162	C2	.784	.241
A3	.780	.041	.286	.268	-.001	B3	.726	.239	.360	.167	C3	.779	.353
A4	.769	.173	.045	.174	.123	B4	.649	.154	.452	.185	C4	.756	.277
A5	-.011	.766	.043	.256	.119	B5	.225	.862	.169	.065	C5	.742	.341
A6	.187	.764	-.028	.033	.239	B6	.250	.798	.227	.138	C6	.681	.456
A7	.186	.705	.068	.198	.251	B7	.112	.767	.332	.135	C7	.607	.435
A8	.167	.655	.288	.311	-.091	B8	.203	.825	.316	.183	C8	.295	.836
A9	.143	.502	.054	.094	.104	B9	.215	.326	.818	.082	C9	.278	.807
A10	.151	.073	.825	.121	.213	B10	.439	.184	.802	.243	C10	.294	.791
A11	.240	.030	.807	.310	.095	B11	-.024	.265	.707	.167	C11	.282	.781
A12	.154	.244	.800	.084	.138	B12	.155	.120	.047	.817	C12	.358	.772
A13	.140	.184	.723	.154	.139	B13	.446	-.047	.093	.650	C13	.442	.730
A14	.355	.173	.202	.804	.130	B14	.382	.011	.247	.621	C14	.421	.709
A15	.255	.328	.188	.742	.293								
A16	.050	.261	.149	.706	.154								
A17	.234	-.053	.220	.138	.778								
A18	.062	.382	-.066	.166	.710								
A19	.051	.297	.290	.260	.628								
A20	.221	.058	.454	-.095	.611								
Eigenvalues	3.010	2.717	2.669	2.532	2.269		2.921	2.516	2.490	2.417		5.048	4.729
Variance(%)	15.18	14.30	14.05	13.33	11.94		20.86	17.97	17.78	17.26		36.05	33.78
α	.810	.865	.797	.853	.783		.794	.865	.857	.881		.914	.6983

Note. 1 = Training management, 2 = Mental power management, 3 = Intrinsic management, 4 = Interpersonal management, 5 = Physical management, 6 = Attention, 7 = Confidence, 8 = Immersion, 9 = Goal setting, 10 = Somatic anxiety, 11 = Cognitive anxiety; Factor loadings higher than .40 are highlighted and in bold.

Hypothesis verification (effects of self-management on concentration)

1) Effects of athletes' self-management on attention

This study evaluated the effects of five independent variables and sub-factors of self-management on the attention of the study participants and found that the developed regression

model was significant ($F=12.692$, $p<.001$). Specifically, training management ($\beta=.206$, $t=2.758$) and mental power management ($\beta=.174$, $t=2.694$) significantly affected attention. Intrinsic, interpersonal, and physical management were not significant variables. The model developed using the self-management variables explained 17.7% of the variance in atten-

Table 3. Effects of athletes' self-management on attention

Variables	Concentration			
	Attention			VIF
	B	β	t-value	
<Constant>	1.558		6.260	
Training	.190	.206	2.758**	1.998
Mental Power	.159	.174	2.694***	1.499
Intrinsic	.023	.027	.405	1.633
Interpersonal	.076	.080	1.243	1.475
Physical	.040	.042	.627	1.635
F=12.692***	R squared =.177(adj.R ² =.163)			

p<.01, *p<.001, β =beta, VIF=variance inflation factor

tion. The standardized coefficients representing the relative contributions of the independent variables showed that the magnitude of the influence on attention was in the order of training and mental power management.

2) Effects of athletes' self-management on confidence

An evaluation of the effects of five independent variables and sub-factors of self-management on the confidence of the study participants found that the developed regression model was significant ($F=54.153$, $p<.001$). Specifically, training

management ($\beta=.279$, $t=4.699$), mental power management ($\beta=.318$, $t=6.184$), and interpersonal management ($\beta=.158$, $t=3.091$) had significant effects on confidence. The model developed using the self-management variables explained 47.9% of the variance in confidence. The standardized coefficients representing the relative contributions of the independent variables showed that the magnitude of the influence on confidence was in the following order: mental power management, training management, and interpersonal management.

Table 4. Effects of athletes' self-management on confidence

Variables	Concentration			
	Confidence			
	B	β	t-value	VIF
<Constant >	.205		2.792	
Training	.057	.279	4.699 ***	1.998
Mental Power	.049	.318	6.184 ***	1.499
Intrinsic	.047	-.014	-.270	1.633
Interpersonal	.050	.158	3.091 **	1.475
Physical	.052	.134	2.495	1.635
F=54.153***		R squared =.479(adj.R ² =.470)		

p<.01, *p<.001, β =beta, VIF=variance inflation factor

3) Effects of athletes' self-management on immersion

Evaluating the effects of five independent variables and sub-factors of self-management on the immersion of study participants revealed that the developed regression model was significant ($F=20.551$, $p<.001$). Specifically, interpersonal management ($\beta=.392$, $t=6.431$) had significant effects on immersion.

The model developed using the self-management variables explained 25.8% of the variance in immersion. The standardized coefficients representing the relative contributions of the independent variables showed that the magnitude of influence on immersion was in the following order: mental power management, training management, and interpersonal management.

Table 5. Effects of athletes' self-management on immersion

Variables	Concentration			
	Immersion			
	B	β	t-value	VIF
<Constant >	.254		6.372	
Training	.070	.069	.974	1.998
Mental Power	.060	.097	1.574	1.499
Intrinsic	.059	-.104	-1.631	1.633
Interpersonal	.062	.392	6.431 ***	1.475
Physical	.064	.110	1.717	1.635
F=20.551***		R squared =.258(adj.R ² =.246)		

****p<.001, β =beta, VIF=variance inflation factor

4) Effects of athletes' self-management on goal setting

Investigating the effects of five independent variables and sub-factors of self-management on the goal setting of the study participants showed that the developed regression model was significant ($F=25.589$, $p<.001$). Specifically, training management ($\beta=.189$, $t=2.752$), mental power management ($\beta=.234$, $t=3.927$), intrinsic management ($\beta=-.130$, $t=-2.090$), interpersonal management ($\beta=.201$, $t=3.401$), and physical management ($\beta=.169$, $t=2.710$) had significant effects on goal setting. The developed model using self-management variables explained 30.3% of the variance in goal-setting. The standardized coefficients presenting the relative contributions of the independent variables showed that the magnitude of influence on goal setting was in the following order: mental power man-

agement, interpersonal management, training management, and physical management.

Hypothesis verification (effects of self-management on competition state anxiety)

1) Effects of athletes' self-management on somatic anxiety

This study evaluated the effects of five independent variables and sub-factors of self-management on somatic anxiety in the study participants and found that the developed regression model was significant ($F=1.910$, $p<.05$). Specifically, mental power management ($\beta=.193$, $t=2.747$) had significant effects on somatic anxiety. The model developed using the self-management variables explained 2.8% of the variance in somatic anxiety.

Table 6. Effects of athletes' self-management on goal setting

Variables	Concentration			
	Goal setting			
	B	β	<i>t-value</i>	VIF
<Constant >	.264		3.883	
Training	.073	.189	2.752**	1.998
Mental Power	.063	.234	3.927***	1.499
Intrinsic	.061	-.130	-2.090*	1.633
Interpersonal	.065	.201	3.401***	1.475
Physical	.067	.169	2.710**	1.635
F=25.589***	R squared =.303(adj.R ² =.291)			

*p<.05, **p<.01, ***p<.001, β =beta, VIF=variance inflation factor**Table 7.** Effects of athletes' self-management on somatic anxiety

Variables	Competition State Anxiety			
	Somatic Anxiety			
	B	β	<i>t-value</i>	VIF
<Constant >	.335		9.266	
Training	.093	.150	1.850	1.998
Mental Power	.080	-.193	-2.747**	1.499
Intrinsic	.077	-.021	-.288	1.633
Interpersonal	.082	.023	.333	1.475
Physical	.085	-.015	-.198	1.635
F=1.910*	R squared =.028(adj.R ² =.012)			

*p<.05, **p<.01, β =beta, VIF=variance inflation factor

2) Effects of athletes' self-management on cognitive anxiety

On evaluating the effects of five independent variables and sub-factors of self-management on the cognitive anxiety of the study participants, this study found that the developed regression model was significant ($F=2.395$, $p<.05$). Specifically, training management ($\beta=.176$, $t=2.179$) and mental power management ($\beta=-.215$, $t=-3.075$) significant-

ly affected cognitive anxiety. The model developed using the self-management variables explained 3.9% of the variance in cognitive anxiety. The standardized coefficients presenting the relative contributions of the independent variables showed that the magnitude of the influence on cognitive anxiety was in the order of mental power and training management.

Table 8. Effects of athletes' self-management on cognitive anxiety

Variables	Competition State Anxiety			
	Cognitive Anxiety			
	B	β	<i>t-value</i>	VIF
<Constant >	.378		9.149	
Training	.105	.176	2.179*	1.998
Mental Power	.090	-.215	-3.075**	1.499
Intrinsic	.087	-.011	-.151	1.633
Interpersonal	.093	-.012	-.175	1.475
Physical	.096	-.053	-.724	1.635
F=2.395*	R squared =.039(adj.R ² =.023)			

*p<.05, **p<.01, β =beta, VIF=variance inflation factor**Discussion**

The results of this study revealed that training and mental power management—the sub-factors of self-management—affect shooting athletes' concentration. These results are in line with those of Kim et al. (2017) and Kim and Kim (2018), who reported similar results. Training management involves regu-

lating personal physical strength, conditions, and intensity of training duration required to meet daily goals in the process of setting and achieving specific training goals. However, mental power management involves maintaining and improving athletic performance (e.g., shooting for shooting athletes). In particular, shooting athletes require strict, systematic, and

continuous management because spontaneous behaviors, such as firing, are directly related to the results of the game. Considering that the training and mental power management of shooting athletes directly affect attention and concentration, developing a long-term plan to improve their attention is necessary. Interestingly, training management was found to be more effective than mental power management in enhancing performance, which is meaningful for coaches (Jun, 2019). As a strategy to improve athletes' concentration within a limited time training management and mental power management should receive attention rather than other factors such as intrinsic management, interpersonal management, and physical management. In particular, investing in training management improves athletes' concentration for a shorter period than making efforts for other elements.

Somatic anxiety in athletes is accompanied by various physiological symptoms; increased heart rate, dry throat, high blood pressure, or sweaty palms (Parnabas, Mahamood, & Parnabas, 2013). Somatic anxiety is a critical factor (Ong & Chua, 2021) that can seriously deteriorate athlete performance in shooting competitions, where a winner is determined in split seconds. Efforts have been made to overcome this problem through training. Notably, the results of this study suggest that self-management, especially mental power management, can reduce somatic anxiety (Ong & Chua, 2021). The speculation that mental power is related to anxiety is convincing from a psychological perspective. Simultaneously, previous studies have also reported cases that could prove this speculation in other sports (Park, 2012; Woon, 2008). In the context of self-management and somatic anxiety, mental power management, a sub-factor of self-management, is believed to reduce cognitive anxiety to a certain degree.

The results of this study invite discourse that will effectively improve the performance of athletes, and also provide useful information for improving athlete support, such as training programs. Although the results of this study are valuable, it had some limitations. First, it is necessary to explore the effects of self-management, concentration, and competitive anxiety on performance through in-depth interviews or qualitative observations rather than a quantitative investigation. More valuable findings can be obtained using various research approaches. Second, there may be factors other than players' self-management, concentration, and competitive anxiety

that affect performance. Thus, subsequent multidimensional studies on other variables are required to comprehensively understand the various factors affecting performance. Finally, there are natural physical abilities beyond mental abilities that also affect athletes' performance; therefore, case and follow-up studies on how the relationships between mental and physical factors change should be conducted in parallel.

Conclusions

This study investigated the effects of self-management behaviors, concentration, and competition state anxiety on athletic performance. Furthermore, this study conducted an in-depth exploration into related phenomena through the content and interpretation of self-management, concentration, and competitive anxiety experienced by athletes. The results showed that athletes' self-management behaviors significantly affected their concentration. Among the sub-factors of self-management behavior, training management and mental strength management had a significant positive (+) effect on attention among the sub-factors of self-management behavior, mental strength management, training management, and interpersonal management have a significant positive (+) effect on confidence among the sub-factors of self-management behavior. In addition, among the sub-factors of self-management behavior, interpersonal management had a significant positive (+) effect on immersion, and training management, mental power management, intrinsic management, interpersonal management, and physical management had a significantly positive (+) effect on goal setting. The results showed that athletes' self-management behaviors had a significant effect on their competition state anxiety. Among the sub-factors of self-management behavior, mental power management had a significant positive (+) effect on somatic anxiety among the sub-factors of competition state anxiety, and training and mental strength management had a significant positive (+) effect on cognitive anxiety among the sub-factors of competition state anxiety. These results help anticipate and produce discourse that will help improve athletic performance. The results also provide useful information for implementing various strategies to improve athletic performance. Finally, this study's findings provide basic data necessary to encourage future research, in turn leading to the growth of discourse on fixed-target sporting events.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Developing an Instrument to Measure the Social Capital in Young Tennis Players

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Abstract

Social capital consists of connections between individuals from which mutual trust and social networks arise. It is a term that encompasses resources that can be defined as social relations and social inclusion. It is an integral social factor that affects the social norms of an individual's behavior. Social components are an important factor for achieving better success in sport. The level of social capital can be assessed with different questionnaires. As there were no existing validated instruments for assessing social capital in young tennis players so far, a new questionnaire in relation to competitive tennis players was made. The goal of this research was to construct and validate a questionnaire for assessing the level of social capital in young tennis players. The social capital questionnaire was filled out by 90 participants (age=16.53±1.18; girls N=31, boys N=59), who were, at the time, participating in an ITF level (International tennis federation) tennis tournament. The reliability of the questionnaire was measured with the Cronbach Alpha coefficient. When analyzing the reliability of the selected 12 questions, the reliability was determined Cr. Alpha=0.714 and the questionnaire validity was determined using the Pearson correlation coefficient. It is understood that many factors impact young athletes during their growth and development, and that social components and social ties play an important role in young athletes' sporting careers. Through the creation of this questionnaire, future researchers will be able to further analyze and investigate social capital in tennis using it as a valid research tool.

Keywords: competitive athletes, questionnaire, validation, sport, adolescents

Introduction

Social components are an important factor in relation to sporting success. Therefore a more detailed insight into social capital and social support in the context of competitive youth sport and tennis is needed. Social capital consists of connections between individuals from which mutual trust and social networks arise (Putnam, 2001). It is a term that encompasses resources that imply social relations and social inclusion, and it is an integral social factor that affects the social norms of an individual's behavior (Novak, Doubova, & Kawachi, 2016). Social capital includes aspects such as moral support, cultural values, guidance, the possibility of playing sports, and logistical and financial support. In an athlete,

this can affect the level of self-confidence, motivation, sense of belonging, and the level of sports performance (Rosso & McGrath, 2012). Social capital is a resource available to individuals or groups through social relations and networks. It can be achieved on an individual or a collective level (Moore & Kawachi, 2017). Social capital is obtained from macro levels (state or regional levels) such as school, neighborhoods, work place and individual levels. Carrilo, Kawachi and Riera (2017) state the need for further social capital investigations. The concept of social capital has a cognitive component (perception of group credibility) as well as structural component (density and frequency of social participation) (Veenstra et al., 2005).



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Social support is a concept similar to social capital, it refers to the exchange of resources between two individuals with the aim of improving the well-being of the supported recipients (Shumaker & Brownell, 1984). Social support can be connected with a decline in sports performance, athlete burnout, competitive stress, and group cohesion (Rees & Hardy, 2000). The influence of coaches, parents and peers is significant for competitive athletes and their influence is characterized as an important resource for athletes. These are key relationships during athlete's growth and development (Sherida, Coffee, & Lavallee, 2014).

A coach has a big impact in giving social support, and their quality feedback and emotional support impact athletes and their performance (Sheridan et al., 2014). Social relationships also play a key role in sport and young adolescence (Villalonga-Olives & Kawachi, 2017).

Until now, researchers focused on studying social capital in recreational sports, sport participation and volunteering (Hayton, 2016; Forsell et al., 2019; Gemar, 2020), whereas fewer studies analyzed social capital in competitive athletes (Forsdike et al., 2017; Novak et al., 2020). Considering the important role social capital plays in sports, research needs to be broadened to include competitive athletes (Novak et al., 2020). Research of social capital in competitive tennis players emphasizes the important role it plays in the athletes' competitive success and why it needs to be further investigated. Athletes need a specific type of social support and help with specific sporting demands like taking care of physical fitness, and any problems they may face with practice or competition (Freeman et al., 2011). There are some claims that engagements and shared cooperative practices that sport consists of enable forming networks of norms, social trust, and social ties. Mutual respect in sport between competitors and the sense of team spirit within a team can also help improve athletic success (Lewandowski, 2018).

The level of social capital is currently assessed with different types of questionnaires. A further review of the literature established that there are validated questionnaires for the general school population (Paiva et al., 2014; Carrillo-Alvarez et al., 2019). However, there are no validated instruments for assessing social capital specifically in young tennis players. Therefore, for this research, a new questionnaire for competitive tennis players was developed. The goal of the research was to construct and validate a questionnaire for assessing the level of social capital in young competitive tennis players.

Methods

Participants

The questionnaire was filled out by 90 participants who were participating at a tennis tournament being held in Čakovec, Croatia. The participants were all aged between 16-18 years old (age=16.53±1.18; girls N=31, boys N=59), and all signed a written consent form prior to participating in the study.

Procedures

The study was conducted at a junior international tennis tournament held by the ITF. Athletes who compete at this level represent quality tennis competitors, all of who are aged under the age of 18. The ITF junior tournament held in Čakovec that was graded J100, which is a mid-level competition for international junior tennis players. To be able to compete at this level, players usually need to start learning

to play the sport at an early age. Participants in the study reported that they started playing tennis at the average age of 6.26±1.84 years old. Players filled out the questionnaire voluntarily after they signed in at the tournament. They filled out the questionnaire on their own and the researchers provided help if necessary. The research was approved by the ethics committee (26/2023, in Zagreb April 3rd 2023) of the Faculty of Kinesiology, University of Zagreb.

The validation of the newly developed questionnaire was conducted in two phases: 1) questionnaire development and 2) the evaluation of psychometric characteristics of the questionnaire. The database search was conducted on PubMed, Scopus and Web of Science databases using information developed within the last twenty years, and the key words were: social capital, tennis, social support, validation, and questionnaire. The database search showed existing questionnaires for social capital assessments within the general adolescent and child population (Paiva et al., 2014; Carrillo-Alvarez et al., 2019). However, to the author's knowledge, there are no validated questionnaires for social capital assessment in competitive tennis players.

Measurements

Items in the social capital questionnaire that was created were defined and adjusted from existing items used in already validated questionnaires (Paiva et al., 2014; Novak et al., 2015; Carrillo-Alvarez et al., 2019). The questionnaire content was subject to conceptual evaluation and following this, key questions were selected. All questions were in English, and the final version was evaluated by a kinesiolgist.

The questionnaire consisted of two parts, the first part consisted of general descriptive questions including: gender, age, current ITF ranking, and a general health self-assessment on a Likert scale: 1= very bad, 2= bad, 3= good, 4= very good, 5= excellent (Fylkesnes, 1993).

The second part of the questionnaire consisted of questions for the assessment of social capital. Questions included topics such as family, sports teams, and school. The first version consisted of 25 questions which are listed below 1) I feel my family understands and gives attention to my tennis career.; 2) I feel my family members work well together.; 3) I feel my family supports me.; 4) I feel my family members criticize me.; 5) I often speak/talk to my family members.; 6) I know when something important happens in my wider family (wedding, birth, illness...); 7) I spend my free time with my family members.; 8) My family members watch me play matches at tournaments. 9) I feel my parents and coaches trust each other.; 10) I feel members of my team collaborate with each other.; 11) I trust my coaches.; 12) I listen and understand when my coach explains something to me.; 13) My coach encourages me to play better.; 14) My coach criticizes me.; 15) I feel I can ask my coach for help if I need it.; 16) My coach watches me play matches at tournaments.; 17) My tennis coach and my strength and conditioning coach trust each other. 18) I feel teachers and students trust each other in my high school.; 19) I feel students trust each other in my high school.; 20) I like attending activities at my school other than regular classes. (school, sport sections, choir, geography...); 21) Classmates send me homework when I am not at school.; 22) I spend time with my school friends outside of school.; 23) Teachers

encourage us to speak to them if we have any problems.; 24) Teachers understand and give attention to my tennis career.; 25) I always know when I have an exam at school.

Participants needed to circle a number next to the question that represents level of agreement: 1= strongly disagree, 2= disagree, 3= neither agree nor disagree, 4= agree, 5= strongly agree.

Statistics

Parameters that were calculated for quantitative variables consist of: arithmetic mean (AS), and standard deviation (SD). The reliability was determined using the Cronbach Alpha coefficient for all questions individually as well as together, and the validity was determined using the Pearson

correlation coefficient. The obtained data were processed in Statistica 13 for the Windows operating system and in Microsoft Excel 2013 (Palo Alto, CA, USA). All conclusions are drawn at the level of statistical error of 5%.

Results

The reliability of the questionnaire was measured with the Cronbach Alpha coefficient. When analyzing the reliability of the selected 12 questions, the reliability was determined to be Cr Alpha=0.714 as shown in table 1 below. Selected variables have satisfying sensitivity, range, standard deviation, skewness, and kurtosis within +/-1.

Descriptive statistics of the analyzed items are shown be-

Table 1. Questionnaire reliability represented by Cronbach's Alpha coefficient

Reliability statistics		
Cronbach's Alpha	N of items	N Valid cases (Excluded)
.71	12	90 (0)

low in table 2 where variables mean and standard deviation can be seen.

To determine the questionnaire's validity, Pearson's cor-

relation was used for the sum of each item, and subsequently their intercorrelation was determined. The correlation was found to be statistically significant for all 12 items.

Table 2. Item descriptive statistics

	Mean	Std. Deviation	N
1. I feel my family understands and gives attention to my tennis career.	4.66	0.58	90
2. I feel my family members work well together.	4.51	0.75	90
3. I feel my family supports me.	4.77	0.56	90
4. I often speak/talk to my family members.	4.42	0.86	90
5. I feel members of my team collaborate with each other.	4.48	0.71	90
6. I listen and understand when my coach explains something to me.	4.71	0.50	90
7. I feel I can ask my coach for help if I need it.	4.76	0.50	90
8. My tennis coach and my strength and conditioning coach trust each other.	4.39	0.91	90
9. I feel teachers and students trust each other in my high school.	3.63	0.94	90
10. I feel students trust each other in my high school.	3.91	0.87	90
11. I spend time with my school friends outside of school.	3.88	1.07	90
12. Teachers understand and give attention to my tennis career.	3.42	1.13	90

Table 3. Questionnaire validity using Pearson correlation coefficient

	Pearson Correlation
1. I feel my family understands and gives attention to my tennis career.	0.52**
2. I feel my family members work well together.	0.58**
3. I feel my family supports me.	0.46**
4. I often speak/talk to my family members.	0.50**
5. I feel members of my team collaborate with each other.	0.55**
6. I listen and understand when my coach explains something to me.	0.38**
7. I feel I can ask my coach for help if I need it.	0.42**
8. My tennis coach and my strength and conditioning coach trust each other.	0.48**
9. I feel teachers and students trust each other in my high school.	0.62**
10. I feel students trust each other in my high school.	0.57**
11. I spend time with my school friends outside of school.	0.45**
12. Teachers understand and give attention to my tennis career.	0.49**

** . Correlation is significant at the 0.01 level (2-tailed); * . Correlation is significant at the 0.05 level (2-tailed)

Discussion

The specificity of the environment that sport and tennis creates requires the construction and validation of a new questionnaire that has the ability to assess the level of social capital in junior tennis players. Adolescents spend most of their time involved in school activities, in their neighborhood, and with their families. However, in comparison athletes spend a lot of their time practicing in their sports clubs with their sports team and coaches. This is one of the reasons why this questionnaire contains specific questions relating to sports teams. The questionnaire was constructed based on social connections related to school, family, and sports teams (Virtanen et al., 2013). As mentioned, to the knowledge of the author this is the first validated questionnaire which looks to assess social capital in young tennis players. In the process of developing the questionnaire, several changes were made before the final version. Originally the number of questions to be asked was 25, but eventually only 12 were selected as appropriate for the research question. This significant reduction is due to the elimination of unreliable questions. The validity scale refers to the level of which the measuring instrument measures the construct it was constructed for (Boyton & Greenhalgh, 2004). The level of Cr Alpha coefficient is satisfactory 0.71. The results presented in this study suggest that the presented questionnaire is valid and reliable for the assessment of social capital in young competitive tennis players.

This study achieved similar results in internal consistency of the instrument as shown through the research by Paiva et al. (2014). The questionnaire in the Paiva et al. (2014) study was primarily constructed for the assessment of social capital within the general youth population. Similarly, it consisted of 12 questions, and some of the statements used were: "I feel I belong in my school," "I believe my friends from school," "My parents get along with my teachers." Subscales identified in the Paiva et al. (2014) questionnaire were: school social cohesion, school friendships, neighborhood cohesion, and trust. These subscales are not appropriate for the assessment of social capital in sport because of the specific nature of an athletes' environment.

Another research questionnaire on the assessment of family social capital in adolescents was constructed by Carillo-Alvarez et al. (2019) which had 26 items assessing two dimensions. The questionnaire from that research focuses on communication in the family, food preparation, spending free time, and family cohesion and conflicts. The social network that surrounds tennis players has a significant influence on the development of the players and their tennis careers. The social network is formed by parents, wider family, coaches, friends, school, and sporting surroundings (Wolfenden and Holt, 2005), which is another reason why the newly constructed instrument used to assess social capital needs to focus on those key roles in an athletes' life.

This above mentioned research shows that some of the important characteristics for success within ten American Olympic champions to be: ability to control and endure anxiety, high self-esteem, resilience to psychological stress, sports intelligence, and work ethics (Gould et al., 2002). Those athletes also concluded that individuals and institutions such as community, family, school, sports staff and sport process, had significant influence on their development and growth. Social aspects of an athletes' sporting career need to be investigated in greater detail to have a better understanding of what makes

young athletes successful.

The beginning of a career in sport is an important period in which social capital can impact an athlete positively or negatively. Financial, logistic, and social support is pointed out to be important in young athletes (Rosso & McGrath, 2012). Social capital in young athletes is not yet well researched, and one of the reasons may be the lack of available and appropriate questionnaires.

According to Gould et al. (2006) coaches think that parents are important factors for success in junior tennis. Results show that most parents (59%) positively influence their child as an athlete. However, 36% of the parents were considered to negatively impact their children's sporting development. Positive influence implies financial, logistic, and social emotional support, with negative parental behavior being unrealistic expectations, criticism, and emphasis on winning. According to Neljak (1998) athletes must have a certain level of everyday life activity, such as school load, training and competition motivation, and quality family relationships that provide social support.

Novak et al. (2020) analyzed the connection of social capital with competitive success in tennis. They concluded that social capital successfully predicts sporting success in junior tennis. A higher level of social capital in family and sports teams helps young tennis players achieve better junior rankings. Quality training supplemented with an adequate level of social support will allow tennis players additional resources that could positively impact their sporting performance. The authors in the aforementioned study analyzed social capital in three different domains: family, school, and sports team social capital. The questionnaire used in the study consisted of seven questions (Furuta et al., 2012; Novak et al., 2015), and the determined CronbachAlpha coefficient was 0.71 and 0.81 respectively for school and sports team social capital scales, with this being similar to the newly constructed social capital questionnaire for young tennis players 0.71.

Social capital attracts attention as a potential influence on young people in different areas. Research in the field of education focused on the role of social capital in academic success (Coleman, 1987). However, research was expanded on health outcomes and behaviors. Research on social capital shows its impact on physical and psychological development in young people (Kawachi et al., 2010), and that social capital also relates to health and well-being (Engstrom et al., 2008).

Achieving sporting and academic success is a hard task, even when the person is extremely academically talented (Carodine et al. 2001). Romar (2012) concluded that 73% of young elite skiers think that sport negatively impacts their academic success. According to Clopton (2012) student athletes are better at setting long term goals than their nonathletic peers. Athletes show higher levels of social involvement and higher level of self-esteem.

Parents play a key role in the early phase of their child's involvement in sport, as they are the ones who enroll their children in sports programs and provide significant logistical, financial, and emotional support. Parental involvement and their support is of crucial importance in the development of young tennis players and their careers. Simultaneously, excessive parental involvement may have a negative impact on young athletes. Furthermore, parental involvement and support are related to self-esteem, quality of sport performance, and sport enjoyment (Gould et al., 2006).

One of the limitations of this study is that social capital assessed by the newly developed questionnaire measures only one dimension. However, it does include sport and tennis specific items that can deepen the understanding of social capital in young tennis competitors. This study can be seen as significant because previously validated questionnaires focus primarily on the general population, with no focus on sport and individuals who compete in tennis. In order to gain even more specific and beneficial results for this research topic, future studies should focus determining the differences in the level of social capital in tennis players of differing levels of suc-

cess and the relationship between social capital and competitive success.

Conclusion

Many factors impact young athletes during their development, with one key aspect being social components. To adequately assess certain social constructs, such as social capital, a validated measuring instrument is needed. By validating the social capital questionnaire created through this research, researchers will be able to further analyze and investigate the relationship between an individual's social capital and their competitive tennis.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Static- vs. Dynamic-Stretching in Development of the Dynamic Balance; Chronic Effects of Specific Training Interventions in Female Football Players

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Abstract

Flexibility is an important determinant of dynamic balance, but studies rarely examined the chronic effects of flexibility training on changes in balance capacity. This study aimed to determine the influence of different forms of flexibility exercises on dynamic balance in female professional football players. Participants were 30 female professional football players (age = 19.0 ± 4.1 years), divided into three groups dynamic stretching group (DSG; N=10, age = 18.9 ± 3.3 years), static stretching group (SSG; N=10, age = 19.5 ± 4.3 years) and control group (N=10, age = 18.5 ± 3.3 years). All groups participated in equal football training, while DSG and SSG performed additional flexibility training 3-times a week (10-15 min of duration) throughout the study course (16 weeks). Dynamic balance was measured by the Y-balance test (YBT), and ANOVA for repeated measurement was applied to identify the effects. Results evidenced the significant influence of both flexibility programs on dynamic balance. Both DSG and SSG were equally effective in the first part of the study (until the 8th week), but between the 8th and 16th study week, only DSG improved their balance capacity significantly. Static stretching should be applied in the first phases of flexibility training aiming at the improvement of dynamic balance, but in later phases, usage of dynamic stretching is suggested.

Keywords: soccer, dynamic balance, flexibility, intervention, females

Introduction

Balance appears as a term in many professions and each defines it within its specifics (Pollock, Durward, Rowe, & Paul, 2000). Viewed from the angle of sports performance, balance can be most simply defined as “keeping the center of gravity of the body within the base of support” (Hrysomallis, 2007). It is a key component of most motor skills and performances, regardless of the degree of complexity (Davlin, 2004). Generally, it can be divided into static, in which the balance is maintained with minimal movement, and dynamic, which implies the ability to perform a task while maintaining a stable position (Winter, Patla, & Frank, 1990; Bressel, Yonker, Kras, & Heath, 2007a).

Balance is an extremely important quality in sports,

not only because it can affect an athlete's efficiency but also because a balance deficit can cause injuries (Zemková, 2014; Viran & Canli, 2022). Although, in some sports, balance manifests in static conditions, dynamic balance is much more present in most team sports, including football. Dynamic balance in team sports is reflected in running, jumping, and many other specific movements (Zemková, 2014). Besides these biotic motoric abilities, during football activity, players perform passing, shooting and dribbling with lower extremities with cleated or noncleated shoes on variable turf conditions (Bressel, Yonker, Kras, & Heath, 2007b). These tasks and environmental conditions put specific demands on the sensorimotor system and affect the balance abilities of football players (Orchard, 2002). In these



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situations, maintaining balance is conditioned by: (i) sensorimotor information collected through the somatosensory, visual and vestibular system and (ii) motor responses to this information which are influenced by coordination, range of motion (ROM) and muscle strength (Grigg, 1994; Palmieri et al., 2003; Bressel et al., 2007a). Therefore, ROM should be observed as being directly related to balance.

The ROM of human joints is directly associated with muscle flexibility (Gleim & McHugh, 1997; Mohammad, Elattar, Elsaïs, & AlDajah, 2021). Indeed, flexibility training affects the improvement of many body functions, including movement control and increasing the elasticity of connective tissue and the manifestation of other motor skills (i.e. speed, strength, power, agility) (Alter, 2004; Fletcher, 2010; Ryan et al., 2014; Su, Chang, Wu, Guo, & Chu, 2017). There are several ways to develop flexibility, while static and dynamic stretching is most commonly used. Static stretching is described as the gradual elongation of a muscle through the static hold of an elongated position to a point of discomfort, while dynamic is defined as controlled movement through an active articular range of motion during which individual limits of flexibility are not exceeded (De Vries, 1962; Fletcher & Jones, 2004). Increased mobility of the joints and the amplitude of movement leads to more precise movement, better motor control and thus improvements in dynamic balance (Behm & Chaouachi, 2011).

The relationship between balance and flexibility so far has been investigated mainly in the elderly population, primarily because it influenced health status. In brief, a mutually positive impact between flexibility and balance has been found (Bird, Hill, Ball, & Williams, 2009; Emilio, Hita-Contreras, Jiménez-Lara, Latorre-Román, & Martínez-Amat, 2014). Moreover, studies have investigated the acute effects of different types of stretching on dynamic balance and the results are quite inconsistent (Costa, Graves, Whitehurst, & Jacobs, 2009; Behm & Chaouachi, 2011; Costa, Ruas, & Smith, 2018). However, most of these studies were short-term and/or cross-sectional and authors either cross-sectionally evaluated associations between flexibility and balance status, and/or determined acute effects of flexibility exercise on balance capacity (Curry, Chengkalath, Crouch, Romance, & Manns, 2009; Chatzopoulos, Galazoulas, Patikas, & Kotzamanidis, 2014). Meanwhile, there is a clear lack of research that has examined changes in the balance due to long-term longer-term specific flexibility treatment.

Collectively, despite the known importance of flexibility on dynamic balance, there is an evident lack of studies that directly examined the effects of flexibility training on changes in balance capacity. The knowledge of such kind would be particularly valuable knowing the importance of both capacities on sports performance and injury occurrence in sports. Although there are some studies on the male football population, studies on female athletes are particularly scarce. Additionally, studies reported different neuromuscular predictors of dynamic balance between genders, with core and hip strength and mobility being the most significant predictors in the female player (López-Valenciano, Ayala, De Ste Croix, Barbado, & Vera-Garcia, 2019). Therefore, the aim of this study was to determine the impact of different forms of flexibility exercises on dynamic balance in a sample of female football players.

Methods

Participants

Participants in this study were 36 female professional football players members of the female football club Široki Brijeg, from Bosnia and Herzegovina, all competing at the highest competitive rank in the country. Inclusion criteria for participating in the study consisted of (i) no injury to the musculoskeletal system within 3 months of the start of the study, (ii) age above 14 years and (iii) voluntary participation in the study. Participants were fully informed about the objectives and procedures of the study using video and oral presentations. All participants received written information on all procedures and gave their signed informed consent to participate. For participants younger than 18, consent was signed by a parent/guardian. Of the 36 football players initially involved, six did not meet the inclusion criteria so 30 players were randomly selected in three groups of 10 players. All three groups trained regularly throughout the study. The first experimental group (E1S; age: 19.5 ± 4.3 years; experience in football 6 years in minimum) participated in a supplementary static stretching/flexibility program (please see later for details), the second experimental group (E2D; age: 18.9 ± 3.3 years; experience in football 6 years in minimum) participated in a dynamic stretching/flexibility training program, while the control group (C; age: 18.5 ± 3.3 years; experience in football 6 years in minimum) did not participate in any kind of additional training program. Study was approved by Ethical Board of University of Split, Faculty of Kinesiology (Ethical board number: 2181-205-02-05-22-0021; June 10th, 2018).

Training procedures

Experimental training programs of E1S and E2D were conducted daily for 15 minutes after training sessions during the 16 weeks. All exercises (presented in Table 1) were explained verbally, and presented in pictures and written text to all participants. The exercises were conducted under the supervision of a strength and conditioning coach, three times per week, after the warm-up, before the main training session (mostly Mon-Wed-Fri) (McGill & Montel, 2019).

Participants were tested at the study baseline (Testing 1), after 8 weeks of intervention (Testing 2), and at the end of intervention (Testing 3). All three testing sessions were done in the same closed facility, at a temperature of approximately 20 degrees, using the same testing equipment, between 16:00 and 19:00. Before testing, participants performed a standardized warm-up consisting of jogging (3 min), and dynamic warm-up exercises (lunging, skipping, jumping, squatting) incorporated in jogging (3 min).

Variables

Variables in this study included anthropometric parameters (body height, body weight, body mass index and lower limb length) and YBT protocol.

In this study, we used the Y Balance test (YBT) as a test of dynamic balance. YBT is a variation of the Star Excursion Balance Test (SEBT) and has been repeatedly confirmed as a valid and reliable diagnostic tool for assessing dynamic balance (Plisky et al., 2009; Gribble, Hertel, & Plisky, 2012; Kim et al., 2020). YBT determines the asymmetries of lower limb movements and the level of balance through single-leg balance

Table 1. Types of stretching exercises over 3 months [sets x repetitions (hold)]

Static exercise (experimental group 1, n=10) [3x5 (15 sec)]	Dynamic exercise (experimental group 2, n=10) [3x10]	None (control group, n=10)
Standing quadriceps stretch	Downward dog to runner's lunge	
Gastrocnemius stretch	Dynamic squat stretch	
Standing split	Crescent to hamstring stretch	
Adductor stretch	Half-kneeling hamstring stretch	
Half-kneeling groin stretch	Side-to-side lunge with reach	
Hamstring stretch on the bench	Standing calf and hamstring stretch	
Prayer squat	Hurdles	
Seated adductor and hamstring stretch	Side-to-side lunge with step	Without stretching exercise
Hurdle seat	IT band stretch	
Seated glute stretch	Walking quad stretch	
Lying glute stretch	Swing the leg forward and back	
Seated groin abduction	Swing the leg cross	
Soleus stretch	Knee pull	
Half-kneeling quad stretch	Standing squat stretch	
Split hamstring stretch	Standing gluteus stretch	

with movements in the anterior (ANT), posteromedial (PM) and posterolateral (PL) direction (Smith, Chimera, & Warren, 2015). It is important to note that researchers have shown that performance on YBT can be a very good indicator of lower-extremity impairments and a predictor of future injuries in various sports (Plisky, Rauh, Kaminski, & Underwood, 2006; Pollock, 2010; Endo & Sakamoto, 2014; Alnahdi, Alderaa, Aldali, & Alsobayel, 2015). YBT consists of a three-part test used to assess lower extremity dynamic balance and neuromuscular control (Coughlan, Fullam, Delahunt, Gissane, & Caulfield, 2012). A YBT kit (Perform Better, FMS) was used, consisting of three connected cylindrical tubular wooden bars marked in centimeters. Each bar has a movable indicator plate, which is pushed with the feet/toes without relying on the weight of the indicator. Before the test, participants conducted three minutes warm up and performed the test without footwear.

The participants were allowed to have six trial attempts on each leg in each of the three directions – anterior, posteromedial and posterolateral - before formal testing. Starting position included standing on one leg (which is not graded) in the center of the platform while the leg being graded needs to remain in the air. While maintaining balance with

one leg, the participants were instructed to reach three attempts with the free leg in the anterior direction, followed by three attempts in the posteromedial direction, and then three attempts in the posterolateral direction, all according to the YBT protocol. The football player was instructed to move the distance indicator as far as possible in the direction being assessed. The researcher monitored testing and did not allow participants to move the indicator by hitting it or accelerating at the end of the push. The maximum reach distance was recorded at the farthest point reached by the foot at the proximal edge of the indicator and was measured at half a centimeter. The trial was rejected for one or more of the following reasons: if the participant (i) lost his balance during the exercise, (ii) raised the heel of the foot which was on the platform, (iii) if the foot did not maintain contact with the distance indicator while the indicator was in motion (e.g. the indicator was kicked), (iv) if the distance indicator was used to maintain posture body (e.g. the athlete leaned his weight on the distance indicator) or (v) there was a loss of balance during the return to the starting position after the distance was marked (Shaffer et al., 2013).

The study design and testing sequences are presented in Figure 1.

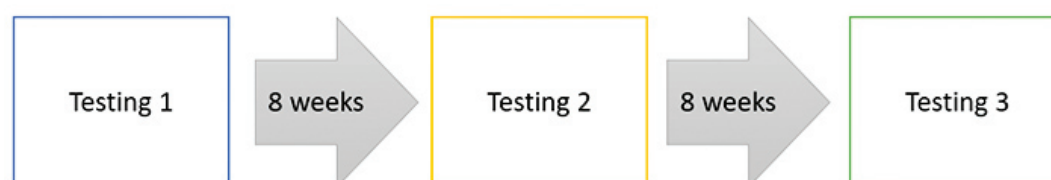


FIGURE 1. Study design with time-frames

Statistics

All data were log-transformed to reduce the non-uniformity of error, and normality was tested using the Kolmogorov-Smirnov test procedure. Homoscedasticity was checked by the Levene test. The statistical analyses were done on log-transformed data, but results in tables and figures are presented as

true-value means and standard deviations.

The ANOVA for repeated measurements (Group x Measurement) was calculated to evaluate the effects of the applied programs, with consecutive, posthoc Scheffé test.

Statistica ver. 13.5 (Tibco Inc. Palo Alto, CA) was used for all calculations, and a p-level of 95% was applied.

Results

Descriptive statistics for general variables are presented in Table 2.

Results of the 2-way analysis of the variance for repeated measurement with “Group” and “Measurement” as main effects and “Group x Measurement” interaction are presented in Table

Table 2. Descriptive statistics of the studied sample of participants for general variables

	Mean	Minimum	Maximum	Std.Dev.
Age (years)	19.13	14.00	30.00	4.52
Body mass (kg)	59.10	40.00	77.00	9.14
Body height (cm)	165.42	155.00	175.00	5.40
BMI (kg/m ²)	21.58	14.30	27.20	3.07
Leg length right (cm)	86.28	76.00	97.00	4.75
Leg length left (cm)	86.28	76.00	97.00	4.75

3. In brief, significant main effects for “Group” were found for Left posteromedial, Right postero-lateral, and Left postero-lateral. Significant effects for “Measurement” were evidenced for

Left posteromedial and Right postero-lateral results. Finally, the “Group x Measurement” effect was significant for the Right posteromedial, and Right leg composite scores.

Table 3. Descriptive statistics and results of the 2-way analysis of the variance for repeated measurements (*denotes statistical significance of $p < 0.05$)

	Test 1	Test 2	Test 3	ANOVA		
				Group	Measurement	Interaction
	Mean±SD	Mean±SD	Mean±SD	F-test	F-test	F-test
Right leg anterior	60.2±7.01	60.83±5.86	60.84±7.07	0.31	2.68	1.34
Left leg anterior	59.71±6.27	60.88±6.94	60.95±7.01	0.11	1.72	1.06
Right leg posteromedial	107.78±12.54	108.9±10.49	108.92±12.65	2.42	2.1	2.85*
Left leg posteromedial	106.89±11.22	109±12.43	109.11±12.54	4.03*	3.54*	1.75
Right leg posterolateral	106.81±12.43	107.92±10.4	107.94±12.54	3.85*	4.01*	2.08
Left leg posterolateral	105.93±11.12	108.02±12.32	108.13±12.43	4.43*	1.53	0.8
Right leg composite score	97.1±11.3	98.11±9.45	98.13±11.4	2.08	1.67	2.91*
Left leg composite score	96.3±10.11	98.2±11.2	98.3±11.3	1.21	1.59	2.41*

A more detailed overview of the changes which occurred in composite scores is evidenced in Figure 2 and Figure 3. Specifically, both experimental programs induced positive changes in Right leg composite scores between Test 1 and Test 2 (posthoc differences within E2D and E1S were statistically sig-

nificant and $p < 0.05$). However, only E2D achieved significant improvement between Test 2 and Test 3 (at $p < 0.05$) (Figure 2). In the left leg composite score, E2D improved results from Test 1 to Test 2 ($p < 0.05$), with no significant posthoc differences between Test 2 and Test 3 for any of the observed groups (Figure 3).

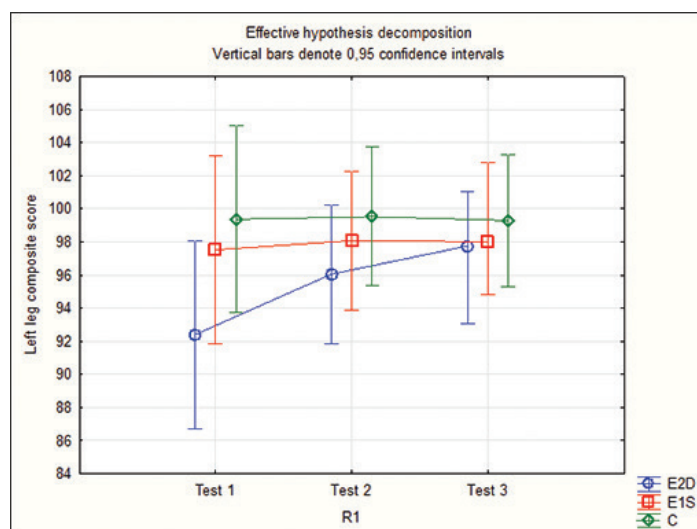


FIGURE 2. Descriptive results (Mean±95%CI) for right leg composite score in each study group – changes over the study course (Test 1, Test 2, Test 3); Legend: E2D – dynamic stretching experimental group, E1S – static stretching experimental group, C – control group

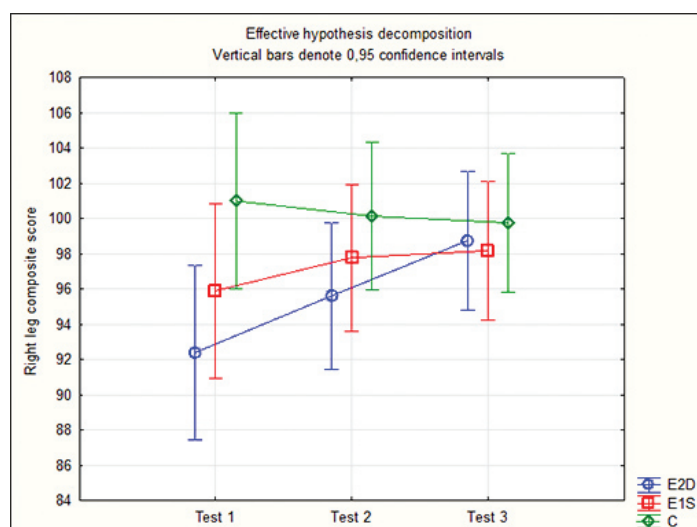


FIGURE 3. Descriptive results (Mean±95%CI) for right leg composite score in each study group – changes over the study course (Test 1, Test 2, Test 3); Legend: E2D – dynamic stretching experimental group, E1S – static stretching experimental group, C – control group

Discussion

This study aimed to evaluate the effects of different types of flexibility training on changes in dynamic balance among female football players. With regard to study aims, we can highlight two most important findings. First, our results showed that flexibility training irrespective of the type (e.g. static or dynamic stretching) has a positive impact on the development of dynamic balance. Second, while both dynamic and static stretching had a similar positive influence on dynamic flexibility in the first period, the impact of static stretching stagnates, while dynamic flexibility exercise improved the dynamic balance over the whole study course.

Flexibility training improves dynamic balance

Both static and dynamic stretching exercises improved dynamic balance significantly. As introduced, balance can be defined as the ability to maintain the center of gravity of the body within the base of support which is performed through neuromuscular actions conditioned by external stimuli (Con Hrysomallis, 2007). This is particularly important knowing the protective influence of improved balance on injury occurrence, and balance is generally considered an important factor contributing to the incidence of injuries in sports, with better balance preventing injury occurrence (Hrysomallis, 2007; Gribble, Hertel, & Plisky, 2012; Han, Anson, Waddington, Adams, & Liu, 2015; Lai et al., 2017; Sekulic, Prus, Zevrnja, Peric, & Zaletel, 2020). Increased mobility and amplitude in the joints leads to more precise movement, better control and thus improvements in dynamic balance (Weirich, Bembem, & Bembem, 2010; Liška, Švantner, & Batalík, 2021). Looking at the movement structure in YBT, it is clear that better mobility of the lower extremities will have a significant impact on the maximal reach and thus on the final result in the test.

Supportively, previous studies evidenced the positive relationship between joint kinematics and performance on the YTB test. Specifically, investigations found that dorsiflexion of the foot, ankle and hip flexion should be observed as significant predictors of YBT results (Robinson & Gribble, 2008; Kang et al., 2015). Therefore, it seems that performing flexibility exercises improves mobility which leads to easier control of the body in the maximal range on balance tests, resulting

in better YBT even in our players. Generally, our results on the positive impact of static stretching on dynamic balance confirmed the results of previous studies (Behm, Bambury, Cahill, & Power, 2004; Costa, Graves, Whitehurst, & Jacobs, 2009; Lewis, Brismée, James, Sizer, & Sawyer, 2009). However, the impact of dynamic stretching is relatively unexplored, and almost exclusively evaluated the acute effects of dynamic stretching on dynamic balance (Wang, 2013; Chatzopoulos, Galazoulas, Patikas, & Kotzamanidis, 2014; Amiri-Khorasani & Gulick, 2015). In general, these studies confirmed the positive acute effect of stretching, especially dynamic, on balance performance and recommend incorporating stretching activities in the warm-up protocols (Amiri-Khorasani & Gulick, 2015). Finally, no study compared the effects of different stretching forms on dynamic balance in professional athletes.

The physiological mechanism behind these results can be explained by the extension of muscle and connective tissue (lengthening) through mechanical and neural adaptations through stretching exercises (Guissard & Duchateau, 2004; Shrier, 2004). The scientific literature has repeatedly confirmed the usefulness of static stretching on ROM development (Bandy, Irion, & Briggler, 1997, 1998; Feland, Myrer, Schulthies, Fellingham, & Measom, 2001), while maintaining the center of gravity within the base of support will certainly be facilitated if the individual can have a greater ROM in the joints. This is especially emphasized in the YBT, where the athlete because of the better hip ROM, can lower her center of gravity of the body more, and consequently is more able to maintain a balance position (Kang et al., 2015).

Differences in the duration of the effect of static and dynamic stretching

Results of our study show that in the first phase of training both stretching methods have approximately equal positive effects on the development of flexibility and consequently the ROM. However, in the second part of the experiment (for the 8th to 16th week of the program) effects of static stretching reached the plateau, and for further progress, it is necessary to move to more aggressive forms of stretching like dynamic stretching. The explanation of such results should be found in specifics of applied stretching programs.

In the flexibility training methodology, it is generally accepted that the static type of stretching should be used in the first phases as it is less intense and risky (Clark & Lucett, 2010). However, it is relatively well known that once an athlete has mastered static stretching techniques, it is necessary to continue to increase the ROM through functional dynamic movements, which preferably mimic the movement structure of a particular sport (Mann & Jones, 1999). This type of training will increase the amplitude of the movement when performing some more or less complex movements. Given that performance in YTB involves a dynamic component, of both balance and flexibility, through reaching the end position with active movement, it becomes clear why dynamic stretching provided better results in developing balance and flexibility than static stretching.

Concerning the superior effects of dynamic stretching, another issue deserves attention. Research has shown that not all muscle groups respond equally to static stretching, and therefore stretching protocols should differ between muscles (Youdas, Krause, Egan, Thernau, & Laskowski, 2003). It has also been found that through more aggressive types of stretching, such as active and neuromuscular stretching, greater improvements in ROM can be achieved (Ettnyre & Lee, 1988; Fasen et al., 2009; Schuback, Hooper, & Salisbury, 2004). This is in accordance with our findings indicating that after a certain period of time (8 weeks) the effect of static stretching on balance cannot be expected.

The main limitation of this study is the fact that only one test was used to assess dynamic balance. Also, in future studies, additional anthropometric measures should be included. One of the limitations of the study was a small sub-sample of the participants, but this was mainly influenced by the number

of players in the observed club. However, this was one of the first studies which investigated the chronic effects of different forms of stretching on dynamic balance. The long duration of the intervention (16 weeks), three groups of athletes and the fact that this is one of the rare intervention studies on female football players are however important strengths of the study.

Conclusion

The results of this study showed the positive effects of both static and dynamic stretching on dynamic balance improvements. This finding is very important in the context of injury prevention in football as better balance is a protective factor for injury occurrence.

Additionally, the impact of both types of stretching over time is also an important finding of this study. In detail, in the first phase of the systematic implementation of flexibility training, static and dynamic stretching achieve almost identical effects. However, while the impact of static stretching weakens over time, progress in balance is possible over a longer period as a result of practicing dynamic stretching. Considering this, we can suggest that football players should start with systematic static stretching training and after a certain time, when adaptations have occurred, move on to more advanced and demanding types of stretching. It is also very important to imitate sport-specific movements in order to have a higher transfer on sports performance.

Regular and systematic flexibility training will result in a greater range of motion, which will enable soccer players to achieve greater movement amplitudes. This will affect a better adaptation to new situations in which the balance is disturbed and will help in both, preventing injuries and in a more optimal and precise sports performance.

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There are no acknowledgments.

Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

The Association between Screen Time and Low Back Pain among Male College Students

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Abstract

Low back pain (LBP) is the most common musculoskeletal complaint. LBP affects all age groups, including college students. In the digital and communication era, computers and gadgets are ordinary. Excessive computer/gadget use can produce musculoskeletal work imbalance leading to LBP. Several risk factors are also linked to LBP, such as flexibility, strength muscle, body mass index (BMI), and standing posture. Thus, our study aims to investigate the association between LBP, screen time, and risk factors in male college students. This study was cross-sectional, involving 132 students. Height and weight were measured, and BMI was obtained. LBP was assessed using the Nordic Musculoskeletal Questionnaire (NMQ), and the duration of computer use was obtained using a questionnaire. Back muscle flexibility was measured using a modified sit and reach test; back muscle strength was measured using Back-Leg Dynamometer. Posture was assessed using the New York Posture Rating Chart (NYPRC). All variables were categorized into two groups. Fifty participants (37.9%) had LBP. Forty-six participants (34.8%) use computer/gadget excessively. The screen time was associated with LBP ($p=0.04$, OR 2.19, 95%CI 1.05-4.56). No relationship was found between flexibility, back muscle strength, BMI, and posture. Screen time is associated with LBP in male colleges. Excessive screen time had a 2.19 times greater risk of having LBP. This study conclude that excessive screen time is associated with LBP in male college students.

Keywords: *flexibility, back muscle strength, body mass index, standing posture, prolonged sitting*

Introduction

Low back pain (LBP) is pain that lies between the lower border of the ribs and the gluteal folds due to known or unknown causes. LBP is rare in children ten years or younger. However, as in adults, the prevalence of LBP in childhood increases with age and is more prevalent in girls than in boys (Brattberg & Wickman, 1992; Troussier, Davoine, de Gaudemaris, Fauconnier, & Philip, 1994; Burton, Clarke, McClune, & Tillotson, 1996; Grimmer & Williams, 2000). Lifetime prevalence of LBP was reported as much as 56.6%, 12-month prevalence 48.8%, and point prevalence of 21.2% in health science college students (Al Shayhan & Saadeddin, 2018). Whereas the year-prevalence of LBP among physiotherapy students in Brazil was about 5.3% (Morais, Silva, & da Silva, 2018), a higher prevalence of LBP was reported among college students in Indonesia, as many as 74.6% (Anggiat, Hon, & Baait, 2018).

LBP is associated with several risk factors. Intrinsic factors have been identified as contributing factors to low back pain. The smaller size (measured by cross-sectional area/CSA) of the lumbar multifidus (MF) and erector spinae were considered more prone to pain in the lumbar area (Goubert et al., 2018). The smaller lumbar muscle CSA was related to LBP's chronicity (Lee, Song, Lee, Kang, Kim, & Ryu, 2011). LBP was also linked with muscle weakness in older women (Kato et al., 2021). Besides, low flexibility was identified as the cause, influencing factor, and consequence of LBP (Ruas & Vieira, 2017).

In addition to intrinsic factors, several external factors are identified as influencing factors of LBP. Impaired neuromuscular control, for example, has been suggested to increase the likelihood of LBP (Cholewicki et al., 2005; Catalá, Schroll, Laube, & Arampatzis, 2018). It has been reported that decreased mobility function (locomotive syndrome) contributed to LBP (Kato et al.,



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2021). Physical activity appears to have an association with LBP. A meta-analysis study indicated that participants with higher physical activity had a low prevalence of LBP (Alzahrani, Mackey, Stamatakis, Zadro, & Shirley, 2019).

Recent studies exhibit that LBP is associated with inappropriate body positions while working for long periods. Sitting had been suspected to increase the likelihood of having occupational LBP (Lis, Black, Korn, & Nordin, 2007). Furthermore, static sitting behavior increases the risk of LBP (Bontrup et al., 2019). Sitting duration has also been acknowledged as a risk factor for LBP (Keskin, Ürkmez, Öztürk, Kepekçi, & Aydın, 2021). A study reported that most adolescents with excessive computer use and video game experienced LBP (Silva, Pitangui, Xavier, Correia-Júnior, & De Araújo, 2016). Therefore, our study purpose to investigate the association between excessive screen time and low back pain among male college students and the role of risk factors.

Material and Methods

Study design and subjects

This study was cross-sectional with a qualitative analytic design, followed by 132 male participants (20.5±1.2 years). The sample selection was performed using a purposive sampling method. The inclusion criteria were male students of the School of Medicine and Health Sciences, Atma Jaya Catholic University of Indonesia. Exclusion criteria were female, history of back trauma, low back pain due to evident causes, history of back surgery, ongoing LBP therapy, and back muscle training. The study was approved by the Ethics Committee of Atma Jaya Catholic University of Indonesia (40/11/KEP-FKUAJ/2019).

Questionnaire Data

Screen time (duration of gadget/computer use) was asked of participants using an open question. The duration of computer use during weekdays and weekend were recorded. The results were averaged in hours per day. Based on the results, screen time was classified into 'normal' use if <7 hours/day and 'excessive' if ≥7 hours/day (Trinh, Wong, & Faulkner, 2015).

Low back pain was assessed using the Nordic Musculoskeletal Questionnaire (NMQ), developed from a project funded by the Nordic Council of Ministers (Kuorinka et al., 1987). NMQ consists of two sections. Section 1 is a general questionnaire with forty items to identify body area-producing musculoskeletal problems. This section was equipped with a body map to locate nine symptoms: neck, shoulders, upper back, elbows, low back, wrist/hands, hips/thighs, knees, and ankles/feet, and a pain scale from 1 to 10. Participants were asked about any musculoskeletal complaints in the last 12 months and seven days that interfered with their regular activity (Kuorinka et al., 1987).

Section 2 is additional questions for further detailed information about the body area. This section consists of twenty-five items that elicit accidents affecting each area, activity disturbances at home and work (change of job or duties), duration of the problem, seeking treatment from a health professional, and musculoskeletal problems in the last seven days (Kuorinka et al., 1987).

The Indonesian version of NMQ had been validated and produced a satisfactory result. The validity items obtained range from 0.501 (min.) to 0.823 (max.), with Cronbach's alpha reliability being 0.726 (Ramdan, Duma, & Setyowati, 2019).

Measurements

Body height and weight were measured using standard procedures. Height was measured using a microtoise in the Frankfort

position, without wearing shoes or socks. Weight was measured using digital scales. A standard formula obtained the body mass index (BMI) value, weight (kg) divided by the square of height in meters (m). BMI was classified as 'obese' if BMI ≥ 25 kg/m² and 'normal' if it < 25 kg/m², according to the standard for Asian people (Pan & Yeh, 2008).

Back muscle flexibility was examined by the modified sit and reach test method using a Takei digital anteflexion meter (TKK 5403, Takei Scientific Instruments Co.Ltd, Niigata, Japan). Subjects were encouraged to warm up, stretch before the examination, and remove their shoes and socks. Back flexibility was measured in a sitting position. Both legs extended forward and feet flat against the standard box. The Toe line (zero point) is at the 15 cm mark (Eurofit, 1993). Both palms face down while the fingertips of both hands touch and slide the display forward with one smooth movement as far as possible. The subject held the position for two seconds. The participants complete the test twice. The best of the two scores was recorded. The results were classified as 'good' if ≥ 14 cm and 'poor' if < 14. This classification was modified based on normative results for males aged 20-29 from the American College of Sports Medicine (ACSM) (ACSM, 2018).

Back muscle strength was assessed using a digital back muscle dynamometer (Takei T.K.K. 5402, Takei Scientific Instruments Co., Ltd, Niigata, Japan). The subject stood upright on the base of the dynamometer with feet shoulder-width apart. Arms straight down to hold the bar, connected to the dynamometer with chain, with both hands. The chain length was adjusted so that the subject's trunks were flexed while the knees were extended. The subject pulls the bar as hard as possible with arms kept straight for 3-5 seconds. Subjects performed the test twice with a 60-second rest between trials. The highest score was recorded. Back muscle strength was classified into two groups based on average value (61.77 kg). 'Good' strength if ≥ 62 kg (rounded from average value), and 'poor' if < 62 kg.

Posture was assessed using the New York Posture Rating Chart (NYPRC) by Howley, and Franks was applied (Howley & Franks, 1992). Ten body segments were observed and checked, five segments from behind (posterior) and five segments from the side (lateral). Three scale was applied to scoring each segment; '10' for 'normal', '5' for 'mild deviation', and '0' for 'marked deviation'. The maximal score indicating normal posture of all segments from both views was 100. Posture was classified into 'normal' if there were no zero scores at one or more body segments and 'poor' if there was a score of zero at least in one segment.

Statistical analysis

Numeric data were presented in the form of mean and standard deviation. Categorical data was displayed in terms of frequency and percentage. Numerical comparisons between LBP and non-LBP were tested by unpaired T-test. The association between risk factors and LBP was tested with Chi-Square. The level of significance was determined when p < 0.05. Data processing and statistical tests were performed using SPSS 19 (Chicago, Illinois).

Results

Table 1. demonstrates the characteristics of the participants. Numeric and categorical data were presented. Obesity was found in sixty-two participants (47%). There were nineteen (14.4%) had 'poor' flexibility, seventy-four (56.1%) had 'poor' strength, and forty-five (34.1%) had 'poor' posture. Excessive use of computers was found in forty-six participants (34.8%), while fifty (37.9%) reported having LBP.

Table 1. Characteristics of participants

Variables	Mean±SD or number (%)
Age (years)	20.5±1.2
Height (kg)	171.4±5.3
Weight (cm)	72.8±12.3
BMI (kg/m ²)	24.7±3.7
Normoweight	70 (53.0%)
Obesity	62 (47.0%)
Back Flexibility (cm)	22.5±8.2
Good	113 (85.6%)
Poor	19 (14.4%)
Back Strength (kg)	61.6±9.0
Good	58 (43.9%)
Poor	74 (56.1%)
Screen Time (hours/day)	5.0±2.3
Normal (<7 hours/day)	86 (65.2%)
Excessive (≥7 hours/day)	46 (34.8%)
Total Score of NYPR score (lateral view)	46.4±5.4
Total Score of NYPR score (posterior view)	41.3±8.6
Overall score of NYPR	87.7±13.9
Normal posture	87 (65.9%)
Poor posture	45 (34.1%)
Low back pain	
No	82 (62.1%)
Yes	50 (37.9%)

Abbreviation: BMI, body mass index; NYPR, New York Posture Rating

The comparison of the variables between participants with and without LBP in numeric data is described in table 2. None of the variables were comparable (all p was >0.05).

The association between LBP and the variables is shown in table 3. LBP was related with posture ($p=0.03$, OR 0.32,

95%CI 0.15-0.68), and screen time ($p=0.04$, OR 2.19, 95%CI 1.05-4.56). Good posture was less likely to have LBP (0.32 times than poor posture), while excessive screen time was more likely to have LBP (2.19 times than normal screen time).

Table 2. The comparison of the variables between LBP vs non-LBP

Variables	No LBP	LBP	p
Height (cm)	171.5±5.7	171.1±4.6	0.07
Weight (kg)	72.8±12.0	72.8±11.1	0.35
BMI (kg/m ²)	24.7±3.9	24.8±3.4	0.31
Flexibility (cm)	22.9±8.8	21.7±7.0	0.07
Back Strength (kg)	62.3±18.9	60.4±17.9	0.39
Screen time (hr/day)	4.7±2.3	5.6±2.3	0.56
NYPR score	87.9±11.2	87.3±10.3	0.20

Abbreviation: BMI, body mass index; NYPR, New York Posture Rating

Table 3. The association between LBP and risk factors

	BMI	Flexibility	Strength	Screen time	NYPR score
Low back pain					
p	0.86	0.54	0.15	0.04	0.07
OR	0.94	1.38	0.59	2.19	0.32
95% CI	0.46-1.90	0.49-3.90	0.29-1.21	1.05-4.56	0.15-1.07

Abbreviation: BMI, body mass index; NYPR, New York Posture Rating

Discussion

Low back pain is one of the most complaints and disorders affecting many people worldwide and all age groups (Wu et al., 2020). LBP has many causes, from muscle origin to more severe diseases such as malignancy (Downie et al., 2013; Kato et al., 2021). The recent study attempted to identify LBP originating from the muscular system and link it with possible risk factors. This study found that LBP was associated with poor posture and excessive screen time. Flexibility, muscle back strength, and BMI were not related to LBP.

Computer use and its association with muscle pain have been investigated. Several previous studies supported our findings. A study on Lithuanian adolescents reported that back pain was more pronounced as the duration of computer use increased (Skemiene, Ustinaviciene, Luksiene, Radisauskas, & Kaliniene, 2012). Moderate to severe musculoskeletal symptoms (of the head, neck, shoulder, low back, eyes, fingers, and wrist) were also linked with computer use among Finnish adolescents (Hakala et al., 2012). Computer use-related pain might be related more to sitting. Previous studies reported the association between sitting and the presence of LBP. A systematic review concluded that sitting was related to LBP. However, they found that sitting in combination with other factors, such as awkward posture, increased the risk of LBP (Lis et al., 2007). Besides the duration, the sitting position also determined the occurrence of LBP (Casas, Patiño, & Camargo, 2016; Zywień, Barczyk-Pawełec, & Sipko, 2022).

The mechanism of LBP due to sitting had been proposed. A sitting position increases the load borne by the spine, which creates pressure on the intervertebral discs greater than standing or lying down (Lis et al., 2007). Further, when sitting, there is a decrease in the degree of lumbar lordosis and sacral slope compared to standing. Lumbar lordosis and sacral slope changes can cause spinopelvic imbalance and lead to chronic low back pain (Casas et al., 2016).

The information about static standing posture and its linkage to LBP is limited due to a lack of investigations. Our findings showed no association between posture and LBP. A review study stated that the evidence for the relationship between posture and LBP was miserable (Kripa & Kaur, 2021). Putri et al. reported that the association between posture and LBP was not evident (Putri, Citrawati, & Astari, 2021). However, when including working or habitual posture and duration of the posture, the linkage with LBP was noticeable. A study reported that standing time was positively associated with LBP. Longer standing time increased the possibility of LBP (de Souza et al., 2022). Longer standing time was hypothesized to increase intervertebral disc compressive force that could promote the occurrence of LBP (Hasegawa, Katsuhira, Oka, Fujii, & Matsudaira, 2018).

Our study did not confirm the association between back muscle strength. However, some previous studies established

the association. Cho et al. reported that trunk muscle weakness was a risk factor for LBP (Cho et al., 2014). The linkage between trunk muscle and LBP was also confirmed in older women (Kato et al., 2021). Besides trunks muscle, limb muscle may have an association with LBP. A study in Brazil found that lower limb muscle strength was weaker than healthy controls (de Sousa et al., 2019). We supposed that our study's cut-off value for strength classification might affect the association with LBP.

The association between flexibility and LBP is still inconclusive. Our recent study did not find a relationship between flexibility and LBP. Some studies supported this finding. The direct association between LBP and low flexibility was unclear (Ruas & Vieira, 2017). In their study, Sweeney et al., which involved female gymnasts, found that limited joint flexibility might not be associated with LBP (Sweeney, Daoud, Potter, Ritchie, & Howell, 2019). Even stretching, an exercise to improve flexibility, was linked to an increased risk of LBP (Sandler et al., 2014). However, a study examining straight leg raise and popliteal angle reported that dysfunction of LBP was related to posterior chain flexibility (da Silva, Ferraz, Ferretti, & Sfredo, 2017). The discrepancy between studies' findings could be caused by many determinants, such as flexibility measurement methods, examination of muscle flexibility, subject characteristics, sample size, etc.

The association between BMI and LBP was established by earlier studies (Keskin et al., 2021; Zywień et al., 2022). However, our findings failed to justify. The distinction could be due to different ages. As in our study, high BMI at a younger age might express high muscle rather than high fat. LBP in older age is linked to lumbar fat infiltration (Goubert et al., 2018). LBP is also connected to increased load in the lumbar in obesity (O'Sullivan, Mitchell, Bulich, Waller, & Holte, 2006).

Several shortcomings of our study existed. First, school bag weight and how to carry it was not recorded. These variables might increase the development of LBP. Second, sitting posture was not included, whereas it considerably impacts LBP. Third, no reference value to determine the cut-off of poor and good posture based on an overall score of the NYPRS. This reference value could help us to distinguish poor posture from good posture. Fourth, the sample size might not be large enough to produce a significant result. Despite these limitations, our study is valuable because it provides further information about the association between excessive screen time and LBP.

Conclusion

The association between screen time and LBP in male college students was confirmed. Excessive screen time (7 hours/day or more) had a 2.19 times greater risk of LBP. However, LBP was not related to back muscle flexibility and strength, static standing posture, and BMI. Further, we suggest that these findings be interpreted cautiously due to several limitations.

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Conflicts of interest

None to declare.

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ORIGINAL SCIENTIFIC PAPER

Relationship between Offensive Modalities and Results in Elite Basketball Games

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Abstract

Basketball is characterized by a set of alternating offensive and defensive phases generally limited by the change in ball possession, also known as conversion. The aim of this research was to evaluate the difference in offensive types between the winning and losing, noting the variance in game result. A sample composed of 2607 entities representing basketball offense was collected by random selection of 15 Euroleague playoff games. The collected data were processed by the Match Analysis System, and the offense was classified into three basic types: set, transition, and early offense. With nearly equal opponents there is evident a high dynamic balance of transitions, set offenses, as well as other forms of offense, therefore the initial results showed no statistical difference in the distribution and success of offensive forms between opponents in balanced matches. There is significant change evident when focusing on the game periods with the highest point differential. In these periods the winning teams showed above 20% more fast breaks, which is proved to be the most effective type of offense, in addition to significantly better 2-point and 3-point shooting efficiency during set plays. Even though the losing team in balanced matches successfully controlled most of the game, and perhaps even outplayed the opponent, the identified critical intervals ("scoring runs") proved to be a decisive factor in the final outcome of the game.

Keywords: basketball, offensive structure, Euroleague, scoring run, set offense, transition offense, early offense

Introduction

The primary task of scientific research related to the analysis of sports activities is a better understanding of the functional principles and a better practical application. In this regard, basketball proves especially noteworthy, as its popularity and complexity have made it one of the most frequently analyzed sports (Mikolajec, Maszczyk, & Zajac, 2013). Basketball is based both on the cooperation of teammates or the integrated roles of all five players with the aim of preventing the critical intervals of play, and the opposition, meaning the cooperative attempt to disrupt the dynamic balance of the opponent (Trninić, Kardum, & Mlačić, 2010). From a structural point of view, the game is characterized by a series of alternating offensive and defensive phases, which are generally limited by the change of ball possession - conversion. The offensive and defensive components reveal specific classifiable profiles and can therefore be divided into two basic game states - position and transition. The position (set) of

fense/defense and the transition (fast break) offense/defense are two basic states are complemented by the classification of other types and variants of offense and other associated modalities.

The classification of offense types in basketball literature is not strictly defined nor generally consistent. Previous research (Tavares & Gomes, 2003; Bazanov et al., 2006; Ortega et al., 2006; Fewell et al., 2012), highlights different classifications of offenses, however their origin is conclusively based on the state of transition and position. Trninić, Perica, and Pavičić (1994), based on a kinematic approach, introduced the "game states analysis system", and presented the initial, intermediate, and final modalities of set and transition offense in order to determine their balance and dependence towards success in a basketball game. However, there is not enough evidence to examine the differences between the winning and losing teams based on the distribution analysis and efficiency of the basic offensive game types.

The quality of implementation of these two game states is



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a valid indicator of efficiency, therefore, the objective of this study is to examine the distribution of offensive types in accordance with the final score and their variance among different game periods between winning and losing teams.

Methods

Subjects sample

The entity in this study is a basketball offense. A sample of 2607 entities representing were collected by randomly selecting 15 Euroleague playoff games from the 2010-11 season. The analysis included only games with balanced opponents because they prove to better identify key game-related statistics associated with winning and losing (Gómez et al., 2014; Lupo et al., 2014). The type of offense was evaluated in relation to the entire game, as well as in relation to the quarters with the highest score difference. The analysis was completed by examining the offensive outcome.

The beginning or end of the offensive phase was not determined by ball possession, but by the rules of basketball (Selmanović et al., 2015), resulting in three different offensive outcomes: positive, negative, and neutral. A positive outcome means at least one point was scored. A negative outcome refers to unsuccessful shot and lost ball possession. Neutral means unsuccessful shot and regaining possession of the ball. For each of these outcomes, the corresponding modalities were tested.

Procedure

In this work, the following four types of offenses are categorized and analyzed: set offense, transition offense, early offense, and other (miscellaneous) types. According to Škegro et al. (2011), the set offense is executed either by running a set play only or by combining the set play with the with the transition, in which case the set play lasts considerably longer than

the transition and involves the defense that fully controls all five offensive players. In the transition offense, the defense has no control over the offensive players due to unfavorable position in the counterbalance of offensive action. The early offense is characterized by a brief play setup and attempted shot before the defenders can establish the proper defensive positions, even though all defensive players are strictly positioned in the front-court. The category of other types does not classify an offense into any of the above categories, therefore, they are not included in the statistical analysis because their realization has remote and no significant impact on the final outcome of the offense. These operational definitions were set as criteria for the video game analysis. The collected data were processed using the program Match Analysis System (MAS), which contains analytic tools for a complete description of the offensive characteristics.

Statistical analysis

The basic features of offensive actions were tested in relative frequency values per game as a whole, and then selected according to the periods (quarters) that showed the greatest difference in results. Differences in qualitative variables between the winning and losing teams were tested using the nonparametric Hi-squared test (χ^2) at $p < 0.05$ significance level. Data was analyzed using the statistical package Statistica ver. 8.0.

Results

According to the set criteria, the study showed that there are about 175 offensive phases per game at the highest level of European competition. Regarding the basic types of offense in the result-balanced matches, there are no statistically significant differences between the winners and the defeated ($X^2=1.915$; $p=0.590$).

Table 1. The difference in the offense distribution in the result-balanced matches

	Set	Early	Transition	Other
Winners	68.28%	5.44%	8.89%	17.39%
Defeated	68.31%	6.37%	9.37%	15.95%

$X^2=1.915$; $df=3$; $p=.590$

Note: X^2 - Chi-square test value; df - degrees of freedom; p - level of significance

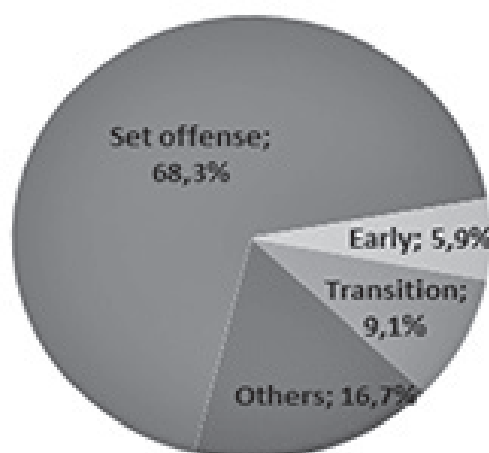


FIGURE 1. Distribution of basic offensive types

Transition offenses assure the highest productivity (Tsamourtzis, Karypidis, & Athanasiou, 2005), but they account for less than 10% of the total share of offense (Figure 1). The set offenses are predominant and account for 2/3 of

the total number of offensive actions, indicating their significant influence on the final outcome of the match. Although the winning teams were expected to be much more efficient, the parameters in Table 2 show no significant difference in the

Table 2. Difference in the outcome of certain types of offenses in the result-balanced matches

	Set offense			Early offense			Transition offense		
Outcome	% POS	% NEG	% NEU	% POS	% NEG	% NEU	% POS	% NEG	% NEU
Winners	38.3	43.1	18,6	43.4	40.9	15.7	53.5	29.3	17.2
Defeated	35.1	46.3	18,6	54.9	33.8	11.3	41.0	37.7	21.3
chi square	$X^2=2.274$; df=2; p=.321			$X^2=2.112$; df=2; p=.348			$X^2=3.719$; df=2; p=.156		

Note: X^2 - Chi-square test value; df - degrees of freedom; p - level of significance

realization of set offense in a match between two equal opponents.

The score difference after each quarter was used as criterion to group the periods and analyze the distribution and out-

come of offenses between winning and losing teams. Although there are no drastic deviations regarding the game as a whole, the quarters with high point differential showed considerably different distribution of offensive types ($X^2=6.709$; $p=0.034$).

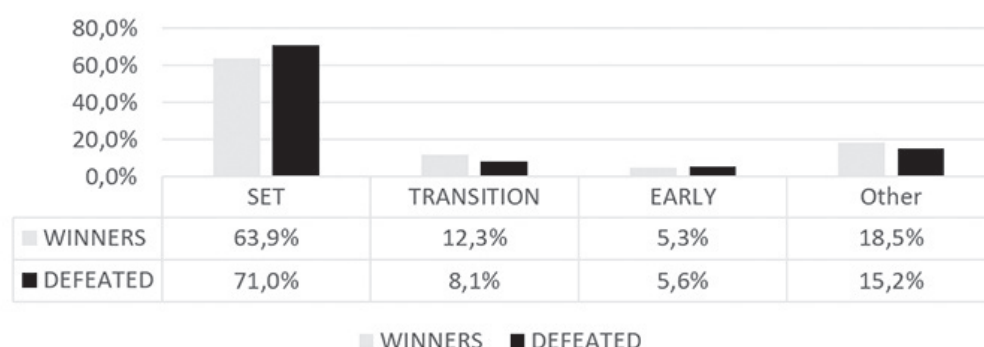


FIGURE 2. Distribution of basic offensive types in unbalanced quarters

Targeted periods showed that winning teams provoked 4% more transitions, while losing teams demonstrated 7% more of set offenses. The parameters of offensive success also showed

significant differences. The statistical indicators presented in Table 3. show that the winning teams had considerably higher efficiency in all observed types.

Table 3. Difference in the outcome of some types of offense after grouping the quarters by difference in the result

	Set offense			Early offense			Transition offense		
Outcome	% POS	% NEG	% NEU	% POS	% NEG	% NEU	% POS	% NEG	% NEU
WINNERS	40.4	42.6	17.0	53.3	33.3	13.3	62.9	25.7	11.4
DEFEATED	30.5	50.6	18.9	31.3	56.2	12.5	39.1	53.5	17.4
chi square	$X^2=8.280$; df=2; p=.016			$X^2=3.609$; df=2; p=.165			$X^2=6.313$; df=2; p=.043		

Note: X^2 - Chi-square test value; df - degrees of freedom; p - level of significance

The biggest difference in the outcomes of the offensive actions is evident in the efficiency of the transition offense, which is more than 20% in favor of the winning teams. More effective performance, that is by 10% is also seen in the set offenses. These findings are extremely remarkable considering the over-

all high quantity of offenses. Despite the apparent difference in the early offense category, a smaller sample of entities did not generate a statistically significant difference between the opposing teams. There are no major discrepancies in the values of the neutral offense outcomes.

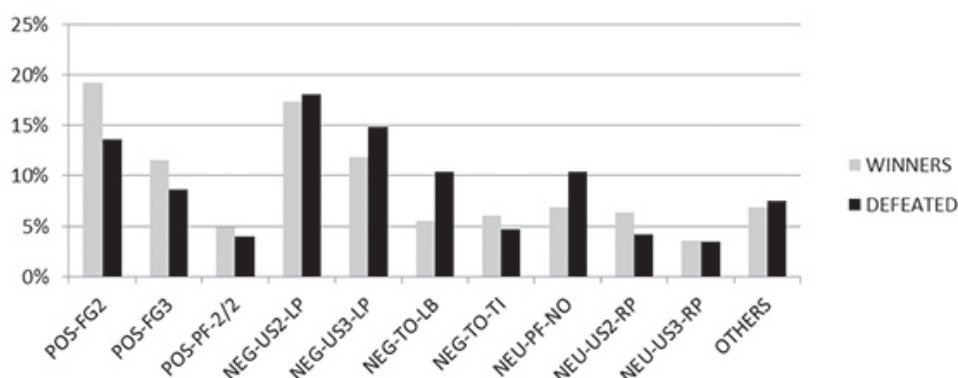


FIGURE 3. Structure of a set offense outcome in unbalanced quarters

Figure 3. shows the modalities of the dominant outcomes on offense, with the winning team plainly having better field goal success (POS-US2; POS-US3) and both free throws after the drawn foul (POS-IOP-2/2). In addition to poor efficiency,

losing teams have more turnovers during critical intervals of the game when the ball remains in play (steals-NEG-IPL-LS) and are only slightly better in terms of turnovers when the offense starts out of bounds (NEG-IPL-LS).

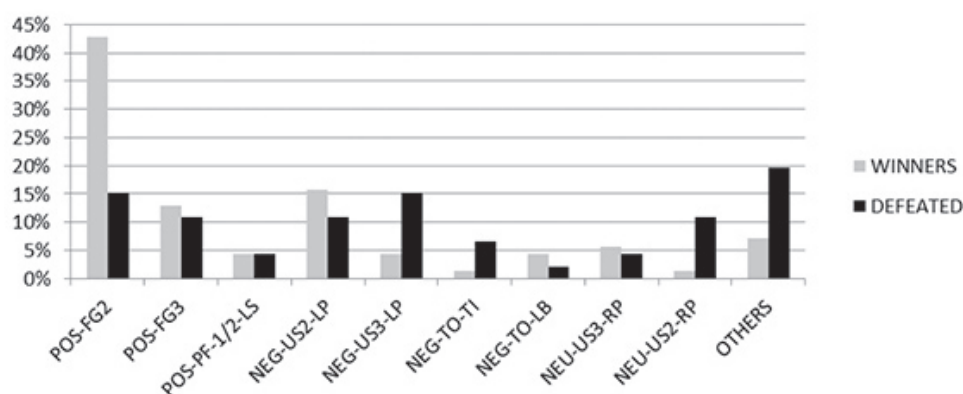


FIGURE 4. Structure of the transition offense outcome in unbalanced quarters

The outcome of the transition offense in the periods analyzed show that a 2-point field goal proved to be a key factor regarding the difference between the winners and the defeated (POS-US2); the difference in efficiency showed to be as much as 28% in favor of the winning team. Although the winning team has a lower scoring rate on 2-point shot success (NEG-US2-IPL), the above parameter annuls this deficit and proves the winning team as capable of taking advantage of the opponent's critical intervals; this is achieved by creating significantly more transitions, in addition to the overall higher success. In the other parameters, the low efficiency of the 3-point shots of the losing team (NEG-IPL-IPL) should be acknowledged.

Discussion

The main findings of the study relate to general offensive structure and efficiency in basketball, as well as the importance of "scoring runs" that generated unbalanced quarters and played a major factor in determining the final score of a basketball game.

Initially, the results indicated a high level of interdependence between offense and defense, therefore confirming that in the game there is a strong dynamic balance of these two phases - it is noticeable that the set offense is mainly followed by the opponent's set offense, and the fastbreak offense is followed by the opponent's fastbreak.

The results show that about 2/3 of the total offensive actions refer to set offense, i.e., 5:5 organized offense with planned tactical solutions. Since a fully formed defense has full control over all offensive players, the probability of a successful defense in the set offense exceeds the efficiency of the offensive players by about 10%. On the contrary, taking advantage of inadequate defense in the early offensive phase is more likely to result in a positive outcome, either in the transition phase or at the opportunal moment of the early stage of set offense (early offense).

The transition offense is most likely to lead to a positive outcome; there is a logical tendency to challenge the opponent to make as many mistakes as possible, which then, in turn, fast breaks and increased scoring opportunities. In general, the transition offense is only 9% of the total offense, however, while combined with early offense, it covers 15% of

the total share of offense. Although transition and, in most cases, early offense, are characterized by unpredictable and spontaneous circumstances (making them less susceptible to schematic practice), they should not be neglected in training, especially due to their high efficiency value (Refoyo et al., 2009). The statistical dominance of the set offense (68.3%) proves its importance in a team's situational preparation. This offense can be more easily prepared in advance because of its structure. Considering the great influence on the final result of a match, the structure of the offense requires extensive technical and tactical (individual and team) preparation.

The following key point of this research refers to a non-linear increase in scoring as evidence of an inconsistent team performance. A "scoring run" of ten or more points is often noticeable and is defined here as a critical interval for the opponent. Furthermore, a team's ability to maintain a high level of intensity and take advantage of an opponent's critical interval makes a major difference in elite basketball. The research also shows that such phases have an unbalanced transition offense and a significant difference in their realization. Although the losing team may have provided an adequate match-up, or possibly even outplayed the winning team for a majority of the game, such phases prove to be a decisive factor in the final defeat.

The critical interval of the losing team in the set offense is mainly characterized by more turnovers with the ball remaining in play (roughly 5%) and poor defense of 3-pointers (3%). A well-organized set offense and better shooting position for the winning team during these intervals is reflected in higher 2-point shooting efficiency (6%), 3-point shooting efficiency (3%), and slightly more successful free throws. These findings coincide with previous research that identified 2-point shooting percentages (Pojskic et al., 2009; Markoski et al., 2012; Russo et al., 2012) and 3-point percentages (Witkos, 2010; Csataljay et al., 2013) as distinguishing indicators between winning and losing teams. During a scoring run, the winning team's 2-point field goal of transition offense is by 28% more efficient. The significantly worse 3-point shooting efficiency (11%) of the losing team reveals improper shot selection and finishing action; a fast-paced game leads to a higher number of turnovers. During critical interval, it is important to note that the winning team had

2% more turnovers due to poor passing, traveling and double dribbling, while the losing team suffered more steals (5%), and this type of conversion is ideal for the effectiveness of the opponent's counterattack. According to Van Wieran (1993), scoring runs are regularly initiated by opportune transition attacks. The fast and easy points are achieved through fast-break and thus the enthusiasm of a team increases in contrast to the low morale of the opponent. Furthermore, it increases the team's scoring rate and creates circumstances situations for more free throws.

From a situational perspective of the game of basketball, the results reveal an importance in neutralizing the opponent's potential scoring run. In order to suppress these critical intervals, it is imperative to emphasize practice tasks or playing methods during individual and group training that ensure the gradual development of athletes' specific phosphagen and glycolytic anaerobic capacities. Under these circumstances, basketball skills are further developed because of the influence of high lactic acid content and accumulated oxygen deficit when, among other things, inhibition occurs in certain parts of the brain. In this way, athletes are trained to efficiently tolerate and/or delay the suppressive processes during situational performance at hormonal, biochemical, and neuropsychological levels (Sekulić, 2007).

Furthermore, in such situations, the coach plays a key role in both perceiving the team's mental and physical decline (and acting accordingly, i.e., using available timeouts), as well as placing emphasis on the consolidation of the team's unstable offensive and defensive efforts – meaning primarily in terms of ball control, disciplined generation of open shots, hindering the opponent's ball advancement (initially through offensive rebound attempts), quick rebounding, and prompt defense that would successfully prevent the opponent's transition. The results also indicate a tactical orientation to aggressive and agile defense, which is a prerequisite for a greater number of prospective fastbreaks. This is mainly reflected in the intense pressure on both the ball and players on the “first pass” by preventing easy ball flow and ultimately securing a defensive rebound by strictly boxing out the offensive player after the shot.

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There are no acknowledgments.

Conflict of Interest

The author declares that there is no conflict of interest.

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This study provides insight into offensive modes, including their ratio and benefit value and helps coaches adjust tactical decisions with the aim of avoiding critical intervals during the game. The added value of this research could contribute the inclusion of sub-modalities of the observed offense types, preferably on a larger game sample. Further research is recommended to evaluate the “scoring run” in more detail, e.g., duration, cause, and effect. Comparison of obtained results with other studies is suitable only if there are assured identical operational definitions of game states, which did causes a certain limitation of this study. The suggested approach of game analysis in this research provided a specific model for evaluation of game states and offensive efficiency in basketball.

Conclusion

The game of basketball clearly demonstrates the coherence of defense and offense. A successful offense depends largely on the preceding defense and vice versa. Dissociated specific offensive profiles - set, transition, early and other (miscellaneous) offenses - are evenly distributed and equally effective between opponents in the resulting balanced matches. However, the dynamic balance of the transition and set offensive and defensive systems changes significantly when focusing on certain game periods (quarters) with greater point differential. The point differential is primarily caused by a scoring run during the match, so in practical terms the study emphasizes the importance of controlling the opponent's potential positive intervals.

It can be concluded that in balanced matches there are crucial short intervals of time when the winning teams benefit from the opponent's physical and psychological decline through a series of successful offenses and defenses. The combination of quality performance, as well as the ability to maintain high intensity levels, will successfully nullify the opponent's scoring runs; doing so would ensure a period of time with a greater number of transitional offenses (the most efficient offense), and a successful implementation of set offenses (the most frequent), which would, in turn, prove to be a decisive factor in the outcome of a match.

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ORIGINAL SCIENTIFIC PAPER

Correlation between Anxiety and Success in Swimming Training Program for Non-Swimmers

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Abstract

The need to determine the factors that positively or negatively affect the acquisition of swimming knowledge arose with the development of various swimming programs. The aim of this research was to determine the connection between the level of anxiety and success in the process of motor learning in swimming training program for non-swimmers. A sample of 77 children, aged (11.00 ± 0.71), participated in a swimming training program for non-swimmers for 20 hours according to the standard method used in the Republic of Croatia. In the first and last lesson, the swimming knowledge was determined by means of a scale of eleven grades that describe the level of acquisition of swimming knowledge. Level of anxiety was assessed by modified CSAI-2CSWIM anxiety level questionnaire. Correlation analysis determined that the level of anxiety has a significant negative correlation with performance in all three observed domains, cognitive anxiety -0.273 and -2.46, somatic anxiety -0.384 and -0.337 and self-confidence -0.420 and -0.308 with results on the final test and with progress in swimming learning. The results of the conducted research showed that the level of anxiety has a significant influence on the children's swimming learning process.

Keywords: elite, non-elite, depression, water sport

Introduction

Swimming as a sport and as a recreational activity has a significant positive influence on the morphological, functional, psychological, motor and intellectual development of children and youth of typical development as well as children with disabilities and various diseases (Gonenc, Acikgoz, Semin & Ozgonul, 2000; Fragala-Pinkham, O'Neil, & Haley, 2010; Jorgić et al., 2014; Fiorilli et al., 2016). According to Rastovski, Tomac, Šumanović, and Filipović (2011), swimming is one of the first sports that children begin to train due to its positive influence on the development of a young child's organism. In order for children to be able to use the positive benefits of swimming, they must first learn to swim. Therefore, trainings for non-swimmers are organized. Different training methods for non-swimmers are applied worldwide (Grčić-Zubčević, 1996; Kapus et al., 2002; Junge, Blixt, & Stallman, 2010). Over time, various tests have been developed to assess swimming knowledge by differ-

ent swimming experts (Getz, Hutzler, & Vermeer, 2006; Tirosh, Katz-Leurer, & Getz, 2008; Šiljeg, Leko, & Sindik, 2016) that can be applied to children of different ages and health conditions.

With the development of various swimming training programs and tests for the assessment of swimming abilities, the need arose to determine the factors that positively or negatively affect the acquisition of swimming knowledge. According to Rastovski (2019a), the external factors that influence the success of swimming training are the working mode or the training model, means and aids, water characteristics and the training coach. Additionally, the internal factors singled out are motor abilities morphological characteristics and the level of anxiety of students during swimming training (Köröglu & Yigiter, 2016; Stanković et al., 2017). The influence of external factors and certain motor abilities and morphological characteristics in the training of non-swimmers is somewhat predictable. These factors definitely need to be determined, consid-



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ering that children with a higher level of anxiety tend to have a greater fear of failure, concern about mistakes, poor performance and defeat (Coreia et al., 2017; Coreia et al., 2018). Results from the research of Burton (1988) show that cognitive anxiety negatively affects success in swimming. Bielec (2007) concluded that, in addition to material conditions and external factors, the greatest issue in swimming training is fear of water or anxiety. Studying the influence of anxiety on swimming abilities, Muhamad, Sattar, Abadi and Haron (2013) concluded that female students with a higher degree of anxiety achieve lower results in the observed swimming abilities.

According to Vujanović and Tišma (2011), athletes lose up to 50% of their total technical-conditioning capacities in important competitions if there is an increased state of anxiety, which affects performance. Those children are also more inclined to feel that their unsuccessful sports performance will be criticized, which results in negative self-evaluation according to Smith, Smoll, and Cumming (2007). One meta-analysis (Woodman & Hardy, 2003) showed a correlation between cognitive anxiety and performance for competitive sports. To the best of our knowledge, no study has looked specifically at the relationship and the influence of the anxiety level on the success of training programs for non-swimmers.

Consequently, the aim of this research is to determine the correlation between the level of anxiety and success in the process of learning swimming skills for non-swimmers.

Methods

Participants

A total of 77 children non-swimmers, with no injury (35 boys and 42 girls) aged (11.0 ± 0.71) were selected from elementary school. All participants were non-swimmers without experience in water environment. Prior to testing, all individuals volunteered and had their parents sign an informed consent form. The advantages and disadvantages of participating in this study were explained to them. The Human Ethics Committee of the Faculty gave its approval to the planned study in accordance with the Helsinki Declaration. Approval number: PZR/16-04-2014-Ad2F.

Procedures

A sample of 77 children's non-swimmers were trained for 20 lessons according to the standard method of Grčić-Zubčević (1996), which is applied in the Republic of Croatia. Lessons were performed 2 time per week lasting one school lesson (45min).

Swimming knowledge

In the first and last lesson, the swimming knowledge was assessed according to the criteria established by Grčić-Zubčević (1996). The degree of success is determined through a scale of eleven grades that describe the degree of acquisition of swimming knowledge.

Anxiety level of questionnaire

The level of anxiety was assessed by a modified CSAI-2CSWIM anxiety level questionnaire (Rastovski, 2019b). It consists of 27 items arranged in three subscales that measure: Cognitive anxiety - cognitive anxiety (w), somatic anxiety (s) and self-confidence (c). On a 4-point Likert scale, participants assessed how they mostly felt, with 1 indicating the absence of a certain feeling, and 4 indicating the intense presence of a certain feeling. Questionnaire has satisfactory reliability of the entire questionnaire (Cronbach $\alpha=0.748$ and average correlation (AVR) $=0.106$), as well as all three subscales.

Data processing methods

In all variables, basic descriptive parameters were calculated: mean, standard deviation (SD), range (Min - Max), and median (Median). Before further data processing, the process of normalization of the variables was carried out. Correlation analysis was used to determine the correlation between the level of anxiety and success in the process of motor learning during the training of non-swimmers. The data analyses were performed by using IBM SPSS Statistics (version 19. IBM Corp. Armonk, NY, USA). Level of significance was set at $p < 0.05$.

Results

Table 1 shows the results of the acquisition of swimming knowledge at the initial and final measurement, as well as

Table 1. Descriptive statistics for the acquisition of swimming knowledge and for the level of anxiety

Variables	Mean	SD	Median	Min	Max
ISWK	1.29	0.70	1.00	1.00	3.00
FSWK	8.79	2.15	9.00	1.00	11.00
PSWK	7.5	2.09	7.00	0.00	10.00
CA	1.77	0.53	1.67	1.00	3.75
SC	3.04	0.58	3.17	1.50	4.00
SA	3.42	0.49	3.56	1.67	4.00
AG	2.60	0.21	2.63	2.07	3.04

Note. ISWK - initial swimming knowledge, FSWK - final swimming knowledge, PSWK - progress in swimming knowledge, CA - cognitive anxiety, SC - Self-confidence, SA - Somatic anxiety, AG - anxiety general

the achieved progress. In addition, the table also shows the results for three subscales of anxiety cognitive anxiety, self-confidence, and somatic anxiety as well as for general anxiety.

The presented results show that significant progress was made in swimming knowledge between the initial and final testing (Mean $\pm$ SD = 7.5 ± 2.1). The result for general anxiety (AG) was 2.60 ± 0.21 , 1.77 ± 0.53 for CA, 3.04 ± 0.58 for SC and

3.42 ± 0.49 for SA.

The results in Table 2 indicate that there is a negative correlation between anxiety levels and performance in each of the three areas that were studied: cognitive anxiety ($r = -0.273$ and -2.46), somatic anxiety ($r = -0.384$ and -0.337), and self-confidence ($r = -0.420$ and -0.308), with the score on the final test results and progress in learning to swim, and all coefficients are low to moderately related.

Table 2. Spearman's rank correlation coefficient of the variables Final and Progress of swimming learning with anxiety domains

Variables	FSWK	PSWK
SA	-.384**	-.337**
SC	-.420**	-.308**
CA	-.273*	-.246*

Note. FSWK – final swimming knowledge, PSWK – progress in swimming knowledge, CA - cognitive anxiety, SC - Self-confidence, SA - Somatic anxiety,

Discussion

In the present study, the correlation between the level of anxiety and success in the process of motor learning during the training of non-swimmers was determined.

The results obtained from this research clearly confirm that the level of anxiety has a significant negative correlation with success in all three observed domains, cognitive anxiety (CA) -0.273 and -0.46, somatic anxiety (SA) -0.384 and -0.337 and self-confidence (SC) -0.420 and -0.308 with the results on the final test (FSWK) and with (PSWK) progress in learning to swim (Table 2). The results coincide with the results of previous research, according to Woodman and Hardy (2003), and Kais and Raudsepp, (2005).

Šilić (2014) points out that an emotional experience, such as anxiety, leads to discomfort, psychological activation and a tendency to escape or avoid an unpleasant situation. Such activation manifests itself differently in the observed domains. However, it is certainly one of the factors that can negatively affect a non-swimmer child, i.e., his success in learning to swim. The obtained results coincide with research by Burton (1998), who applied the CSAI-2 to swimmers and concluded that cognitive anxiety negatively affects success. Negative thoughts before a swimming competition reduce swimming capacity (Lin et al., 2021). Determined anxiety reduced capacities in children with a higher level of cognitive anxiety. Therefore children with higher anxiety were not able to complete the assigned tasks or concentrate on the non-swimmer training lessons. Instead, they tried to overcome their cognitive anxiety, which resulted in poorer progress and performance.

Salovey and Sluyter (1997) points out that the group of primary emotions includes, among others, fear and shame. This group of emotions (fear and shame) are dominant for a child in a newly created situation, such as the training of non-swimmers. Fear of a new environment and fear of endangering one's own life are certainly some of the strongest emotions. Furthermore, cognitive anxiety, or fear for one's own life, when training non-swimmers can be the result of a bad experience, especially the one related to the fear of drowning the child has previously experienced in a swimming pool. This was confirmed by Irwin, Irwin, Ryan and Drayer (2009) who stated that the fear of drowning is the strongest predictor of low swimming competence. This research also confirmed the connection between self-confidence and success in teaching non-swimmers. Although the correlation coefficients were low to moderately related ($r = -0.273$ to -0.420), the current results show a negative correlation between anxiety levels and performance in each of the three areas that were studied: cognitive anxiety, somatic anxiety and self-confidence. Persons with high self-confidence will interpret the symptoms of competitive anxiety as an incentive, a challenge, an opportunity to achieve their goal

and success. On the contrary, persons with lower self-confidence will see such symptoms as a hindering and threatening factor to success (Hanton, Mellalieu, & Hall, 2004). According to Šilić (2014), in order for self-efficacy to develop, a person must believe that he/she has everything under control and that what he/she does is done with the intention of achieving a certain goal. Consequently, even in the training of non-swimmers, it is necessary to work on increasing self-confidence in children so that they can master the tasks set more easily. This is achieved by gradually assigning tasks through games and respecting the principle from an easier to a harder task.

The research results showed that somatic manifestations are negatively related to success in training non-swimmers. Somatic anxiety occurs due to excitement and/or unpleasant feelings such as tension and stress and is manifested through a series of physiological symptoms (McNally, 2002; Kais & Raudsepp; 2005). Physiological symptoms that occur in the training of non-swimmers are specific and often related to the environment in which non-swimmers are trained. These external factors are related to water temperature and chlorine. It often happens that children have a "reduced" tolerance to lower temperature and chlorine evaporation. In this way, they actually try to justify their discomfort in the new situation. Any somatic manifestation can distract the students' attention from the lesson. Therefore, the students' attention will partly be focused on the symptoms, that is, on concealing them, while the manifestations themselves can significantly affect the quality of the lesson. Somatic manifestations must also be noticed by swimming coaches and are one of the basic elements for recognizing an increased level of anxiety in children who are learning to swim. The results of the conducted research showed that the level of anxiety in it all domains (cognitive, somatic anxiety and self-confidence) has a significant impact on children's learning to swim.

The main limitation of the present study is not objectively measuring swimming knowledge of the participants. Moreover, children's participation in some other extra-curricular activities may contribute to the difference in social anxiety prior to engaging in swimming classes. Furthermore, maturation level was not determined. Future studies should include younger children without engagement in other sport activities.

Conclusion

In conclusion, there is a negative correlation between anxiety levels and performance in each of the three areas that were studied: cognitive anxiety, somatic anxiety, and self-confidence, with the score on the final test results and progress in learning to swim. The findings of the research revealed that anxiety levels are significantly connected with how effectively children learn to swim.

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There are no acknowledgments.

Conflict of Interest

The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Relationship between Maximum Oxygen Consumption and Lactate Metabolism with Situational Efficiency of Highly Selected Young Judoists

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Abstract

Judo is often referred to as an explosive sport, which requires great anaerobic strength and capacity, with a well-developed aerobic system. Attaining a high level of physical fitness, strength, and fatigue tolerance is essential for achieving success in competition. Fatigue leads to decreased muscle strength, prolonged reaction time, reduced agility, neuromuscular coordination, overall body speed, concentration, and agility. The intensity at which this phenomenon occurs is known as the lactate threshold. The primary objective of the planned research was to establish a link between maximal oxygen consumption (VO_{2max}), lactate metabolism, and the situational efficiency of selected young judoists. The sample consisted of 30 cadet and junior judo athletes from the national teams of Serbia (average age of 16.43 ± 0.76 years, body height of 176.94 ± 5.15 cm, and body weight of 69.71 ± 10.64 kg). The research employed precisely standardized protocols and modern equipment to determine anthropometric characteristics, and the values of maximal oxygen consumption, lactate thresholds, and the index of a specific judo fitness test among the selected young judo athletes. Based on the obtained results, there was a moderate negative correlation between VO_{2max} and the index of the special judo fitness test, as well as a low correlation between the first lactate threshold (PLAP) and the second lactate threshold (DLAP) with the index of the special judo fitness test (ISJFT). Additionally, a moderate negative correlation was found between VO_{2max} and anthropometric parameters, while PLAP and DLAP exhibited low correlations with anthropometric parameters. The research results quantitatively illustrate the physiological adaptation of the top young judo athletes to the physical demands encountered during years of specific training. The proposed battery of tests can be utilized to assess the functional status of competitors more accurately and determine the competition profile for elite judo athletes.

Keywords: maximal oxygen uptake, VO_{2max} , aerobic capacity, anaerobic capacity, lactate thresholds, specific judo fitness test, elite judokas

Introduction

The functional properties of the organism show a direct influence on the size and character of the manifestation of working ability, during physical activity (Heimar & Medved, 1997). There are two general functional properties depending on the biochemical nature of energy processes: aerobic capacity and anaerobic

capacity. During physical activity, skeletal muscles utilize the energy created through anaerobic and aerobic processes (Bradarić, 1991).

Anaerobic energy is released through the breakdown of adenosine triphosphate, creatine phosphate, or carbohydrates into pyruvate by the process of glycolysis, whereby lactates are formed.



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In practical terms, the term “anaerobic potency” is used when assessing the alactic and lactic anaerobic capacity of an organism (Radovanović, 2009; Radaković, 2016). Anaerobic power means the energy capacity of an organism, in which energy is created from energetic substances, without the simultaneous consumption of oxygen (Kinderman, Simon & Keul, 1979). With programmed multi-year training, anaerobic capacity parameters exhibit significant improvements (Dopsaj, 2008).

Modern judo demands high intensity and includes a wide range of tactical and technical elements (Franchini & Sterkowicz, 2000; Franchini, Takito, & Bertuzzi, 2005). The increased requirements in modern judo, particularly in the field of periodization, aim to optimize training effects by varying different training parameters, utilizing analytical-diagnostic approaches, and incorporating specific situational training.

Aerobic processes involve the resynthesis of adenosine triphosphate with simultaneous oxygen consumption. Under normal conditions, 90% of the total adenosine triphosphate is resynthesized through aerobic processes with the involvement of enzymes located in mitochondrial cells. The amount of energy created by aerobic processes is 10 times larger than the energy obtained by anaerobic glycolysis (Baxter-Jones, Goldstein & Helms, 1993). During physical activity, in addition to muscle glycogen reserves, liver glycogen, fats, and amino acids can be utilized as energy substrates in skeletal muscle cells. The final products of aerobic reactions are water and carbon dioxide, which are easily eliminated and do not cause significant changes in the body (Baxter-Jones et al., 1993; Radovanović, 2009).

To assess the organism's ability to supply the required amounts of energy through aerobic processes, the term “aerobic power” is used in practice. It represents the maximum level of metabolic processes that occur in the presence of oxygen. This process is primarily limited by the quantity of oxygenated blood that the cardiorespiratory system can deliver to the working muscles and, to a lesser extent, by the muscles' ability to extract the delivered oxygen from the blood (Radovanović, 2009).

In research, during the last century, a decrease in the number of mitochondria in muscles was observed, which were subjected to a program for the development of muscle strength (McDougall, Sale, Moroz, Elder, & Sutton, 1979). Since mitochondria are necessary for aerobic energy production processes, any reduction in mitochondrial numbers theoretically leads to a decrease in muscle oxidative capacity. However, at that time, it was not known that strength training provided other benefits such as injury prevention and relief from exhaustion syndrome. High-intensity endurance training can lead to protein loss from type 1 and 2C muscle fibers, which can result in injury and decreased performance over time (Kraemer, Fleck, & Evans, 1996). During muscle strength training, muscle fibers are stimulated, leading to changes in protein quality and the expression of different types and combinations of myosin ATPase enzymes (Kraemer, Fleck, & Evans, 1996; Tanaka & Swensen, 1998). One of the common reasons why athletes, who compete in endurance-dominated disciplines, avoid training to develop muscle strength is the fear of muscle hypertrophy, and thus weight gain. However, using a force development program (Behm & Sale, 1993; Almasbakk & Hoff, 1996) that incorporates heavy loads and focuses on maximum force mobilization can have a greater impact on muscle coordination and nerve adaptation rather than muscle hypertrophy.

In nearly all studies that have examined concurrent strength and endurance training, no detrimental effects on endurance were observed, and markers of aerobic capacity even increased

(Hickson, Dvorak, Gorostiaga, Kurowski, & Foster, 1988).

This indicates that the hypothesis stating that strength training increases body weight and negatively affects performance is not entirely accurate. In almost all studies, which studied simultaneous training for the development of muscle strength and endurance, no negative effects on endurance were observed, and markers of aerobic capacity even increased (Hickson, Dvorak, Gorostiaga, Kurowski, & Foster, 1988). Neurological adaptations, as well as the change of the relationship between the types of muscle fibers, in favor of more oxidative muscle fibers during training for the development of strength, lead to increased economy of performing movements. High-intensity training, with a small number of repetitions and the maximum manifestation of muscle strength in the concentric phase of the movement, is considered the most effective for increasing the economy of movement. It is also recommended to emphasize the rapid manifestation of force. After such training, the greatest improvements were observed in the rate of force development (RFD), maximum force output, and the one-repetition maximum test (1RM; Heggelund, Fimland, Helgerud, & Hoff, 2013). Increased RFD results in shorter contraction times, allowing more time for relaxation during cyclic movements. Longer rest periods also facilitate better blood circulation in the capillaries, thus delaying the onset of fatigue.

Judo is often referred to as an acyclic, explosive sport, which requires great anaerobic strength and capacity, with a well-developed aerobic system (Bratić, Nurkić & Stanković, 2011). From a sports physiology perspective, competitive success largely depends on the ability of a judoist to, within his weight category, reach high values of anaerobic capacity and show great muscle strength, with a quick recovery between consecutive matches. Judoists are characterized by a high level of development in both the anaerobic and aerobic bioenergetic systems (Thomas et al, 1989). However, studies of judo competitors have revealed significant differences in anaerobic capacity and maximum oxygen consumption, which can be partly explained by differences in anthropometric characteristics (Callister, Callister, Fleck, & Dudley, 1991). Skirowski et al. (1987) categorized judoists' activities during a match into four duration categories: 0-10 s, 11-20 s, 21-30 s, and longer than 30 s. Activities lasting 11-20 s have the highest frequency (39%), while resting periods and/or breaks typically last 0-10 s in about 80% of cases. Similar studies (NCCP, 1990) have shown that seizures occur every 10-15 s. Therefore, the anaerobic component of the bioenergetic capacity is predominantly engaged in judoists (Pulkkinen, 2001). A high level of physical fitness and strength, with good fatigue tolerance, are necessary prerequisites for competitive success since judo is characterized by changing activities of maximum intensity with an average duration of 15-30s and rest lasting about 10s (Sterkowicz & Franchini, 2000).

Gariod et al. (1994) determined two dominant judo profiles by the analysis of high-energy phosphates, nuclear magnetic resonance spectroscopy, using radioactive ^{31}P : aerobic and anaerobic. The analysis of creatine phosphate dynamics during maximal voluntary contraction revealed a less pronounced decrease in judokas belonging to the aerobic profile, along with faster creatine phosphate resynthesis compared to judokas in the anaerobic profile (Gariod et al., 1994).

Despite the significance of functional abilities in judo athletes and the existing research on this topic, it remains unknown whether studies have established a correlation between maximum oxygen consumption ($\text{VO}_{2\text{max}}$), lactate metabolism, and situational efficiency in young judoists. Therefore, the objective of this research was to investigate the relationship between maximum

oxygen consumption, lactic metabolism, anthropometric characteristics, and the situational efficiency of selected young judoists.

Methods

Sample of participants

The research was conducted on a sample of 30 highly selected young judoists, members of a wider list of cadet and junior national teams of Serbia, aged 14 to 18 (average age 16.43 ± 0.76 yrs, body height 176.94 ± 5.15 cm, and body weight 69.71 ± 10.64 kg). The participants were required to meet specific criteria, including being on the list of potential representatives, having no organic or somatic diseases, and falling within the 14 to 18 years age range. Ethical review and approval required for the study on human participants was obtained in accordance with local legislation and institutional requirements. Prior to participation, all participants were informed about the experimental procedures. The research was conducted in accordance with the Helsinki Declaration.

Measurements

Anthropometric Characteristics

Standardized anthropometric instruments were used to measure morphological characteristics. Three variables were measured to assess morphological characteristics: body height (ATVI), body mass (ATMS), and fat tissue percentage (APMT). Body height was measured using the GPM anthropometer (Switzerland) with a measurement accuracy of 0.1 cm. Body mass was measured using the Tefal 6010 electronic scale (France) with a measurement accuracy of 0.1 kg. The ratio of fat and non-fat tissue in the participants was indirectly estimated using the laboratory method of bioelectrical impedance analysis (BIA). For bioelectrical impedance analysis, the Omron BF 300 device (Japan) was employed. The data on the percentage of body fat were recorded from the device's display with an accuracy of 0.1% (Eston & Reilly, 2001).

Cardiorespiratory Endurance

The assessment of cardiorespiratory fitness involved the use of a six-stage maximal protocol test on a manual bicycle ergometer, specifically the maximal multistage test on the Monark Rehab 881 E manual bicycle ergometer (Sweden). Oxygen consumption was measured using the Hans Rudolph mask (Kansas City, USA), and Cosmed's FitMate system. The test aimed to measure maximal oxygen consumption ($\text{VO}_{2\text{max}}$) in ml/kg/min.

Lactate Metabolism

For determining lactate thresholds, data on lactate levels in

capillary blood (expressed in mmol/L) were used at the end of each stage of the incremental continuous treadmill test. Capillary blood samples were obtained from the fingertip using specialized test strips. Immediately after sampling, the lactate concentration was enzymatically determined using the Lactate Scout analyzer, EKF SensLab (Germany). The sensitivity and validity of lactate concentration determination using the Lactate Scout analyzer, EKF SensLab, have been scientifically confirmed (McNaughton et al., 2002; Von Duvillard et al., 2005). The assessed parameters included the first lactate threshold (PLAP; mmol/L) and the second lactate threshold (DLAP; mmol/L).

Situational Efficiency of Judoists

The situational efficiency of judoists is determined by the index of the Special Judo Fitness Test (SJFT), which is extensively described in the study by Radovanovic, Bratic, and Milovanovic (2008). The index of the SJFT (ISFJT) is calculated using the following equation:

$$\text{ISFJT} = (\text{HR}_{\text{reff}} + \text{HR}_{\text{res}}) \cdot (A + B + C) - 1$$

In the equation, HR_{reff} represents the heart rate value immediately after exercise (after performing series A, B, and C), HR_{res} represents the pulse value one minute after the test, and $A + B + C$ represents the total number of throws performed in all three series. A lower index indicates better results.

Statistics

To determine the levels in the manifest space of the examined anthropological dimensions of judoists, the basic statistical parameters will be applied: arithmetic mean ($\bar{X}$), standard deviation (SD), minimum (Min) and maximum (Max), range of results (Rang). To determine the normality of the distribution, the following will be calculated: skewness (Skw) for symmetry, kurtosis (Krt) for elongation, and P-P plot graphics. Pearson correlation analysis was applied to determine overall quantitative relationships between groups in the motor, situational motor, and functional space. The strength of correlation was determined according to Cohen (1988), where a weak correlation is defined as $r=0.1-0.29$, moderate correlation as $r=0.3-0.49$, and strong correlation as $r=0.5-1.0$. Data analysis was conducted using the IBM SPSS Statistics program (version 19.0, Armonk, NY: IBM Corp.).

Results

Insight into the basic parameters of the distribution of the applied variables, measured in young judoists (Table 1), reveals that the results are well grouped and normally distributed around the arithmetic mean. Additionally, there is no significant deviation in

Table 1. Basic statistical parameters of applied variables.

	N	Mean	SD	Med	Min	Max	Range	Skw	Kurt
GSTA	30	16.43	0.76	16.50	15.00	17.50	2.50	-0.38	-0.76
ATVI	30	176.94	5.15	178.65	168.30	185.10	16.80	-0.26	-1.23
ATMS	30	69.71	10.64	71.10	52.10	87.70	35.60	-0.08	-1.09
APMT	30	9.18	4.06	7.50	4.00	16.80	12.80	0.68	-0.74
FVO2M	30	41.27	2.67	41.40	35.90	45.90	10.00	-0.35	-0.73
PLAP	30	3.43	0.55	3.50	1.90	4.40	2.50	-0.58	0.95
DLAP	30	6.85	0.67	6.85	5.80	8.90	3.10	1.16	2.62
ISFJT	30	13.63	2.05	13.65	10.60	17.10	6.50	0.03	-1.37

Note: N-number, Mean-arithmetic mean, SD-standard deviation, Med-mediana, Min-minimum, Max-maximum, Skw-skewness, Krt-Kurtkurtosis, GSTA-age, ATVI-body height, ATMS-body mass APMT-fat tissue percentage, FVO2M-maximal oxygen consumption, PLAP-first lactate threshold, DLAP-second lactate threshold, ISFJT-index of the special judo fitness test.

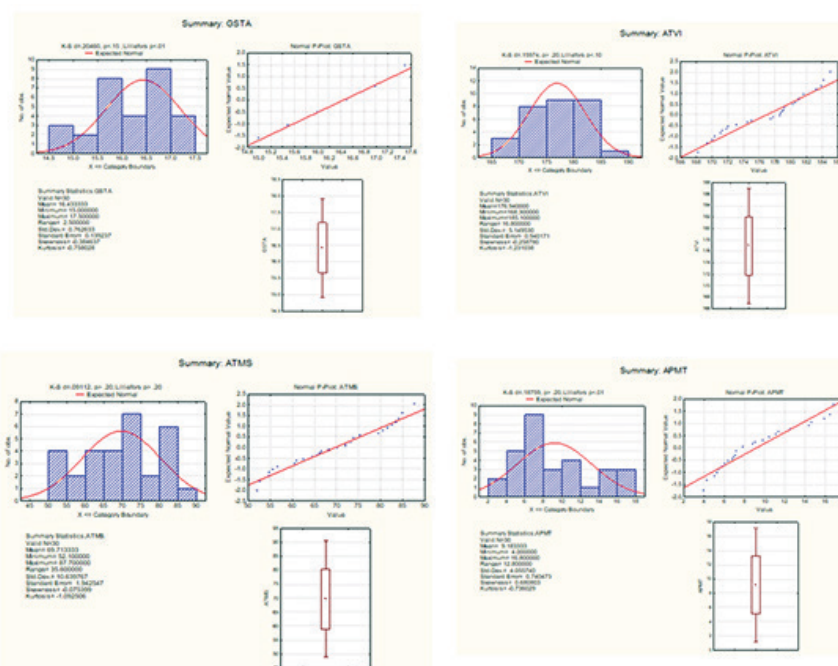


FIGURE 1. Normality of data distribution, skewness and kurtosis, and P-P plot graphs for the variables GSTA, ATVI, ATMS, and APMT.

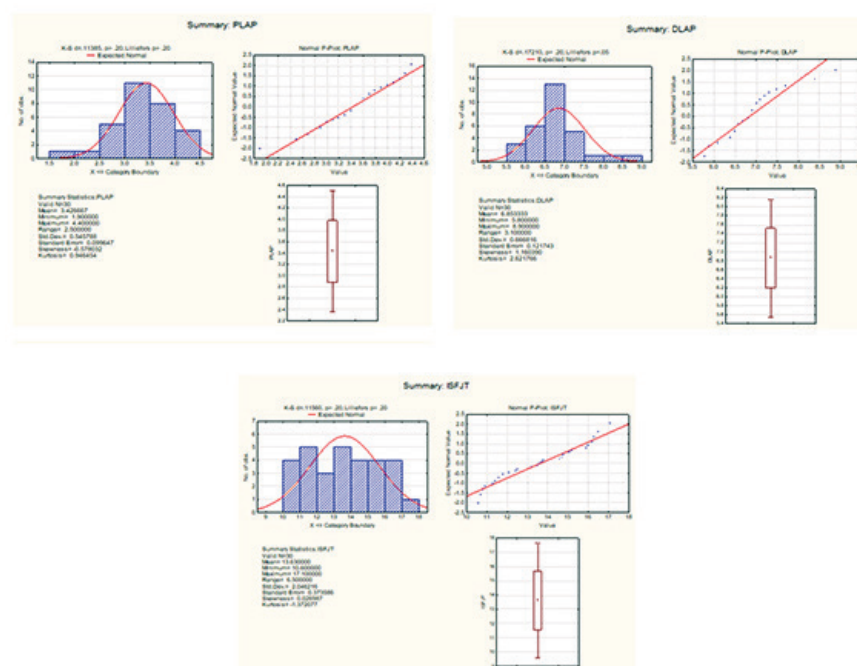


FIGURE 2. Normality of data distribution, skewness and kurtosis, and P-P plot graphs for the variables PLAP, DLAP, and ISFJT

the results (Figure 1 and 2).

Correlations between the analyzed variables are presented in Table 2. Strong correlations were observed between APMT

and ATMS (0.90), ATMS and ATVI (0.75), and APMT and ATVI (0.53). On the other hand, VO_{2max} showed a moderate negative correlation with body composition variables.

Table 2. Correlation matrix between anthropometric characteristics and cardiorespiratory fitness

	GSTA	ATVI	ATMS	APMT
ATVI	0.26			
ATMS	0.32	0.75		
APMT	0.21	0.53	0.90	
FVO2M	-0.21	-0.35	-0.48	-0.49

Note: GSTA-age, ATVI-body height, ATMS-body mass APMT- fat tissue percentage, FVO2M-maximal oxygen consumption.

Table 3. Correlation matrix between anthropometric characteristics and lactate thresholds

	GSTA	ATVI	ATMS	APMT	PLAP
ATVI	0.26				
ATMS	0.32	0.75			
APMT	0.21	0.53	0.90		
PLAP	0.05	-0.16	-0.02	0.01	
DLAP	-0.06	-0.10	0.09	0.01	0.33

Note: GSTA-age, ATVI-body height, ATMS-body mass APMT- fat tissue percentage, DLAP-second lactate threshold, ISFJT-index of the special judo fitness test.

Meanwhile, PLAP and DLAP variables exhibited weak associations with body composition parameters (Table 3).

The analysis of the correlation between the variables of anthropometric characteristics and functional abilities with

specific motor tests reveals a moderate negative association between VO_2max and ISFJT (Table 4). However, a weak correlation was found between lactate tolerance and ISFJT variables (Table 5).

Table 4. Correlation between anthropometric characteristics, and cardiorespiratory fitness with situational efficiency

	ISFJT
GSTA	0.21
ATVI	-0.11
ATMS	0.16
APMT	0.34
FVO2M	-0.49

GSTA-age, ATVI-body height, ATMS-body mass APMT-fat tissue percentage, FVO2M-maximal oxygen consumption, ISFJT-index of the special judo fitness test.

Table 5. Correlation between anthropometric characteristics, and lactate thresholds with situational efficiency

	ISFJT
GSTA	0.21
ATVI	-0.11
ATMS	0.16
APMT	0.34
PLAP	0.27
DLAP	-0.21

GSTA-age, ATVI-body height, ATMS-body mass APMT-fat tissue percentage, PLAP-first lactate threshold, DLAP-second lactate threshold, ISFJT-index of the special judo fitness test.

Discussion

Judo is a dynamic and physically demanding sport characterized by high-intensity activity within a short duration. The final outcome in judo is influenced by various variables, including physiological, motor, technical, tactical, and psychological factors (Todorov, 2014). Both anaerobic and aerobic capacities play a role during fights (Thomas et al., 1989). Anaerobic capacity enables intensive, but a short-term manifestation of the maximum muscular strength that characterize this type of sport. The aerobic capacity allows for extended efforts during the potential, five-minute duration of the match. Increasing anaerobic capacity while reducing the percentage of adipose tissue allows for more attacks to be performed during combat while improving aerobic capacity allows for a faster recovery process between fights. Multiple studies have demonstrated the negative impact of adipose tissue percentage on motor skills in judoists, regardless of gender (Nakajima et al., 1998; Frachini et al., 2005). A negative relationship has

been observed between adipose tissue percentage and general physical fitness parameters such as isometric strength, flexibility, and balance (Frachini et al., 2005). Similarly, in judoists and other athletes, body weight and adipose tissue percentage have a negative influence on maximum oxygen consumption. It is believed that a lower adipose tissue percentage in elite judoists enables better metabolic adaptation to the diverse technical and tactical demands during matches. The results of the conducted research (Table 2) indicate a strong correlation between body weight and adipose tissue percentage in the subjects. This finding is expected, considering that in the respondents aged 14-18, the chronological age is very close and without significant influence on the body composition. Also, the results of the conducted research confirmed the previously known fact that the percentage of adipose tissue in judoists is higher in competitors of higher weight categories.

To obtain an accurate understanding of the contribution of aerobic metabolism in judoists, it is necessary to measure

the consumption of oxygen in combat, which is not possible for technical reasons. Judo matches involve quasi-static positions with raised upper arms, leading to increased heart rate while reducing oxygen consumption (Todorov, Dopsaj, Radovanovic, & Bratic, 2014). Previous research has shown that in static muscle contractions at a certain percentage of maximum heart rate, the percentage of maximum oxygen consumption was lower than the percentage in dynamic contractions (Collins, Cureton, Hill & Ray, 1991). This non-linearity between heart rate and oxygen consumption complicates conclusions regarding energy metabolic pathways during judo. Short periods of high intensity of effort, with even shorter rest intervals, result in a high percentage of lactate anaerobic metabolism. However, the increase in the maximum concentration of lactate in the blood indicates that in high-intensity alternating activities before the onset of fatigue, the largest source of energy is aerobic metabolic processes, while the share of anaerobic is significantly lower. The results of targeted scientific research indicated that during the match there is a need for intensive energy production from glucose because after the fight high levels of lactate concentration in the blood of judoists were registered (Callister, Staron, Fleck, Tesch & Dudley, 1991; Gupta, Goswami, Sadhukhan & Mathur, 1996).

The analysis of the correlation matrix results (Table 4) showed a moderate correlation between VO_2max and Special Judo Fitness Test (SJFT). The SJFT as a field test has less sensitivity and validity compared to the Wingate test, which is performed in laboratory conditions on a hand-held bicycle ergometer. Counting of performed throws during a special judo fitness test gives less precise indicators, but all movements are characteristic of judo. The ability of subjects to maintain a higher intensity, before fatigue and/or test interruption, indicates a higher value of aerobic endurance (Cosgrove, Wilson & Watt, 1999; Bentley, McNaughton & Thompson, 2001). This is particularly important as the metabolic response during non-maximal intensity physical activities can vary depending on the training level of the subject (Baldwin, Snow & Febbraio, 2000). Modification of the multistage progressive test protocol and lactate analysis procedures may affect physiological parameters. These changes can be significant, in terms of interpreting physiological variables in order to monitor the effects of particular training or to predict the level of ability development (Bentley, Newell & Bishop, 2007).

The basic distribution parameters (Table 1.) reveal a wide range of test results for the second lactate threshold (DLAP), indicating the presence of subjects with higher scores. This suggests variations in aerobic metabolism among the subjects included in the study. The correlation matrix showed the lowest correlations between the age and the first lactate threshold (PLAP), which confirmed that chronological age does not affect the level of the first lactate threshold, but it is a consequence of the degree of physical fitness. It is important to highlight that PLAP and DLAP have low correlations with all anthropometric parameters (Table 3), indicating that anthropometric characteristics do not significantly influence lactate threshold. Additionally, the analysis of the correlation matrix between variables (Table 5) shows moderate correlations between the first and second lactate threshold (PLAP) with the judo fitness test variable. Total lactate levels can be increased due to various factors such as decreased glycogen reserves in skeletal muscle cells, low-carbohydrate diets, or previous exhaustive physical activities (Reilly & Woodbridge,

1999). Furthermore, factors like muscle fiber type, activity of glycolytic and lipolytic enzymes, as well as capillary and mitochondrial density, may also influence the blood lactate concentration curve (Midgley, McNaughton, & Jones, 2007). It is important to note that a particular multistage progressive test protocol can vary significantly concerning initial and subsequent load intensity and stage duration.

From a methodological point of view, the place from which the blood sample is obtained to measure lactate concentration (ear shell, fingertip), the type of blood vessel (venous, arterial, capillary), and laboratory methods used (the type of lactate analyzer) also affect the test result. Samples taken from the earlobe showed a lower result of blood lactate concentration than samples taken from the finger. The most accurate approach would be to sample blood from the working muscles (Forsyth & Farrally, 2000). Most published studies on the relationship between the lactate threshold and endurance have shown a strong correlation, meaning that training-induced improvements in cardiorespiratory are significantly associated with improvements in the lactate threshold. Although processes in the central nervous system may have regulatory and decisive effects on endurance (Draper, Brent & Hale, 2006; Noakes, 2007), peripheral metabolic adaptations associated with the lactate threshold are a necessary prerequisite for maintaining aerobic endurance (Midgley, McNaughton, & Jones, 2007). It can be said that there is a large number of evidence (Faude, Kindermann, & Meye, 2009) that the concept of the lactate threshold is of great importance for diagnosis as well as the prediction of aerobic endurance levels. The concept of the aerobic-anaerobic transition can serve as a valuable tool for diagnosing aerobic abilities and guiding training protocols.

Lactate produced in active muscles is subsequently metabolized during high-intensity physical activity in the recovery phase (Westerblad, Allen, & Lannergren, 2002). According to Saltin (1990), there is evidence that a return to values that approximately correspond to the basal values of lactate in the blood can be expected at the earliest 30 to 60 minutes after high-intensity physical activity followed by high lactate accumulation. Since the average accumulation of lactate in the blood after a judo match ranges from 10 to 17 mmol/L, the time interval between matches during tournament competitions, which is typically 15 to 30 minutes, is insufficient for adequate lactate removal. As a consequence, athletes enter the next match in a fatigued state, which may occur earlier than in the previous match. Judo competitors often participate in multiple fights on the same day, sometimes with very short time intervals between them. Therefore, it is highly likely that if lactate accumulation is the cause of muscle fatigue, an athlete who can clear lactate faster will begin the next match with less fatigue and thus have a better chance of achieving a superior competitive result.

The strength of this study lies in the fact that it comprehensively assessed the anthropological status of young elite judo athletes, which has not been done before. The lactate threshold, maximal oxygen consumption in conjunction with anthropometric characteristics, as well as a specific judo test tailored to the judo athletes themselves, were determined. Furthermore, the study established the correlations between the mentioned factors.

However, the limitations of the study are reflected in the relatively small sample size when it comes to correlations, as well as a limited variable system for each individual factor.

Therefore, future research in this field should include a larger sample size, encompassing a broader range of functional and anthropometric variables. Additionally, incorporating precise match analysis would contribute to providing more comprehensive insights.

Conclusion

Improving the ability to recover after a judo fight can have a significant impact on success in multiple fights within the same competition day. Objective measurement data can be used to control, guide, and regulate the training process. The findings of this research can be of great importance for judo schools, clubs, and all experts and coaches involved in judo athlete selection.

In practical terms, one approach could be to incorporate alternating series of high-intensity exercises during training, interspersed with technical actions against opponents, while allowing for short breaks between subsequent series. The theoretical application of the results is evident in the potential for

comparing the obtained findings with those of other authors. These findings provide motivation for further research aimed at addressing a wide range of issues related to adaptive changes resulting from long-term training.

Based on the obtained results and the statistical data processing and interpretation, the following conclusions were drawn: There is a moderate negative correlation between VO_2max and the index of the special judo fitness test, as well as a low correlation between PLAP and DLAP with the index of the special judo fitness test. Additionally, a moderate negative correlation between VO_2max and anthropometric parameters was found, while PLAP and DLAP showed a low correlation with anthropometric parameters.

The proposed test battery can be utilized to assess the functional status of competitors and more accurately determine the competitive profile for elite judo. Future studies could be designed to link physiological measurements with precise match analysis, aiming to establish an even stronger relationship between physiological and technical characteristics in judo.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Somatotypes of Top Croatian Male Volleyball Players

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Abstract

Situational efficiency of volleyball players depends on many factors, including specific ones in the area of morphology. The specialization of playing roles implies the performance of specific motor tasks, and therefore a specific combination of morphological factors necessary for the player's efficiency. This study aimed to determine the somatotypes of groups of top-level volleyball players divided according to their playing role. The research was conducted on 40 (body height 192.22 ± 8.58 cm, body weight 86.93 ± 11.22 kg, age 24.18 ± 4.97 years) male volleyball players from the Croatian A1 volleyball league. According to the playing role, 10 setters (body height 191.21 ± 6.39 cm, body weight 84.12 ± 8.76 kg, age 24.12 ± 6.77 years), 10 middle blockers (body height 199.97 ± 4.41 cm, body weight 94.82 ± 6.67 kg, age 26.05 ± 26.05 years), 10 liberos (body height 181.71 ± 5.71 cm, body weight 74.97 ± 5.59 kg, age 22.67 ± 2.88 years) and 10 outside hitters (body height 196.06 ± 4.34 cm, body weight 93.80 ± 10.29 kg, age 23.89 ± 5.53 years). Heath-Carter method was used for the somatotype determination. A statistically significant difference between the groups was detected in Body height, Body weight, Flexed arm girth circumference, Flexed calf girth circumference and Calf skinfold. The determined somatotypes for setters (2.8 - 3.8 - 3.4) and middle blockers (2.3 - 3.8 - 3.6) are mesomorph-ectomorph, and liberos (2.4 - 4.6 - 3.0) and outside hitters (2.5 - 4.3 - 3.1) are ecto-mesomorph. The results are in line with previous research. This confirms the importance of the longitudinal dimensionality of the skeleton in volleyball. In the future, we can expect the growth of the ectomorph component, with a stagnation or slight increase in the mesomorph component of the somatotype in top level volleyball.

Keywords: volleyball, morphology, Heath-Carter, somatogram, playing roles

Introduction

Due to its simplicity and variability, volleyball is one of the most popular team sports, and so is the FIVB (FIVB, 2023) federation one with the largest number of members (222). Considering the specificity of volleyball ("game on the net"), certain factors of the morphological area are particularly important for player selection, both in younger age categories, as well as at the top level (Đurković, 2009). Specialization of playing roles implies the performance of specific motor tasks, and therefore also a specific combination of morphological factors required for players' efficiency (Marques, Van den Tillaar, Gabbett, Reis & Gonzalez-Badillo, 2009; Palao, Manzares & Valades, 2014). Generally, a player's body height is of significant relevance for playing professionally in this sport, apart from the role of libero. Upon studying mor-

phological characteristics among groups with different playing roles (setters, middle blockers, outside hitters and libero players), the results can present information on the tendency of a certain playing role towards a specific body constitution type (Palao et al., 2014). Previous research on a sample of senior male volleyball players confirmed the hypothesis on the presence of four latent dimensions of the morphological area (Momirović, 1966; Strahonja & Matković, 1983; Đurković, Marelić & Rešetar, 2011), which were also confirmed on a sample of young and active persons (Momirović, Medved & Horvat, 1969; Medved, 1987; Marković, 2004). The factor of longitudinal dimensionality of the skeleton (which is defined by body height, arm span and the length of the arms, legs, and feet) is registered as the one in which the groups of players significantly differ (Đurković, Marelić & Rešetar, 2012).



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One of the systems for body constitution assessment is the Heath-Carter method (Carter & Heath, 1990), where the somatotype is defined by using three indexes – indexes of the endomorph, mesomorph, and ectomorph component. The endomorph component is an indicator of subcutaneous fat tissue presence, the mesomorph of joint robustness and proportion of lean mass, while the ectomorph point to the elongation of the body. Considering the aforementioned, it can be assumed that male volleyball players who were singled out in the process of selection as higher athletes shall be dominant in the last component. The tallest players, middle blockers, are also the heaviest players (Marques et al., 2009), which can affect the results of the ectomorph component. The setters, who are tall, however do not have the need for large muscle mass, i.e., body robustness, can demonstrate characteristics of an asthenic constitution. Spikers, due to the need for fast spikes and constant displays of maximum force through jumps and spikes, have the most pronounced results in the mesomorph component, as well as the libero because of his quick reactions, and agile and explosive movements (Petroski et al., 2013). In top-level volleyball (Janković, Đurković & Rešetar, 2009) the following playing roles are recognized: setters, middle blockers (central players), liberos, opposite (diagonal) hitters and receiver-attackers. Especially interesting are the results of longitudinal studies of somatotypes in top-level male volleyball players selected into national teams (Petroski et al., 2013 – Brazil / 11 years; Carvajal & Serviat Hung, 2014 – Cuba / 40 years) where there were changes registered in terms of a reduction of subcutaneous fat tissue of the examinees, however, also in terms of constant increase of the ectomorph component which is manifested through the increase of the average body height of the competitors of approximately 1 cm per decade, which shows us that taller volleyball players with less subcutaneous fat tissue passed the selection process, however, with constant, and even somewhat increased results in the mesomorph component. The results obtained on a sample of top-level club male volleyball players (Gualdi-Russo & Zaccagni, 2001; Toselli & Campa, 2018) who compete in the world's strongest volleyball league – the Italian league, confirm the fact that better quality players are taller and have a lower percentage of subcutaneous fat tissue, identically as the ones conducted on a sample of Greek first-league volleyball players (Giannopoulos, Vagenas, Noutsos, Barzouka & Beregles, 2017), where in addition to the previously mentioned trends, there was also a significantly greater body mass of quality volleyball players which is related to active muscle mass, as well as to a lower endomorph component. Greater body mass of individual playing roles is related to body height, and therefore taller volleyball players are also heavier (Marques et al., 2009). The results of a research by D'isanto, Di Tore and Altavilla (2018) confirmed a significant and high correlation between morphological indicators and the height of reach in volleyball players tested by using the Vertec system, which is relevant, as the height of reach is very important for the efficiency of the technical-tactical element of spike, which is used to win the greatest number of points in volleyball (Bellendier, 2003; Marelić, Rešetar, Zdražnik & Đurković, 2005; Đurković, Ban & Krmpotić, 2021). Challoumas and Artemiou (2018) determined a statistically significant, but negative correlation between the percentage of subcutaneous fat tissue and the speed of the volleyball spike. Although there are certain studies related to morphol-

ogy and volleyball in Croatia, this is the first one that has top senior volleyball players as respondents. In addition to the somatotype of the entire sample, the somatotypes for each playing role will be determined. This information can help coaches as model values for selection processes in the future. The aim of this study is to determine the somatotypes of different groups of male volleyball players, depending on their playing role (setter, middle blocker, libero, outside hitter). In addition, it will be verified if there is a statistically significant difference between the groups of volleyball players in 10 anthropometric morphological measurements in accordance with the Heath-Carter method for assessing somatotypes.

Methods

Sample of examinees

The research was conducted on a sample of forty senior male volleyball players (body height 192.22 ± 8.58 cm, body weight 86.93 ± 11.22 kg, age 24.18 ± 4.97 years) all members of A1 league volleyball clubs. The tested volleyball players compete at the highest level in Croatia, and most of them were members of senior national team in some part of their career. In the presentation of the obtained results and data processing, the players are distributed into four groups according to their roles: 10 setters (body height 191.21 ± 6.39 cm, body weight 84.12 ± 8.76 kg, age 24.12 ± 6.77 years), 10 middle blockers (body height 199.97 ± 4.41 cm, body weight 94.82 ± 6.67 kg, age 26.05 ± 26.05 years), 10 liberos (body height 181.71 ± 5.71 cm, body weight 74.97 ± 5.59 kg, age 22.67 ± 2.88 years) and 10 outside hitters (body height 196.06 ± 4.34 cm, body weight 93.80 ± 10.29 kg, age 23.89 ± 5.53 years). For this study, the roles of the opposite (diagonal) hitter and receiver-attacker have been combined into a group of outside hitters, as it was done in several previous research (Paz et al., 2017; Bisch et al., 2020; Rush, Littlefield, McInnis & Donahue, 2022). All measurements of the volleyball players were conducted at the Faculty of Kinesiology University of Zagreb. The research related to human use was complied with all the relevant national regulations and institutional policies, has followed the tenets of the declaration of Helsinki and has been approved by the research Ethics Committee of the Faculty of Kinesiology, University of Zagreb, Croatia (approval No.22./2023. as of March 13, 2023).

Measurements procedure

For the purpose of this research, 10 morphological measurements were estimated for determining the somatotype in accordance with the Heath-Carter method: Body height, Body weight, Humerus breadth diameter, Femur breadth diameter, Flexed arm girth circumference, Flexed calf girth circumference, Triceps skinfold, Subscapular skinfold, Supraspinal skinfold and Calf skinfold. The measurements were performed by qualified measurers who are members and external associates of the Sports Diagnostics Centre at the Faculty of Kinesiology in Zagreb. All participants gave written consent. The measurements were taken according to the anthropometric protocol of the International Society for the Advancement of Kinanthropometry – ISAK (Stewart, Marfell-Jones, Olds & De Ridder, 2011). The somatotypes were calculated and categorized by using the Heath-Carter method for assessing somatotypes.

Statistical analysis

The collected data was entered and processed by using the

statistical programme Statistical package Statistica, version 13.5.0.17 (TIBCO Software Inc., Palo Alto, CA, USA). The basic descriptive parameters were calculated for assessing the normality of distribution by using the Kolmogorov-Smirnov test. For verifying the statistical significance of the measured variables among the groups of tested volleyball players the Kruskal-Wallis test was used, whereas for further analysis of the detected significantly different results the Mann-Whitney U post-hoc test with the Bonferroni correction was used, and a p-level of 95% was applied.

Results

The normality of distribution was verified with the Kolmogorov-Smirnov test. The results of the variables for the assessment of skin folds were statistically significantly deviated from normal distribution, and therefore non-parametric methods were chosen for the continuation of statistical data processing. The descriptive indicators of the overall sample (AM±SD), as well as separately for each playing role are presented in Table 1. This table also shows the results of the Kruskal-Wallis test with the Mann-Whitney U post-hoc test

Table 1. Results of Kruskal-Wallis test with the Mann-Whitney U post-hoc test and the Bonferroni correction

Variables	Total (n=40)	Setters (n=10)	Middle blockers (n=10)	Liberos (n=10)	Outside hitters (n=10)	p
Body height (cm)	192.22±8.58	191.21 ± 6.39ml	199.97 ± 4.41sl	181.71 ± 5.71smo	196.06 ± 4.34sl	0.00
Body weight (kg)	86.93±11.22	84.12 ± 8.76ml	94.82 ± 6.67sl	74.97 ± 5.59smo	93.80 ± 10.29l	0.00
Humerus breadth diameter (cm)	7.30±0.36	7.25 ± 0.53	7.37 ± 0.29	7.12 ± 0.19	7.45 ± 0.29	0.10
Femur breadth diameter (cm)	10.17±0.55	10.08 ± 0.64	10.46 ± 0.43	9.83 ± 0.46	10.29 ± 0.48	0.06
Flexed arm girth circumference (cm)	34.42±2.15	33.92 ± 1.65	35.12 ± 2.37l	33.08 ± 1.20m	35.55 ± 2.46	0.04
Flexed calf girth circumference (cm)	38.72±2.42	37.29 ± 1.76mo	40.56 ± 1.64sl	36.61 ± 1.52m	40.40 ± 1.77sl	0.00
Triceps skinfold (mm)	9.46±3.39	10.22 ± 3.05	9.57 ± 2.63	8.24 ± 2.38	9.82 ± 5.06	0.39
Subscapular skinfold (mm)	10.67±2.67	11.37 ± 3.83	10.26 ± 1.75	10.18 ± 1.64	10.87 ± 3.06	0.99
Supraspinal skinfold (mm)	8.95±4.50	10.68 ± 5.15	8.23 ± 2.64	7.60 ± 4.89	9.29 ± 4.95	0.11
Calf skinfold (mm)	7.30±2.67	8.70 ± 1.75m	6.87 ± 2.00s	5.87 ± 1.69s	7.73 ± 4.03	0.03

Significantly different from ssetters, mmiddle blockers, lliberos, ooutside hitters

and the Bonferroni correction.

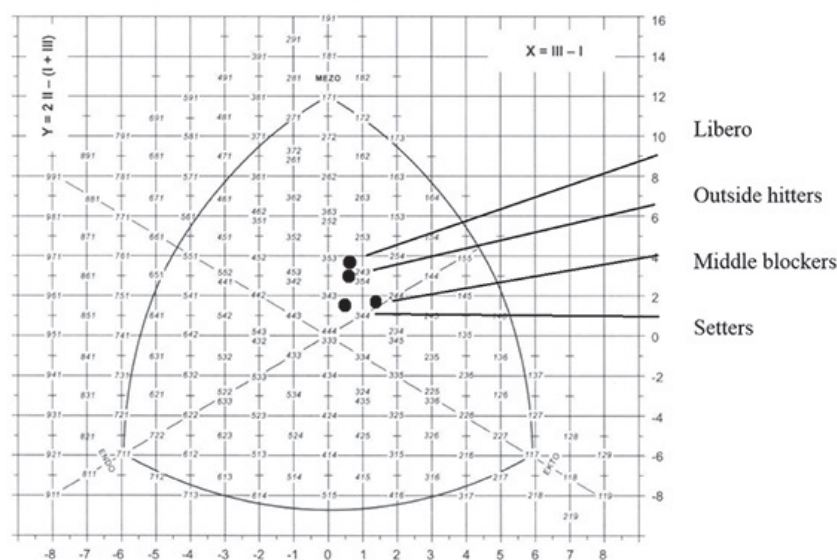
The Kruskal-Wallis test revealed a statistically significant difference in 5 out of the 10 measured variables: Body height, Body weight, Flexed arm girth circumference, Flexed calf girth circumference and Calf skinfold. The additional Mann Whitney post-hoc test with the Bonferroni correction (as this was an analysis between 4 different groups, the level of statistical significance was set at $p=0.05/4=0.0125$) was used for analysis of the variables for which the Kruskal-Wallis test registered statistically significant differences. Upon analysis of the Body height variable, a statistically significant difference was recorded between setters and middle blockers ($U=10.00$, $z=-3.02$, $p=0.00$). The average height of middle blockers is 199.97 ± 4.41 , while of setters it is 191.12 ± 6.39 . Significant difference exists between libero players and setters ($U=8.00$, $z=-3.17$, $p=0.00$), liberos and middle blockers ($U=0.00$, $z=-3.78$, $p=0.00$), as well as between liberos and outside hitters ($U=0.00$, $z=-3.78$, $p=0.00$). The average height of libero players is 181.71 ± 5.71 cm, whereas the heights of setters, middle blockers and outside hitters are respectively 191.12 ± 6.39 cm, 199.97 ± 4.41 cm and 196.06 ± 4.34 cm. By analysing the variable of Body weight, a statistically significant difference was found between setters and middle blockers ($U=13.00$, $z=-2.80$, $p=0.01$). The average body weight for the group of setters is 84.12 ± 8.76 kg, while for the middle blockers it is 94.82 ± 6.67 kg. Significant difference exists between middle blockers and libero players ($U=3.00$, $z=-3.55$, $p=0.00$), with a considerably lower average body weight of libero players 74.97 ± 5.39 kg, as well as between liberos

and outside hitters whose average weight is 93.80 ± 10.29 kg ($U=3.00$, $z=-3.55$, $p=0.00$). In the analysis of the variable Flexed arm girth circumference the only recorded significant difference was found between the groups of libero players and outside hitters ($U=16.00$, $z=-2.58$, $p=0.01$). A smaller average value of 33.08 ± 1.20 cm was measured for libero players, as opposed to the group of outside hitters with the average of 35.55 ± 2.46 cm. Upon analysis of the variable Flexed calf girth circumference a significant difference was found between setters and middle blockers ($U=8.00$, $z=-3.18$, $p=0.00$), as well as between setters and outside hitters ($U=10.50$, $z=-2.99$, $p=0.00$). Furthermore, a significant difference was recorded between the groups of libero players and middle blockers ($U=2.50$, $z=-3.60$, $p=0.00$) and between liberos and outside hitters ($U=4.00$, $z=-3.48$, $p=0.00$). The analysis of the variable Calf skinfold showed a significant difference between the groups of setters and libero players ($U=14.00$, $z=-2.72$, $p=0.01$). The group of setters showed higher average values of 8.70 ± 1.75 mm, while the group of libero players showed values of 5.87 ± 1.69 mm. The calculated somatotypes of the groups of players are presented in Table 2. Considering the obtained data, the conclusion can be made that the somatotypes of setters (2.8 - 3.8 - 3.4) and middle blockers (2.3 - 3.8 - 3.6) are mesomorph-ectomorphs, while of libero players (2.4 - 4.6 - 3.0) and outside hitters (2.5 - 4.3 - 3.1) are ectomorph-mesomorph.

The results of somatotypes for groups of volleyball players calculated by using the Heath-Carter method on the somatogram are presented in Figure 1.

Table 2. Somatotypes of tested volleyball players

Somatotype	Total (n=40)	Setters (n=10)	Middle blockers (n=10)	Liberos (n=10)	Outside hitters (n=10)
Endomorphy	2.5	2.8	2.3	2.4	2.5
Mesomorphy	4.16	3.8	3.8	4.6	4.3
Ectomorphy	3.28	3.4	3.6	3	3.1

**FIGURE 1.** Somatogram of volleyball players according to playing roles

Discussion

Upon analysis of the Body height variable, a statistically significant difference was recorded between setters and middle blockers. The longitudinal dimensionality of the skeleton is a key morphological factor for middle blockers (Janković et al., 2009), as their primary task is in the block phase, as well as attacks by quickly setting balls (so-called first tempo attack). Defensive plays of middle blockers are very limited, and that is to the rotation in which they serve. Although it is preferable for setters to also be taller because they participate in plays on the net (a taller setter can be more efficient while blocking and attacking – most often with the so-called second touch, i.e., dumping the ball, however also sometimes with a spike), the focus of their performance nevertheless remains on tactically justified and precise sets (Janković & Marelić, 2003) for the hitters. Significant difference between the mentioned two playing roles in this variable is also often found during other research on samples of top-level senior male volleyball players (Grgantov, 2002; Marques et al., 2009; Sattler, Sekulić, Hadžić, Uljević, & Dervišević, 2012; Palao et al., 2014). Significant difference was also determined between the liberos and all the other playing roles (Đurković, Marelić, & Zekić, 2020). In the Body height variable, there is a statistically significant difference between libero players and setters, liberos and middle blockers, as well as between liberos and outside hitters. Libero players do not participate in plays on the net, and therefore body height is not a crucial factor for being a successful player in that position. Their primary role is service reception, defending the field and covering the attackers, however, more and more often also setting the ball.

By analysing the variable of Body weight, a significant difference was found between setters and middle blockers. Identical as for the variable of Body height, this variable al-

so indicated a statistically significant difference. The reason for this is in the correlation between body height and body weight, as taller players are generally also heavier (Ciesla et al., 2015). The same conclusion can also be made in relation to the difference between middle blockers and libero players, and outside hitters and libero players, with a considerably lower average body weight of libero players.

The only recorded significant difference in the variable Flexed arm girth circumference was found between the groups of libero players and outside hitters. A smaller average value was measured for libero players, as opposed to the group of outside hitters. The upper arm circumference is partially related to the cross-section of upper arm muscles (Eaton-Evans, 2013). The probable cause for this is the high number of repetitions of the spike and serve techniques which are performed by outside hitters. The mentioned difference in fact shows the specificities of these two playing roles – outside hitters, whose priority is to perform strong and efficient spikes and serves, and libero players, who according to the rules of volleyball are not allowed to serve and spike.

Upon analysis of the variable Flexed calf girth circumference a significant difference was found between setters and middle blockers, as well as between setters and outside hitters. A potential reason for this lies in the specific performance of middle blockers and outside hitters, as they generally perform jumps with maximum intensity, i.e., maximum engagement of muscle groups (particularly in the phases of attack spike and counter-attack spike). Although according to some research setters perform the highest number of jumps during games (Fontani, Ciccarone, & Giulianini, 2000; Bahr & Bahr, 2014), most of them are performed in the preparatory phase for setting and thus most often do not de-

mand maximum intensity. Furthermore, a significant difference was recorded between the groups of libero players and middle blockers, and between liberos and outside hitters. A likely reason for this significant difference in the mentioned variable is the specificity of tasks which the two playing roles perform during a volleyball match. During trainings and games, middle blockers and outside hitters execute many vertical rebounds, whereas libero players do not. The analysis of the variable Calf skinfold showed a significant difference between the groups of setters and libero players. The group of setters showed higher average values than liberos. The possible reason for this is in the fact that certain setters were older (three of them are in their late thirties). Even though they were not in their top-level sports form (which was potentially also manifested in the results of this skinfold), they played at a very high level and were selected for the team in their older veteran age because of some other advantages – experience, calmness, responsibility, and tactical decision-making. Similar data was also found by Marques and Marinho (2009) and Palao et al. (2014) on a sample of senior male volleyball players. The variables for assessing the skinfolds provide information on the proportion of subcutaneous fat tissue and thus define endomorphy (Đurković, 2009). In volleyball, subcutaneous fat tissue should be reduced to a minimum due to the need for speed, agility and jumping (Nikolaidis, 2013), however also as prevention from chronic injuries because of a high number of landings (Vanderlei et al., 2013). This feature is independent of other morphological characteristics, and it shows the highest result variability in Table 1. Regardless of a higher proportion of fat tissue, technical-tactical maturity and player experience can be an advantage in the selection of players who are in poorer sports form in relation to the team. A confirmation of this lies in the players who in their older veteran age play at a very high level, which is also certified by different research (Marques & Marinho, 2009; Palao et al., 2014).

The variables that show a low variability of results are diameters and circumferences which define the mesomorph component of the somatotype. Within each playing role, as well as within the overall sample, very similar results were found with low variations of standard deviation. Mesomorphy is particularly pronounced in playing roles of outside hitters and libero players (Giannopoulos et al., 2017; Toselli & Campa, 2017). A bigger calf circumference in middle blockers and outside hitters is potentially the result of performing maximum intensity jumps, so anatomical adaptation to specific loads is thus possible (Cengizel & Cengizel, 2021). The upper arm circumference is defined by the cross-section of upper arm muscles, while muscle hypertrophy is partially an indicator of flexor strength and elbow extensors (Toselli & Campa, 2017). The extensors are of particular importance, dominantly *musculus triceps brachii*, whose activity is extremely important (Teoh et al., 2021) in preparation and main phases when performing overhead passes, spikes and serves.

The variables of Body height and Body weight are also highly correlated as, generally, taller athletes are heavier as well (Gualdi-Russo & Zaccagni, 2001). Without libero players, the average height is 195.75 cm and average weight is 90.91 kg. The variables of Body height and Body weight cause the values of the ectomorph somatotype index, which shows high results for volleyball players. It is somewhat higher

among middle blockers (3.6), whose key role is in the block phase, as well as among setters (3.4), as opposed to outside hitters (3.1) and libero players (3.0).

Comparison with similar samples points to the fact that Croatian volleyball players match players from other countries in terms of their constitution (Petroski et al., 2013; Toselli & Campa, 2018). A low endomorph component was calculated, which is an indicator of the percentage of subcutaneous fat tissue in the body. The mesomorph component is most pronounced among players in the libero position, whereas the ectomorph component is highest among the tallest volleyball players – middle blockers. When comparing the characteristics of volleyball players from Brazil in the 2000's, the most common somatotype was mesomorph-ectomorph, which is also the case among Croatian players in this research as the somatotype of setters and middle blockers. The somatotype components match the ones that were measured on Italian volleyball players in the 90's (Gualdi-Russo & Zaccagni, 2001), however, differ from the results of a more recent research (Giannopoulos et al., 2017). It is to be expected that in the future the ectomorph component shall be increased, while the mesomorph somatotype component shall be reduced.

The limitation of this research is a relatively small sample, considering that the goal was related to calculating the somatotype of each playing role separately. The value of this sample is represented by the fact that they are top volleyball players, most of whom played for the national senior team in some part of their career. Longitudinal monitoring of the somatotypes of national team members is recommended in order to additionally confirm the correct criteria during the selection process by comparing them with the most successful volleyball nations.

Conclusion

The obtained results are similar to other results obtained on a sample of top-level senior male volleyball players, which indicates a quality-level player potential in the area of the tested characteristics. A statistically significant difference was found in morphological characteristics between the tested Croatian top-level volleyball players distributed according to their playing roles. The explanation lies in the specificity of volleyball and the requirements which must be met by players in a particular playing role. The efficiency in the performance of technical-tactical elements depends, among other things, on individual morphological measurements, as since the tasks for each playing role are also specific. As libero players do not take part in plays over the net, thus body height is not a key factor for achieving success for players in that position, however, due to the required speed of reaction and acrobatic landings after defending hard spikes, the mesomorph component is therefore an important one. Contrary to libero players, the height of reach in the block, as well as the ability to quickly set blocks present priorities for the position of middle blockers, and thus body height and the length of extremities is crucial. Outside hitters form in a way a link between liberos and middle blockers, as in addition to defensive techniques of service reception and defense, for which the mesomorph component is important, they also need longitudinal dimensionality of the skeleton which is relevant for playing on the net (spikes and blocks). The priority for setters is creating situations for the easiest possible attack by hitters. Although during three rota-

tions they participate in blocks and body height is important, their focus is on quality ball distribution towards the hitters. Constitution is a variable feature during growth and development, it is measurable and can be influenced by a planned and programmed training process. Numerous attempts to predict the development of morphological characteristics in younger age categories have been more or less successful, and it is thus important to continuously monitor competitors from the level of younger age categories to senior nation teams in order to gain an overall insight into the development of volleyball players from their entrance into competitive volleyball until

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ORIGINAL SCIENTIFIC PAPER

Differences in Motor Abilities among Basketball Players in Relation to Biological Maturity

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Abstract

Basketball is a popular team sport with a combination of alternating and high-intensity activities. Physical characteristics related to age and maturity can be quite important for developing the career of a young athlete. The aim of this study was to determine the differences in motor skills among basketball players with regard to their biological maturity. Using the Mirwald equation, the total sample of participants was divided into three groups based on their biological maturity: early-maturing (postPHV), normally-maturing (midPHV), and late-maturing (prePHV). The sample of participants consisted of 51 basketball players (body height 171 ± 12 cm; body weight 63 ± 15.7 kg; BMI 21.3 ± 3.8). Tests of speed (5, 10, 20m sprint), explosive power of the lower extremities (the countermovement jump with arm swing (CMJwas), the countermovement jump (CMJ), and the squat jump (SJ)), and agility (the t-test, zig-zag, and slalom test) were evaluated. The one-factor ANOVA showed a statistically significant difference between groups in the slalom, t-test, and zig-zag test ($p < 0.05$). On the sprint tests and tests assessing explosive power no difference was found. We can conclude that this group of participants showed equal performance in motor skills on most tests, regardless of their maturity status. The primary significance of this study is that it shows coaches the importance of understanding their players' biological maturation for optimal selection, suitable training design, and a reduction in the risk of injury.

Keywords: maturation, youth sport, maximum height gain, growth rate, motor performance

Introduction

Basketball is a well-known team sport that combines alternating and high-intensity actions (McKeag, 2008). In addition, it is a sport that includes plenty of body contact and an aggressive approach. Although it is a team sport, the individual quality of the athletes can decide the winner of the match in most situations (Mujika et al., 2018). Basketball was primarily invented as a five-on-five game, but then other types were added, such as the three-on-three game that later emerged as an independent sport (Erčulj et al., 2020). Both aerobic and anaerobic movements are involved in the game of basketball, where the predominant anaerobic movements are sprints, changes of direction, and vertical jumps.

Precisely because of the manner of play, with plenty of body contact and demanding physical activities, the athlete should always be mentally and physically prepared for every match and training session (Alemdaroğlu, 2012). Strength is known to be an ability that helps us overcome resistance and is important in almost every sport, especially contact sports (Siff, 2000). The dominant type of strength used in the game of basketball is explosive strength, whether it is for shooting at the hoop or taking advantage of the offense. Assessment of strength performance in basketball players is performed through the vertical jump and vertical jump tests (Cole et al., 2020). Vertical jumps are the most frequently repeated motor ability in a basketball game, occurring per



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minute (Stojanović et al., 2018). Therefore, analyzing peak height velocity can be crucial in working with young basketball players when it comes to training (Živković et al., 2022). Basketball players aged 14 have a lower jump height than those aged 16 (Stanković et al., 2022). It is known that athletes who develop per their generation, especially those who reach maturity less progressively, later achieve better results compared to those who developed ahead of time (Chelladurai & Carron, 1983). That is why calculating biological maturity is important during the team selection process and even more important for the development of athletes.

Radiography and the method of sexual characteristics are just some of the methods for calculating biological maturity, but an equally accurate and non-invasive method is the calculation of peak height velocity using anthropometric characteristics (Mirwald et al., 2002). In youth, motor skills are strongly related to growth (Malina et al., 2004). Interestingly, during their growth spurt, male athletes of the same age can differ in biological age (Ramos et al., 2021). Static strength and explosive power develop very quickly after a growth spurt (Te Wierike et al., 2014), and this is the period in which individual players stand out from their peers in terms of quality and talent. One study (Coelho e Silva et al., 2008) showed that height has a positive correlation with passing skills in basketball. Motor and morphological development as well as maturity should be synchronized to stop the stagnation of motor abilities or even injuries (Jakovljević et al., 2016). Physical parameters related to

maturity and chronological age can be crucial in the career of a young athlete (Torres-Unda et al., 2013). Some authors have concluded that physical performance is influenced by maturity status, while technical skills are influenced by years of training (Guimarães et al., 2019).

Although the mentioned studies (Torres-Unda et al., 2013; Guimarães et al., 2019) examined the impacts of maturity status in male basketball players aged 11 to 14, no study has looked specifically at male basketball players aged between 14 and 16. Moreover, little is known about the differences in motor abilities related to maturity status in the population of Serbia. Therefore, the aim of this study was to determine the differences in the motor skills of basketball players in relation to biological maturity.

Methods

Participants

Consent for participation was provided by the parents/guardians. All of the participants were fully informed about all the testing procedures, possible risks, and the general purpose of the study before participating. Fifty-one young basketball players aged 12 to 16 volunteered to participate in this study. The total sample of participants was divided into three groups based on their biological maturity: early-maturing (postPHV), normally-maturing (midPHV), and late-maturing (prePHV). The participants were familiar with the testing protocol, were healthy, and had no musculoskeletal disorders. The descriptive statistics of the participants are shown in Table 1.

Table 1. Descriptive statistics of the participants

Groups	Mean±SD	Height (cm)	Body mass (kg)	Chronological Age	Maturity Age@PHV	Body mass index
prePHV	23	162.2±8.5	52.4±8	13±0.6	14.3±0.9	19.9±1.9
midPHV	9	171.9±7.9	64.3±8.5	13.5±0.9	13.6±0.8	21.9±3.1
postPHV	19	182±7.6	75.3±16.5	14.7±0.9	13±0.7	22.7±5.1

Test Procedures

Assessment of anthropometry

Body height was assessed using the Martin anthropometer, model 101 (GPM, Switzerland). The participants stood barefoot on a firm and leveled surface. The standardized command given by the measurer was “Stand still, put your feet together, and look straight ahead”. Body mass, the body mass index, fat percentage, and muscle percentage were measured using a portable scale (Vasold et al., 2019) (Omron BF511, Kyoto, Japan). The participants stood on the scales barefoot, in their underwear.

Maturation assessment

Peak height velocity (PHV) was assessed using the method proposed by Mirwald et al. (2002) due to its non-invasive nature and relatively reliable measurement accuracy. The following equation, unique to boys, was used to calculate the number

of biological years.

Male offset = $-9.236 + ((0.0002708 \times (\text{leg length} \times \text{sitting height})) + (-0.001663 \times (\text{age} \times \text{leg length})) + (0.007216 \times (\text{age} \times \text{sitting height})) + (0.02292 \times ((\text{weight}/\text{height}) \times 100))$. The ranges that determine the maturity of the group of participants are: (prePHV = <-0.5; midPHV = -0.49 – 0.49; post-PHV = >0.5).

Physical performance assessment

The warm-up protocol developed by Jeffreys et al. (2007) was used. The coach led the participants through a series of exercises to raise their body temperature before activating the muscles involved and then mobilizing the muscles. The warm-up was completed by using the last phase (potency) as a preparation for more intensive work. All of the participants repeated the test protocol they had learned the day before the test.

The warm-up is shown in Table 2.

Table 2. Warm-up protocol

Aim	Exercise	Sets	Distance (m)
Lifting	knees high, skips, and lateral changes	3	20
Activation	lunges, alternating squats, and “Inchworms”	2	20
Potentialiation	Squat jumps, one leg jumps, and pogo jumps	3	20

Speed

Photocells were used to estimate the speed for the 5, 10, and 20-meter sprint (Witty, Microgate, Italy). Each participant repeated the speed test three times, while the pause between attempts was determined by the time it took the participant to return to the starting position by walking at a normal pace. The fastest attempt was included in the statistical analyses. The photocells were placed at hip height, and the participants started the linear sprint at a distance of 5 centimeters from the first gate.

The explosive power of the lower extremities

Photoelectric cells (Optojump, Microgate, Bolzano, Italy) were used to assess the explosive strength of the lower extremities, the reliability and validity of which were already confirmed (Healy et al., 2016). The participant places his hands on his hips. For the starting position of the squat jump, he is in the upright position. The participant needs to perform a quick counter movement - a push-off action. When performing the lowest position, it should be from a semi-squat (knees $\sim 90^\circ$ and torso/hips in a bent position) as in the case of a squat jump. The jump must be performed as quickly and explosively as possible to ensure the highest jump is made in the shortest time possible. The participants performed three attempts each for the CMJwas, CMJ, and the SJ. The participants performed the jumps voluntarily. The break between each repetition was 30 seconds, while the break between different types of jumps was two minutes. The highest vertical jump value was used for statistical analyses.

Assessment of agility

Agility was assessed using the slalom, t-test, and zig-zag test. The same equipment used for speed assessment was used to assess agility (Witty, Microgate, Italy). In all three tests, the participants voluntarily began the testing procedure. They had three attempts for each test, with one-minute breaks between attempts, while the breaks between tests lasted from 3 to 5 minutes. The slalom test is described as follows: the participant is in a high start with both feet behind the photocells. Six cones are placed 2 m apart. The first one is 1 m away from the starting line. Running from the first cone to the second cone, the participant goes around

the second cone with the right side of his body and continues running as fast as possible, constantly changing direction from right to left, until he reaches the last cone. At the last cone, the participant makes a 180° turn and continues the slalom run to the finish/start position. In the t-test, three cones are placed in the same plane at a distance of 4.57 meters. The participant runs in a straight line from the gate to the middle cone (9.14m) and touches the top of the cone with his right hand, then laterally moves to the right cone (4.57 m from the middle cone), touches the cone with his right hand, touches the left cone with a lateral movement to the left, touches the middle cone again and then runs back through the gate. The reliability and validity were confirmed in a previous study (Pauole et al., 2000). For the zig-zag test, four cones placed in the corners of a 3-meter by 5-meter rectangle, with another cone placed in the center. If the cones placed in the corners of the rectangle are marked 1 to 4, starting from the longer side, and the central cone is C, the test starts from cone 1 and continues around the cones in the following order C, 2, 3, C, 4, then back to 1. The reliability and validity were confirmed in a previous study (Kutlu & Doğan, 2018).

Statistical data processing

Descriptive statistics were calculated and presented. The Kolmogorov-Smirnov test was used to calculate the data distribution. The one-way ANOVA test and Kruskal-Wallis test were used to determine the differences between groups, and a post hoc analysis was used to determine which group contributed to that difference. The statistical level of significance was set at the level of 0.05, while the statistical processing of the data was done using a statistical package IBM SPSS, version 20 (Inc., Chicago, IL, USA).

Results

Our results provide strong evidence that young basketball players do not differ in motor abilities in most of the variables. The slalom, t-test, and zig-zag test showed a statistically significant (0.013, 0.029, and 0.004 respectively) difference between the groups. The Kolmogorov-Smirnov test showed normal distribution for all the variables excluding the t-test, where the Kruskal-Wallis test was used. The results of the motor abilities are shown in Table 3.

Table 3. P-values of motor ability tests

Tests	p-value
5m (s)	.72
10m (s)	.68
20m (s)	.14
CMJ (cm)	.74
CMJwas (cm)	.83
SJ (cm)	.78
Slalom(s)	.013
T-test (s)	.029
Zig-zag (s)	.004

Legend: 5m - sprint test, 10m - sprint test, 20m - sprint test, CMJ - countermovement jump, CMJwas - countermovement jump with arm swing, SJ - squat jump, T-test - agility test, Slalom test - agility test, Zig-zag - agility test

Table 4. Descriptive statistics for each test group

	PHV Groups	Mean	Std. Deviation
prePHV	5m	1.22	.13
	10m	2.06	.17
	20m	3.66	.26
	CMJ (cm)	25.68	4.80
	CMJwas(cm)	31.48	6.02
	SJ (cm)	24.70	4.82
	Slalom(s)	10.19	1.97
	T-test (s)	9.93	2.59
	Zig-Zag(s)	7.87	.83
midPHV	5m	1.15	.11
	10m	1.96	.13
	20m	3.48	.19
	CMJ (cm)	29.15	4.64
	CMJwas(cm)	33.31	6.22
	SJ (cm)	26.91	4.32
	Slalom(s)	10.11	1.35
	T-test (s)	9.36	2.61
	Zig-zag(s)	8.05	.63
postPHV	5m	1.20	.11
	10m	2.01	.15
	20m	3.47	.25
	CMJ (cm)	30.30	7.08
	CMJwas (cm)	36.40	7.60
	SJ (cm)	26.13	5.22
	Slalom(s)	11.49	.98
	T-test (s)	7.48	1.18
	Zig-zag(s)	8.59	.56

Legend: prePHV – late-maturing, midPHV – normally-maturing, postPHV – early-maturing, 5m - sprint test, 10m - sprint test, 20m - sprint test, CMJ - countermovement jump, CMJwas - countermovement jump with arm swing, SJ - squat jump, T-test - agility test, Slalom test - agility test, Zig-zag - agility test

The homogeneity of variance test showed that equal variance was assumed and the Tukey post hoc test was used. In contrast to this, the homogeneity of variance test showed a p value below 0.05 for the t-test; therefore, the Games-Howell test was calcu-

lated. The results of the post hoc analysis are shown in Table 5.

The post hoc analysis showed that prePHV and postPHV contribute to statistically different outcomes in the slalom, t-test, and zig-zag test.

Table 5. Post Hoc Analysis for Differences between Maturation Stage

Variable	Sig	PHV
Slalom	0.009	prePHVa
		midPHVab
		postPHVb
T-test	0.001	prePHVa
		midPHVab
		postPHVb
Zig-zag	0.002	prePHVa
		midPHVab
		postPHVb

Legend: prePHV – late-maturing, midPHV – normally-maturing, postPHV – early-maturing
different letters denote significant differences

Discussion

The goal of this study was to determine whether there are differences between the motor skills of basketball players with regard to biological maturity. Our results showed significance in the slalom test, t-test, and zig-zag test, while there was no significance in the tests of speed and explosive power of the lower extremities.

Explosive power is considered the most important characteristic of basketball players (Santos & Janeira, 2008). Our results showed no significant difference in vertical jump height between prePHV, midPHV, and postPHV. The results are similar to those of Coelho e Silva et al. (2008b), which showed no statistical difference for the CMJ and SJ in fourteen-year-olds. Also, the results of Ramos et al. (2020) showed that there are no significant differences in the vertical jump performance between participants divided by PHV. Contrary to our findings, Jakovljević et al. (2016) found results showing that normally-maturing young basketball players showed the best CMJ performance, hypothesizing that matched biological and chronological age is the best combination for good motor performance. Another study (Sekine et al., 2019) proved that post-PHV basketball players had better jump performance (Cloud jumps) than midPHV and prePHV basketball players. It can be said that the speed and tempo of anaerobic ability improvement depend on biological maturity, not on chronological age.

The change of direction speed is the mechanical basis that supports agility in basketball players (Popowczak et al., 2021). In our findings, a difference was found between the prePHV and postPHV groups on the agility tests, the slalom, t-test, and zig-zag agility test. One study had different results, i.e., there was no difference between groups of basketball players who had different PHV on the agility tests (Arede et al., 2019). In contrast to our results, where prePHV and postPHV had the best results on the agility tests, Jakovljević et al. (2016) found that midPHV basketball players had the best results on agility tests (the t-test, zig-zag test).

During a basketball game, a player's speed ranges from 0 to 8 m/s (Oba & Okuda, 2008). Our results showed that players whose timing of maturation varies did not differ on speed

tests (5, 10, and 20m). In contrast, normally-maturing basketball players had the best results on the 20 m sprint (Jakovljević et al., 2016). Another study had different results; specifically, postPHV basketball players were better on the 20m speed test than the midPHV and prePHV groups. Also, Sekine et al. (2019) confirmed that a group of postPHV basketball players had the best results on the 10 and 20m sprints. These results were confirmed by Guimarães et al. (2019), indicating that the early-maturing group was taller and heavier, and had better results not only on speed tests, but also on strength and agility tests. The results of these authors have a logical basis, which is that early maturation leads to the overall development of the body and, therefore, better physical performance on tests of motor abilities (Malina et al., 2015).

The limitation of this study is that it is cross-sectional and that some type of intervention is missing. Also, the participants were exclusively basketball players and it is not possible to generalize these results to other populations. Moreover, the number of participants in each subgroup is small, and this might be a problem with the generalization of the data. Future research should be global, with a focus on the playing position, case study, follow-up method, and a greater understanding of the effect of biological maturity on basketball players' motor ability.

Conclusion

Basketball players aged 12 to 16 showed that groups divided by maturity differ on the slalom, t-test, and zig-zag agility tests, while tests of speed and explosive power of the lower extremities showed similar results. Also, prePHV and postPHV had a significant impact on the differences between groups. The primary significance of this study is that it shows coaches the importance of understanding their players' biological maturation for optimal selection, suitable training design, and a reduction in the risk of injury. The results of this study emphasize the significance of the maturity status effect on youth basketball players' physical performance and technical skill development. In general, our findings show that physical performance is affected by maturity status.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Construction of a Specific Test for Flexibility Assessment in Rhythmic Gymnastics

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Abstract

Rhythmic gymnastics is a combination of sport and art, in which flexibility is one of the most important abilities and plays an important role in the performance specification equation. However, there is limited systematic research on specific tests of flexibility in rhythmic gymnastics for recognising children's potential and tracking the training process. The primary purpose of this research was to construct a measurement instrument for evaluating flexibility that would be applied specifically to rhythmic gymnastics. The sample consisted of 41 female rhythmic gymnasts aged 10-12 ($X=11.05$, $SD=0.84$). All of them are members of two rhythmic gymnastics clubs from Croatia and registered with the Croatian Gymnastics Federation. We used three familiar flexibility tests: - forward bend on a bench, frontal split with hand support, split in a supine position - and one new test - frontal split on an elevated surface. Results of the statistical analyses have shown that the new test measures flexibility, which is required for success in rhythmic gymnastics. There is a significant correlation between individual measurements ($p<0.001$), so this test is considered reliable. It would be good to implement the new test when selecting children for rhythmic gymnastics as well as during transitive measurements in the training process.

Keywords: *ability, performance, newly developed test, success, rhythmic gymnasts*

Introduction

Rhythmic gymnastics is a sport that contains its own artistic component. In addition to the five apparatuses that the gymnasts use, the body performs natural forms of movement following the rhythm and tempo of music. Body movements, apparatus work and music all come together in the beauty of performance. Countless different movements and combinations classify rhythmic gymnastics as one of complex acyclic sports (Wolf-Cvitak, 2004). It affects the development and improvement of a range of motor skills and the ability to follow the rhythm and tempo of music (Furjan-Mandić et al., 2010). Motor abilities affect the efficiency of human movement. Various studies have shown that motor skills cannot be defined by a single, universal feature, but rather, an analysis of human mobility requires division into several quantitative and qualitative motor skills (Cvenic, 2007). One of the most important motor skills in rhythmic gymnastics is flexibility (Lafranchi, 2005). It is the ability to perform the maximum amplitude of movement in a particular joint. Not all joints are equally flexi-

ble, which is conditioned by the structure of the articular bodies and the elasticity of the ligaments, tendons and muscles that perform the movement (Benjamin et al., 2006). Flexibility also depends on age, gender, room and body temperature (Breslauer et al., 2014). The ability to efficiently perform various elements of different techniques is very important in many sports, especially in rhythmic gymnastics. According to Ljevakovic (1982), a reduced range of motion can cause injuries in case of sudden movements and lead to excessive energy expenditure in movements that are difficult to perform due to inelasticity of antagonists or excessive muscle tone in agonists. Performance of each element requires certain flexibility which is often crucial in the selection of children for this sport (Hume et al., 1993). Due to early specialization, flexibility training begins at the age of 4-5, which is essential in addition to developing other abilities such as proper muscle activation and movement control that can prevent many injuries (Kristicevic et al., 2017). There are three types of flexibility: static, dynamic and passive (Kos, 1966). Static flexibility refers to the maximum range of motion



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without an emphasis on speed, for example a split. On the other hand, passive flexibility is a measure of the maximum range of motion without voluntary muscle contraction with the help of a trainee (Alter, 2004). According to Kos (1966), dynamic flexibility is the ability to perform the maximum range of motion in fast - moving activities, such as various jumps. Donti, Tsolakis and Bogdanis (2014) indicate that this sport requires high level of flexibility and good compromise between strength and flexibility, proving that in their study which examined the effects of baseline flexibility and vertical jump ability on straight leg raise range of motion (ROM) and counter-movement jump performance (CMJ) following different volumes of stretching and potentiating exercises. According to Laffranchi (2005), these performances are reached through a detailed planning and training organization. Santos et al. (2015) concluded that the best suggestion for the flexibility training is the implementation of symmetrical work to accomplish the correct and balanced development the gymnasts. However, participants in the study were high level Portuguese juniors and it does not include the rest of gymnasts. Also, the tests applied in the study had as limitation the type of evaluation with comparison of images and an assessment chart, but they are the FIG suggested tests, and they are close to rhythmic gymnastics reality. Since there is no adequate specific test for assessing the static flexibility in rhythmic gymnastics and the need for it can be seen from the given literature, in this research we will propose a new one. The aim was to construct a measurement instrument for evaluating flexibility that would be applied specifically to rhythmic gymnastics. The test could serve not only to facilitate the assessment of flexibility in the selection of children as successful rhythmic gymnasts in the future, but also to control the training progress and provide an insight into the current state of a gymnast's flexibility.

Methods

Participants

The sample consisted of 41 female rhythmic gymnasts aged 10-12 ($X=11.05$, $SD=0.84$) who are registered with the Croatian Gymnastics Federation. The gymnasts are members of two rhythmic gymnastics clubs from Zagreb (GK Aura) and Osijek (KRSR Rondo). The participants were measured voluntarily and the parents of the gymnasts gave their written consent for the study. It should be pointed out that the gymnasts

were free of injuries, warmed up before the testing and were familiar with the protocol of this research. All participants are competitors in rhythmic gymnastics and they started participating in the research the same level of learning. The Ethics Committee of the Faculty of Kinesiology concluded that the research respects ethical and professional principles and approved it on December 1, 2020, with decision 101/2020.

Measurements

To test flexibility, three familiar tests with verified metric characteristics were used – forward bend on a bench, split in a supine position (Metikos et al., 1989) and frontal split with hand support (Santos et al., 2015). Another new test – frontal split on an elevated surface was used as well. These tests measure flexibility and involve holding a static position for a minimum of 30 seconds, with measurements made in centimeters.

Protocol of the research

Tests were performed twice in the same way at two-month intervals, from the beginning of January to the end of February 2022. The study was conducted at the sports hall of St. Anne's Elementary School in Osijek and at the Arena Sports Hall in Zagreb.

Forward bend on a bench

The forward bend on a bench (FBB) test evaluates the flexibility of the lower back and back of the legs. Validity and reliability are described in the book "Measurement of Basic Motor Dimensions of Athletes" by Metikos et al. (1989). The duration of testing is up to 15 seconds per gymnast. To perform the test, a 40-cm-high bench is needed, on which a board with an 80-cm-long centimeter strip is attached to the front. The fortieth centimeter of the strip is at the level of the upper surface of the bench. In the starting position the gymnast is standing barefoot on a bench with her feet together. Her legs are fully extended and the tips of her feet are placed to the edge of the bench. The gymnast stretches out her arms and places the palm of the right hand on the back of the left one so the middle fingers overlap. She performs the maximum forward bend by slow descent (Figure 1). The task is completed when the gymnast touches the maximum possible value on the centimeter tape with her fingertips and maintains the position until the result is read.



FIGURE 1. Forward bend on a bench test

Frontal split with hand support

The test evaluates the maximum passive and active flexibility through the dimensionless method which compares the gymnast's joint range of motion against an assessment chart. Validity and reliability are described in the paper "Active and Passive Lower Limb Flexibility in High Level Rhythmic Gymnastics" by Santos et al. (2015). The limb which effectively performs the task is considered the preferred lower limb (PLL) and the one which functions as

support is considered the non-preferred lower limb (NPLL). There are five classification values attributed to each movement, referring to the maximum possible range of motion and on the scale from 0 to 4 points, in which 0 = very poor, 1 = poor, 2 = average, 3 = good and 4 = excellent. Only whole numbers are attributed to the results. For movements with a range of motion between two points of the assessment chart, the next lower value is registered (Santos et al., 2015). The test is presented in Figure 2.



FIGURE 2. Frontal split with hand support test

Split in supine position

The test was performed in a hall on a flat surface. Two mats were placed in the middle so that the longer sides touched. An adhesive tape was placed along the width of one

mat and a protractor was placed on it so that its straight edge was to the edge of the mat. The 90° mark was on the "drawn" line. The scale was numbered so that the 0° mark is on the line that divides the mat into two equal parts. The task was



FIGURE 3. Split in supine position

performed so that the gymnast lay on her back on the line that divided the mats. The arms were spread and the legs parallel and fully extended. The gymnast stretched to the maximum and maintained in this position until the meter read degrees that the gymnast achieved with maximum stretching (Figure 3). Validity and reliability are described in the book

"Measurement of Basic Motor Dimensions of Athletes" by Metikos et al. (1989).

Frontal split on an elevated surface

The gymnast assumes the front split position with the extended front leg elevated at 40 cm and the other leg extended



FIGURE 4. Frontal split on an elevated surface

backwards on the floor. The distance between the pubic bone and the floor is measured with a centimeter tape. The measurement is made by performing a split on the dominant and non-dominant leg (Figure 4).

Statistical analysis

Microsoft Excel 365 was used for data entry and preparation and visualization of measurements collected. The statistical analysis was performed on all data using SPSS Statistics 14.0 (Dizdar, 2006, p. 63-145). This included examining percentage changes from pre-to post-tests for both groups in terms of all outcome measures. Kruskal-Wallis was used to determine the normality of the distribution, on the basis of which the use of parametric or nonparametric tests

was determined in the further processing of the results. All the tests were conducted at the significance level of $p=0.05$. Reliability of the frontal split on an elevated surface test was assessed using Fleiss' kappa (Fleiss, 1971) for ordinal data. Additionally, the correlation between individual measurements was assessed using Spearman's Rank correlation coefficient (Daniel, 1990).

Results

The simplest way to determine whether the distribution of results was normal was analysing normality using the Kruskal-Wallis test. The results indicate that the distribution of none variable deviated statistically significantly from the normal distribution. Table 1 shows descriptive indica-

Table 1. Descriptive statistics

	Mean	Std. Dev.
R Frontal split on elevated surface1	9.95	7.60
L Frontal split on elevated surface1	11.60	6.20
Split in supine position1	159.75	15.40
Forward bend on a bench1	58.89	4.66
R Frontal split with hand support1	2.25	1.50
L Frontal split with hand support1	2.08	1.25
R Frontal split on elevated surface2	9.36	7.61
L Frontal split on elevated surface2	11.07	6.23
Split in supine position2	163.70	14.71
Forward bend on a bench2	59.87	4.65
R Frontal split with hand support2	2.70	1.32
L Frontal split with hand support2	2.75	1.03

Table 2. Connectivity and reliability of individual test measurements - Frontal split on an elevated surface

Spearman's Rank Correlation	Frontal split on a elevated surface		Reliability	
	1.Measurement	2.Measurement	Fleiss Kappa	p- value
1. Measurement	0.025	0.034		
2. Measurement	0.034	0.025	0.870	<0.001

tors, where the arithmetic mean and standard deviation of the sample is visible.

There is a statistically significant correlation between individual measurements ($p < 0.001$), so this test is considered reliable. For the test to be reliable, the expected Fleiss Kappa value should be above 0.810 (Landis & Koch, 1977) and in this test it was 0.870 (Table 2).

Discussion

The problem of measuring flexibility with an object is the topic of numerous kinesiological and medical studies. In this study, the possibilities of the maximum range of motion were analyzed and the results showed that the test is reliable, so it should be implemented in future studies, since, according to Laffranchi (2005) flexibility is the main motor ability in rhythmic gymnastics. Lebre (1993) pointed out the importance of coaches devoting great attention to the training of motor abilities. Numerous measurement procedures have been developed where the amplitude of the movement is measured either by a linear scale or by angular degrees. Gymnasts are mainly characterized by their flexible joints and compliant muscles (Donti et al., 2014). Due to the changes made in the Code of Points in the last three decades, flexibility has become the most required physical ability and currently makes up the framework of fundamental rhythmic gymnastics movements (Laffranchi, 2001). That is the reason why it is necessary to work on developing specific tests for this motor ability in rhythmic gymnasts. During training, knowing each gymnast individually is crucial for drawing up and planning a training program focused on their needs and physical and technical shortcomings, while also respecting their limits. The „frontal split on an elevated surface“ test showed high reliability, which is in line with other test that assess the flexibility in rhythmic gymnastics,

for example frontal split with hand support which compares the gymnast's range of motion against an assessment chart.

It would be interesting to construct specific tests for measuring flexibility of the upper and lower dominant and nondominant extremities. In addition to the previously mentioned indicators, a greater number of researches on the considered issue should certainly be carried out. Every research has its limitations and so does this one. There is a time gap of two months between measuring and the longer the time interval between tests, the lower reliability of the measurement will be, because during this time period many changes in variables can occur.

It would be even more complete if it included larger number of participants and tests specifically for each topological region. Therefore, for future studies it is necessary to construct a larger battery of tests which would measure the flexibility of the shoulders, back and other body extremities. The review of previous research that is offered in this study with guidelines for further research could, at least in part, provide help in the work and selection of children for rhythmic gymnastics.

Conclusion

These results confirm the hypothesis that the newly developed test measures flexibility, but only in the observed sample, in this case rhythmic gymnasts, because the test includes specific movements that would be difficult to perform correctly with other athletes. It would be good to implement this newly developed test when selecting children for rhythmic gymnastics as well as during transitive measurements in the training process. For future research, it is recommended to include a larger number of participants from all Croatian clubs and especially for every level of competition.

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Conflicts of interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Cognitive Functions of Youth Water Polo Players

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Abstract

Some previous studies suggested that children's engagement in sports has been associated with the improvement of cognitive functions, especially working memory, visual-spatial memory, motor coordination and cognitive flexibility. The aim of this study was to compare cognitive functions between female and male youth water polo players and the control subjects – school children of the same chronological age who did not participate in organized sports activity. There were 23 female (14.65±1.11 years old) and 23 male (14.52±0.51 years old) water polo players (WP), members of Croatian National team enrolled in this study for cognitive testing (Stroop test). The control group consisted of 8 female (14.75±0.89 years old) and 18 male (14.28±0.89 years old) school students who did not participate in organized sports activity. There were no significant differences in psychomotor speed (Stroop Off), while youth water polo players showed better response inhibition (Stroop On) (WP 65.33±9.09 s vs. control subjects 72.78±11.97 s, $P=0.030$) and psychomotor ability (OnTime minus OffTime) (WP 7.22±5.18 s vs. control subjects 14.13±9.22 s) ($P<0.001$) than control subjects. The findings of this study suggest that children who participate in high-level water polo develop better cognitive flexibility than school students who do not participate in organized sports activity.

Keywords: team sport games, school children, psychomotor ability, inhibition

Introduction

Regular physical activity is known to have numerous physical benefits including prevention of the most prevalent lifestyle chronic diseases: coronary heart disease, stroke, obesity, type II diabetes and specific types of cancer, as well as enhancing cognitive function (World Health Organization, 2010). Participating in physical activities and exercise can improve the physical, perceptual, and cognitive well-being (Mandolesi et al., 2018). Moreover, superior performance of athletes with the result of the long-term practice has also been observed in some perceptual motor skills, like reaction time (Mann et al., 2007; Best & Miller, 2010; De Waelle et al., 2021). To date, the literature supports the causal link between regular physical activity and brain development particularly in the prefrontal cortical area (Best & Miller, 2010). It is well known that children's engagement in sports has been associated with the improvement of cognitive functions (De Waelle et al., 2021). Also, it has been proven that

playing high-level team sport games demands well-developed cognitive functions (Kamijo et al., 2011; Bizdan-Bluma & Lipowska, 2018; De Waelle et al., 2021).

Cognitive functions include memory, attention, visual-spatial and executive functions, while language and thinking (abstract, cause and effect, creative thinking and planning) represent complex cognitive processes. Executive functions (EF) include specific mental skills such as inhibition (inhibitory control, including self-control and interference control – selective attention and cognitive inhibition), working memory and cognitive flexibility (Halligan et al., 2003; Bizdan-Bluma & Lipowska, 2018). They represent skills essential for mental and physical health; success in school and in life; and cognitive, social, and psychological development. Inhibitory control involves being able to control one's attention, behavior, thoughts, and/or emotions to override a strong internal predisposition or external stimulus, and instead do what is more appropriate or needed.



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Inhibitory control of attention (interference control at the level of perception) enables us to selectively attend, focusing on what we choose and suppressing attention to other stimuli, which might be useful in sports (Best & Miller, 2010; Diamond, 2013). Another EF is working memory, divided in verbal and non-verbal – visual-spatial working memory, also involved during sports activities (Diamond, 2013). And third component of EF is cognitive flexibility, meaning being able to change perspectives spatially or interpersonally. EF make it possible for us to mentally play with ideas, quickly and flexibly adapt to changed circumstances, take time to consider what to do next, resist temptations, stay focused, and meet novel, unanticipated challenges which is necessary for playing many complex team sport games on high level (Diamond, 2013).

Interestingly, it has been shown that elite athletes outperform non-elite athletes on more complex nonsport-specific neurocognitive functions such as inhibition (Mann et al., 2007; Vestberg et al., 2012). In the study of Vestberg and colleagues (2012), it was reported that EF even predicted later performance in soccer in terms of scoring goals and providing assists. Also, it has been proven that team sport games players have better EF compared to players from other sports (Kamijo et al., 2011).

One of the sports that requires a high level of perceptual and motor performance is water polo, as a highly demanding physical contact team sport taking place in water, not natural environment for human being, mainly including opened skills (Smith, 1998; Botonis et al., 2015). Water polo players need to process real-time actions with regard to teammate positions and ball path while constantly updating that information in working memory. They need to be able to inhibit planned actions, for example passing the ball instead of scoring themselves and they need to have well developed cognitive flexibility for better and faster adaptation to the game dynamics of one specific high-level game (Botonis et al., 2015; Diamond & Ling, 2016; De Waelle et al., 2021).

Water polo as a game includes cognitive components such as anticipation, problem solving and decision making, similar as in other team sport games (Smith, 1998; Botonis et al., 2015). In the study of Falk et al. (2004) those cognitive functions were subjectively evaluated by the coaches during only 2-3 games each season, indicating better scores by the players selected to the youth national team. Also, Kovačević et al. (2023) showed better cognitive functions of the selected players, objectively evaluated using the Stroop test. Still, both studies evaluated only cognitive functions of youth water polo players, without the comparison with the control subjects, presenting the research gap and the need for more studies about cognitive functions of youth water polo players.

Water polo has been chosen in this study because it challenges physical, perceptual and cognitive performance and the literature on the association of the cognitive performance and development in children practicing water polo is limited. Therefore, the aim of this study was to compare cognitive functions between female and male youth water polo players and the control subjects – school children of the same chronological age who did not participate in organized sports activity. It was hypothesized that children who play water polo at a high level would show better cognitive functions, especially concerning cognitive flexibility and inhibition.

Methods

Participants

There were 46 water polo players (23 female / 50%) between 12 and 16 years old ($M_{age}=14.59\pm0.86$) and 26 school children who did not participate in organized sports activity (8 female / 30.8%) between 13 and 16 years old ($M_{age}=14.42\pm0.90$) included in this study. Water polo players self-reported mean of 5 years of training experience with at least 5 training sessions per week, lasting approximately 2 hours, while control subjects self-reported no training experience in organized sports activity. The inclusion criteria for water polo group was selection to the national water polo team and for the control group no training experience in any sports. Measurements for this study were conducted from the season 2020/2021 until 2022/2023.

This study was conducted in full accordance with the ethical principles, including the World Medical Association Declaration of Helsinki and it was approved by the Ethical Committee of the University of Split School of Medicine, Split, Croatia (No: 2181-198-03-04-19-0053). Informed consent was obtained from parents or legal guardians of children participating in the study after they were introduced to the background and the aim of the study.

Measurements and procedures

Anthropometric variables included body mass and height which were measured using a stadiometer and a digital scale, respectively, while the subjects wore only light clothes. Body mass index (BMI) was calculated as body mass (kg) divided by height squared (m^2).

The detail of the experimental setup was previously explained (Kovačević et al., 2023). Cognitive functioning of the participants was measured using The EncephalApp_Stroop application (MacLeod, 1991; Homack & Riccio, 2004; Bajaj et al., 2013) which was downloaded from the Google Play app store (EncephalApp Stroop) on the 7" HD C80 MeanIT tablets (MeanIT, China). The reliability and validity of the application has been well reported previously (Bajaj et al., 2015; Luo et al., 2020). In addition, the reliability of The EncephalApp_Stroop application has been confirmed in our previous study, in a sample of 24 water polo players and test-retest showed correlation coefficient for OffTime $r=0.872$ (95% CI 0.723-0.944) and OnTime $r=0.890$ (95% CI 0.760-0.952), respectively (Kovačević et al., 2023).

The tablet screens were used to administer the task to all subjects. Tests were performed in the quiet, bright room with separate desks and it was performed in groups of 8 to 10 participants. Before the test, participants received a detailed explanation of the test battery. A trained researcher was present to ensure the test was executed correctly, without any noise, and to answer any additional questions. The participants did not report prior experiences in Stroop test.

The Stroop test comprises two components: "Off" state presenting congruent stimuli and "On" state presenting incongruent stimuli. Both components were administered after two training runs. In the easier "Off" state, the participants viewed a neutral stimulus, hashtag signs (###) presented in red, green or blue, one at a time and had to respond as quickly as possible by touching the matching color of the stimulus to the colors displayed at the bottom of the screen. The colors at the bottom of the screen were

randomized and not fixed to their respective positions. In more challenging “On” state, nine of the ten presented stimuli were incongruent. The participants had to accurately touch the color of the word presented on the screen which was contradictory to the actual color, for example the word “blue” was displayed in green color and the correct response was “green”, not “blue”. Both parts of the test consisted of five runs with 10 different tasks in each run and if the participant made a mistake, pressed a wrong color, the run stopped and had to restart again until completing all five runs correctly (Bajaj et al., 2013; Scarpina & Tagini, 2017).

The specific outcomes of the Stroop test were OffTime – total time for five correct runs in the “Off” state, primarily assessing psychomotor ability; OnTime – total time for five correct runs in the “On” state which is a measure of response inhibition and motor speed. OnTime minus OffTime presented a measure of cognitive processing controlling for psychomotor speed and OffTime plus OnTime showed a composition measure of psychomotor speed and response inhibition (Bajaj et al., 2013; Scarpina & Tagini, 2017).

Statistical analyses

Data analyses were performed using statistical software MedCalc for Windows, version 19.4. (MedCalc Software, Ostend, Belgium). Continuous data were presented as

mean±standard deviation while categorical variables were presented as whole number and percentage. The Kolmogorov-Smirnov test was used to assess normality of data distribution. Differences in cognitive performance between youth water polo players and school children who did not participate in organized sports activity were tested using Mann-Whitney *U* test. Additionally, a multiple regression analysis was used to determine a relationship between selected independent variables (participation in sports coded as 0-no participation and 1-currently participating in sports, age, gender, BMI) with the outcomes of the Stroop test as dependent variables (StroopOn time, StroopOff time, Ontime minus Offtime, Offtime plus Ontime). The statistical significance was set at $p<0.05$.

Results

There were 46 water polo players (WP), 23 females (50%) and 18 males (50%), with the mean age 14.59 ± 0.74 years (median 14, 12-16 years) and 26 control subjects (C) – school children who did not participate in sports regularly, 8 females (30.8%) and 18 males (69.2%), with the mean age 14.37 ± 0.93 years (median 15, 13-16 years) ($p=0.567$). Descriptive statistics of the whole study sample is presented in Table 1. Water polo players performed faster in StroopOn Time (65.33 ± 9.09 s) than control subjects (72.78 ± 11.97 s) ($p=0.030$), as well as in OnTime minus OffTime (WP 7.22 ± 5.18 s vs. C 14.13 ± 9.22 s) ($p<0.001$) (Table 1).

Table 1. Descriptive Statistics for Total Sample of Participants and Comparison of Results of Anthropometric Variables and Cognitive Performance between Youth Water Polo Players and Control Subjects

Variables		Total study sample N=72	WP N=46	Control subjects N=26	p
Anthropometric characteristics	Age (years)	14.53±0.87	14.59±0.86	14.42±0.90	0.708
	Body height (cm)	172.13±8.88	174.33±8.60	167.40±7.68	0.003*
	Body mass (kg)	65.28±10.08	67.15±9.95	61.28±9.36	0.029*
	Body mass index (kg/m ²)	21.98±2.59	22.05±2.63	21.82±2.55	0.939
Cognitive functions	StroopOff time (s)	58.31±6.14	58.12±6.39	58.65±5.80	0.673
	StroopOn time (s)	68.02±11.16	65.33±9.09	72.78±11.97	0.030*
	StroopOff+StroopOn time (s)	126.33±16.33	123.45±14.84	131.42±17.84	0.098
	Ontime minus Offtime (s)	9.71±7.62	7.22±5.18	14.13±9.22	<0.001*

Note. Data are presented as mean±standard deviation. *Mann Whitney *U* test; $p<0.05$. WP - water polo players.

A multiple regression analysis showed that independent variables (participation in sports – water polo, gender, age and BMI significantly predicted StroopOn Time ($F=3,3784$, $p=0.015$, $R^2=0.181$) and OnTime minus OffTime ($F=5,4258$, $p<0.001$,

$R^2=0.262$). Out of independent variables included in the model, only participation in sports (water polo) contributed significantly to the prediction of higher cognitive performance measured by StroopOnTime minus OffTime ($p<0.001$, $R^2=0.262$) (Table 2).

Table 2. Multiple Regression Analysis Showing the Predictive Status of Participation in Sports (Water Polo), Age, Gender and Body Mass Index on Cognitive Performance

	β coefficient	SE	t	p
StroopOff time				
Participation in sports	1.395	1.686	0.828	0.411
Gender	-0.476	1.633	-0.037	0.772
Age	-0.676	0.895	-0.755	0.453
Body mass index	0.222	0.304	0.093	0.468
$R^2=0.030$; R^2 -adjusted=-0,033; $F=0,474$; $p=0.755$				

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Table 2. Multiple Regression Analysis Showing the Predictive Status of Participation in Sports (Water Polo), Age, Gender and Body Mass Index on Cognitive Performance

StroopOn time				
Participation in sports	10.047	2.886	3.482	<0.001*
Gender	-1.713	2.796	-0.613	0.543
Age	-1.130	1.533	-0.737	0.464
Body mass index	0.313	0.520	0.601	0.550
$R^2=0.181$; R^2 -adjusted=0.128; $F=3.378$; $p=0.015^*$				
StroopOff+StroopOn time				
Participation in sports	11.443	4.331	2.642	0.010*
Gender	-2.189	4.196	-0.522	0.604
Age	-1.806	2.300	-0.785	0.435
Body mass index	0.535	0.781	0.685	0.496
$R^2=0.121$; R^2 -adjusted=0.064; $F=2.107$; $p=0.091$				
StroopOn minus StroopOff time				
Participation in sports	8.652	1.894	4.569	<0.001
Gender	-1.237	1.835	-0.674	0.503
Age	-0.454	1.006	-0.451	0.653
Body mass index	0.091	0.342	0.266	0.791
$R^2=0.262$; R^2 -adjusted=0.214; $F=5.426$; $P<0.001$				

Note, β coefficient – multiple correlation coefficient; SE – standard error; t – test statistic; R^2 – coefficient of determination; R^2 -adjusted – coefficient of determination adjusted for the number of independent variables in the regression model; F – F-statistic. *Significant difference between the groups, $p<0.05$.

Discussion

There are three main findings of this study. First, youth water polo players showed a better performance in psychomotor speed, response inhibition and motor speed than school children not participating in organized sports activity. Second, participation in water polo was found to be a significant predictor of cognitive functions in school children at the age of 12 to 16 years. And third, youth water polo players had higher values of body height and weight than their sedentary peers.

Cognitive functions

Coordinative activities, such as walking, running, jumping, kneeling, throwing, grasping, have been hypothesized to train and increase the activity in the cerebellum and prefrontal cortex and to improve such executive functions as working memory, focused attention and cognitive flexibility (Budde et al., 2008). In such activities, children must quickly integrate available information to make split-second decisions, as well as to plan and perform a sequence of habitual or unfamiliar actions (Stratton et al., 2004). Considering the nature of water polo, which stimulates components of visual-spatial attention, players need to be able to search and select targets from a spatial field requiring not only visual selective or focal attention, and peripheral visual acuity to move the attentive focus. In addition, water polo also requires more sophisticated metacognitive strategies (Stratton et al., 2004). During a match, players have to process, more or less at the same time, cues including opponents' and teammates' positions and the ball's location and direction of movement. Water polo is categorized as mostly open skills sport, characterized by activities rich in distractors, played in water, an unusual environment for human being. So, this sport should act as a powerful challenge to the ability to inhibit distraction

and to stay focused for better and faster adaptation to the game dynamics of one specific high-level game (Smith, 1998; Botonis et al., 2015; Sekulić et al., 2016). Current study reported faster OnTime of water polo players in comparison to control subjects. This is consistent with categorization of water polo among the "thinking games", where the environment constantly changes and movements have to be continuously adapted (Smith, 1998). Actions must be continually updated and changed due to teammates' actions, the ball's movement, the opponents in the swimming pool. These factors are relevant to adapting, planning, problem-solving, the use of cognitive control and inhibition of first inclinations (Diamond, 2013). Youth water polo players showed better performance in psychomotor speed and response inhibition compared to school children who did not participate in organized sports activities, which is consistent with other studies that have examined the difference between athletes and non-athletes in this age group (Alesi et al., 2016; Ishihara et al., 2016; Ishihara et al., 2017). For example, Alesi et al. (2016) showed improvement of executive functions (agility, visuo-spatial working memory, attention, planning and inhibition) by children practicing football compared to the sedentary group, although children in their study were evaluated before and after only 6 months of a Football Exercise Program compared to a control group of sedentary peers.

Ishihara et al. (2016) provided evidence for the relationship between cognitively engaging exercise (i.e., game-based tennis training and coordinative exercise) and improved executive functions in 6 to 12 years old children. A longer duration of game-based exercise was positively correlated with inhibitory control and physical fitness. Coordination training was associated with improved working memory. Non-physical activity was inversely correlated with inhibitory control, working mem-

ory, and physical fitness. The results suggest that game-based tennis lessons have beneficial effects on inhibitory control and physical fitness levels, and a longer duration of coordination training is associated with better working memory (Ishihara et al., 2016). In another study of Ishihara et al. (2017) findings suggest that tennis play is associated with the development of three foundational aspects of executive function – inhibitory control, working memory and cognitive flexibility, in children 6 to 15 years old. Especially, frequent participation in tennis play is related to better inhibitory control and working memory, while longer experience of tennis play is associated with better cognitive flexibility (Ishihara et al., 2017). Although tennis belongs to self-paced sports including mostly closed skills, the results of the present study are in accordance to these findings, so additional research is needed to compare executive functions development among youth water polo players and children practicing other team sport games, as well as self-paced sports.

It has been suggested that water polo may affect the development of EF through multiple pathways: as an exciting and enjoyable activity (increasing enjoyment); by producing an “enriched environment” that places demands on multiple EF, thereby challenging and improving them; by creating opportunities for social interaction, cooperation with others, sharing leisure experiences, and receiving the encouragement of peers, increasing a sense of belonging and social support and a sense of ability and competence; and by improving motor skills and physical fitness (Alesi et al., 2016).

To the best of our knowledge, there are only few studies investigating cognitive functioning in children practicing water polo, but those studies considered only influence of cognitive functions on the selection and identification of the talented players (Falk et al., 2004; Kovačević et al., 2023). Specifically, water polo as a complex team sport employs exercises in a cognitively-engaging context and implies sophisticated and complex moves which are regulated by the prefrontal neural network. It requires high cognitive engagement and impacts many of the same processing components as those implied in executive functioning tasks: involving in a relevant task, shifting the focus of processing activities, updating of information, monitoring mental representations and inhibiting irrelevant responses. Moreover, water polo stimulates strategic and goal-directed behaviours when faced to constantly changing situations and movements (Best & Miller, 2010). The impact of water polo on the cognitive development has been explained through multiple pathways: 1) neuro-physiological by inducing changes in the brain structure; 2) contextual by producing an “enriched environment”; 3) social by creating opportunities of social interaction, cooperation with teammates, respecting rules, sharing leisure experiences, receiving the encouragement of peers; 4) motivational by increasing self-concept and self-awareness (Barenberg et al., 2011; Alesi & Pepi, 2013).

Many previous studies demonstrated the positive relationship of sports participation to multiple aspects of executive functions in children, suggesting that the frequency of participating in sports activity may be an important factor for development of inhibitory control and working memory (Ishihara et al., 2017). Such results may also be supported by our findings showing better cognitive functions of high-level water polo players, members of youth national teams, in comparison to children of the same age with no training experience in organized sports activity. Sensorimotor learning in sports and cognitive engagement during exercise has been postulated as a key mechanism linking training and cognitive enhancement,

and water polo requires both, cognitive engagement and sensorimotor learning (Best & Miller, 2010; Diamond, 2013).

Anthropometric characteristics

Concerning the anthropometric variables, water polo players showed higher values of body height and weight which might be explained by specificity of the game. It has been indicated in previous studies that physical abilities and anthropometric characteristics are important factors to achieve a good level of quality in technical-tactical actions, both for young and adult water polo players, especially considering constant physical contact during the game (Sekulić et al., 2016, Vieiro et al., 2020, Kovačević et al., 2023, Kovačević et al., 2023). Greater body height and longer extremities allow players to reach for the ball more easily, to shoot and to perform blocks more efficiently (Idrizović et al., 2014, Dimitrić et al., 2022). Similar results have been presented in different sports such as volleyball or football (Tatar & Cupic, 2011).

Strengths and limitations

Considering that studied variables are strongly influenced by the age of the subjects, given the existing developmental differences in functioning of the cognitive-motor areas, unique age distribution in both studied groups is one of the main strengths of this study, showing benefits of sports activities on cognitive development of children. The main limitation is the inclusion of unequal sample of participants (46 water polo players and 26 school children who did not participate in organized sports activity). Also, there is a need for use a larger battery of cognitive tests for assessment of different EF components as in the study of Alesi et al. (2016) or Ishihara et al. (2017). Future studies using longitudinal or interventional designs and more diverse measures (e.g. age and gender differences, socioeconomic status, different sports) should investigate the associations among EF, sports activity, and gender specificity of these relationships. Additionally, since it has been shown that more frequent tennis play is associated with better working memory, the importance of increasing the frequency of sports activity participation in the development of EF has been emphasized even in physically active children (Ishihara et al., 2017). Therefore, future studies should assess the relevance of the full timeline of training experience in the investigated outcomes regarding water polo players. Finally, since a gender specificity of the relationships between certain sports training exposure and EF has also been suggested (Ishihara et al., 2017), larger samples including both genders are necessary in order to assess this issue in water polo players.

Conclusion

The present study supported the hypothesis that a more frequent and longer experience of participating in sports activity is associated with better inhibitory control, working memory and cognitive flexibility. High-level youth water polo players showed better results in psychomotor speed, inhibitory control and motor speed than sedentary children, taking both age gender into account. Still, well-developed cognitive functions may serve only as one of the crucial factor along with other components of athlete's development in order to become an elite water polo player. Despite the limitations of this study, the presented results contribute to the issue of sport activities as a tool in the stimulation of cognitive development. Considering that EFs are skills essential for mental and

physical health, success in school and in life, and cognitive, social, and psychological development based on the results of this study it would be advantageous to encourage children to participate in organized sports activities. Still, additional research is needed to compare executive skills development among youth water polo players and children practicing in-

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Conflict of interests:

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ORIGINAL SCIENTIFIC PAPER

The Effect of Combined Treatment with Passive Therapy, Physical Exercises, Lumbar Traction, and Walking Program on Chronic Low Back Pain

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Abstract

Chronic low back pain (CLBP) is one of the most common musculoskeletal disorders worldwide with far-reaching implications for social, economic, and public health. The study aimed to compare the effect of combined physiotherapy treatment in patients with CLBP. This observational retrospective randomized controlled study involved reviewing medical records of patients who were treated during a period of 6 weeks in the physiotherapy department of the Special Hospital for General Rehabilitation "Banja e Klokotit," Klokot, Republic of Kosovo. The observational research was conducted for 60 patients, divided into two groups: Group 1: experimental group (n=30), with mean age of 41.7 years, average height of 1.68 cm, and average body mass index (BMI) of 21.7 for both sexes; Group 2: control group (n=30) with mean age of 43.1 years, average height of 1.66 cm, and average body mass index (BMI) of 21.5 for both sexes. In Group 1, combined treatment with passive therapy [thermotherapy and transcutaneous electrical nerve stimulation (TENS)], physical exercises, lumbar traction, and walking program was applied, while in Group 2, passive therapy like thermotherapy and TENS was applied. Outcome measurements included pain intensity, lumbar flexibility, revised Oswestry disability index (ODI), and self-confidence, which were assessed pre-treatment, at 3 weeks, and after 6 weeks of clinical treatment. The results showed significant improvement in both groups. However, the improvement was greater and with a significant difference only after 6 weeks in Group 1 compared to Group 2. The differences were observed in various outcome measures, including the visual analog scale for pain (VAS; $p < 0.0001$), fingertip-to-floor distance (FTF; $p < 0.0001$), ODI ($p < 0.0001$), and Rosenberg self-esteem scale (RSE; $p = 0.0002$). According to our data, combined treatment with thermotherapy, therapeutic exercises, lumbar traction, TENS, and walking program was more effective and can be considered as a treatment protocol for patients with CLBP. However, further research is recommended on the efficacy of combined physiotherapy treatment, especially lumbar traction for longer periods.

Keywords: back pain, modalities, therapeutic exercise, lumbar flexibility, traction

Introduction

Low back pain (LBP) has a high prevalence in many countries around the world and is a major health problem in modern society (Zheng et al., 2012; Ibrahim-Kaçuri, 2015). LBP is a complex condition; it is one of the most common musculoskeletal complaints and causes of disability global-

ly. It can be attributed to various causes, including trauma, degenerative conditions, inflammatory disorders, oncologic, spinal infections, and occupational factors (Vos et al., 2010; Patrick et al., 2014). Several factors increase the risk of developing back pain: smoking, obesity, longevity, female gender, physical work, sedentary job, high levels of stress,



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job dissatisfaction, and psychological factors such as anxiety or depression (Hartvigsen, 2018; Shiri et al., 2019). In addition to these factors, the rapid development of the industrial world and modern lifestyle has further increased the likelihood of developing LBP, particularly among young people. Physiotherapy, with a combination of various interventions such as electrotherapy, different therapeutic exercise protocols, and lumbar traction, represents one of the primary treatment approaches for CLBP. It has shown promising results in terms of pain reduction, improved flexibility, prevention of disability progression, and restoration of patient's ability to engage in daily activities (Eleswarapu et al., 2016; Choi et al., 2016; Foster et al., 2018; Shipton, 2018; Hayden et al., 2021; Tanabe et al., 2021).

There is scientific evidence that the application of different physiotherapeutic methods can reduce pain and disability due to LBP (Moseley, 2002; Hayden et al., 2005; Petersen et al., 2011). This study aimed to compare the effect of different combinations of physiotherapy treatment methods like thermotherapy, physical exercises, lumbar traction, electrotherapy TENS, and walking program on the treatment of patients with CLBP.

Methods

Participants

In this retrospective observational randomized study, 60 patients, who were treated at the Special Hospital for General Rehabilitation "Banja e Klllokotit" in Klllokot, Republic of Kosovo, were selected through the review of medical documentation. These patients were referred by specialists in different fields during the period between September 2020 and November 2021. All patients included in the study met the research criteria, such as age between 18 and 65 years, male or female, LBP with or without iridescence along the leg, and duration of symptoms of more than 12 weeks. Patients with the following conditions were excluded from the study: compression of the nerve roots (positive neurological tests), osteomyelitis of the spine, ankylosing spondylitis, vertebral fracture, malignant disease, structural scoliosis, spinal instability, previous spine surgery, spondylolisthesis, and retrolisthesis. The sample was divided into two groups. Group 1 was the experimental group ($n=30$); it included 15 women and 15 men with mean age of 41.7 years ($SD\pm 9.0$ years), average height of 1.68 cm, and average body mass index (BMI) of 21.7 for both sexes. Group 2 was the control group ($n=30$); it included 15 women and 15 men, with mean age of 43.1 years ($SD\pm 10.7$ years), average height of 1.66 cm, and average body mass index (BMI) of 21.5 for both sexes.

Assessment

General data were obtained through subjective examination. Pain intensity was observed and evaluated using the visual analog scale (VAS; Delgado et al., 2018). Lumbar flexibility or fingertip-to-floor distance (FTF; Perret et al., 2001) was assessed with a measuring tape. Functional disability was evaluated with the Oswestry disability index (ODI), also known as the Oswestry Low Back Pain Disability Questionnaire (Fairbank & Pynsent, 2000). The Rosenberg self-esteem scale (RSE) was used to assess self-worth by measuring both positive and negative feelings patients had about themselves (Rosenberg, 1965). All these parameters were measured before treatment, at 3 weeks, and after 6 weeks of treatment.

Physiotherapy treatment

The duration of treatment for both groups was 3 weeks in the hospital, and after 3 weeks, they continued to come to the hospital for treatment for 3 more weeks (3 times a week). In the first group, combined treatment with thermotherapy, physical exercises, lumbar traction, TENS, and walking program was applied, while in the second group (control) passive modalities, such as thermotherapy and TENS, were applied.

For 90% of patients, the physiotherapeutic treatment started from the first day of hospitalization, while for 20%, it started one day after hospitalization, after all general examinations were completed. Treatment lasted for 6 weeks. The physiotherapeutic program was modified and the application was conducted as follows:

During the first 3 weeks, thermotherapy was applied 5 days a week with hot pieces in the lumbosacral area and along the affected leg for a duration of 15–20 minutes. Then the application continued for another 3 weeks, 3 times a week with the same procedure and duration.

During the first 3 weeks, the exercises were performed 3 times a week for 15–20 minutes until the limit of pain. It was continued for another 3 weeks, 3–5 times a week for 20–30 minutes. The physical exercise regimen consisted of static and dynamic exercises for strengthening the muscles of the back, with a specific focus on the lumbar region, abdominal region, pelvic muscles, and the lower extremities in general (Koumantakis et al., 2005; Ferreira et al., 2006).

Lumbar traction treatment was administered 3 times a week for 6 weeks, with each session lasting for up to 30 minutes (around 3–30 minutes). Weight-bearing was applied intermittently by progressively increasing the weight to about one-third of the patient's body weight, based on their reported tolerance (Rasmussen & Kyle, 2017; Oh et al., 2018).

TENS current was applied through four cutaneous electrodes of medium size (2.5 cm) along the pain points, with a frequency of 7 Hz and pulse duration of 0.1 ms, progressively increasing the intensity until the patient indicated that they could tolerate it. TENS was applied 3 times a week for a duration of 20–30 minutes from the first day of treatment up to 6 weeks.

The walking program was conducted 5 days a week. The patients first started with walking for 10 minutes (1200 steps per day) and then reached 30 minutes of moderate-intensity physical activity. The patients had the option to choose to walk up to the limit of pain, walk with minimal pain (1–3/10) for 15–20 minutes, or walk 4 times a day for 4–5 minutes (Park et al., 2019).

Physical exercises, lumbar traction, and walking program were applied to Group 1 only, while thermotherapy and TENS were applied similarly to both groups. So the treatment for both groups lasted a total of 6 weeks, 3 weeks inside the Special Hospital and another 3 weeks on an outpatient basis in the same hospital. Additionally, during the treatment period, the patients were given recommendations for different standing positions and advice and instructions for engaging in daily life activities.

Ethics

The study was approved by the Ethical Committee of Special Hospital for General Rehabilitation "Banja e Klllokotit," (Nr. 123) Klllokot, Republic of Kosovo.

Statistical analysis

Data processing was done with the statistical package SPSS 22.0 (SPSS Inc., Chicago, IL, USA). Data are presented as mean and standard deviation. Data testing was conducted using the Chi-square test, Mann–Whitney test, Friedman test, and Dunn's multiple comparison test. A p-value <0.05 was

considered significant.

Results

The general characteristics of patients are presented in Table 1, which shows the significant differences according to age (p=0.608), weight (p=0.918), and height (p=0.231).

Table 1. Distribution of demographic data among groups

	Group1 n = 30	Group2 n = 30	P-value
Gender			
Men	15 (50.0%)	15 (50.0%)	1.000
Women	15 (50.0%)	15 (50.0%)	
Age (years)	41.7 ± 10.7	43.1 ± 9.8	0.608
Weight (kg)	71.7 ± 7.3	71.5 ± 5.1	0.918
Height (cm)	168.3 ± 5.8	166.8 ± 5.0	0.231

Note: Group 1 was given Thermotherapy, physical exercises, lumbar traction, TENS, and walking program; Group 2 was given thermotherapy and TENS

Table 2 presents the outcome measurements. After 6 weeks, both groups demonstrated a significant improvement in the pain level. However, the improvement was greater and

with a significant difference only after 6 weeks in Group 1 compared to Group 2 in the VAS (p<0.0001), FTF (p<0.0001), ODI (p<0.0001), and RSE (p=0.0002).

Table 2. Outcome measures for subjects by groups and times

	Group1 n = 30	Group2 n = 30	P-value
VAS			
Baseline	6.7 ± 0.8	6.6 ± 1.0	0.237
3 Week	4.4 ± 0.9	4.4 ± 0.9	0.100
6 Week	2.4 ± 1.1	3.9 ± 1.1	< 0.0001
P-value ¹	< 0.0001	< 0.0001	
OSW			
Baseline	43.9 ± 4.6	42.9 ± 5.1	0.573
3 Week	32.4 ± 6.0	34.9 ± 5.8	0.073
6 Week	17.7 ± 6.8	27.2 ± 6.4	< 0.0001
P-value ¹	< 0.0001	< 0.0001	
FTF			
Baseline	53.0 ± 5.7	51.5 ± 5.1	0.166
3 Week	39.9 ± 7.1	43.0 ± 5.7	0.058
6 Week	24.0 ± 7.4	34.4 ± 7.0	< 0.0001
P-value ¹	< 0.0001	< 0.0001	
RSE			
Baseline	11.6 ± 2.5	12.4 ± 2.8	0.279
3 Week	17.4 ± 3.0	16.5 ± 3.2	0.347
6 Week	24.0 ± 3.0	20.5 ± 3.1	0.0002
P-value ¹	< 0.0001	< 0.0001	

Note *Mann-Whitney test, ¹Friedman test and Dunn's multiple comparison test; Group 1 was given thermotherapy, physical exercises, lumbar traction, TENS, and walking program; Group 2 was given thermotherapy and TENS; VAS - Visual Analogue Scale; FTF- Fingertip-to-Floor Distance; OSW - Oswestry Disability Questionnaire; RSE - Rosenberg Self Esteem Scale.

Discussion

The success of different methods of physiotherapy treatment in producing symptomatic and functional change by decreasing pain, increasing lumbar flexibility, and improving quality of life in LBP patients and other outcomes have

been documented in numerous studies all over the world (Moseley, 2002; Garcia, 2011; Şahin et al., 2017). Studies have reported that using thermotherapy and different electrical modalities like TENS in combination with various exercise protocols has a positive effect on pain reduction in patients

with LBP (Hayashi, 2004; Rojhani-Shirazi, & Rezaeian, 2015; Sanjana & Yatish 2021). Besides pain reduction, studies have reported that therapeutic exercise also has a positive effect on improving back flexibility and reducing functional disability, thus enabling patients to perform daily life activities (Bronfort et al., 2011; Geneen et al., 2017; Hayden et al., 2021; Hrkać et al., 2022). Thus, the results of this study are consistent with existing research. This research indicates that the use of thermotherapy and TENS had a positive impact on LBP patients in terms of pain reduction, but applied thermotherapy and TENS in combination with therapeutic exercises, lumbar traction, and walking program proved to be more effective in pain management, functional disability reduction, and improvement of quality of life. Regarding the application of lumbar traction, previous systematic reviews have concluded that it has little or no value on the clinical outcomes like pain intensity and functional status of patients with LBP (Clark et al., 2005, 2006; Alrwaily et al., 2018; Wegner et al., 2013). Nevertheless, many clinicians and physical therapists continue to use lumbar traction, primarily as an additional modality, and it is accepted as a treatment for patients with CLBP in many countries. Tanabe et al. (2021) reported that lumbar traction can improve pain and functional status immediately in patients with CLBP. Beyki et al. (2007) also found that the application of traction in a prone position for 4 weeks showed improvements in pain intensity and ODI in a sample of patients with activity-limiting LBP. However, to understand the long-term effect of the relationship between traction and their findings, they suggest a long-term follow-up. Positive results regarding the application of traction therapy in CLBP have also been reported by / and Rasmussen and Kyle (2017).

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Conflict of interest

The authors declare that there are no conflicts of interest.

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In this study, it was found that Group 1 (where therapeutic exercise, walking program, and lumbar traction were applied besides passive modalities) exhibited significant improvements in the degree of pain, lumbar flexibility, and functional disability. It is worth noting that in addition to exercises, traction, and passive modalities, walking is one of the most frequent recommendations preferred by patients with this problem. Walking is easy to perform and highly accessible and can be recommended for the prevention or management of CLBP; it has been found to be as effective as other interventions for pain and disability reduction (Sittipornvorakul et al., 2018; Shiri et al., 2019). Moreover, in our study, the group of patients that was recommended the walking program reported more improvement in disability and pain relief.

The evidence cited above shows that physical therapy treatment should be routinely used in the long term. While the treatment program suggested in this study was effective, it is important to acknowledge that CLBP requires long-term treatment and ongoing follow-up. So the existing literature often includes longer follow-up periods for patients with CLBP, which were not incorporated in this study (Goldby et al., 2006; Oh et al., 2018). Thus, this can be considered a limitation of this study.

Conclusions

According to this study, a combination of thermotherapy, therapeutic exercises, lumbar traction, TENS, and walking program was effective and can be considered a treatment protocol for patients with CLBP. However, it is recommended that future studies should examine the efficacy of combined physiotherapy treatment, especially lumbar traction, for longer periods.

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ORIGINAL SCIENTIFIC PAPER

Video Analysis of Technical and Tactical Behavior in Elite Climbers

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Abstract

Climbing performance is considered multifactorial and is characterized by the interaction of technical, tactical, physical, as well as psychological components. However only few studies have investigated these components in elite climbers. The present study aimed to identify elite athletes' technical and tactical behavior regarding the following parameters: a) type of climbing handhold, b) type of climbing foothold and c) torso motion through video analysis. The sample consisted of the final competitions of IFSC Climbing World Cup Kranj (2019). In total, 543 actions were recorded through SportScout video-analysis program, taking into account the parameters a) type of climbing handhold, b) type of climbing foothold and c) torso motion. According to the results, and concerning handholds, it was observed that the majority of the participants selected to use Incuts / Mini-jugs, regardless of whether they performed the next movement with the left or the right hand. Concerning the footholds, it was found that participants mostly selected to use Big footholds regardless of whether they performed the next movement with the left or the right leg. Finally, regarding the correlation between Torso motion and Handholds and the correlation between Torso motion and Footholds, it was found that the majority of the climbers selected to perform movements without Torso twisting. Only the techniques of stepping with the right foot had a significant effect on the motion of the torso (Chi-Square=15.53, $p<0.05$). In conclusion, the selection of smaller and more difficult handholds and big footholds, as well as the climber's trunk maintenance towards the wall are highly important performance elements in elite climbers.

Keywords: task analysis, performance, climbing

Introduction

During the last decades, the activity of sport climbing not only has emerged as a recreational and challenging activity but at the same time has evolved into a competitive mainstream sport since the 1st World Championships held in Germany in the early 90s (Mittelstaedt, 1997; Sanchez et al., 2019). Nowadays, sport climbing is a fast-growing sport which was included for the first time in the 2020 Summer Olympics in Tokyo. Also, the International Olympic Committee (IOC) Executive Board officially confirmed Sport Climbing's inclusion in the program of 2024 in Paris (Sanchez et al., 2019).

Sport climbing is an intermittent activity that consists of

single attempts at a climbing route of a minimum of 15 meters in length that takes from 2 to 7 minutes to be implemented, during which participants must combine their motor skills with both their cognitive and perceptual abilities (Watts, 2004; UIAA, 2006; López & Sitko, 2019; Whitaker, Pointon, Tarampi, & Rand, 2020). The climbers' aim is to reach the end-point of a natural or artificial wall while climbing up, across or down through a pre-oriented route and returning to the base successfully (Mermier et al., 1997). The climbing attempt requires a discontinuous effort in mixed aerobic and anaerobic work (Bertuzzi et al., 2007) due to the length and inclination of the wall surface, the ascent speed, the psychological state, the physical condition, the anthropometric characteristics of the



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athlete, and the climbing methods and techniques (Mermier et al., 1997; de Geus et al., 2006; Bertuzzi et al., 2007; Matsouka et al., 2019).

According to Horst (2012), during their attempt, climbers use a plethora of climbing methods and techniques of which the most widespread are the “Three-Holds-Rule” and the “Left-Right Rule”. The technique “Three-Holds-Rule” demands from the participant to climb by using three holds, in order to maintain his balance on a wall. As for the “Left-Right Rule», the participants climb the wall by using the contra lateral hand-foot in pairs (Horst, 2012). Furthermore, the sequence of movements and body positioning, in terms of grasping handholds with different features such as size and shape, the position of hands and foot, as well as torso twisting during the attempt, seem to strongly affect climbing performance (Marino & Kelly, 1988; Tucker & Ellis, 1998; Phillips, Sassaman, & Smoliga, 2012; Garrido-Vásquez & Schubö, 2014). Regarding torso twisting, when the slope of the wall surface becomes steeper, it is important for the climber to bring the body as close as possible to the wall. As the wall becomes overhanging it is much easier to reach the next handhold and move by twisting the torso sideways. Also, the torso twisting technique allows the participant to reach further (Phillips, Sassaman, & Smoliga, 2012).

The regular monitoring of the above climbing performance factors in elite climbers plays a decisive role in increasing the likelihood of success in a competition. Among various performance evaluation methods, video analysis is one of the most preferable methods for assessing the performance level of the athletes and enhancing training techniques (White & Olsen, 2010). It has been proved that movement analysis has been used extensively in team sports, such as soccer (Papadimitriou et al., 2001; Patton et al., 2020; Papadopoulos et al., 2021). However, less research has been undertaken on individual-based activities, like climbing.

What is more, to date, climbing research has mainly focused on the physiological and psychological aspects of performance (Mermier et al., 1997; Sheel et al., 2003; Aykora, 2019), while at the same time, the development and the use of observational instruments for analyzing and evaluating climbing performance elements have been limited (Taylor et al., 2020). Additionally, research to date has not identified the technical and tactical behavior of elite climbers in competition conditions and specifically regarding the parameters a) type of climbing handhold, b) type of climbing foothold and c) torso motion. Taking into consideration all the above, we strongly believe that through our findings we will enhance the knowledge of trainers and physical education teachers, so as/ in order to plan specific climbing training programs for elite

climbers. Moreover, we want to underline the importance of video analysis as an effective way of improving their athletes’ performance, and therefore give them the chance to review their technique and make better decisions on what they have to focus on.

The present study aimed to identify elite athletes’ technical and tactical behavior regarding the parameters a) type of climbing handhold, b) type of climbing foothold and c) torso motion during their climbing attempt, through video analysis, as perceived by experts.

Materials and methods

Participants

The sample of the present research was the final competitions of men and women of the IFSC Climbing World Cup Kranj 2019 / Lead finals (8 competitions for each sex of athletes). In total, 543 actions of 16 climbers (N=16) were recorded and the criterion for selecting the climbers (elite athletes) was their advancing to the final climbing round.

Moreover, for the conducting of the present study ethics approval was not required, due to the fact that no experimental analysis involving human studies was performed. Also, according to the Belmont Report (United States National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979), the use of public images for research purposes does not require informed consent or the approval of an ethical committee.

Measurements

The SportScout (SportScout STA) video-analysis program for PC was used for the data recording regarding the following parameters: a) type of climbing handhold, b) type of climbing foothold and c) torso motion. This software is generally used to analyze both technical and tactical behavior of team and individual sports.

Procedure

Each athlete’s technical and tactical behavior during his climbing attempt on the wall was examined based on Sport Climbing Assessment Tool (CM-PAT) (Taylor et al., 2020). In our study, the observation protocol was created in collaboration with a certified climbing coach by the Hellenic Ministry of Culture and Sport and contained three categories of technical-tactical actions with their respective parameters (Table 1). In total, two raters evaluated the actions and the inter-rater agreement was confirmed. In case of disagreement, consensus was reached. The observation of each ascent started from the beginning of the athlete’s attempt and stopped when the athlete completed his attempt or had a fall.

Table 1. Observation protocol

1 st Category - Type of climbing handhold (right hand vs. left hand)
Incuts/Mini-jugs: In general, incuts/mini-jugs consist a type of climbing handhold which is only deep enough to fit up to the second knuckle of the fingers; however, they still provide a very solid handhold.
Jugs: The term “Jugs” basically encompasses every large, easy to grab handhold. They can be held with the entire hand and sometimes with both hands.
Palming: In palming you push against the climbing wall with an open palm. This type of climbing handhold can help you to maintain balance while you reposition your feet. Also, it comes in handy if no good handholds are available and it allows you to apply counter pressure to a blank face.

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Table 1. Observation protocol

2nd Category - Type of climbing foothold (right leg vs. left leg) Small footholds: Small footholds are small blocks that aren't big enough in order to use them for your hands but have enough surfaces to step on with the first phalanx of the big toe. In this type of foothold it is important to keep the heels low and not above the ledge level. Big footholds: Big footholds are blocks that can bear your weight on its vertical axis and are large enough to ensure that the surface of the sole of your shoe will stay in contact with the hold. What is more, this type of foothold allows you to set yourself up for the next set of movements without extraordinary effort. Smearing: Smearing happens when you don't have an actual foothold, so you rely on your shoe's rubber for friction against the wall surface. It is important to have as much surface contact between the sole of your shoe and the wall as possible, therefore maximizing friction.
3rd Category – Torso motion (Twisting vs. Without Twisting) Twisting" to "Twisting": Two consecutive hand moves with simultaneous torso twisting. "Without Twisting" to "Without Twisting": Two consecutive hand moves without torso twisting. "Twisting" to "Without Twisting": Two consecutive hand moves. More specifically, initially, at the same time with the 1st hand move the climber performs a torso twisting motion and immediately afterwards, at the same time with the 2nd hand move he performs a motion without torso twisting. "Without Twisting" to "Twisting": Two consecutive hand moves. More specifically, initially, at the same time with the 1st hand move the climber performs a motion without torso twisting and immediately afterwards, at the same time with the 2nd hand move he performs a torso twisting motion.

Statistics

Data were analyzed using the statistical package SPSS version 20.0 (IBM, Armonk, USA). The type of analysis used was the Crosstabs analysis and the criterion of the Chi-square test significance value $p < .05$. It was checked whether the frequency of occurrence of the type of climbing handhold (right hand vs. left hand) and the type of climbing foothold (right leg vs. left

leg) were independent of the torso motion tactic (Twisting vs. Without Twisting).

Results

"1st Category - Type of climbing handhold (right hand vs. left hand)"

According to the results, the most frequent type of climbing handhold, both for the right (72%) and the left hand (63%),

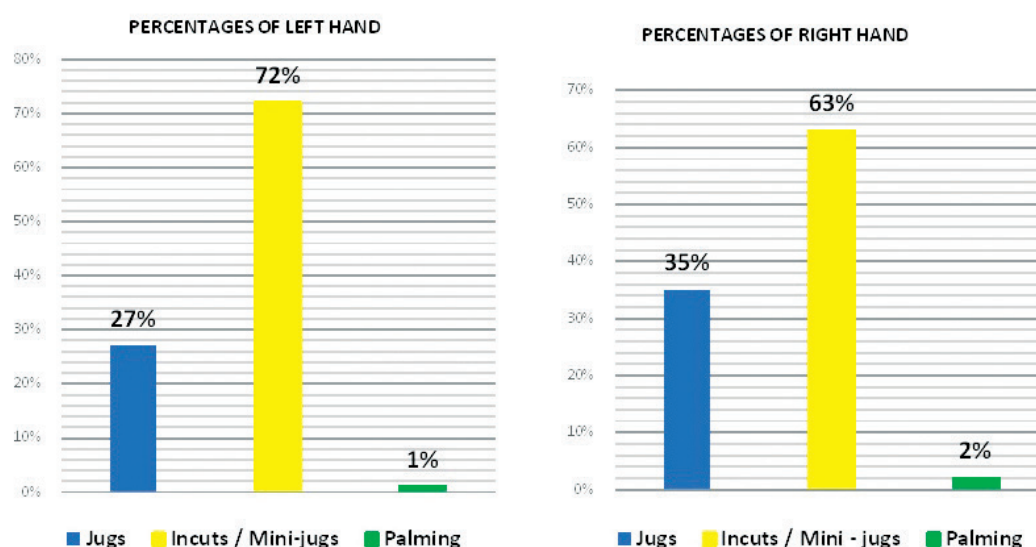


FIGURE 1. 1st Category - Handholds (Left hand vs. Right hand)

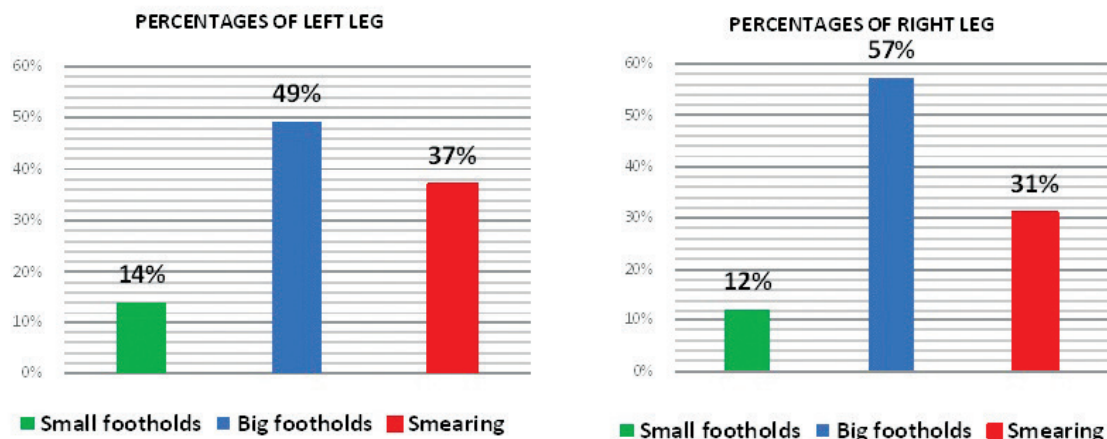
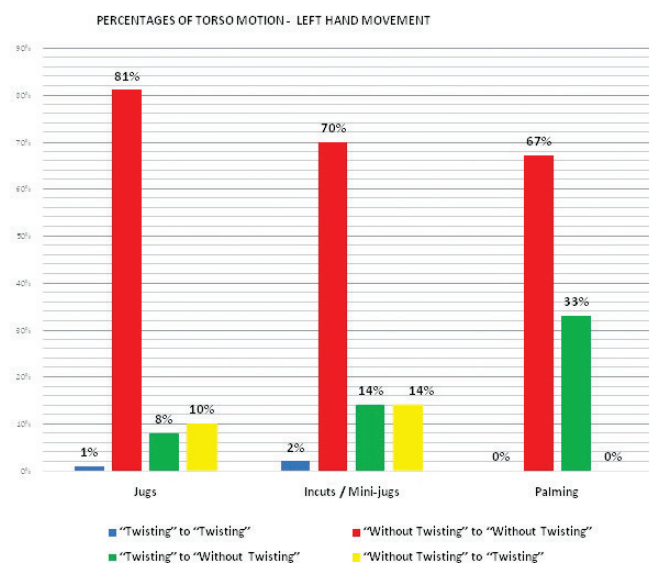
was Incuts/Mini-jugs. Jugs appeared with a smaller percentage (right hand 35%, left hand 27%) and palming with a minimum, 2% & 1% respectively (Figure 1).

"2nd Category - Type of climbing foothold (right leg vs. left leg)"

Figure 2 shows that the most frequent technique for both feet was big footholds (right 57% and left 49%). Smearing had a rate of 31% with the right leg and 37% with the left, while small footholds had a rate of 12% and 14% respectively.

"Relation between 3rd Category – Torso motion (Twisting vs. Without Twisting)" and "1st Category - Type of climbing handhold" regarding Left-hand movement

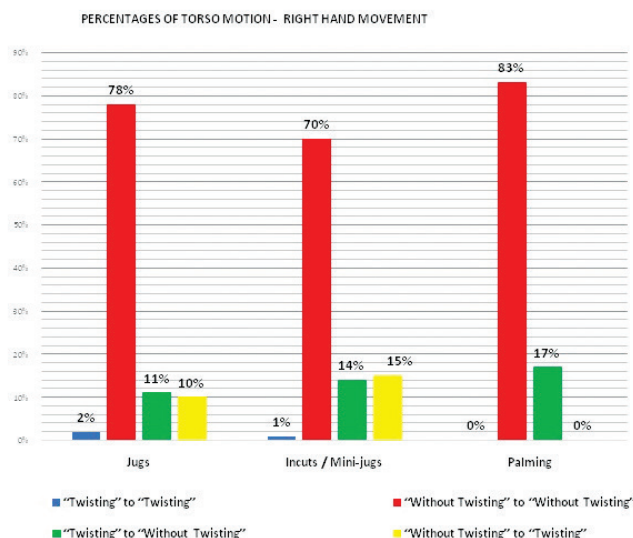
According to the results, the techniques of the left-hand grips did not significantly affect the torso motion (Chi-Square=6.33, $p > .05$). In particular (Figure 3), the techniques with the left hand were performed mainly without torso twisting (Incuts/Mini-jugs 70%, jugs 81%, palming 67%).

FIGURE 2. 2nd Category - Footholds (Left leg vs. Right leg)FIGURE 3. 3rd Category – Torso motion and Left hand movement

Relation between "3rd Category – Torso motion (Twisting vs. Without Twisting)" and "1st Category - Type of climbing handhold" regarding Right-hand movement

Also, the right-hand grip techniques did not significantly

affect torso motion (Chi-Square=4.43, $p>.05$). In particular (Figure 4), the techniques with the right hand were performed mainly without torso twisting (Incuts/Mini-jugs 70%, jugs 78%, palming 83%).

FIGURE 4. 3rd Category – Torso motion and Right hand movement

Relation between "3rd Category – Torso motion (Twisting vs. Without Twisting)" and "2nd Category - Type of climbing foothold" regarding Left leg movement

It was also found that the techniques of stepping with

the left leg did not significantly affect torso motion (Chi-Square=6.43, $p>.05$). All techniques (Figure 5) were mainly performed without torso twisting (big footholds 72%, smearing 78%, small footholds 65%).

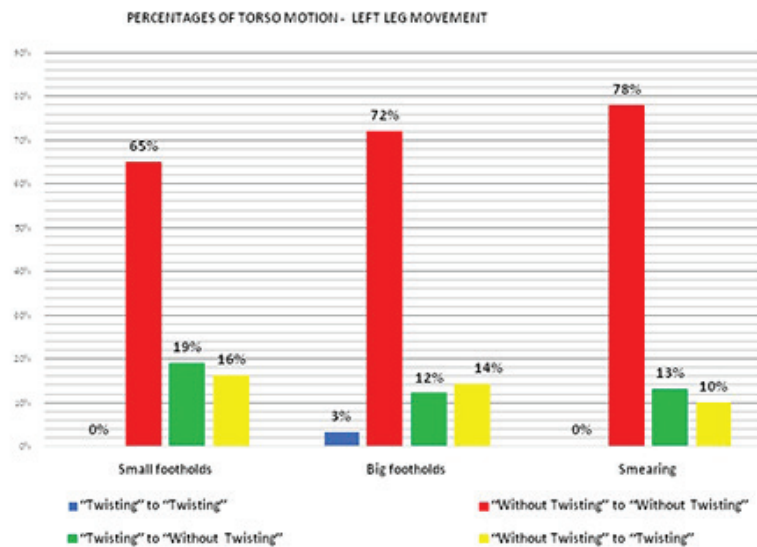


FIGURE 5. 3rd Category – Torso motion and Left leg movement

Relation between 3rd Category – Torso motion (Twisting vs. Without Twisting) and 2nd Category - Type of climbing foothold regarding Right leg movement

On the contrary, the techniques of stepping with the right foot had a statistically significant effect on torso motion (Chi-

Square=15.53, $p<.05$). In particular, although the majority of the techniques with the right foot were mainly performed without torso twisting (big footholds 65%, smearing 80%, small footholds 91%), 19% of the big footholds were performed with "Without twisting" to "Twisting" (Figure 6).

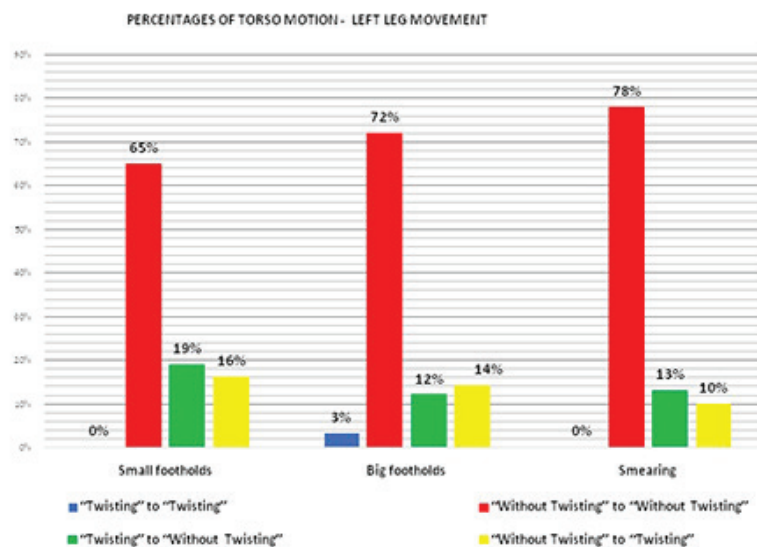


FIGURE 6. 3rd Category – Torso motion and Right leg movement

Discussion

Through video analysis, it was revealed that the selection of smaller and more difficult handholds and big footholds, as well as the climber's trunk maintenance towards the wall, are highly important performance elements in elite climbers. However, to the best of our knowledge, as this is the first study which aimed to identify elite athletes' technical and tactical behavior regarding the parameters: a) type of climbing handhold, b) type of climbing foothold and c) torso motion through video analysis, we cannot compare our results to other studies. Nevertheless, we strongly believe that our findings

provide knowledge of the technical skills, as well as tactical knowledge in elite climbing athletes, which is fundamental for the understanding of individual sports performance and consequently for the adaptation of the teaching-learning training processes to competition demands.

Regarding the 1st Category - Handholds (left hand vs. right hand), it was observed that the majority of the participants selected to use "Incuts / Mini-jugs", regardless of whether they performed the next movement with the left or the right hand, as the percentages appear to be similar (72% left hand – 63% right hand). Taking into account the results of the present

study, these could be partly explained by the fact that climbers tend to grasp handholds that have ambiguous features and are usually incongruent with their responding hand, regardless of their size and shape. However, it has been proved that climbers respond significantly faster and more accurately when the functional feature is congruent with their responding hand (Tucker & Ellis, 1998). What is more, according to Garrido - Vázquez and Schubö (2014), in their attempt to implement quickly the next movement, climbers often select to grasp handholds which are closer to them, although in some cases the geometry of the object does not prove its graspness. In addition, according to Pezzulo and his colleagues (2010), elite climbers are able to hold small and difficult holds, while at the same time, they can simulate sequences of actions that are too complex, “much like how expert chess players ‘see’ complex strategies” (Figure 1).

Regarding the 2nd Category - Footholds (left leg vs. right leg), it was found that participants mostly selected to use “Big footholds”, regardless of whether they performed the next movement with the left or the right leg (49% left leg – 57% right leg). Big footholds are blocks that can bear climbers weight on its vertical axis and are also large enough to ensure that the surface of the sole of the shoe will stay in contact with the hold. Additionally, this type of foothold allows the climber to set himself up for the next set of movements without extraordinary effort, which, according to Marino and Kelly (1988), is essential in order for the climber to conserve his upper body strength for the most difficult parts of the climb attempt. Finally, given the results of the present study, it seems that climbers selected big footholds in order to enhance their body balance and equilibrium, as it is widely known that during a large proportion of the climbing time most of the body weight is supported by the legs (Marino & Kelly, 1988) (Figure 2).

Regarding the correlation between “3rd Category – Torso motion (Twisting vs. Without Twisting)” and “1st Category - Handholds (left hand vs. right hand)”, as well as the correlation between “3rd Category – Torso motion (Twisting vs. Without Twisting)” and “2nd Category - Footholds (left leg vs. right leg)”, it was found that the overwhelming majority of the climbers selected to perform in all cases movements without “Torso twisting”, regardless if they used a) Left or Right

hand, b) Left or Right leg, c) a specific type of handhold (Jugs, Incuts/Mini jugs and Palming) or d) a specific type of foothold (Small footholds, Big footholds and Smearing). On the contrary, only the techniques of stepping with the right foot had a statistically significant effect on torso motion. In particular, although the majority of the techniques with the right foot were mainly performed without twisting the torso, 19% of the big footholds were performed with “Without twisting” to “Twisting”. Similarly to our findings, Seifert and his colleagues (2015), found that the majority of the climbers during their climbing attempt spent about 55 to 75% of the time with the trunk in a face-to-wall orientation, while the rest of the time was mainly spent with the trunk in an oblique position, and less than 2% of the time was spent with the trunk side to the wall. According to Newell and McDonald (1992), these could be partially explained by the fact that the design of a climbing route has a significant impact on the climber’s hip rolling motion variability and it seems that this is happening due to the meta-stability appearance in movement systems during a climbing attempt (Seifert, Boulanger, Orth, & Davids, 2015). This fact is also verified from other researchers who argued that the texture, surface, shape, size and orientation of holds invite various hand-grasping patterns and body positions (Phillips, Sassaman, & Smoliga, 2012). On the other hand, except for the effect of the design of the climbing route on trunk orientation, it has been proved that a climber’s improved physical fitness factors such as postural stability and flexibility, as well as anthropometrics like height, weight and body fat, are important components for the precise movement of hands and feet, in order to gain a constant body position during the climbing attempt and achieve a better performance through the implementation of limbs’ movements without torso twisting (Draper, Brent, Hodgson, & Blackwell, 2009; Stephan et al., 2011; Watts, 2004) (Figure 3 – Figure 6).

Finally, it is important to note that the small number of analyzed parameters and the insufficient testing of protocol validity and reliability constitute the limitations of the present study. Future research should focus on the evaluation of more technical and tactical parameters in elite climbers and on the development of reliable – high-profile athlete performance tools.

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Conflict of Interest

The authors have no conflict of interest to declare.

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ORIGINAL SCIENTIFIC PAPER

A Comparative Study of Motivation in Sports Training among Senior and Junior Professional Football Players in Bosnia and Herzegovina during the COVID-19 Pandemic

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Abstract

The Covid-19 pandemic has brought significant changes to the lives of athletes, particularly regarding the uncertainty surrounding the continuation of sports training and the postponement of sporting events. Given these disruptions to athletes' routines, the role of motivation in sports participation is considered even more crucial. Therefore, this study aimed to examine the differences in motivation for sports participation during the Covid-19 pandemic between senior and junior professional football players. The sample consisted of 358 elite football players from the Premier League of Bosnia and Herzegovina. The players were divided into two groups: senior ($n=193$, 24.87 ± 3.78 years) and junior ($n=165$, 18.08 ± 1.00 years) football players. The Participation Motivation Questionnaire (PMQ) was utilized, comprising 19 items classified into six motivational dimensions: "Sports success", "Social status", "Friendship", "Physical health", "Sports activity", and "Support". A five-point Likert scale was used in the questionnaire, which was administered electronically via Google Forms. Results revealed that junior football players achieved significantly higher values in only three out of the 19 items: "Want to be popular" (.005), "Want to get status" (.001), and "Wished for friendly games" (.000). However, there were no differences between the senior and junior football player groups in the remaining 16 items. Also, both groups displayed exceptionally high motivation levels in the dimensions of "Sports success", "Physical health", and "Friendship", while the weakest results were observed in the dimension of 'Support' for both groups. Despite the specific challenges posed by the Covid-19 pandemic, the motivation to play football remains very high among both senior and junior football players. This study is essential as it provides insights into football players' motivation and identifies differences in motivation levels between junior and senior players. The findings can be valuable for practical purposes, aiming to enhance motivation among football players, especially during circumstances where training conditions are disrupted, such as the Covid-19 pandemic.

Keywords: soccer, sports motivation, training, Covid-19 lockdown, PMQ, questionnaire, elite athletes

Introduction

The Covid-19 pandemic has had a significant impact on athletes' lives, particularly in terms of the uncertainty surrounding the continuation of sports training and the postponement of sporting events (Mon-Lopez, de la Rubia, Hontoria, & Refoio, 2020; Parm, Aluoja, Tomingas, & Tamm, 2021). The negative consequences of social isolation are the absence of organized training and competitions, lack of adequate commu-

ment of sporting events (Mon-Lopez, de la Rubia, Hontoria, & Refoio, 2020; Parm, Aluoja, Tomingas, & Tamm, 2021). The negative consequences of social isolation are the absence of organized training and competitions, lack of adequate commu-



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nication between athletes and coaches, the inability to move freely, and unsuitable training conditions (Chen et al., 2020). These challenges have also affected football, leading to match cancellations and players training in isolated home environments (Brooks et al., 2020). During this period, professional football clubs devised home training programs for their players (Pena et al., 2021). These efforts aimed to minimize the impact of the pandemic on sports careers and enable athletes to remain active and ready for the return of competitions.

However, despite coaches' efforts to devise appropriate training plans and programs, athletes mostly had to exercise at home without supervision (Sarto et al., 2020), which made training more challenging. Several researchers have suggested the harmful effects of isolated individual training, indicating that it can reduce the volume and quality of training and, consequently, decrease physical fitness (Girardi et al., 2020; Sarto et al., 2020; Mohr et al., 2022). Athletes who previously relied on their daily routines and enjoyed their teammates' company were forced to stay at home. This feeling of isolation significantly impacted their well-being, and they needed support. It is noticeable that home training faces multiple challenges, such as inadequate training conditions, poorly organized workouts, restricted movement, and lack of communication with coaches (Jukic et al., 2020), which can negatively impact football players' motivation to exercise and engage in sports activities.

Motivation involves initiating human activity and directing that activity toward specific objects and goals (Lazarević, 2001). It is a complex process that encompasses the following elements: activated needs, identification of needs, situation analysis, selection of one of the possible goals, planning activities that lead to the goal, and goal achievement (Tubić, 2004). Wigfield and Eccles (2000) emphasize that motivation influences our choices, behavior, perseverance, and achievements, thus playing a crucial role in our daily lives.

Motivation is the critical factor that drives athletes to participate in sports activities (Roychoudhury, 2018) and influences their success in sports (Bollok, Takacs, Kalmar, & Dobay, 2011). Therefore, more and more authors are paying

attention to this problem (Ruffault, Bernier, Fournier, & Hauw, 2020; Leyton-Román, de la Vega & Jiménez-Castuera, 2021; Katanic, Bjelica, Corluka, Preljevic & Osmani, 2022; Posavac, Katanic, Sebic, & Bjelica, 2023).

Home training presents more significant challenges due to the lack of equipment and space, limited opportunities for interaction with other athletes, and a lack of motivation (Tingaz, 2021). However, some athletes have adapted to the new circumstances and found ways to maintain motivation and regular training (Parm et al., 2021).

Although motivation plays a crucial role in athletes' training, a limited number of studies have examined the motivation of football players to train during the Covid-19 pandemic (Washif et al., 2022; Posavac et al., 2023). However, studies that explored differences in motivation between junior and senior football players are yet to be found. Therefore, this study aimed to determine whether there are differences in the level of motivation for sports training during the Covid-19 pandemic between junior and senior football players. If differences in between younger and older football players' motivation are found, this study could also have practical implications by highlighting different approaches to motivation for engaging in sports across age categories.

Methods

Participants

The sample consisted of 358 football players from Bosnia and Herzegovina (B&H), who were divided into two groups (table 1), senior football players ($n=193$, 24.87 ± 3.78 years) competing in the Premier League and junior players ($n=165$, 18.08 ± 1.00 years) competing in the top junior league in B&H. All participating football players were healthy, without serious injuries, and had been training for over five years. Ethical review and approval required for the study on human participants were obtained in accordance with local legislation and institutional requirements. Prior to participation, all participants were informed about the experimental procedures. The research was following the Helsinki Declaration.

Table 1. Descriptive parameters of the groups.

	Senior	Junior
n	193	165
Age (Mean $\pm$ SD)	24.87 $\pm$ 3.78	18.08 $\pm$ 1.00

Note: n – number, Mean – Arithmetic mean, SD – Standard deviation

Measurements

The questionnaire on motivation for sports and sports training during the COVID-19 pandemic was used. The questionnaire consisted of 19 items classified into six dimensions (table 2) and it was based on the modified Participation Motivation Questionnaire (PMQ) (Gill, Gross, & Huddleston, 1983; Zahariadis & Biddle, 2000). In previous research (Katanic et al., 2022), the internal structure of the questionnaire was examined using factor analysis, and variables were grouped into factors based on their calculated factor weights. Factor analysis identified dimensions such as "Sports Success", "Support", "Social Status", "Friendship", "Physical Health", and "Sports Activities" which were used in this study. The questionnaire used a 5-point Likert scale, and responses were identified as "very important", "important", "somewhat important", "not important", and "not at all important" (Joshi, Kale, Chandel, &

Pal, 2015). The questionnaire was in electronic form and was sent to the football players via Google Form.

Statistics

Data were analyzed using the SPSS statistical analysis program (IBM SPSS Statistics for Windows, version 26.0. Armonk, NY: IBM Corp.). Descriptive analysis was used to distribute the data based on the number of participants, means, and standard deviation. An independent-sample T-test was used to determine differences in the level of motivation for sports training between junior and senior football players. Statistical significance was assumed at $p>0.05$.

Results

Participant motivation for exercise was assessed using 19 items divided into six factors (sporting success, social status,

friendship, physical health, and sports activity). A significant difference was found in only 3 out of the 19 items, specifically “Want to be popular” (.005), “Want to get status” (.001), and “Wished for friendly games” (.000). However, there was no difference between the senior and junior football player

groups in the remaining 16 items (Table 2).

It is also worth noting that both groups achieved exceptionally high scores in the dimensions of “Sports success,” “Physical health,” and “Friendship,” while the weakest results were obtained in the dimension of “Support” for both groups.

Table 2. Differences in motivation for sports training between senior and junior football players.

			Senior (Mean±SD)	Junior (Mean±SD)	t	p
Participant Motivation Questionnaire (PMQ)	Sport success	Like to compete	4.93±0.28	4.92±0.36	.366	.714
		Want to play at a higher level	4.96±0.22	4.93±0.29	.917	.360
		Like the rewards	4.89±0.35	4.88±0.47	.167	.867
		Like the sport challenge	4.90±0.41	4.89±0.37	.132	.895
		Need support from colleagues	3.23±1.24	3.13±1.20	.778	.437
	Support	Need support from a coach	3.43±1.25	3.46±1.16	-.238	.812
		Wished for socializing with colleagues	3.95±1.07	4.03±1.13	-.659	.510
		Want to be popular	3.66±1.17	3.99±1.06	-2.801	.005*
	Social status	Want to gain status	4.49±0.89	4.76±0.62	-3.440	.001*
		Like to meet new friends	4.33±0.88	4.41±0.81	-.827	.409
	Friendship	Like the teamwork	4.74±0.50	4.79±0.49	-1.017	.310
		Like being on a team	4.89±0.36	4.88±0.40	.157	.875
		Like the excitement	4.75±0.50	4.73±0.63	.214	.831
	Physical health	Want to be healthy	4.95±0.23	4.95±0.24	.073	.942
		Want to stay in shape	4.93±0.34	4.97±0.17	-1.330	.185
		Get rid of excessive energy	4.27±0.93	4.08±1.17	1.733	.084
	Sport activity	Wished of group training	4.85±0.51	4.90±0.34	-1.141	.255
		Wished of friendly games	4.16±0.81	4.55±0.65	-5.122	.000*
		Wished of official games	4.93±0.28	4.93±0.28	.006	.995

Note: Mean - Arithmetic mean; SD - Standard deviation; t - t test value; p - Statistical significance; * - Significant differences.

Discussion

Motivation involves directing human activity toward specific objects and goals, and in sports is expressed through persistence and perseverance (Lazarević, 2001). Athletes face various challenges and obstacles, but their motivation drives them to overcome these difficulties and continue with training and competitions. This resilience in sports can result from a combination of biological, psychological, sensory, and situational factors that influence athletes' determination to achieve their goals (Lazarević, 2001; Tubić, 2004).

This study examined the difference in the level of motivation for sports training during the Covid-19 pandemic between junior and senior football players. A significant difference was found in only 3 out of 19 items, specifically “I want to be popular” (.005), “I want to gain status” (.001), and “I want friendly games” (.000). However, in the remaining 16 items, there was no difference between the senior and junior groups of football players (Table 2).

The structure of motivation is divided into six factorial dimensions (“Sports success,” “Social status,” “Friendship,” “Physical health,” “Sports activity,” and “Support”) and is in line with previous research, which mostly identified a basic structure of 5 to 8 factors (Fabrigar, Wegener, MacCallum, & Strahan, 1999; Katanic et al., 2022). Regarding motivational dimensions, differences were observed in 2 out of 6 dimensions between the younger and older football player groups,

specifically in the factors of “Social status” and “Sports activity.” However, since there were no studies of this kind, comparing these results is impossible. On the other hand, a previous study (Posavac et al., 2023) demonstrated no difference in motivation for training based on the level of competition among football players. By examining ratings given on a Likert scale (from 1 to 5), the average ratings for each dimension were relatively high for both groups, consistent with previous studies (Mladenović & Marjanović, 2011; Posavac et al., 2023). When analyzing individual factors, the lowest values were observed in the Support factor (3.13-4.03), while all other factors had values above 4. Both groups achieved exceptionally high results in the dimensions of “Sport success,” “Physical health,” and “Friendship,” while the weakest results were obtained in the dimension of “Support” for both groups. These findings indicate that motivation for training among football players was very high under the given conditions. In the study by Mladenović and Marjanović (2011), which included young football players from these regions (Serbia and Montenegro), it was observed that these athletes demonstrated higher motivation driven by the intrinsic (internal) desire to improve their sports skills compared to their peers from Russia. This suggests that intrinsic motivation, i.e., personal satisfaction and enjoyment that athletes experience while engaging in sports is a crucial to their involvement. Although we did not categorize motives into internal and external in our study, the motivations were

predominantly internal, so our results align with the findings of the mentioned authors, indicating that internal motivation for sports participation is also high among junior and senior football players. Similar results were obtained by Malčić (2012) in his research, which also identified intrinsic motivation as the primary driver of youth participation in sports.

When it comes to the structure of motivation, the research conducted by Bačanac, Lazarević, and Arunović (1994) identified several key factors: "Achievement of a higher level," "Competition," "Being in good physical condition," "Training," and "Doing something to learn," which align with the factors in our study. Similar conclusions were reached by the authors (Bacanac, Radovic, & Veskovic, 2007), who point out that among young athletes there is a high level of awareness regarding the contribution of sports to health improvement and preservation. In addition to socializing, the love for sports is identified as one of the most stable and frequently mentioned motives (Bacanac et al., 2007). When examining the factors individually, physical health emerged as the most important motivator, indicating an awareness of the significance of physical well-being and maintaining health through sports activities (Thøgersen-Ntoumani, Fox, & Ntoumanis, 2005; Stojmenović, Katanić, & Ugrinić, 2021). Love for the sport is also essential because passion for a particular sport often drive athletes to dedicate themselves to training and competitions. Success, as a motivation, represents the desire to achieve results and reach goals. Friendship and social interaction also play a crucial role in athletes' motivation, as many young football players enjoy socializing and forming relationships with their teammates (Sheridan, Coffee, & Lavalley, 2014). Popularity and good appearance can be motivational factors for some athletes, as social acceptance and attention can be considered rewards for achievements in sports. In our study, high results were achieved for these mentioned factors, and similar factors were identified in other studies (Bačanac et al., 1994;

Katanić et al., 2022).

Thus, we conclude that motivation is fundamental in sports and manifests through persistence and perseverance. Persistence in sports is influenced by biological, psychological, sensory, and situational factors (Abernethy, Hanrahan, Kippers, Mackinnon, & Pandey, 2012). On the other hand, it is crucial to set realistic goals and strive to achieve them, as each accomplished goal can strengthen athletes' motivation. This study is significant because it is one of the few that compared the difference in motivation for training between junior and senior football players in a large sample of professional footballers. It is also important because the motivation for training of football players was examined during a specific period of altered training conditions due to the Covid-19 pandemic.

Like any study, this one also has certain limitations, and one of the main limitations is the fact that the players filled out the questionnaire from home, raising the possibility that someone may have assisted them. In any case, due to the Covid-19 pandemic, it was not possible to administer the questionnaire in person. Additionally, it would have been beneficial to precisely categorize the statements into intrinsic and extrinsic factors to obtain a more complete picture. Overall, suggestions for further research should consider having respondents complete the questionnaire in person, ensuring precise item categorization, and creating a questionnaire with more items for more comprehensive answers. Furthermore, conducting interviews with players using a mixed-methods approach could lead to more comprehensive conclusions.

Despite the limitations, this study is crucial as it provided insights into the motivation of football players and the difference in motivation levels between juniors and seniors. Therefore, the findings can be used for practical purposes and work towards strengthening motivation among football players, especially in specific circumstances when training conditions are disrupted, such as during a pandemic.

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Conflict of interest

The authors declare that there is no conflict of interest

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ORIGINAL SCIENTIFIC PAPER

Analysis of the Offensive Tactics Used by High-Level Basketball Players of Different Age Groups

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Abstract

The rapid and continuous development of basketball has necessitated the long-term training of young basketball players, enabling them to meet the ever-changing demands of the sport. The aim of the present study was to record the offensive competitive behaviour of men and boys by analysing world championship matches in order to determine whether the choice of tactics in them depended on the age category they belonged to. The sample of the research consisted of 2,997 actions performed by basketball national teams in Men's (seniors) and juniors' World Championships. The juniors' category includes athletes 14-15 years old (U16). Offenses were recorded with the Sportscout game observation software. The recorded parameters are the following: age group, types of initiating an offense, offensive tactics, type of offensive, attacking from the perimeter & from inside the paint, screening in offense, pick and roll in offense, offensive inbound plays, inbound position, attack efficiency, shooter, shooting area/position. Summarizing, the juniors' group showed a higher percentage in offenses fast breaks, in motion offense, in the attacks that were starting after a steal and an offensive rebound. In addition men's group had higher percentage in inside game and in pick and roll. According to the above the offensivetactics depended significantly on the age group. This confirms the tendency of coaches to use continuity offences in order to train young athletes in all competitive positions rather than in a single one. Their next step would be to adopt the defensive transition tactic, found to have been adopted by the men's group.

Keywords: performance analysis, match analysis, basketball tactics, senior and juniors athletes

Introduction

The impressive development of tactics in basketball has created the need for continuous update and reflection. Year after year, new trends are emerging, mainly due to changes in regulations. Obradovic (2012) states that the reduction of the attack duration to 24 seconds, the team fouls, the new three-point line, the observed flexibility in terms of fouls, and the tendency not to stop the game for minor reasons are some of the factors that have had a significant impact on the development of the game and led to more powerful offenses and defenses.

The need for well-documented scientific theories by coaches and researchers has made the demand for the recording and analysis of matches imperative (Mavridis, Laios, Taxildaris, & Tsiskaris, 2003). Research related to the analy-

sis of players' technical-tactical behaviour has been carried out for several years (Ribas, Navarro, Tavares, & Gómez, 2011; García, Ibáñez, Gómez, & Sampaio, 2014; Vaquera, García-Tormo, Gómez, Ruano, & Morante, 2016; Zhang et al., 2017; Remmert, 2019). Apart from identifying the talented player and generally selecting the right players, coaching science is also interested in studying the opponent through video analysis. Its ultimate goal is to provide information and help coaches at every competitive level do their job more effectively (Hughes & Franks, 2004). The selection of a game situation and the precise analysis of the corresponding individual actions performed by players are of particular interest in order to evaluate and subsequently improve tactical skills (Clay & Clay, 2014; Ibáñez, McRobert, Toro, & Vélez, 2016; Hojo, Fujii, & Kawahara, 2019).



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In the case of basketball, observational studies have been conducted by a considerable number of researchers through video analysis of matches. While some of them aimed at analysing the technical-tactical characteristics of high-level players (Gómez et al., 2015; Vaquera, et al., 2016; Zhang, et al., 2019), others tried to draw conclusions about the frequency of occurrence and use of various technical-tactical partnerships in defence and attack (Polykratis, Tsamourtzis, Mavridis & Zaggelidis, 2010) while there are also researchers who tried to find the parameters that are related to the outcome of the match through the corresponding statistics (Csataljay, O' Donoghue, Hughes, & Danks, 2009; Zhang et al., 2017).

A review of both older and relatively recent literature showed that there are few studies relating to the analysis of basketball games with players in the developmental stage (juniors/U16). Although there are studies on the competitive behaviour of the specific age group (Ortega, Cardenas, Sainz, & Palao 2006; Lorenzo, Gomez, Ortega, Ibanez, & Sampaio, 2010), none of them compares the competitive behaviour among different age groups (men, under 16, under 18 men and women). Moreover, as there are no previous research studies on the competitive behaviour of U16 male players, following the changes in rules implemented in 2010 (extension of the 3-point line from 6.25m to 6.75 m/in place from 1984, and the change of the trapezoid restricted area/in place since the 1950s to a rectangular one), there are no relevant data on which to base any further analysis, in order to study its evolution in terms of playing and coaching. The resulting research gap prompted the implementation of the present study. Therefore, we started by recording the in-game offensive behaviour of the U16 group in the 2010 World Championship. The primary objective of the study was to gather data that characterized the offensive technical-tactical trends of the specific group at that time. At the same time, the recording and analysis of the same data in games in the men's group and their further comparison with the U16 group outlined the offensive behaviour of men as well as the "distance" that separates the two groups. The aim of the present study was to record the offensive competitive behaviour of seniors (men) and juniors (U16) by analysing world championship matches in order to determine whether the choice of tactics in them depended on the age group they belonged to.

Method

Sample

The research sample consisted of 20 randomly selected 2010 World Championship matches (10 from each age group). The above number of matches was the limitation of this study, as they were the only ones available on the website platform, www.fibatv.com. However, it was found that the number of actions studied (a total of 2.997 actions / 1391 for seniors and 1606 for juniors) did not have a negative impact on the statistical processing of the data.

Data collection and measuring instruments

The Sportscount game observation software was used in the recording of the attacks and the following observation protocol was created with the participation of a professional basketball coach (a member of the Hellenic Basketball Federation): 1) Age group: Seniors (men), Juniors (U16), 2) Types of initiation

ing an offense: Defensive rebound, Offensive rebound, Steal, point, Out back, Out front, 3) Offensive tactics: Fast break, Quick attack, Organized attack/Set play, Offensive transition, 4) Type of offensive play: Set-play offense, Continuity offense, Free flowing offence, 5) Attacking from the perimeter & from low post: Inside game, Outside game, 6) Screening in offense: Screen offense, offense without screen, 7) Pick and roll in offense: pick and roll offense, offense without pick and roll, 8) Offensive inbound plays: Out play, pass followed by Set play, 9) Inbound position: Sideline, Baseline, 10) Attack efficiency: Shoot +2, Shoot -2, Shoot +3, Shoot -3, Foul awarded, Turnover, 11) Shooter: Point guard, Shooting guard, Small forward, Power forward, Center, 12) Shooting area/position: Low post, Middle post, High post, Left forward, Right forward, Guard area, long shot from the center of the field, long shot from the opposing field.

Data analysis

Data were processed with the SPSS statistical software suite using the Crosstabs statistical analysis (Kent State University, 2023). The above analysis tested the relationship between the age groups (seniors and juniors) and the corresponding technical-tactical choices during a match. The criterion used to test for homogeneity or independence of variables was Chi-Square with $p < .001$.

Results

Types of offense initiation

According to the results (Table 1a), there was a difference between the two age groups in the initiation of offense after an "offensive rebound" (56% juniors and 44% seniors) as well as in the initiation of offense after a "steal" (60% & 41%). A differentiation was also found in "out back" (55% & 45%). According to the value ($\text{Chi-Square}_{(5)} = 30.05$, $p < .001$) it was found that the mode of initiation of the attack was significantly dependent on the age group.

Offensive tactics

The main difference in "offensive tactics" was found in the cases of "fast break" (74% & 27%) and "offensive transition" (57% & 43%). The statistical test ($\text{Chi-Square}_{(3)} = 102.37$, $p < .001$) showed that offensive tactics were significantly dependent on age group.

Type of offensive tactics

Regarding the types of offensive tactics, it was found that they were significantly dependent on the age group ($\text{Chi-square}_{(2)} = 90.51$, $p < .001$). In particular, juniors performed more "continuity offenses" (65%) than seniors (35%), as well as more "set-play offences" (46% & 35%).

Attack with peripheral play & with play inside the paint:

According to the results, the juniors mainly performed "inside game" offenses (52%) while the seniors preferred "outside game" offenses (52%). $\text{Chi-Square}_{(1)} = 7.898$, $p < .05$ showed that these types of offense were significantly dependent on the league the games were played in.

Use of offensive screens

Regarding the use of "screen", it was found that the rate of occurrence did not depend on the age group ($\text{Chi-Square}_{(1)} = 2.40$, $p > .05$).

Use of offensive pick and rolls

In contrast, the occurrence rate of pick and roll was significantly dependent on age group (Chi-Square₍₁₎=26.80, $p<.001$). In particular, “pick and roll” was used more by seniors (55%) and less by juniors (45%).

Offensive inbound plays

The results showed that the type of offensive inbound plays used did not depend on age group (Chi-Square₍₁₎=0.47, $p>.05$). Thus, regarding the offensive inbound plays, it was observed that most of them were executed with an inbound pass

Table 1a. Relationship between offensive parameters and age groups

CATEGORIES	PAPAMETERS	SENIORS	JUNIORS
Types of offense initiation	Defensive rebound	49%	51%
	Offensive rebound	44%	56%
	Steal	41%	60%
	Point	51%	49%
	Out back	45%	55%
	Out Front	48%	52%
$\chi^2_{(5)}=30.05, p=.000$			
Offensive tactics	Fast break	27%	74%
	Quick attack	52%	48%
	Organized attack/Set play Set play	50%	50%
	Offensive transition	43%	57%
$\chi^2_{(2)}=102.37, p=.000$			
Type of offensive play	Set-play offense	54%	46%
	Continuity offense	35%	65%
	Free flowing offence	50%	50%
$\chi^2_{(2)}=90.51, p=.000$			
Attack with peripheral play & with play inside the paint	Inside game	48%	52%
	Outside game	52%	48%
$\chi^2_{(1)}=7.89, p=.160$			
Screening in offense	Offense with screen	50%	50%
	Offense without screen	52%	48%
$\chi^2_{(1)}=2.40, p=.329$			
Use of offensive pick and rolls	Offense with pick and roll	55%	45%
	Offense without pick and roll	49%	52%
$\chi^2_{(1)}=26.80, p=.000$			

followed by an organised offense (setplay). The above competitive behaviour was similar for seniors (62%) and juniors (64%). Also, both seniors (38%) and juniors (36%) chose out play after an inbound pass (Table 1b).

Offensive Effectiveness

According to the results, the biggest differences in offensive effectiveness, were found in the case of “successful three-point shots”, where seniors had a higher percentage (55%) compared to juniors (46%). In contrast, in the case of turnovers, most of them were made by the juniors (60%). Based on Chi-Square₍₁₎=47.68, $p<.001$, a significant relationship was found between age group and offensive effectiveness.

Shooter

The test showed that the relationship between the parameters “shooter” and age group was significant (Chi-Square₍₄₎=28.06, $p<.001$). More specifically, there were obvious differences in the case of point guard and shooting guard posi-

tions, where the highest percentage of participation in attacks (shooting) appeared in juniors (55% of point guards and 56% of shooting guards). Differentiation was also observed in the small forward position where the highest percentage occurred in seniors (52%) as opposed to juniors who had a lower percentage (48%).

Shooting area/position

Regarding the area from which most shots were taken, it was found that low post shots were taken mainly by seniors (53%), in contrast to the “middle post” and “high post” areas where most shots were taken by juniors (58% from the midlepost and 54 from the highpost). Also, the juniorstook more shots from the “right forward” position (55%), while the seniors took more shots from the guard position (56%).

Moreover, juniors took shots from the “advancing area” with a rate of 58% while the seniors with a rate of 42%. According to the value (Chi-Square₍₇₎=87.87, $p<.001$) the position from which the shots were taken was significantly dependent on the age group.

Table 1b. Relationship between offensive parameters and age groups

CATEGORIES	PAPAMETERS	SENIORS	JUNIORS
Offensive inbound plays	Out play	38%	36%
	Pass followed by set play	62%	64%
$\chi^2_{(1)}=0.47, p=.691$			
Offensive Effectiveness	Shoot+2	50%	50%
	Shoot-2	48%	52%
	Shoot+3	55%	46%
	Foul υπέρ	50%	50%
	Turnover	40%	60%
$\chi^2_{(1)}=47.68, p=000$			
Shooter	Point guard	45%	55%
	Shooting guard	44%	56%
	Small forward	52%	48%
	Power forward	50%	50%
	Center	49%	51%
$\chi^2_{(4)}=28.06, p=000$			
Shooting area/position	Low post	53%	47%
	Middle post	43%	58%
	High post	46%	54%
	Left forward	51%	49%
	Right forward	45%	55%
	Guard area	56%	44%
	Long shot from the center of the field	42%	58%
	Long shot from the opposing field	36%	64%
$\chi^2_{(7)}=87.87, p=000$			

Discussion

The aim of the present study was to correlate the competitive behaviour of seniors and juniors in coaching corresponding high-level matches in order to determine any possible differences in the tactics followed. In this way, it could be documented whether the used tactics depended on the age group, so that researchers and coaches could gain a more thorough insight into the tactics they should probably follow depending on the age category they are coaching.

The way offenses are initiated varies, but it seems that in the developmental ages of children, many of them start after a steal or an aggressive rebound, a characteristic that coaches should take into account when focusing on the reduction of mistakes and teaching blockout, which is one of the basic technical-tactical elements of basketball. The considerable number of mistakes made by juniors is probably due to the fact that they have not yet perfected their tactics (Canadas et al., 2013). Winston (2009) states that one of the most important elements in the development of the offense is going for offensive rebounds without making turnovers.

As far as tactics is concerned, it is observed that in the developmental age group, emphasis is placed on fast break in an attempt to impose fast pace and increase the number of ball possessions. According to Ory (2009) the quintessence of the offense is to create an advantage in fast break, while Vargas (2009) believes that the offense should always create conditions for scoring before the opponent's defense is organised. At

men's level, there are fewer surprises because teams emphasize the defensive transition, the transition of teams from offense to defense (Messina, 2007).

In the offensive system there was a tendency to use the set-play offense more often. However, it should be noted that there was a great disparity in continuity offense recorded in the case of juniors. At these ages, coaches tend to ask their players to try all positions, taking into account their biological age (GarciaToledo, 2017).

In modern basketball, perimeter play seems to be used more than play near the hoop. Cardenas (2006) states that from a young age athletes need to train in one-on-one perimeter play in order to be able to perfect it in the future. As far as play near the hoop is concerned, although it appears to be less frequent nowadays, it is the most effective (Romaris-Duran, 2016).

Regarding offensive screens, it seems that they were used by both age groups. This is probably due to the fact that in modern basketball the learning methodology starts with individual offense and continues with the cooperation of several players, either through movement without the ball or through screens. In particular, pick and roll cooperation, which aims to deliver the ball in an advantageous position near the hoop, is less common at the developmental level of children than that of men (Conte et al., 2013).

In the case of offensive tactics, it appeared that the two age group had similar inbound play tactics, either from the side-

line or the baseline. As it was found, most of inbound plays were done with a pass. This was probably aimed primarily at initiating a new offensive cooperation rather than a specific offensive inbound play. It appears that, as their primary goal is to limit the number of turnovers, coaches tend to avoid the use of any particular inbound play. After all, at critical points of the game, where athletes perform under pressure, making the right decisions is very important (Barragan, 2015).

In the men's age group, most of the attacks were executed by players playing in the forward and center positions. Indeed, the goal of the offense is to create an advantage near the hoop. It therefore seems that the accuracy rate of field shots is one of the most important factors associated with winning (Winston 2009; Martinez & Martinez, 2013). On the contrary, most of the offenses in the seniors' group were executed by players playing in guard position and far from the hoop. This is probably due to the fact that at the beginning of the coaching process coaches implement methods that can help their players to better understand their technique and tactics. For this reason, they also take into account other parameters such as the biological age of their players, which seems to be very significant in the case of players under 15 years old (GarciaToledo, 2017). As mentioned above, this research was the starting point for the study of the U16s in the year that the basketball regulations were changed. So, future studies could study a recent World Championship to check if, after a period of about ten years, the offensive behaviour of these age groups (seniors/juniors), and therefore the priorities of the coaches, have changed. Also, future research could use or improve the observation protocol of this study by studying the other age groups (women's

U16, men's and women's U18), both in 2010 and in the recent World Championship, in order to determine their own offensive profile as well as their differentiation from the offensive profile of seniors. Thus, the younger age groups and their coaches will adjust the content of their training according to the prevailing trends.

Conclusions

To summarise, the conclusions drawn are the following: In the case of juniors, most offenses started after a steal and after an offensive rebound. Most fast breaks occurred in the juniors' age group. Motion offense is more common among juniors. Outside game occurs more with juniors, while inside game with seniors. At both age levels, screen attacks are common. Pick and roll is used more frequently by seniors. Most inbound plays are made with a pass and an organized attack by both seniors and juniors.

By recording and analysing the offensive behaviour of high-level players, the present study is, on the one hand, believed to have filled a research gap that dealt primarily with the identification of differences between individual age groups (seniors and juniors). In this way, the prevailing trends in each category as well as their distinctive characteristics were highlighted. Therefore, it is believed to have provided significant help to coaches in lower divisions in order to prepare their players in accordance with the standards of higher divisions. On the other hand, the created observation protocol can be used by coaches to evaluate the offensive tactics of their teams and adjust their training sessions according to the resulting data.

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Conflict of Interest

The author declares that there is no conflict of interest.

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REVIEW PAPER

Effects of Neuromuscular Electrical Stimulation on Peripheral Muscle Strength and Exercise Tolerance in Chronic Obstructive Pulmonary Disease Patients: A Systematic Review

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Abstract

Currently, Chronic Obstructive Pulmonary Disease (COPD) is perceived not only as a pulmonary disease, but more as a systemic disease affecting the functioning of other organs including the locomotor system. The application of physical therapy in patients with COPD has been found to be safe and beneficial. Neuromuscular Electrical Stimulation (NMES) treatment is particularly beneficial for patients who are unable or unwilling to participate in daily activities or regular physical exercise. The efficacy of this technique has not yet been fully demonstrated in patients with COPD. Therefore, this study aimed to evaluate the effects of NMES on peripheral muscle strength and exercise tolerance in COPD patients. As part of the literature review for this study, seven studies over the last 19 years have been taken into consideration. The review included the following databases: Physiotherapy Evidence Database (PEDro), Google Scholar, and PubMed. Based on our literature review, we concluded that NMES has positive effects on increasing peripheral muscle strength and exercise tolerance in COPD patients. Future research should examine whether the therapeutic effects are long-lasting, whether they are experienced by all COPD patient groups, and which frequency or type of NMES delivers the most effective results.

Keywords: *pulmonary rehabilitation, neuromuscular electrical stimulation, physiotherapy, chronic obstructive pulmonary disease*

Introduction

Chronic obstructive pulmonary disease (COPD) is defined as a preventable and treatable condition/disease characterized by airflow limitation. Although COPD affects the lungs, it also produces significant systemic consequences (Viegi et al., 2007). COPD produces inactivity, which leads to deconditioning, mainly caused by breathing. This leads to an increased fear of exertion and avoidance of physical activities and social activities, trapping the patient within a vicious circle leading to further isolation and depression, accompanied by reduced quality of life (Corhay et al., 2014). Currently, COPD is perceived not only as a pulmonary disease, but more as a systemic

disease that affects the functioning of other organs, including the locomotor system. Skeletal muscle dysfunction is among the most common extra pulmonary manifestations of COPD (Kucio et al., 2016).

Pulmonary rehabilitation (PR) is a comprehensive intervention. It is based on a detailed patient assessment. The assessment is followed by therapies that include, but are not limited to, exercise treatment, education, and behavior modification, which are designed to improve the physical and emotional states of people with chronic respiratory disease (Gloeckl et al., 2013). The main components of PR include treatment with exercises, psychosocial and nutritional support, as well as



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education. The exercise program includes two main components: endurance treatment (continuous or with intervals) and strength treatment (Gloeckl et al., 2013; Corhay et al., 2014).

The application of physical therapy in patients with COPD is safe and beneficial (Neder et al., 2002). Recently, Neuromuscular Electrical Stimulation (NMES) has been successfully used as a localized training modality in severely disabled patients, who are unable to follow formal PR and/or tolerate higher training intensities (Nápolis et al., 2011). When COPD symptoms worsen or patients have movement disorders making it impossible for them to perform traditional physical exercises, NMES of the lower limbs can be used as an alternative to traditional physical training. NMES can be used safely on COPD patients when their symptoms get worse, when their disease is advanced, or even while they are receiving respiratory treatment (Kucio et al., 2016). NMES training is more likely to be beneficial for inactive patients with advanced illness than for active patients with stable symptoms and healthy individuals. This is due to the quadriceps muscles, which NMES primarily targets. Being extensively involved in various everyday tasks makes NMES less responsive to improvements in active people. Therefore, NMES treatment is especially advantageous for patients who are unable or unwilling to engage in daily activities or consistent physical activity (Veldman et al., 2016). The efficacy of this technique has not yet been fully demonstrated in patients with COPD (Neder et al., 2002). Implementation of NMES has been associated with increase in muscle strength, functional capacity, and health status in COPD patients (Vieira et al., 2014). NMES appears to be an effectual means of enhancing quadriceps strength and exercise capacity in moderate-to-severe COPD patients (Chen et al., 2016). Based on a meta-analysis reporting equivocal findings about the effects of NMES on COPD patients (Pan et al., 2013) we believe there is a need for further highlighting the effect of NMES on peripheral muscle strength and exercise tolerance in COPD patients. Therefore, the main purpose of this review paper was to evaluate the effectiveness of NMES in

the management of COPD patients. In particular, the effect of NMES on peripheral muscular strength and exercise tolerance needs to be investigated.

Methods

Search Strategy

As part of this literature review, seven studies conducted between 2004 and 2022 were considered. All these studies primarily focused on the use of NMES in COPD patients. Figure 1 describes the different phases of the search process. These studies included the following databases: Physiotherapy Evidence Database (PEDro), Google Scholar, and PubMed. They used keywords such as pulmonary rehabilitation, neuromuscular electrical stimulation, physiotherapy, and chronic obstructive pulmonary disease. This search was adapted for searching other databases as well. Based on the information found in the titles and abstracts, a first selection of articles was made using some generalized inclusion criteria. Then, a final selection was made according to an individual critical assessment of the quality of the studies. The assessment was carried out using the PEDro scale, which assesses the validity of each study (Table 1).

Inclusion Criteria

Study inclusion criteria included the following: RCT studies, studies with adult participants (>18 years) diagnosed with COPD, studies involving NMES treatment, English-language studies, free studies, and studies with specific measurements.

Exclusion Criteria

Exclusion criteria included the following: literature review studies, studies carried out prior to 2002, and case studies.

Results

Quality of the Studies

Of all seven studies included in this systematic review, 4 studies showed fair quality (4-5), and 3 studies showed good quality (6-8), which is shown in Table 1.

Table 1. Quality Assessment Using the PEDro Scale

Article	Items by Number on the PEDro Scale											Total Score
	1	2	3	4	5	6	7	8	9	10	11	
Giavedoni et al. (2012)	Y	Y	N	N	N	N	N	Y	N	Y	Y	4
Vieira et al. (2014)	Y	Y	N	Y	N	N	N	N	N	Y	Y	4
Abdellaoui et al. (2011)	Y	Y	N	Y	N	N	N	Y	N	Y	Y	5
Nápolis et al. (2011)	Y	Y	N	N	Y	N	Y	Y	N	Y	Y	6
Neder et al. (2002)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	7
Kucio et al. (2016)	Y	Y	N	Y	N	N	N	Y	N	Y	Y	5
Maddocks et al. (2016)	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8

Note. N: criterion not fulfilled; Y: criterion fulfilled; 1: eligibility criteria were specified; 2: subjects were randomly allocated to groups or to a treatment order; 3: allocation was concealed; 4: the groups were similar at baseline; 5: all subjects were blinded; 6: all therapists were blinded; 7: all assessors were blinded; 8: measures of at least one key outcome were obtained from over 85% of the subjects who were initially allocated to groups; 9: intention-to-treat analysis was performed on all subjects who received the treatment or control condition as allocated; 10: the results of between-group statistical comparisons are reported for at least one key outcome; 11: the study provides both point measures and measures of variability for at least one key outcome; total score: each satisfied item (except the first) contributes 1 point to the total score, yielding a PEDro scale score that can range from 0 to 10

Selection and Characteristics of Studies

A search of electronic databases and scanning the reference lists yielded 65 studies. Lastly, a total of 7 full-text studies were included in the systematic review. The study selection process is shown in Figure 1.

These seven studies included in the systematic review were presented in Table 2. Scientific studies selected for detailed analysis usually had one experimental and one control group. The study results are presented in a table 2.

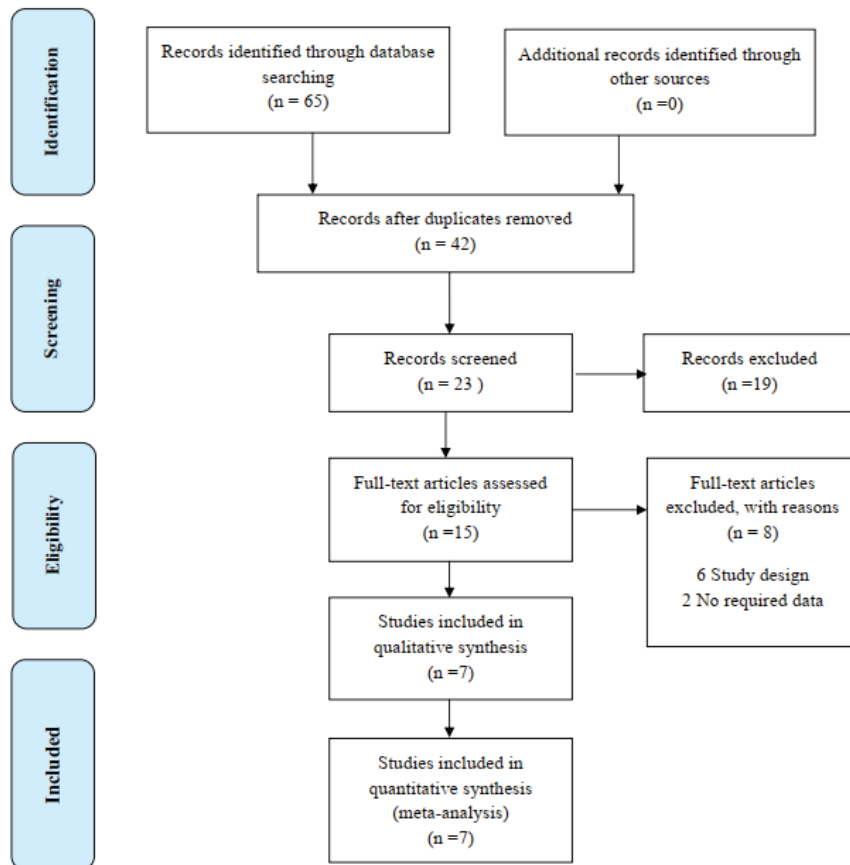


FIGURE 1. PRISMA Study Flow Diagrams

Table 2. Review of Studies

Author / Year /	Population	Experimental Group	Control Group	Measurements	Results
Giavedoni et al., (2012)	N = 11, patients <75 years old with COPD	Standard treatment + NMES on one leg, once a day for 14 days, 50 Hz, pulse duration of 400 ms, 30 minutes	Standard treatment without NMES	QMVC	Mean quadriceps muscle strength increased in the stimulated legs ($p < 0.05$).
Vieira et al., (2014)	N = 20, patients <69 years old with stable COPD	NMES 5 times a week, twice a day for 8 weeks, 60 min per session + respiratory therapy, and stretching exercises	Respiratory physical therapy and regular stretching exercises for 8 weeks	Tlim and 6-minute walk test	NMES improves exercise tolerance in patients with COPD.
Abdellaoui et al., (2011)	N = 15, patients <75 years old with acute exacerbated COPD	Weekly sessions of therapeutic education, daily active-passive mobilization, and 35 Hz NMES for 1 hour per day; 5 days per week for 6 weeks	Weekly sessions of therapeutic education, daily active-passive mobilization, and sham electrostimulation	QMVC, 6-minute walking test	QMVC was improved with a significant increase only in the NMES group. Changes in the 6-minute walking distance were also noted, with a significant increase in the NMES group.
Nápolis et al., (2011)	N = 30, patients <70 years old with stable COPD	NMES, with frequency 50 Hz, pulse duration 300-400ms, initial amplitude 15-20 mA, increasing up to 60 mA; Duration: 15 minutes the first week, 30 minutes the second week and then 60 minutes, 5 times a week, for 6 weeks	Sham NMES, with a frequency of 50 Hz, pulse duration of 200 ms, sham stimulation was applied to each leg for 15 minutes, 3 times a week for 6 weeks	Tlim	Exercise capacity was significantly improved in the NMES group.

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Table 2. Review of Studies

Author / Year /	Population	Experimental Group	Control Group	Measurements	Results
Neder et al., (2002)	N = 15, patients <73 years with advanced COPD	Group 1 first received NMES. It was assessed twice (before and after NMES). The treatment period was consistent with the duration shown to be effective in the NMES studies.	Group 2 received NMES after a control period. It was evaluated three times (before and after the 6-week control period and after another 6-week period of NMES).	Peripheral muscle strength and endurance; Tlim	Improvements post NMES were found in the endurance capacity of Tlim and maximal isokinetic strength; but, in isometric mean force and other indices of muscle endurance, they did not reach statistical significance ($p>0.05$).
Kucio et al., (2016)	N = 30, <75 years old with COPD	NMES 35Hz. + pharmacological treatment+ PR	In addition to pharmacological treatment, the patients were treated only with traditional PR.	6-minute walk test	Post-rehabilitation, a significant increase of 6MWT was observed only in the NMES+PR group.
Maddocks et al., (2016)	N = 52, <70 years with COPD.	Self-administered, 6-week program consisting of 30 minutes of daily bilateral NMES in the quadriceps, with a frequency of 50 Hz in 350 μ s pulses, range 0–120 mA	Self-administered, 6-week program consisting of 30 min daily bilateral NMES in the quadriceps, with a frequency of 50Hz in 350 μ s pulses, range 0–20mA	6-minute walk test, QMVC	Greater improvements were found in the 6MWT distance, in the active NMES group. Also, positive changes in outcomes in QMVC were found at 6 weeks.

Note. N- total number of respondents, Hz- Hertz, ms- millisecond, mA-milliamperes, μ s- microsecond, Tlim - the time to the limit of exercise tolerance of constant work rate, QMVC- maximal voluntary quadriceps contraction

Discussion

Peripheral Muscle Strength

In the study by Abdellaoui et al. (2011), the group receiving NMES provided more substantial and meaningful results related to improvement in quadriceps muscular strength after a 6-week program. The control group also saw a little spontaneous recovery in quadriceps strength, albeit the results were not statistically significant ($r=0.79$, $p<0.001$). According to Maddocks et al. (2016), treatment with NMES was linked to improvements in the quadriceps muscle's maximum voluntary contraction (QMVC) ($p<0.5$). Giavedoni et al. (2012) showed that after 14 days of research, muscle power decreased ($p=ns$) in the control group's legs, whereas it rose ($p<0.01$) in legs stimulated with NMES. Between the treatment and control groups, there were statistically significant variations in QMVC ($p<0.05$). Additionally, Nader et al. (2002) concluded that home-based NMES can enhance peripheral muscle strength and endurance, in the form of a new rehabilitative technique.

Exercise Tolerance

According to Neder et al. (2002), implementation of NMES was associated with an increase in whole-body exercise tolerance. The greatest effects, however, were found in resistance capacity and in time to the limit of exercise tolerance of constant work rate (Tlim); these two values were found to be significantly greater post NMES training. The study by Nápolis et al. (2011) concluded that training systematically with the highest NMES intensities improves the exercise capacity of COPD patients. Kucio et al. (2016) observed that, only the NMES + PR group showed a significant increase in walking distance as part of the walking test (6MWT) following reha-

bilitation. In the NMES + PR group, the increase in walking distance was 24.1 meters, whereas it was only 10.3 meters in the PR group. The study by Abdellaoui et al. (2011) discovered that differences in 6-minutes walking distance were significantly different between groups ($p=0.008$), with the NMES-treated group covering significantly more distance ($p=0.03$), and the control group showing only a modest and non-significant improvement ($p=0.24$). In the study by Maddocks et al. (2016), the group treated with NMES showed a considerably larger improvement in the distance assessed with the 6MWT test, compared to the control group ($p<0.05$). The results of this literature review are detailed below, specifying the publications that belong to each of the five categories analyzed (Table 2).

Benefits and Drawbacks of NMES in COPD patients

Various studies have tested the effect of NMES on peripheral muscle strength and exercise tolerance. Most of the studies reconfirm the positive effect of NMES on peripheral muscle strength and exercise capacity. However, studies focused not only on these two goals, but also on other benefits of NMES in COPD patients. According to Wu et al. (2020), NMES was found to be helpful in enhancing exercise capacity. Further, functional capacity rose following NMES therapy in COPD patients, as evidenced by an improvement in dyspnea feeling. In patients with moderate to severe COPD, Chen et al. (2016) found that NMES is beneficial in boosting quadriceps muscle strength and boosting exercise capacity. Regardless of the severity of airway obstruction, Coquart et al. (2016) observed that NMES significantly improves exercise tolerance, depression, and overall health-related quality of life in COPD

patients. The 6-minute walking distance and the modified Borg's dyspnea scale had significant pre- and post-test values between and across groups, according to Mahendiran et al.'s (2019) findings. Despite considerable changes in the physical activity index (PAI), it is obvious that the patient's condition significantly improved following NMES combined with traditional PR. Therefore, the importance of combining NMES with traditional PR is being emphasized.

However, some studies question the efficacy of NMES. According to Latimer et al. (2013), six weeks of NMES training in individuals with stable COPD resulted in modest gains in quadriceps strength and thigh mass that, while statistically significant, had little therapeutic importance. According to a research study by Gerovasili (2016), NMES helps patients with severe COPD, who are unable to finish a traditional rehabilitation program, although with temporary benefits. According to Pan et al. (2013), existing scant data suggest that NMES may not be helpful for COPD therapy in terms of enhancing lower limb functioning. However, given the restricted data availability and potential heterogeneity, the findings of this meta-analysis should be evaluated with care.

Limitations of the Study

Finally, our study too has its limitations. First, clinical studies with a low risk of bias were included. There exists heterogeneity among the studies, including the manner of

presentation of treatment protocols and tests. Assessment, comparisons between control and intervention groups, as well as study design are the main methodological limitations in the validity of the current findings. Second, studies that focus on a small number of subjects and studies with larger groups are required to confirm and extend our findings. Third, only English-language studies and free studies were reviewed, excluding non-English-language and paid articles. As a future research direction, several studies can be conducted to provide useful evidence of the effectiveness of NMES across different stages of the disease, as well as for the development of standardized NMES treatment protocols for COPD patients.

Conclusions

Evidence from the above-mentioned literature review suggests that NMES treatment for COPD patients has positive effects on peripheral muscle strength and exercise tolerance. The effectiveness of NMES in improving COPD patients' exercise tolerance and peripheral muscle strength has been highlighted in this review. RCT and other studies with high-quality, evidence-based data are necessary for drawing firmer conclusions about the efficacy of NMES. Future research should examine whether the therapeutic effects are long-lasting, whether they are experienced by all COPD patient groups, and which NMES treatment protocol delivers the best results.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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REVIEW PAPER

Isokinetic Hamstrings-to-Quadriceps Peak Torque Ratio in Combat Sports: A Systematic Review and Meta-analysis

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Abstract

Combat sports are very demanding in terms of physical fitness as well as the risk of injury. Elite athletes during daily training due to specific movements can develop significant muscle asymmetry that can affect injury. The aim of this study was to systematically review the existing literature with respect to total average value of hamstrings-to-quadriceps (H/Q) 60°·s⁻¹ and H/Q 180°·s⁻¹ in judo, karate, taekwondo and wrestling. We placed special emphasis on sex-based differences and possible asymmetries between legs in terms of isokinetic performance. This document was developed and reported in accordance with the Guidelines for Preferred Reports for Systematic Reviews and Meta-Analyses (PRISMA). Web of Science, PubMed and Google Scholar were searched for the relevant studies. Studies were included in the review if they were original and written in English and had to have H/Q ratio measured as an outcome. In total, data from 243 athletes from eleven studies included studies was analysed. The Mean of H/Q 60°·s⁻¹ for all sports were: 58.33% for the right leg and 58.91% for the left leg. Women had higher values: 59.38% vs. 57.42% for the right leg and 61.35% vs. 56.56% for the left leg. The Mean of H/Q 180°·s⁻¹ for all sports were: 71.44% for the right leg and 71.35% for the left leg. Women had higher values: 75.76% vs. 69.04% for the right leg and 73.39% vs. 70.33% for the left leg. All these values, combined for all sports, and divided into subgroups according to gender, show that they are within the recommended values. Combined results for all included studies showed that there was no asymmetry between the left and right leg. According to the results of the balance of the mentioned agonists and antagonists, which are crucial for knee joint stabilization during dynamic muscle contractions, we can conclude that athletes in combat sports do not represent a risk group.

Keywords: H/Q ratio, biomechanics, judo, taekwondo, wrestling, karate combat sports

Introduction

Modern combat sports are very demanding in terms of physical fitness not only to achieve sporting success, but also to reduce the risk of injury. For instance, during high-in-

tensity action in judo, competitors try to throw each other on their backs or control the opponent in the ground phase (Thomas et al., 1989; Franchini et al., 2005). Similarly, wrestling (Greco-Roman style) is one of the most demanding



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sports from a metabolic standpoint, in which superior strength and power are critical (Kraemer, 2002; Drid & Dokmanac, 2007). In Taekwondo, muscle functions such as muscle strength and endurance of the lower extremities are important factors for performing a strong and precise blow (Cools et al., 2007; Markovic et al., 2008; Kim et al., 2015). According to the rules imposed by the World Karate Federation, the main goal of an athlete is to score a point by touching the opponent; therefore, high-intensity actions in striking and shooting techniques are required before the opponent's response (Mori et al., 2002). Also, karate practitioners spend a lot of time training in positions that could put a lot of stress on joints such as the hips, ankles, and especially the knees (Sorensen et al., 1996; Probst et al., 2007).

The hamstrings to quadriceps H/Q peak torque ratios have received much attention in connection with their use to quantify muscle imbalance, as well as in rehabilitation and physical fitness (Rosene et al., 2001; Kong & Burns, 2010). The H/Q peak torque ratio is critical for providing muscle stability in the knee and for preventing anterior cruciate ligament (ACL) injuries (Aagard et al., 1998). Low back muscle strength relative to the quadriceps increases the risk of non-contact knee injuries (Hewett et al., 2008), and some studies have also shown that asymmetries between dominant and non-dominant limbs may also be associated with an increased risk of injury in athletes (Knapik et al., 1991). Posterior contraction helps prevent excessive quadriceps contraction, thus improving the stability of the knee joint (Weiss et al., 2015). Thus, deficiencies in posterior locus strength and activation limit the potential for muscle contraction to protect ligaments (Hewett et al., 2008).

Elite athletes can develop significant muscle asymmetry during daily training, and the specificity of sports movements can lead to muscle imbalance that could affect injuries (Magalhães et al., 2004). As a result of repeating the same movement patterns over a number of years, a negative adaptation of the body to a particular sport can occur, which often manifests as a contralateral and ipsilateral imbalance of strength and flexibility (Nofal, 2003). Indicators that show the occurrence of contralateral and ipsilateral imbalance are as follows: the ratio of the maximum torque of the hamstrings-to-quadriceps and bilateral deficit (BD). Maintaining a balance between the hamstrings and the quadriceps is necessary for the general stabilization of the knee joint (Croisier et al., 2008). Balance exercises are quite effective in terms of neuromuscular and functional performance (Polat et al., 2018). The risk of injury decreases with increasing leg muscle strength (Ugur et al., 1999).

The H/Q ratio is particularly significant in acyclic sports, as well as in sports in which there is a dominant muscle group in terms of strength (Scoville et al., 1997). Steindler (1955) was the first to suggest that the absolute muscle strength of the knee extensor should be greater than the strength of the knee flexor at 3:2 (66%). In the 1960s, the introduction of isokinetic dynamometers began (Hislop & Perrine, 1967). Isokinetic reports published up to the 1980s assessed only concentric muscle actions and recommended an optimal H/Q ratio in the range of 50 to 80% (Kannus, 1994). The limit value is 60% (Klein & Allman, 1969; Kellis & Baltzopoulos, 1995). This value has become normative and has been widely used to identify muscle imbalance through the so-called "H/Q conventional relationship" (Heiser et al.,

1984). Andrade et al. (2012) reach 57% of average values, while Struzik and Pietraszewski (2015) reach results from 48.8% to 58.8%, depending on the angular velocity.

It is complex to assess whether higher or lower H/Q values are more desirable, because those that are within the mentioned limits can be there most often due to quadriceps weakness. The study was conducted in order to combine the results of several studies, which is a great advantage when we draw conclusions, then on the basis of only one study. Individual studies often have limitations due to the small number of respondents, so we combined the results of several studies to obtain a reference number of respondents. Therefore, the obtained results will represent a better picture of the research subject, and we will be able to conclude whether the obtained values are within the recommended framework. This is the uniqueness of this study in relation to those published so far. The aim of this study was to systematically review the total average value of H/Q 60°·s⁻¹ and H/Q 180°·s⁻¹ of the conventional ratio for judo, karate, taekwondo, and wrestling, and also to compare the values by gender and to compare the values of the left and right leg in order to determine whether there is an asymmetry in the muscle strength of the thighs.

Materials and Methods

Study Design

This document was developed and reported in accordance with the Guidelines for Preferred Reports for Systematic Reviews and Meta-Analyses (PRISMA) (Moher et al. 2009).

Data sources and searches

The search strategy was developed to identify all relevant studies that measure the H/Q ratio in sports: Judo, Karate, Taekwondo, and Wrestling. Our systematic search included Web of Science, PubMed, and Google Scholar databases. The date limit for the publication period was set from the year 2005 until August 2022. The inclusion/exclusion of studies was done by two investigators by consultation and consensus. We used the following combinations of the search terms via the Boolean operator: ("Hamstring" OR "Posterior Thigh Muscles" OR "Quadriceps") AND ("Ratio" OR "Peak Torque" OR "Torque Ratios" OR "Strength Ratios") AND ("Martial Arts" OR "Judo" OR "Karate" OR "Taekwondo" OR "Wrestling"). The search strategy is shown in Fig. 1. We also manually searched for reference citations of identified studies and selected original research that met our inclusion criteria. Two investigators (V. D. and B. R.).

Study selection

To be included in our analysis, the original study had to meet the following criteria: 1) H/Q ratio had to be measured as an outcome 2) athletes had to be from one of four combat sports: Judo, Karate, Taekwondo, and Wrestling. The inclusion of studies in our analysis was limited to English only. Studies excluded were systematic reviews, meta-analyses, study protocols, books, book reviews, and conference publications. Also excluded were those studies that measured dominant and non-dominant leg as an outcome. There was no age limit for the respondents included. The analysis included studies that had both women and men as respondents and athletes elite and non-elite.

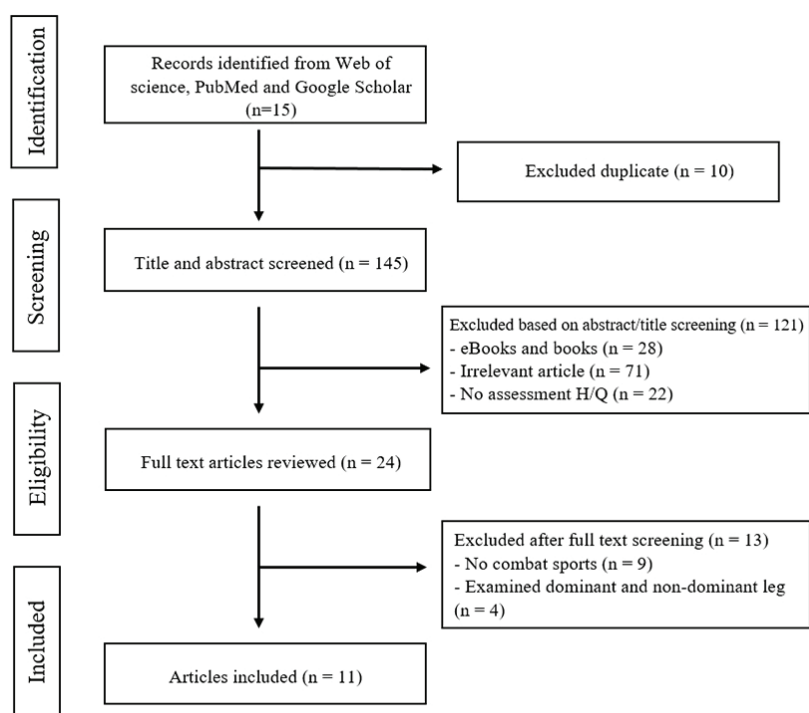


FIGURE 1. Flow diagram of the process of study selection

Data extraction and quality assessment

After selecting studies based on inclusion and exclusion criteria, the two investigators conducted data extraction jointly. The following variables were abstracted into a pre-formatted table: authors, year of publication, characteristics of study participants (number of participants, age), sex, angular velocity, sport, isokinetic dynamometer, H/Q values, and mode.

Two researchers independently assessed the quality of the studies involved, if there were disagreements, they were resolved by consensus. Methodological qualities and risk of bias were evaluated by the methodological index for non-randomized studies (MINORS) (Slim et al., 2003). MINORS score ≥ 10 was set as the level of inclusion. MINORS involves

12 items: A clearly stated aim, Inclusion of consecutive patients, Prospective collection of data, Endpoints appropriate to the aim of the study, Unbiased assessment of the study endpoint, Follow-up period appropriate to the aim of the study, Loss to follow up less than 5%, Prospective calculation of the study size, An adequate control group, Contemporary groups, Baseline equivalence of groups, and Adequate statistical analyses. Disagreements were resolved by a discussion to reach a consensus.

Statistical analysis

Statistical analysis was performed using MetaAnalyst software version 3.2 (Brown University, USA) (Wallace et al., 2009). For the included studies, the Mean of H/Q ratio 60°-s

Table 1. Risk of bias for each study

Study	Methodological items												Total score
	1	2	3	4	5	6	7	8	9	10	11	12	
Blach 2021	2	2	2	2	0	0	0	0	2	2	2	2	16
Drid 2009	2	1	0	2	0	0	0	0	2	2	2	2	13
Drid 2011	2	2	0	2	0	2	2	0	0	0	0	2	12
Jung 2017	2	2	2	2	0	0	0	0	1	2	2	2	15
Kim 2015	2	1	2	2	0	2	1	0	0	0	0	2	12
Kotrljanović 2016	2	2	2	2	0	0	1	0	2	0	2	2	15
Lisowska 2020	2	2	2	2	0	0	0	0	2	0	2	2	14
Malović 2020	2	2	2	2	0	0	0	0	2	2	2	2	16
Polat 2018	2	2	2	2	0	0	0	0	0	0	0	2	10
Radjo 2011	2	2	1	2	0	0	0	0	2	2	2	2	15
Trajković 2018	2	2	1	2	0	0	0	0	2	2	2	2	15

Note, (MINORS score): 1 - A clearly stated aim, 2 - Inclusion of consecutive patients, 3 - Prospective collection of data, 4 - Endpoints appropriate to the aim of the study, 5 - Unbiased assessment of the study endpoint, 6 - Follow-up period appropriate to the aim of the study, 7 - Loss to follow up less than 5%, 8 - Prospective calculation of the study size, 9 - An adequate control group, 10 - Contemporary groups, 11 - Baseline equivalence of groups, 12 - Adequate statistical analyses.

1 and 180°·s⁻¹ for the right and left leg was measured, and a subgroup analysis by gender was performed. The asymmetry between the left and right legs was estimated using the Mean Difference (MD) and 95% confidence interval (CI) for the measured outcomes H/Q ratio 60°·s⁻¹ and 180°·s⁻¹. Statistical significance was set at $p < 0.05$.

Results

Based on the search strategy, a total of 155 studies were selected from the initial database search. Of that number, 10 studies were excluded as duplicates, so 145 studies were selected for further analysis. Following the presentation of the abstract and title, 121 studies were excluded because they did not meet the inclusion criteria. The remaining 24 studies were fully reviewed. When the full-text articles were reviewed, 13

studies were excluded. The remaining 11 studies are included in this review article. The flow diagram of the study selection process is shown in Fig. 1.

Risk of bias

Table 1. shows a summary of the risk of bias for each study included. Of the eleven included studies, no study was randomized, so MINORS was used as a risk of bias assessment tool for non-randomized studies. The maximum possible score for comparative studies is 24, while for non-comparative studies it is 16. The lowest score was the study by Polat et al. (2018) with 10 points, while the highest score was the study by Blash et al. (2021), and Malović et al. (2020) with 16 points.

Table 2 shows the main characteristics of the included

Table 2. Characteristics of the included studies

Study	N	Age	Sex	Sport	Isokinetic dynamometer	Angular velocity	H/Q (%) right leg	H/Q (%) left leg	Mode
Blach 2021	25	21.02±3.11	Female	Judo	Humac Norm	60°·s ⁻¹	(52kg) 63±1 (63kg) 55±1.6 (70kg) 58±1.3	62±1.4 54±2 61±1.6	concentric
Drid 2009	20	19.6±2.6 20.8±2.8	Male	Judo and Wrestling	Easy-Tech	60°·s ⁻¹	(Jud) 47.8±10.8 (W) 39.5±4.9	45.9±9.1 39.3±6	concentric
Drid 2011	15	22.86±2.8	Female	Judo	Easy-Tech	60°·s ⁻¹	47.49±5	47.09±4.2	concentric
Jung 2017	30	19.4 ± 0.99 19.1 ± 1.13	Male/ Female	Taekwondo	770 Norm; Cybex	60°·s ⁻¹	(M) 62.8±8.88 (F) 52.4±7.1	59.4±7.52 52.8±5.76	concentric
Kim 2015	12	18.9	Female	Taekwondo	770 Norm; Cybex	60°·s ⁻¹ 180°·s ⁻¹	(2004) 61.7±3.65 (2005) 63.1±13.89 (2004) 69.9±6.45 (2005) 81.2±11.86	62.2±3.03 66.5±9.46 64.1±5.21 78.3±13.06	concentric
Kotrljanović 2016	18	24.1±3.72 21.25±2.76	Male/ Female	Karate	Humac Norm	60°·s ⁻¹ 180°·s ⁻¹	(M) 68.89±6.81 (F) 66.56±9.44 (M) 74.89±8.1 (F) 78.11±11.91	66.56±12.14 71.22±4.68 76.67±11.44 78.67±6.54	concentric
Lisowska 2020	30	25.9±6.9 22.6±3.8	Male	Karate	Biodex	60°·s ⁻¹ 180°·s ⁻¹	(K) 58.63±9.13 (Contr) 52.23±6.9 (K) 58.95±8.94 (Contr) 55.66±6.15	54.75±6.9 51.59±5.99 59.38±5.94 53.28±7.77	concentric
Malović 2020	25	22.50 ± 1.85 18.33 ± 0.50 16.50 ± 0.53	Male	Judo	Humac Norm	60°·s ⁻¹ 180°·s ⁻¹	(S) 63.77±9.2 (J) 65.7±20.6 (C) 69.3±11.1 (S) 77.1±14 (J) 72.7±18.5 (C) 78.5±12.6	66.4±10.5 68±15.9 69.3±9.3 74.3±12.4 77.6±15 85.6±15.5	concentric
Polat 2018	18	24.27±3.18	Male	Wrestling	Isomed 2000	60°·s ⁻¹	62.85±6.23	60.02±7.98	concentric
Radjo 2011	20	24.8±3.29 20.95±3.25	Male	Judo	Biodex	60°·s ⁻¹	(J-A) 55.67±10.9 (J-B) 44.88±9.15	55.88±7.2 46.97±10.74	concentric
Trajković 2018	30	21.25±2.76 16.87±0.99 15.00±0.81	Female	Karate	Humac Norm	60°·s ⁻¹	(S) 61.22±22.33 (J) 62.55±8.05 (C) 70.1±10.44	71±4.47 61.18±9.69 68.4±9.26	concentric

Note, (Wrest) - Wrestling; (Jud) – Judo; (2004) – season 2004; (2005) – season 2005; (F) – female; (M) – male; (Contr) – control; (K) – karate; (C) – cadet; (J) – junior; (S) – senior; (J-A) – Judo A group; (J-B) – Judo B group

ed studies. A total of 243 respondents participated in the eleven included studies, and the sample size of the included studies ranged from 12 to 30. The age of the respondents ranged between 14 and 29 years. In a study by Blach et al. (2021), we took results from three categories (up to 52 kg, up to 63kg, and up to 70 kg) from seven categories of respond.

Quantitative analysis

All eleven studies measured the H/Q 60°·s-1 ratio. The Mean H/Q 60°·s-1 ratio for the right leg was 58.33% (95% CI = 55.67%, 60.98%, $p < 0.001$) (Fig. 2). The Mean H/Q 60°·s-1 ratio for the right leg for males and females was: Male 57.42% (95% CI = 51.28%, 63.57%, $p < 0.001$); Female 59.38% (95% CI = 56.19%, 62.58%, $p < 0.001$) (Fig. 2).

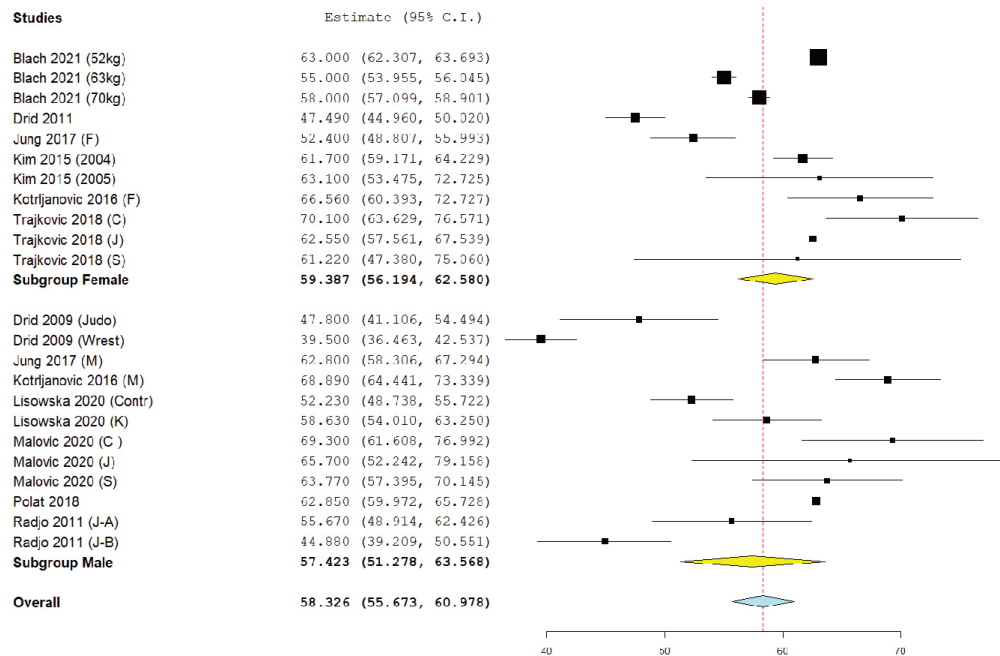


FIGURE 2. H/Q 60°·s-1 Right leg Subgroup Gender: (52kg) – up to 52kg; (63kg) – up to 63kg; (70kg) – up to 70kg; (Wrest) - wrestling; (2004) – season 2004; (2005) – season 2005; (F) – female; (M) – male; (Contr) – control; (K) – karate; (C) – cadet; (J) – junior; (S) – senior; (J-A) – Judo A group; (J-B) – Judo B group.

The Mean H/Q 60°·s-1 ratio for the left leg was 58.91% (95% CI = 56%, 61.81%, $p < 0.001$) (Fig. 3). The Mean H/Q 60°·s-1 ratio for the left leg for males and females was: Male

56.56% (95% CI = 51.63%, 61.49%, $p < 0.001$); Female 61.35% (95% CI = 57.58%, 65.12%, $p < 0.001$) (Fig. 3).

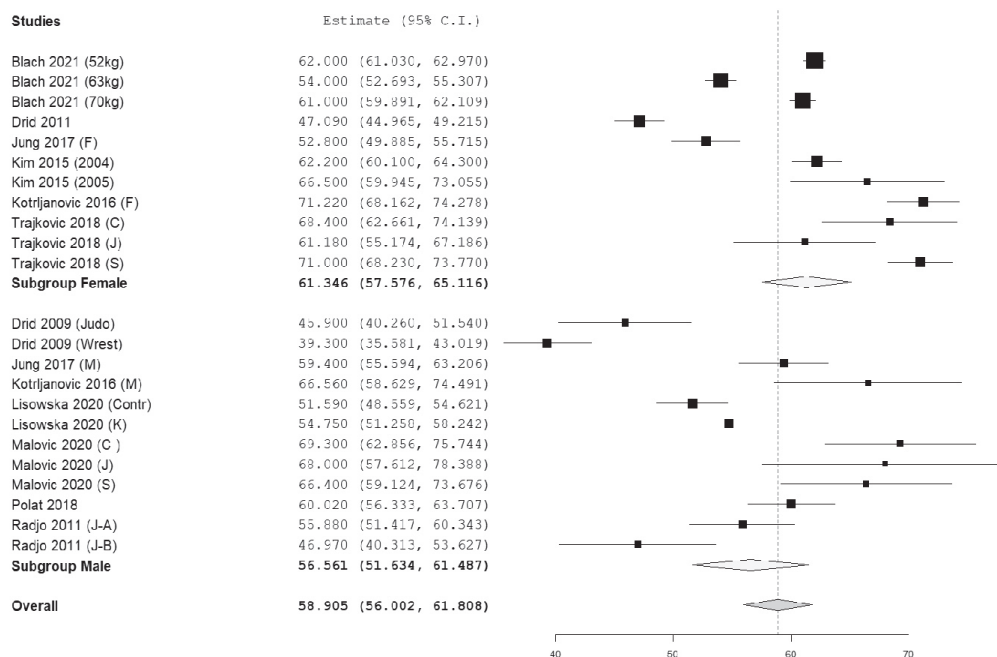


FIGURE 3. H/Q 60°·s-1 Left leg Subgroup Gender: (52kg) – up to 52kg; (63kg) – up to 63kg; (70kg) – up to 70kg; (Wrest) - wrestling; (2004) – season 2004; (2005) – season 2005; (F) – female; (M) – male; (Contr) – control; (K) – karate; (C) – cadet; (J) – junior; (S) – senior; (J-A) – Judo A group; (J-B) – Judo B group.

Four studies measured the H/Q 180°·s⁻¹ ratio. The Mean H/Q 180°·s⁻¹ ratio for the right leg was 71.44% (95% CI = 64.41%, 78.46%, $p < 0.001$) (Fig. 4). The Mean H/Q 180°·s⁻¹

ratio for the right leg for males and females was: Male 69.04% (95% CI = 60.26%, 77.82%, $p < 0.001$); Female 75.76% (95% CI = 68.38%, 83.14%, $p < 0.001$) (Fig. 4).

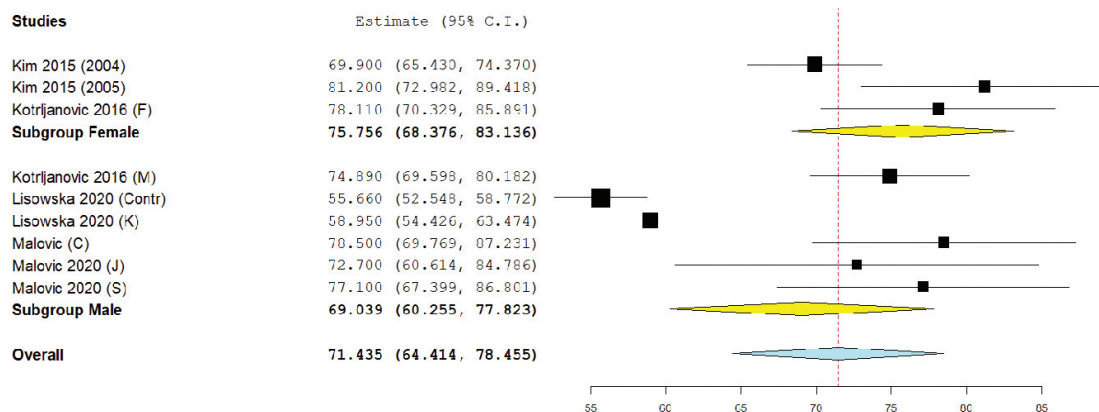


FIGURE 4. H/Q 180 Right leg Subgroup Gender: (2004) – season 2004; (2005) – season 2005; (F) – female; (M) – male; (Contr) – control; (K) – karate; (C) – cadet; (J) – junior; (S) – senior.

The Mean H/Q 180°·s⁻¹ ratio for the left leg was 71.35% (95% CI = 64.22%, 78.48%, $p < 0.001$) (Fig. 5). The Mean H/Q 180°·s⁻¹ ratio for the left leg for males and females was: Male 70.33% (95% CI = 61.12%, 79.54%, $p < 0.001$); Female 73.39% (95% CI = 62.32%, 84.47%, $p < 0.001$) (Fig. 5). The results show

that there was no asymmetry, ie a statistically significant difference between the right and left leg, neither for the outcome of H/Q 60°·s⁻¹, nor for the outcome of H/Q 180°·s⁻¹. Outcome H/Q 60°·s⁻¹: (MD= 0.03; 95% CI=-1.04, 1.11, $p=0.95$); outcome H/Q 180°·s⁻¹: (MD= 1.37; 95% CI=-1.21, 3.95, $p=0.3$).

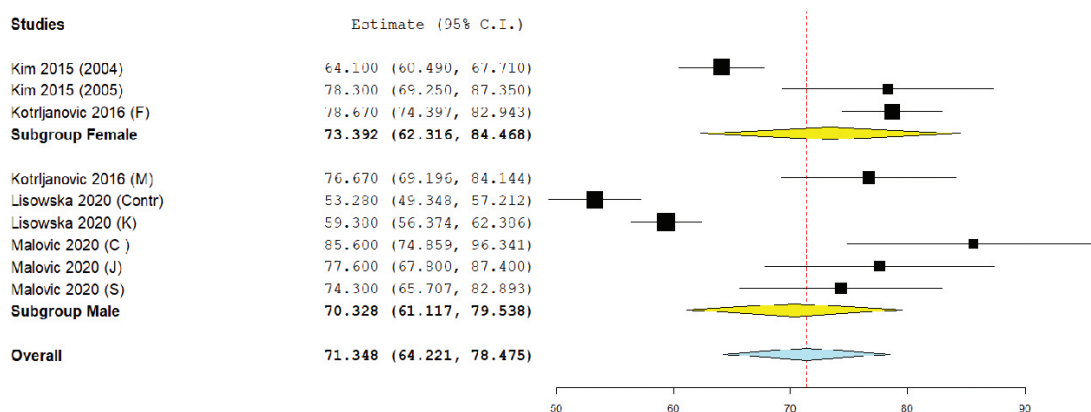


FIGURE 5. H/Q 180 Left leg : (2004) – season 2004; (2005) – season 2005; (F) – female; (M) – male; (Contr) – control; (K) – karate; (C) – cadet; (J) – junior; (S) – senior.

Discussion

The balance between the strength of the muscles of the hamstrings and the quadriceps is an important factor in athletes in combat sports. In this case, the balance of the mentioned agonists and antagonists is crucial for the stabilization of the knee joint during dynamic muscle contractions (DonTigny, 2005). Also important is the symmetry between the opposite legs, whose shortcomings can lead to an increased possibility of injury (Drid et al., 2009; Malović et al., 2020). This systematic review aimed to, based on the results of the included studies, arrive at average values of the conventional H/Q ratio for angular velocities of 60°·s⁻¹ and 180°·s⁻¹ and also to compare the values by gender. Also, based on the obtained results, it was determined whether there is an asymmetry between the left and right leg.

The Mean value of the conventional ratio for an angular velocity of 60°·s⁻¹ for all included studies was 58.33% for the right leg and 58.91% for the left leg (Fig. 2 and 3). From

these results, it can be concluded that there is no asymmetry between the left and right leg, and the analysis shows that there is no statistically significant difference: (MD=0.03; $p=0.95$). These values correspond to a recommended limit of 60% (Klein & Allman, 1969; Kellis & Baltzopoulos, 1995). Women had higher H/Q 60°·s⁻¹ than men, 59.38% vs. 57.42% for the right leg and 61.35% vs. 56.56% for the left leg (Fig. 2 and 3). The Mean H/Q 180°·s⁻¹ for all included studies was 71.44% for the right leg and 71.35% for the left leg. Women had higher H/Q 180°·s⁻¹ than men, 75.76% vs. 69.04% for the right leg and 73.39% vs. 70.33% for the left leg (Fig. 4 and 5). Even at this angular velocity, no asymmetry was observed between the left and right leg, and the analysis shows that there is no statistically significant difference: (MD=1.37; $p=0.3$).

This is the only systematic review that analyzes the average values of the conventional ratio for angular velocities H/Q 60°·s⁻¹ and H/Q 180°·s⁻¹, in these combat sports. An extensive

systematic review was done by Baroni et al. (2018) including studies that had football players for the respondents. The average value for angular velocities from $12^{\circ}\cdot s^{-1}$ to $180^{\circ}\cdot s^{-1}$ was about 60%, and for speeds from $240^{\circ}\cdot s^{-1}$ to $360^{\circ}\cdot s^{-1}$ was between 70% and 80%. Comparing the results with our study, although we measured only angular velocities of $60^{\circ}\cdot s^{-1}$ and $180^{\circ}\cdot s^{-1}$, we can say that the results are approximate. Aagaard et al. (1998) reported karate athletes showed H/Q ratios greater than 60% at $60^{\circ}\cdot s^{-1}$. Risberg et al. (2018) reach values of 57% -58% for handball players and 59% -60% for football players. Baroni et al. (2018) recommend the use of an angular velocity of $60^{\circ}\cdot s^{-1}$ to examine the conventional H/Q ratio. Our study also came to the result that the values of H/Q increase with increasing angular velocity, as well as the study of Baroni et al. (2018). We did not deal with the differences in the results of the included studies here, but only with the obtained average values.

In combat sports, joint injuries mainly affect the knee (19-28%). Injuries to the meniscus and ligament joints are a major consequence of throwing, grasping, and maneuvering activities that often result in extreme joint positions that cause excessive stretching of muscles and ligaments (Hammani et al., 2017). In judokas, 28% of injuries are knee injuries, in wrestlers 20.5% (Pocecco et al., 2013; Hammani et al., 2017), and in taekwondo, the lower extremities are the most commonly injured part of the body (32%) (Kazemi & Pieter, 2004), while in karate, knee injuries are the most common (Kurland, 1980). As a consequence of hyperextension of the leg during impact and twisting or violent external rotation, the knee is prone to acute injuries of the medial collateral ligament (MCL), anterior cruciate ligament (ACL), meniscus, and patella. Although the most common knee lesions, ACL and MCL injuries, have been reported in all combat sports (Burke, 1981), meniscus injuries are most common in taekwondo (16%), judo (24%), and karate (50%); and patellar instability is common in wrestling (50%) (Nicolini et al., 2014). In general, there is no significant difference between men and women in injuries reported in combat sports, although some studies suggest that female athletes have a much higher risk of ACL injuries than men (Stevenson et al., 2000; Biene, 2011; Koshida et al., 2010). With the increased participation of women in combat sports competitions, similar injury rates and locations are more prevalent in both men and women (Pocecco et al., 2013; Hammani et al., 2017). Isokinetic exercise can significantly improve results in certain sports fields and reduce the frequency of injuries (Shelbourne & Gray, 1997; De Carlo et al., 1999). Moreover, the power imbalance is one of the strongest predictors of injury in highly competitive sports (Croisier et al., 2008; Yeung et al., 2009). The use of isokinetic devices provides an effective tool for assessing power imbalances and applying isokinetic strength exercises. Isokinetic testing is valuable information about strength asymmetry between muscle groups that may be a potential site for injury, but more im-

portantly, for an injury prevention program (Drid et al., 2011; Maly et al., 2017; Blach et al., 2021). Similarly, when the difference in strength between the right and left quadriceps and hamstrings exceeds 10-15%, it is considered that there is an asymmetry between the two sides (Elliott, 1978). According to the Kannus study (1994) the risk of injury is high when this difference exceeds 20%. The results we obtained in our statistical analysis coincide with some recommended values (Steindler, 1955; Klein & Allman, 1969; Kannus, 1994; Kellis & Baltzopoulos, 1995), although Tengman et al. (2014), conclude that it is debatable how useful the value of H/Q, when there are deficits in the quadriceps. Their results show that the injured legs had higher H/Q values than the uninjured ones, which according to them represents a deficit in the strength of the quadriceps.

The results of our research indicate that the subjects from the four included combat sports have values of H/Q $60^{\circ}\cdot s^{-1}$ and H/Q $180^{\circ}\cdot s^{-1}$ within the recommended range and the pooled results from our analysis showed that there is no asymmetry between the left and right leg and that according to studies by Elliott (1978) and Kannus (1994) athletes in combat sports do not represent a risk group. We did not exclude outliers in our analysis. If we did that, the results would confirm the recommended values even more. This study had several limitations. First, a relatively small number of studies had inclusive criteria. Some studies that met the inclusive criteria measured the values of the dominant and non-dominant leg, and could not be included in the statistical processing. Second, all included studies measured only two angular velocities of $60^{\circ}\cdot s^{-1}$ and $180^{\circ}\cdot s^{-1}$. Third, our search included only studies written in English. Future studies can compare H/Q peak torque ratio values of combat sports with team sports and/or water sports, and analyze the percentage occurrence of injuries by different types of sports.

Conclusion

This study examined the H/Q peak torque ratio at two angular velocities in athletes from four combat sports. The results show that their values are within the recommended values, and also show that there is no asymmetry between the thigh muscles of the left and right leg. By analyzing the subgroups, we came to the results by which we were able to compare the values of H/Q according to gender. In both H/Q $60^{\circ}\cdot s^{-1}$ analyzes and H/Q $180^{\circ}\cdot s^{-1}$ analyzes, women had higher values than men for both the right and left leg. This is the first systematic review examining the overall mean hamstring/quadriceps ratio for judo, karate, taekwondo and wrestling, which also compares values by gender and compares left and right leg values to determine asymmetry in the muscle strength of the thighs. The study could raise awareness of the importance of muscle symmetry which is focused on preventing injuries in combat sports and improving performance through strengthening.

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Conflict of Interest

This research was supported by the Provincial Secretariat for Higher Education and Scientific Research of AP Vojvodina, grant number: 142-451-2595. The authors declare that there is no commercial or financial relationship that could be interpreted as a potential conflict of interest.

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Guidelines for Authors

Revised September 2019

*** Please use the bookmark function to navigate within the guidelines. ***

When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

1. UNIFORM REQUIREMENTS

1.1. Overview

The *Sport Mont* (SM) applies the Creative Commons Attribution (CC BY) license to articles and other works it publishes.

The submission with SM is free of charge but author(s) has to pay additional 215 euros per accepted manuscript to cover publication costs. If the manuscript contains graphics in color, note that printing in color is charged additionally.

SM adopts a double-blind approach for peer reviewing in which the reviewer's name is always concealed from the submitting authors as well as the author(s)'s name from the selected reviewers.

SM honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

Manuscripts must be provided either in standard UK or US English language. English standards should be consistent throughout the manuscripts accordingly.

Format the manuscript in A4 paper size; margins are 1 inch or 2.5 cm all around.

Type the whole manuscript double-spaced, justified alignment.

Use Times New Roman font, size eleven (11) point.

Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

Include line numbers (continuous) for the convenience of the reviewers.

Apart from chapter headings and sub-headings avoid any kind of formatting in the main text of the manuscripts.

1.2. Type & Length

SM publishes following types of papers:

Original scientific papers are the results of empirically- or theoretically-based scientific research, which employ scientific methods, and which report experimental or observational aspects of sports science and medicine, such as all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side. Descriptive analyses or data inferences should include rigorous methodological structure as well as sound theory. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒ Open Submissions

☒ Indexed

☒ Peer Reviewed

Original scientific papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

Review papers should provide concise in-depth reviews of both established and new areas, based on a critical examination

of the literature, analyzing the various approaches to a specific topic in all aspects of sports science and medicine, such as all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Review papers should be:

- Up to 6000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 100.

Editorials are written or commissioned by the editors, but suggestions for possible topics and authors are welcome. It could be peer reviewed by two reviewers who may be external or by the Editorial Board.

☐Open Submissions

☒Indexed

☒Peer Reviewed

Editorials should be:

- Up to 1000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 10.

Short reports of experimental work, new methods, or a preliminary report can be accepted as two page papers. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Short reports should be:

- Up to 1500 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Peer review - fair review provides authors who feel their paper has been unfairly rejected (at any journal) the opportunity to share reviewer comments, explain their concerns, and have their paper reviewed for possible publication in SM.

☒Open Submissions

☒Indexed

☐Peer Reviewed

Peer review - fair review should be:

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- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Invited papers and award papers include invited papers from authors with outstanding scientific credentials. Nomination of invited authors is at the discretion of the SM editorial board. SM also publishes award papers selected by the scientific committee of the International Scientific Conference on Transformation Processes in Sport.

☐Open Submissions

☒Indexed

☐Peer Reviewed

Invited papers and award papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
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- Maximum number of references is 30;
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1.3. Submission

SM only accepts electronic submission to the e-mail of the Journal Office: **sportmont@ucg.ac.me**.

Submitted material includes:

- A manuscript prepared according to the Guidelines for the Authors;
- A signed form that states the study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere, that states that all of the authors are in agreement with submission of the manuscript to SM, and that, for studies that use animal or human individuals, authors must include information regarding their institution's ethics committee, and which identifies the official approval number;
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Name the files according to the family name of the first author. Authors submitting revised versions of the manuscript can use the identification number of their manuscript as provided by the Journal Office. *See example:*

- ✓ FAMILY NAME-manuscript.doc – (main manuscript file)
- ✓ FAMILY NAME-statement.PDF – (authorship statement)
- ✓ FAMILY NAME-declaration.PDF – (declaration of potential conflict of interest)
- ✓ FAMILY NAME-fig1.tiff – (Figure 1)

1.4. Peer Review Process

A manuscript submitted for publication will be submitted to the review process as long as it fits the following criteria:

- The study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere;
- All persons listed as authors approved its submission to SM;
- Any person cited as a source of personal communication has approved the quote;
- The opinions expressed by the authors are their exclusive responsibility;
- The author signs a formal statement that the submitted manuscript complies with the directions and guidelines of SM.

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1.6. After Acceptance

After the manuscript has been accepted, authors will receive a PDF version of the manuscripts for authorization, as it should look in printed version of SM. Authors should carefully check for omissions. Reporting errors after this point will not be possible and the Editorial Board will not be eligible for them.

Should there be any errors, authors should report them to the Office e-mail address **sportmont@ucg.ac.me**. If there are not any errors authors should also write a short e-mail stating that they agree with the received version.

1.7. Code of Conduct Ethics Committee of Publications



SM is hosting the Code of Conduct Ethics Committee of Publications of the **COPE** (the Committee on Publication Ethics), which provides a forum for publishers and Editors of scientific journals to discuss issues relating to the integrity of the work submitted to or published in their journals.

2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. See example:

Body Composition of Elite Soccer Players from Montenegro

Original Scientific Paper

Elite Soccer Players from Montenegro

Dusko Bjelica¹

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University of Montenegro

Faculty for Sport and Physical Education

Narodne omladine bb, 81400 Niksic, Montenegro

E-mail: sportmont@t-com.me

Word count: 2,946

Abstract word count: 236

Number of Tables: 3

Number of Figures: 0

2.1.1. Title

Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: “and”, “or”, “between” etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

2.1.2. Type of publication

Authors should suggest the type of their submission.

2.1.3. Running head

Short running title should not exceed 50 characters including spaces.

2.1.4. Authors

The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

In case all authors belong to the same institution affiliation numbering is not needed.

Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

2.1.6. Corresponding author

Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

2.1.7. Manuscript information

All authors are required to provide word count (excluding title page, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References), the Abstract word count, the number of Tables, and the number of Figures.

2.2. Abstract

The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

Since abstract is independent part of your paper, all abbreviations used in the abstract should also be explained in it. If an abbreviation is used, the term should always be first written in full with the abbreviation in parentheses immediately after it. Abstract should not have any special headings (e.g., Aim, Results...).

Authors should provide up to six key words that capture the main topics of the article. Terms from the Medical Subject Headings (MeSH) list of Index Medicus are recommended to be used.

Key words should be placed on the second page of the manuscript right below the abstract, written in italic. Separate each key word by a comma (and a space). Do not put a full stop after the last key word. *See example:*

Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

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Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements

(optional), Conflict of Interest (optional). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

2.3.1. Headings

Main chapter headings: written in bold and in Title Case. *See example:*

- ✓ **Methods**

Sub-headings: written in italic and in normal sentence case. Do not put a full stop or any other sign at the end of the title. Do not create more than one level of sub-heading. *See example:*

- ✓ *Table position of the research football team*

2.3.2 Ethics

When reporting experiments on human subjects, there must be a declaration of Ethics compliance. Inclusion of a statement such as follow in Methods section will be understood by the Editor as authors' affirmation of compliance: "This study was approved in advance by [name of committee and/or its institutional sponsor]. Each participant voluntarily provided written informed consent before participating." Authors that fail to submit an Ethics statement will be asked to resubmit the manuscripts, which may delay publication.

2.3.3 Statistics reporting

SM encourages authors to report precise p-values. When possible, quantify findings and present them with appropriate indicators of measurement error or uncertainty (such as confidence intervals). Use normal text (i.e., non-capitalized, non-italic) for statistical term "p".

2.3.4. 'Acknowledgements' and 'Conflict of Interest' (optional)

All contributors who do not meet the criteria for authorship should be listed in the 'Acknowledgements' section. If applicable, in 'Conflict of Interest' section, authors must clearly disclose any grants, financial or material supports, or any sort of technical assistances from an institution, organization, group or an individual that might be perceived as leading to a conflict of interest.

2.4. References

References should be placed on a new page after the standard title written in upper and lower case letters, bold.

All information needed for each type of must be present as specified in guidelines. Authors are solely responsible for accuracy of each reference. Use authoritative source for information such as Web of Science, Medline, or PubMed to check the validity of citations.

2.4.1. References style

SM adheres to the American Psychological Association 6th Edition reference style. Check "American Psychological Association. (2009). Concise rules of APA style. American Psychological Association." to ensure the manuscripts conform to this reference style. Authors using EndNote® to organize the references must convert the citations and bibliography to plain text before submission.

2.4.2. Examples for Reference citations

One work by one author

- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

Works by two authors

- ✓ Duffield and Marino (2007) studied
- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.

- ✓ First citation: Bangsbo, Iaia, and Krstrup (2008) stated that
- ✓ Subsequent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year

- ✓ Krstrup et al. (2003) studied
- ✓ In one study (Krstrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)

- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

- Nepocatych, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine*, 6(2), 57-65. doi: 10.26773/mjssm.2017.09.008
- Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology*, 100(6), 727-735. doi: 10.1007/s00421-007-0468-x
- Krstrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise*, 35(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

- Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies*, 50(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

- Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia*, 4. Retrieved November 15, 2005, from <http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html>

Conference paper:

- Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from <http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf>

Encyclopedia entry (print, with author):

- Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

- Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from <http://www.bartleby.com/65/et/ethnolog.html>

Thesis and dissertation:

- Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport*. Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

- Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

- Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from *Routledge Handbook of Applied Sport Psychology* ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

- Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See example:*

- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

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Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. ^{a,b,c}), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript ^a.

- ✓ ^aOne participant was diagnosed with heat illness and n = 19.^bn = 20.

Probability notes provide the reader with the results of the tests for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

- ✓ *P<0.05, †p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See example:*

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. SM prefers TIFF, EPS and PNG formats.

If a figure has been published previously, acknowledge the original source and submit a written permission from the copyright holder to reproduce the material. Permission is required irrespective of authorship or publisher except for documents in the public domain. If photographs of people are used, either the subjects must not be identifiable or their pictures must be accompanied by written permission to use the photograph whenever possible permission for publication should be obtained.

Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of SM.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written below the figure, in sentence case. *See example:*

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, * $p > 0.01$, † $p > 0.05$.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See example:*

- ✓ Figure 1
- × figure 1
- × Figure 1.
- ✓ ...exhibit greater variance than the year before (Figure 2). Therefore...
- ✓ ...as shown in Figures 1 and 3. (citing more figures at once)
- ✓ ...result has shown (Figures 1-3) that... (citing more figures at once)
- ✓ ...in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See example:*

- ✓ Figure 1a
- ✓ ...in Figures 1a and b we can...
- ✓ ...data represent (Figures 1a-d)...

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

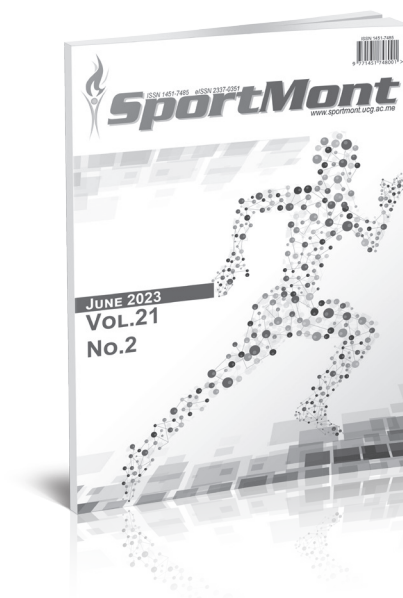
Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10°	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 °	× 10kg	× 12 : 2	× .056
Signs should be placed immediately preceding the relevant number.				
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age		
× 45 ± 3.4	× p < 0.01	× males > 30 years of age		

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. See example:

✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*



ISSN 1451-7485

Sport Mont Journal (SMJ) is a print (ISSN 1451-7485) and electronic scientific journal (eISSN 2337-0351) aims to present easy access to the scientific knowledge for sport-conscious individuals using contemporary methods. The purpose is to minimize the problems like the delays in publishing process of the articles or to acquire previous issues by drawing advantage from electronic medium. Hence, it provides:

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SMJ covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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Prospective authors should submit manuscripts for consideration in Microsoft Word-compatible format. For more complete descriptions and submission instructions, please access the Guidelines for Authors pages at the MJSSM website: <http://www.mjssm.me/?sekcija=page&p=51>. Contributors are urged to read MJSSM's guidelines for the authors carefully before submitting manuscripts. Manuscripts submissions should be sent in electronic format to office@mjssm.me or contact following Editors:

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Spring issue – March 2024



MONTENEGRIN SPORTS ACADEMY

Founded in 2003 in Podgorica (Montenegro), the Montenegrin Sports Academy (MSA) is a sports scientific society dedicated to the collection, generation and dissemination of scientific knowledge at the Montenegrin level and beyond.

The Montenegrin Sports Academy (MSA) is the leading association of sports scientists at the Montenegrin level, which maintains extensive co-operation with the corresponding associations from abroad. The purpose of the MSA is the promotion of science and research, with special attention to sports science across Montenegro and beyond. Its topics include motivation, attitudes, values and responses, adaptation, performance and health aspects of people engaged in physical activity and the relation of physical activity and lifestyle to health, prevention and aging. These topics are investigated on an interdisciplinary basis and they bring together scientists from all areas of sports science, such as adapted physical activity, biochemistry, biomechanics, chronic disease and exercise, coaching and performance, doping, education, engineering

and technology, environmental physiology, ethics, exercise and health, exercise, lifestyle and fitness, gender in sports, growth and development, human performance and aging, management and sports law, molecular biology and genetics, motor control and learning, muscle mechanics and neuromuscular control, muscle metabolism and hemodynamics, nutrition and exercise, overtraining, physiology, physiotherapy, rehabilitation, sports history, sports medicine, sports pedagogy, sports philosophy, sports psychology, sports sociology, training and testing.

The MSA is a non-profit organization. It supports Montenegrin institutions, such as the Ministry of Education and Sports, the Ministry of Science and the Montenegrin Olympic Committee, by offering scientific advice and assistance for carrying out coordinated national and European research projects defined by these bodies. In addition, the MSA serves as the most important Montenegrin and regional network of sports scientists from all relevant subdisciplines.

The main scientific event organized by the Montenegrin Sports Academy (MSA) is the annual conference held in the first week of April.

Annual conferences have been organized since the inauguration of the MSA in 2003. Today the MSA conference ranks among the leading sports scientific congresses in the Western Balkans. The conference comprises a range of invited lecturers, oral and poster presentations from multi- and mono-disciplinary areas, as well as various types of workshops. The MSA conference is attended by national, regional and international sports scientists with academic careers. The MSA conference now welcomes up to 200 participants from all over the world.

It is our great pleasure to announce the upcoming 21th Annual Scientific Conference of Montenegrin Sports Academy "Sport, Physical Activity and Health: Contemporary Perspectives" to be held in Dubrovnik, Croatia, from 18 to 21 April, 2024. It is planned to be once again organized by the Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, University of Montenegro and other international partner institutions (specified in the partner section).

The conference is focused on very current topics from all areas of sports science and sports medicine including physiology and sports medicine, social sciences and humanities, biomechanics and neuromuscular (see Abstract Submission page for more information).

We do believe that the topics offered to our conference participants will serve as a useful forum for the presentation of the latest research, as well as both for the theoretical and applied insight into the field of sports science and sports medicine disciplines.





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We have expanded the quality of our journals considerably over the past years and can now claim to be the market leader in terms of breadth of coverage.

As we continue to increase the quality of our publications across the field, we hope that you will continue to regard MSA journals as authoritative and stimulating sources for your research. We would be delighted to receive your comments and suggestions, mostly due to the reason your proposals are always welcome.

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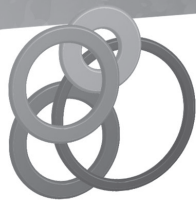
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